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Single or multiple nerve stimulation to treat sleep disordered breathing

Abstract

Devices and/or methods for treating sleep disordered breathing may include stimulation of a single nerve, multiple nerves, as well as other tissues relating to upper airway patency.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This 35 U.S.C. § 371 National Phase application claims priority to International Application No. PCT/US2021/033639, filed May 21, 2021, which claims the benefit of U.S. Provisional Patent Application No. 63/029,446, filed May 23, 2020; which are both incorporated herein by reference in their entirety.

BACKGROUND
(1) Sleep disordered breathing, such as obstructive sleep apnea, may cause significant health problems and is common among the adult population. Some forms of treatment of sleep disordered breathing may include electrical stimulation of nerves and/or muscles relating to upper airway patency.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram schematically representing an example device and/or example method for stimulating airway patency-related tissue.
(2) FIG. **2** is a diagram schematically representing patient anatomy and an example device and/or example method for stimulating an ansa cervicalis-related nerve and/or hypoglossal nerve.
(3) FIG. **3**A is a diagram schematically representing patient anatomy and an example device and/or example method for stimulating airway patency-related tissue, including an implantable pulse generator (IPG) and associated stimulation elements.
(4) FIG. **3**B is a block diagram schematically representing an example device including an IPG and patient remote control.
(5) FIG. **3**C is a diagram schematically representing patient anatomy and an example device and/or example method like FIG. **3**A, while explicitly illustrating sensing elements.
(6) FIG. **4** is a diagram schematically representing an example device and/or method for implanting stimulation elements into stimulating relation to target nerve stimulation locations.
(7) FIGS. **5**A-**5**B are diagrams including a side view schematically representing an example stimulation lead with connection features.
(8) FIGS. **6**A-**6**B are diagrams including a side view schematically representing example anchor elements.
(9) FIGS. **7**A-**8** are diagrams including a side view schematically representing example stimulation leads with bifurcation features and/or related delivery tools.
(10) FIGS. **9-11**A are diagrams schematically representing an example device and/or method for implanting stimulation elements into stimulating relation to target nerve stimulation locations, such as the hypoglossal nerve and ansa cervicalis-related nerve.
(11) FIG. **11**B is a diagram like FIGS. **9-11**B including implantation of a microstimulator in the neck region for connected stimulation elements.
(12) FIGS. **12-13** are diagrams schematically representing an example device and/or method for implanting stimulation elements via an implant-access incision in proximity to ansa cervicalis-

related nerve.
(13) FIGS. **14**A-**14**G are diagrams schematically representing an example device and/or method for implanting stimulation elements and an IPG via implant-access incisions in proximity to ansa cervicalis-related nerve and/or a hypoglossal nerve.
(14) FIGS. **14**H-**14**K are diagrams schematically representing an example device and/or method for implanting stimulation elements and a microstimulator via implant-access incisions in proximity to ansa cervicalis-related nerve and/or a hypoglossal nerve.
(15) FIGS. **14**L-**14**R are diagrams schematically representing an example device and/or method for implanting stimulation elements and associated leads, IPG or microstimulator via implant-access incision(s) for at least bilateral stimulation locations for the hypoglossal nerve.
(16) FIGS. **15**A-**15**C are diagrams schematically representing an example device and/or method for implanting stimulation elements via intravascular access and delivery.
(17) FIG. **16** is a diagram schematically representing an example device and/or example method, relative to patient anatomy, for stimulating an ansa cervicalis-related nerve and/or hypoglossal nerve.
(18) FIG. **17** is a diagram including a top view schematically representing an example stimulation element as paddle electrode in stimulating relation to target nerve stimulation locations.
(19) FIGS. **18**, **20** are diagrams including a sectional view schematically representing example cuff electrodes.
(20) FIG. **19** is a diagram including a side view schematically representing the example cuff electrodes of FIG. **18**.
(21) FIG. **21** is a diagram including a side view schematically representing an example device and/or example method for implanting a microstimulator within a neck region, along with wireless power delivery to an external power element.
(22) FIGS. **22**A-**23** are diagrams schematically representing patient anatomy and an example device and/or example method for stimulating to an ansa cervicalis-related nerve, including anchor elements.
(23) FIGS. **24**A-**25**B are diagrams including top and side views schematically representing example stimulation elements including a linear electrode array.
(24) FIG. **26**A is a sectional view as taken along lines **26**B-**26**B of FIG. **26**B of an example cuff electrode.
(25) FIG. **26**B is a side view schematically representing an example cuff electrode.
(26) FIGS. **27**A-**28** are diagrams including top views schematically representing an example paddle electrode including anchor elements.
(27) FIGS. **29**A-**29**C are diagrams including top views schematically representing an example axial stimulation portion including a linear electrode array and anchor elements.
(28) FIG. **30**A is a flow diagram schematically representing an example method of implantation.
(29) FIGS. **30**B-**30**U are diagrams including a side view schematically representing example devices and/or example methods of implantation, including access and delivery tools for stimulation elements, some of which include anchor elements.
(30) FIGS. **30**V-**30**W are diagrams including a top view and a side view, respectively, schematically representing example anchor structures.
(31) FIGS. **31**A-**31**G are diagrams including a side view schematically representing example axial stimulation elements including examples anchor elements.
(32) FIGS. **32**A-**32**C are diagrams schematically representing patient anatomy and an example device and/or example method for stimulating various locations of an ansa cervicalis-related nerve, including some intravascular delivery pathways and other delivery pathways.
(33) FIGS. **33**A-**37**D are diagrams including graphs schematically representing example respiratory cycles and example methods of stimulation for upper airway patency-related nerves.
(34) FIG. **38**A is a flow diagram schematically representing an example device and/or example

method for stimulation therapy.
(35) FIGS. **38**B, **38**C, and **38**D are block diagrams schematically representing examples of a sensing engine, sensing tools, and a stimulation engine, respectively.
(36) FIG. **39** is a flow diagram schematically representing an example device and/or example method for stimulation therapy.
(37) FIGS. **40**A-**51**B are block diagrams schematically representing example methods, or portions thereof, of sleep disordered breathing care.
(38) FIGS. **52**A-**52**C are diagrams including front and side views schematically representing a neck region and an example device and/or example method for sensing impedance.
(39) FIGS. **53**A-**53**D are diagrams including front and side views schematically representing patient anatomy and example methods relating to collapse patterns associated with upper airway patency.
(40) FIGS. **53**E-**53**F are block diagrams schematically representing example devices and/or example methods relating to collapse patterns associated with upper airway patency.
(41) FIG. **54**A is a block diagram schematically representing an example care engine.
(42) FIGS. **54**B-**54**E are block diagrams schematically representing example control portions, a user interface, and associated devices.
(43) FIGS. **55-59**B are diagrams schematically representing patient anatomy and example devices and/or example methods for implanting stimulation elements and applying stimulation therapy for a phrenic nerve and/or ansa cervicalis-related nerve.
(44) FIGS. **59**C-**59**E are diagrams including a side view schematically representing example stimulation elements incorporating anchor structures, with FIG. **59**F including a sectional view of FIG. **59**E.
(45) FIG. **60** is a diagram schematically representing patient anatomy and an example device and/or example method for transvenous stimulation of target nerve locations relating to upper airway patency.
DETAILED DESCRIPTION
(46) In the following detailed description, reference is made to the accompanying drawings which form a part hereof, and in which is shown by way of illustration specific examples in which the disclosure may be practiced. It is to be understood that other examples may be utilized and structural or logical changes may be made without departing from the scope of the present disclosure. The following detailed description, therefore, is not to be taken in a limiting sense. It is to be understood that features of the various examples described herein may be combined, in part or whole, with each other, unless specifically noted otherwise.
(47) At least some examples of the present disclosure are directed to example devices for, and/or example methods of, therapy for sleep disordered breathing (SDB). In some examples, the sleep disordered breathing may comprise obstructive sleep apnea, while in some examples, the sleep disordered breathing may comprise multiple-type sleep apneas including obstructive sleep apnea and/or central sleep apnea.
(48) Moreover, the general principles associated with the example arrangements of the present disclosure relating to sleep disordered breathing may be applied in other areas of a patient's body to treat conditions other than sleep disordered breathing. For instance, at least some aspects of the example arrangements of the present disclosure may be deployed within a pelvic region to treat urinary and/or fecal incontinence or other disorders, such as via stimulating the pudenal nerve, which may cause contraction of the external urinary sphincter and/or external anal sphincter.
(49) Other body regions and/or disorders also may be suitable candidates for an example arrangements in which multiple nerve targets are available to be stimulated to treat one type or class of physiologic conditions. It will be further understood that example sensing arrangements of the present disclosure (for sensing physiologic data relative to the condition of interest) may be deployed in association with the various example arrangements for stimulating single nerve targets

or multiple nerve targets.

(50) As shown in FIG. **1**, some example methods may comprise stimulating, via at least one stimulation element **110**, at least one upper airway patency-related tissue **120**, which may comprise nerve(s) **130** and/or muscle(s) **140**. In some examples, nerve **130** may comprise an ansa cervicalis-related nerve and/or a hypoglossal nerve, as further shown in FIG. **2**. Moreover, as further described later throughout various examples, nerve(s) **130** may comprise nerves in addition to, or instead of, the hypoglossal nerve and/or the ansa cervicalis-related nerve. Meanwhile, in some examples the muscle **140** may comprise a genioglossus muscle (innervated by the hypoglossal nerve), while in some examples the muscle **140** may comprise one or more muscle groups (e.g. omohyoid, sternothyroid, sternohyoid) innervated by the ansa cervicalis-related nerve. Moreover, as further described later throughout various examples, muscle **140** may comprise muscles in addition to, or instead of, the genioglossus muscle and/or the muscle groups innervated by the ansa cervicalis-related nerve.

(51) In one aspect, stimulation of one or more of such example nerves and/or muscles may serve to increase or maintain patency of the upper airway of the patient, and hence may sometimes be referred to as upper airway patency-related tissue(s).

(52) FIG. **2** is a diagram **300** including a side view schematically representing an ansa cervicalis nerve **315**, in context with a hypoglossal nerve **305** and with cranial nerves C**1**, C**2**, C**3**. As shown in FIG. **2**, portion **329**A of the ansa cervicalis nerve **315** extends anteriorly from a first cranial nerve C**1** with a segment **317** running alongside (e.g. coextensive with) the hypoglossal nerve **305** for a length until the ansa cervicalis nerve **315** diverges from the hypoglossal nerve **305** to form a superior root **325** of the ansa cervicalis nerve **315**, which forms part of a loop **319**. A portion of the hypoglossal nerve **305** extends distally to innervate the genioglossus muscle **304**. As further shown in FIG. **2**, the superior root **325** of the ansa cervicalis-related nerve **315** extends inferiorly (i.e. downward) until reaching near bottom portion **318** of the loop **319**, from which the loop **319** extends superiorly (i.e. upward) to form an lesser root **327** (i.e. inferior root) which joins to the second and third cranial nerves, C**2** and C**3**, respectively.

(53) As further shown in FIG. **2**, several branches **331** extend off the ansa cervicalis loop **319**, including branch **332** which innervates the omohyoid muscle group **334**, branch **342** which innervates the sternothyroid muscle group **344** and the sternohyoid muscle group **354**. Another branch **352**, near bottom portion **318** of the ansa cervicalis loop **319**, innervates the sternohyoid muscle group **354** and the sternothyroid muscle group. In some examples, the entire ansa cervicalis nerve **315** (including loop **319**) and its related branches (e.g. at least **332**, **342 352**) when considered together may sometimes be referred to as an ansa cervicalis-related nerve **316**. It will be further understood that one such ansa cervicalis-related nerve is present on both sides (e.g. right and left) of the patient's body.

(54) In one aspect, stimulation of an ansa cervicalis-related nerve **316** may comprise stimulation of the superior root **325** of the ansa cervicalis nerve **315** (e.g. loop) and/or any one of the branches **331** extending from the loop **319**, which may influence upper airway patency. However, in some examples, upper airway patency also may be increased by directly stimulating the above-identified muscle groups, such as the omohyoid, sternothyroid, and/or sternohyoid muscle groups.

(55) Among other effects, stimulation of such nerves and/or muscles act to bring the larynx inferiorly, which may increase upper airway patency.

(56) Various example implementations of stimulating different portions of the ansa cervicalis-related nerve **316** to activate different associated muscle groups are described later in association throughout various examples of the present disclosure. Moreover, some of these FIGURES illustrate various example implementations of stimulating the hypoglossal nerve and/or of stimulating other nerves in addition to, or instead of, the ansa cervicalis-related nerve and the hypoglossal nerve.

(57) While stimulation of just the hypoglossal nerve **305** (or some branches thereof) may be

effective in increasing upper airway patency to a sufficient degree to ameliorate obstructive sleep apnea in at least about seventy percent of appropriate patients when using certain types of implantable neurostimulation devices, some patients may benefit from stimulation of an ansa cervicalis-related nerve **316** in addition to, or instead of, stimulation of the hypoglossal nerve **305**. Moreover, for a single patient, certain positions of the head-and-neck and/or of their body (e.g. supine, lateral decubitis, etc.) may be treated more effectively by stimulating an ansa cervicalis-related nerve **316**, with or without stimulation of the hypoglossal nerve **305**. In some such examples, upon detecting that a patient is in a certain body position (e.g. supine), stimulation of the ansa cervicalis-related nerve **316** may be implemented.

(58) In addition, because the ansa cervicalis-related nerve **316** innervates several different muscle groups which may influence upper airway patency, stimulation may be applied at several different locations of the ansa cervicalis-related nerve **316**. Such stimulation at the respective different locations may occur simultaneously, sequentially, alternately, etc., depending on which nerves (or muscles) are being stimulated, depending on when the stimulation occurs relative to the respective respiratory phases (or portions of each phase) of a respiratory period of the patient's breathing, and/or based on other factors.

(59) Many different examples of various stimulation locations of an ansa cervicalis-related nerve **316**, example timing, patterns, etc. are described below throughout the present disclosure. Of these various potential stimulation locations, FIG. **2** (and FIGS. **16**, **32**A) generally illustrates three example stimulation locations A, B, and C. A stimulation element may be placed at all three of these locations or just some (e.g. one or two) of these example stimulation locations. At each location, a wide variety of types of stimulation elements (e.g. cuff electrode, axial array, paddle electrode) may be implanted depending on the particular delivery path, method, etc. At each example stimulation A, B, C, a stimulation element may be delivered subcutaneously, intravascularly, etc. At each stimulation location, in some examples the stimulation element may comprise a microstimulator. Various aspects of these example implementations are further described below.

(60) In some example implementations, a stimulation element may be percutaneously delivered to a position to be in stimulating relation to the upper airway patency-related muscle. In some such examples, a percutaneous access point may be formed and located intermediate between a hyoid bone and a sternum and lateral to a midline. As with other implantation methods described herein, the implantation may comprise monitoring nerves during the percutaneous delivery and doing so via a nerve integrity monitor (NIM) in some examples.

(61) In some examples, further example implementations for these stimulation location are described in association with at least FIG. **16** (at least stimulation location A), FIGS. **22**A, **32**B (at least stimulation location B), FIG. **32**B (stimulation location C). It will be understood that these example stimulation locations A, B, C are not limiting and that other portions of the ansa cervicalis-related nerve may comprise suitable stimulation locations, depending on the particular objectives of the stimulation therapy, on the available access/delivery issues, etc.

(62) Among the different physiologic effects resulting from stimulation of the various branches of the ansa cervicalis-related nerve **316**, in some examples stimulation of nerve branches which cause contraction of the sternothyroid muscle and/or the sternohyoid muscle may cause the larynx to be pulled inferiorly, which in turn may increase and/or maintain upper airway patency in at least some patients. Such stimulation may be applied without stimulation of the hypoglossal nerve or may be applied in coordination with stimulation of the hypoglossal nerve **305**.

(63) Other physiologic effects of stimulating the ansa cervicalis-related nerve **316** and/or other nerves may be described later in the context of particular examples of the present disclosure.

(64) FIG. **3**A is a diagram **500** including a front view schematically representing an example arrangement **501** including one or more stimulation elements forming part of an example device and/or example method for increasing and/or maintaining upper airway patency or other purposes.

As shown in FIG. **3**A, in some examples, a first stimulation element **510**A is positioned at a hypoglossal nerve **505**R on a first side (e.g. right side) of a head-and-neck portion **520** of a patient's body and a second stimulation element **5106** is positioned at a hypoglossal nerve **505**L on an opposite second side (e.g. left side) of the head-and-neck portion **520**, and therefore spaced apart from first stimulation element **510**A. As further shown in FIG. **3**A, in some examples, a third stimulation element **513**A is positioned at an ansa cervicalis-related nerve **515**R on a first side (e.g. right side) of the head-and-neck portion **520** and a second stimulation element **513**B is positioned at an ansa cervicalis-related nerve **515**L on an opposite second side (e.g. left side) of the head-and-neck portion, and therefore spaced apart from the third stimulation element **513**A. As shown in FIG. **3**A, the stimulation elements **513**A, **513**B are depicted as being in stimulating relation to the first and second ansa cervicalis-related nerve **515**L, **515**R at a position just superior to the clavicles **522**. However, for illustrative simplicity, it will be understood that this depiction is also representative of the stimulation elements **513**A, **513**B being positioned at any portion of the ansa cervicalis nerve loop and/or related branches, etc. with at least FIGS. **2**, **16**, **22**A, **32**A-**32**D, etc. providing more detailed illustrations of an ansa cervicalis-related nerve **316**.
(65) As apparent from FIG. **3**A, the stimulation element **510**A is also spaced apart from stimulation element **513**A, while stimulation element **5106** is spaced apart from stimulation element **513**B.
(66) While FIG. **3**A depicts stimulation elements for both the hypoglossal nerve (e.g. elements **510**A, **510**B) and for the ansa cervicalis-related nerve (e.g. elements **513**A, **513**B), it will be understood that in some examples, stimulation of an upper airway patency-related tissue may comprise stimulation of solely of the hypoglossal nerves **505**R and/or **505**L. In such arrangements, stimulation of the ansa cervicalis-related nerve **515**R, **515**L does not occur at all or at least does not occur during specified time periods, situations, etc.
(67) Stimulation of just one hypoglossal nerve (e.g. **505**R or **505**L) may sometimes be referred to as unilateral stimulation, while stimulation of both such hypoglossal nerves (e.g. **505**R and **505**L) may sometimes be referred to as bilateral stimulation. It will be further understood that in some instances of unilateral stimulation, just one of the respective stimulation elements **510**A, **510**B has been implanted. However, in other examples of unilateral stimulation, both stimulation elements **510**A, **510**B may be implanted, but just one of them is stimulated to provide unilateral stimulation.
(68) In some examples of bilateral stimulation of the hypoglossal nerves (e.g. **505**R, **505**L), the stimulation may be implemented simultaneously, alternately, and/or in other patterns.
(69) Furthermore, in some examples in which one or both of stimulation elements **510**A, **510**B are implanted (to stimulate hypoglossal nerve(s)), neither of the stimulation elements **513**A, **5138** (for stimulating an ansa cervicalis-related nerve) are implanted.
(70) However, in some examples in which one or both of stimulation elements **510**A, **5108** are implanted (to stimulate hypoglossal nerve(s)), one or both of the stimulation elements **513**A, **513**B (for stimulating an ansa cervicalis-related nerve) may be implanted. In some such examples, even though such stimulation elements **513**A, **513**B may be implanted, such stimulation elements **513**A, **513**B may be not activated in some examples in which just stimulation of one or both of the hypoglossal nerve(s) **505**R, **505**L is to be provided.
(71) While FIG. **3**A depicts stimulation elements for both the hypoglossal nerve (e.g. elements **510**A, **510**B) and for the ansa cervicalis-related nerve (e.g. elements **513**A, **513**B), it will be understood that in some examples, stimulation of an upper airway patency-related tissue may comprise stimulation of solely one or both of the ansa cervicalis-related nerves **515**R, **515**L. In such arrangements, stimulation of the hypoglossal nerve **505**R, **505**L does not occur at all or at least does not occur during specified time periods, situations, etc.
(72) Stimulation of just one ansa cervicalis-related nerve (e.g. **515**R or **515**L) may sometimes be referred to as unilateral stimulation, while stimulation of both such nerves (e.g. **515**R and **515**L) may sometimes be referred to as bilateral stimulation. It will be further understood that in some instances of unilateral stimulation, just one of the respective stimulation elements **513**A, **513**B has

been implanted. However, in other examples of unilateral stimulation, both stimulation elements **513**A, **513**B may be implanted, but just one of them is stimulated to provide unilateral stimulation.
(73) In some examples of bilateral stimulation of ansa cervicalis-related nerves (e.g. **515**R, **515**L), the stimulation may be implemented simultaneously, alternately, and/or in other patterns.
(74) Furthermore, in some examples in which one or both of stimulation elements **513**A, **513**B are implanted (to stimulate the ansa cervicalis-related nerve(s)), neither of the stimulation elements **510**A, **510**B (for stimulating a hypoglossal nerve) are implanted.
(75) However, in some examples in which one or both of stimulation elements **513**A, **513**B are implanted (to stimulate ansa cervicalis-related nerve(s)), one or both of the stimulation elements **510**A, **510**B (for stimulating a hypoglossal nerve) may be implanted. In some such examples, even though such stimulation elements **510**A, **510**B may be implanted, such stimulation elements **510**A, **510**B may be not activated in some examples in which just stimulation of one or both of the ansa cervicalis-related nerve(s) **515**R, **515**L is to be provided.
(76) In some examples, stimulation of just the ansa cervicalis-related nerve(s) **515**R and/or **515**L may be implemented for particular collapse patterns of the upper airway or less than complete collapse behaviors.
(77) With further reference to the example arrangement **501** in FIG. **3**A, in some examples just one stimulation element is implanted at a left side of the head-and-neck portion **520** to stimulate a first type of nerve (e.g. hypoglossal, ansa cervicalis-related, or other) and just one stimulation element is implanted at right side of the head-and-neck portion **520** to stimulate a different second type of nerve (e.g. hypoglossal, ansa cervicalis-related, other). For instance, in some examples just stimulation element **510**A is implanted to stimulate a right hypoglossal nerve **505**R and just stimulation element **513**B is implanted to stimulate a left ansa cervicalis-related nerve **515**L, or vice versa.
(78) Alternatively, all stimulation elements **510**A, **510**B, **513**A, **513**B of example arrangement **501** may be implanted, but stimulation is implemented solely via stimulation element **510**A for right hypoglossal nerve **505**R and solely via stimulation element **513**B for the left ansa cervicalis-related nerve **515**L, or vice versa. In some such examples, the stimulation elements (e.g. a combination of **510**A and **513**B, or a combination of **510**B and **513**A) may be activated to deliver stimulation simultaneously to the respective hypoglossal and ansa cervicalis-related nerves. However, in some examples, the stimulation elements (e.g. a combination of **510**A and **513**B, or a combination of **510**B and **513**A) may be activated to deliver stimulation alternately to the respective hypoglossal and ansa cervicalis-related nerves In yet other examples, various stimulation patterns may be implemented in which one stimulation element (e.g. **510**A) is activated multiple times within a selectable period of time and then the other stimulation element (e.g. **513**A) is activated one or more times. In further examples, stimulation applied via the respective stimulation elements **510**A, **510**B, **513**A, **513**B may be implemented in an interleaving manner.
(79) It will be further understood that the various stimulation elements **510**A, **510**B, **513**A, **513**B illustrated in FIG. **3**A may be embodied as part of a lead, a microstimulator, etc., and may be anchored to a non-nerve tissue or structure within the patient's body via various anchor elements, as described more fully below in association with at least FIGS. **6**A-**6**B, **22**A-**23**, and/or **27**A-**30**B. For example, with reference to FIG. **3**A, in some examples such anchor elements may be secured relative to a non-nerve tissue, such as but not limited to, the illustrated clavicles **522**, manubrium **524** of the sternum, etc.
(80) Similarly, the respective stimulation elements **510**A, **510**B, **513**A, **513**B may be embodied as one of the various electrode arrays, cuff electrodes, paddle electrodes, etc. as described more fully below in various example arrangements of the present disclosure. The respective stimulation elements may be embodied in a unipolar configuration, a bipolar configuration or multi-polar configuration.
(81) In some examples, the various stimulation arrangements described in association with at least

FIG. **3**A may be implemented and stimulation performed without any sensing at all or with limited sensing, such as (but not limited to) just sensing to evaluate effectiveness of the stimulation but not using the sensing to time or trigger the stimulation. In either case, in some examples stimulation may be applied simultaneously to both an ansa cervicalis-related nerve and a hypoglossal nerve. Further details are described throughout various examples of the present disclosure.

(82) FIG. **3**B is a diagram schematically representing an example arrangement **571** comprising an example device for, and/or example method of, communication between an implantable medical device (IMD) **570** and a patient remote control **572**. In some examples, the implantable medical device **570** may comprise an implantable pulse generator (IPG) (e.g. **533** in FIG. **3**A, etc.), microstimulator (e.g. **1313**A in FIGS. **10**A, **100**; **6575** in FIG. **14**H). In some examples, the implantable medical device **570** (with the patient remote control **527**) may comprise one example implementation of the IPG **533** in FIG. **3**A, and is therefore applicable to example implementations throughout the present disclosure.

(83) The patient remote control **572** comprises inputs to change stimulation strength settings, activate or deactivate therapy, etc. The patient remote controls **572** also may receive control data, sensed data, therapy data, and/or other data from the IMD **570**. The patient remote control **572** may communicate wirelessly with the IMD **570** via telemetry or other wireless communication protocols. At least some aspects of initiating, terminating, adjusting stimulation settings and/or other settings of the IMD **570** will be further described later in association with various examples throughout the present disclosure.

(84) In some examples, the example arrangement **571** may comprise one example implementation of the care engine **10000** (FIG. **54**A), control portions **10500**, **10528**, **10600** (FIG. **54**B, **54**C), and/or user interface **10520** (FIG. **54**D), as described later. With this in mind, the patient remote control **572** may comprise one example implementation of the patient remote control **10530** in FIG. **54**C and/or of the patient remote control **10640** in FIG. **54**E.

(85) FIG. **3**C is a diagram on an example arrangement **575** comprising at least some of substantially the same features and attributes as example arrangement **500** in FIG. **3**C, except including various examples of sensors which may form part of the IPG **533** and/or may be independent of the IPG **533**. In general terms, the sensors described in association with FIG. **3**C may comprise any one or more of the sensing types, modalities, parameters, etc. as later described in association with at least FIGS. **38**B-**38**C, and at least FIGS. **40**A-**51**B, **54**A (care engine **10000**). In some examples, the example arrangement **575** may comprise one example implementation of the care engine **10000** (FIG. **54**A), control portions **10500**, **10528**, **10600** (FIG. **54**B, **54**C), and/or user interface **10520** (FIG. **54**D), as described later.

(86) In some examples IPG **533** may comprise an on-board sensor **560** which is incorporated within a housing of the IPG **533** and/or exposed on an external surface of the housing of the IPG **533**. In some examples, the sensor **560** may comprise an accelerometer (e.g. **8754** in FIG. **38**C), which may comprise a single axis accelerometer or a multiple axis (e.g. 3 axis) accelerometer. As noted in association with at least FIGS. **38**C-**38**D, the accelerometer may be used to sense various physiologic information, such as but not limited to body position (e.g. **8722** in FIG. **38**B), respiration (e.g. **8274** in FIG. **38**B), sleep (e.g. **8728** in FIG. **38**B), disease burden (e.g. **8726** in FIG. **38**B). In some examples, the sensed respiration may be used for timing application of stimulation to treat sleep disordered breathing, evaluate the severity of the sleep disordered breathing or other disease burdens, the effectiveness of the stimulation therapy, and/or other physiologic information.

(87) In a manner similar to sensing body position, the accelerometer may be used to sense posture and/or activity based on gross body movements. The accelerometer also may be used to sense at least ballistocardiography (**8762** in FIG. **38**C), seismocardiography (**8764** in FIG. **38**C), heart rate (HR) (**8766** in FIG. **38**C), sleep (**8728** in FIG. **38**C), disease burden (**8726** in FIG. **38**C), as further described later in association with at least FIG. **38**C. In some examples, via at least such

accelerometer sensing, the disease burden may comprise a cardiovascular burden and/or be determined via a cardiac output and/or cardiac waveform morphology.

(88) In some examples, the on-board sensor **560** may comprise an electrode formed on the external surface of a housing of the IPG **533**, and may be used for sensing impedance (e.g. **8752** in FIG. **38**C) in combination with other implanted sensors, such as but not limited to sensors **568**A, **568**B, which may be located on the torso of the patient. As further described later, sensor **560** also may be used in combination with sensing elements such as electrodes implanted in the head-and-neck region **520**. In some examples, a stimulation element (e.g. **510**A, **510**B, **513**A, **513**B) may comprise electrodes which may serve in combination with sensor **560** (as an electrode) to sense impedance. In some such examples, the sensed impedance may be used to determine respiration, which may be used for at least some of the above-identified purposes and/or other purposes.

(89) In some examples, sensed impedance may indicate a degree of upper airway patency. For example, a smaller cross-sectional upper airway, which reflects less upper airway patency, may be sensed as a lower impedance. Conversely, a larger cross-sectional upper airway, which reflects more upper airway patency, may be sensed as a higher impedance. Accordingly, maximal patency (measured as a higher impedance) may general correspond to periods of stimulation (HGN and/or ACN) or correspond to peak expiration of a respiratory cycle. Meanwhile, minimal patency (measured as a lower impedance) generally corresponds to inspiration, just prior to inspiration, or the onset of stimulation (e.g. HGN and/or ACN).

(90) In some examples, the on-board sensor **560** may comprise an ECG sensor or may comprise an electrode, which when used in combination with other electrodes (e.g. **568**A, **568**B), may be used to sense electrocardiogram (ECG) information, such as per ECG parameter **8760** in FIG. **38**C.

(91) As further shown in FIG. **3**C, in some example implementations the example arrangement **575** may comprise a sensor lead **564** which supports a sensor **566**, which in turn in some examples may comprise a pressure sensor (e.g. differential pressure) (e.g. **8756** in FIG. **38**B). Among other physiologic parameters, the pressure sensor may be used to sense respiration, which may be used for at least some of the above-identified purposes and/or other purposes. In some examples, the sensor **566** may sense physiologic parameters other than pressure.

(92) In some examples, the example arrangement **575** may be implemented via at least some external sensors relating to at least some of the sensing types, modalities, physiologic parameters, etc. which were described above as being implemented via implantable sensors.

(93) FIG. **4** is a diagram **600** including a front view schematically representing an example arrangement **601** including one or more stimulation elements forming part of an example device and/or example method for increasing and/or maintaining upper airway patency and/or other purposes. In some examples, the example arrangement **601** may comprise at least some of substantially the same features as, or comprise one example implementation of, the examples as previously described in association with at least FIGS. **1-3**.

(94) In some examples, an example method may comprise implanting stimulation element (e.g. **510**A and/or **510**B) at a hypoglossal nerve (e.g. **505**R and/or **505**L). As shown in FIG. **4**, in some examples this implantation may involve tunneling (T**3**) between a first incision **609**C and a second incision **609**A, wherein an implantable pulse generator (IPG) **533** is implanted via first incision **609**C and the stimulation element (e.g. **510**A) is implanted via second incision **609**A. In some examples, this portion of example arrangement **601** may be operated to treat sleep disordered breathing (SDB) without amendment or supplementation, either indefinitely or for at least a period of time during which the treatment is deemed satisfactory.

(95) However, in some examples, the example arrangement **601** may be supplemented to enhance treatment of sleep disordered breathing. In such instances, such as after a time period following the implantation of the first stimulation element (e.g. **510**A), a second implant procedure is performed while leaving the first stimulation element (e.g. **510**A) implanted relative to the hypoglossal nerve (e.g. **505**R). In the separate, second implant procedure, a second stimulation element (e.g. **513**A) is

implanted to be in stimulating relation to the ansa cervicalis-related nerve (e.g. **515**R). In some examples, the method comprises performing the implanting of the second stimulation element (e.g. **513**A) upon a determination of a patient exhibiting sleep disordered breathing (SDB) despite treatment via the first stimulation element (e.g. **510**A and/or **510**B) of the hypoglossal nerve (e.g. **505**R and/or **505**L). In some such examples, the patient may exhibit symptomatic AHI despite the stimulation of the hypoglossal nerve(s). In some examples, the baseline stimulation therapy involving the hypoglossal nerve may reduce collapsibility of the upper airway as measured in Pcrit by 5 cm of water pressure. However, upon adding stimulation of the ansa cervicalis-related nerve to be concomitant with the stimulation of the hypoglossal nerve, the collapsibility of the upper airway is changed by or to 8 cm of water pressure, in some examples.
(96) There are several example methods by which the first stimulation element **510**A and by which the second stimulation element **513**A may be implanted to supplement the already implanted stimulation element **510**A.
(97) It will be understood that the first stimulation element **510**A may be implanted via a first stimulation lead, on which the first stimulation element **510**A is supported, in a position to extend between the implanted pulse generator **533** (at or near access incision **609**C) and the position of the first stimulation element at the hypoglossal nerve (at or near access incision **609**A), as shown in FIG. **4**. One example arrangement **700** for doing so is illustrated in FIG. **5**A.
(98) FIG. **5**A is a diagram including a side view schematically representing an example device **700** which comprises a stimulation lead **740** comprising a body **741** extending between a proximal portion **744** and an opposite distal portion **742**, which supports a first stimulation element **710**. In some examples, the first stimulation element **710** may comprise one example implementation of stimulation element **510**A in FIG. **4**. With further reference to FIG. **5**A, the first stimulation element **710** may comprise a linear array of electrodes **716** adapted to stimulate hypoglossal nerve **505**R (or **505**L). However, it will be understood that the first stimulation element **710** may comprise other types of electrode configurations (e.g. cuff, paddle, etc.). Meanwhile, the proximal portion **744** of lead **740** is connectable to a port in header **735** of implantable pulse generator (IPG) **533**. As further represented by the dashed lines in FIG. **5**A, the IPG **533** is implanted via implant access-incision **609**C, which may be in the pectoral region **532** as shown in FIG. **4**.
(99) With this in mind, the first stimulation lead **740** may be implanted subcutaneously via implant access-incisions **609**A, **609**C, and via appropriate tunneling, stimulation element **710** may be placed in stimulating relation to hypoglossal nerve **505**R, with body **741** of lead **740** extending between the hypoglossal nerve **505**R and the IPG **533** in the pectoral region **532**.
(100) As noted previously, the first stimulation lead **740** may be operated to treat sleep disordered breathing via stimulation of the hypoglossal nerve **505**R.
(101) However, upon a determination that the patient exhibits an unsatisfactory level of sleep disordered breathing despite stimulation via stimulation lead **740** at the hypoglossal nerve **505**R, an example method schematically represented via FIG. **5**A (in context with FIG. **4**) comprises implanting a second stimulation lead **760** supporting a second stimulation element **713** to be positioned in stimulating relation to the ansa cervicalis-related nerve **515**R. In some examples, the second stimulation element **713** may comprise one example implementation of stimulation element **513**A in FIG. **4**, and may comprise a linear array of electrodes **716** as shown in FIG. **5**A, in some examples.
(102) As further shown in FIG. **5**A, the second stimulation lead **760** comprises a proximal portion **764** for connection to an intermediate portion **745** of the first stimulation lead **740**. In particular, in some examples, the intermediate portion **745** of lead **740** comprises a port interface **750** including an extension arm **752** including a connection port to receive the proximal portion **764** of second stimulation lead **760** in order to establish electrical connection (and mechanical connection) of the lead **760** to the IPG **533**.
(103) As further shown in FIG. **4** (in context with FIG. **5**A), in some examples tunneling T**1** may be

performed between the implant access-incision **609**C and a new implant access-incision **609**B, via which the second stimulation element **713** (FIG. **5**A) is to be positioned, with the tunnel T**1** providing a path to implant second stimulation lead **760**. As represented via FIG. **5**A, the second stimulation lead **760** may be substantially shorter than first stimulation lead **740**. While the second stimulation lead **760** is depicted as being relatively short, it will be understood that the second stimulation lead **760** may have a greater relative length than shown in FIG. **5**A and that its length may depend on the location of port interface **750** along the stimulation lead **740**.

(104) Via this arrangement, once it is determined that stimulation of the ansa cervicalis-related nerve **515**R is desirable in view of insufficient treatment of sleep disordered breathing, then at a time period after implantation of the first stimulation lead **740**, the second stimulation lead **760** may be implanted for connection to the IPG **533** via releasable connection of second stimulation lead **760** to the port interface **750** of stimulation lead **740**.

(105) In some examples, the sequence of implantation described in association with example device **601**, **700** in FIGS. **4**, **5**A may be reversed such that a first stimulation lead is implanted to stimulate the ansa cervicalis-related nerve **515**R, and then at a later point in time, a second stimulation lead may be implanted to stimulate the hypoglossal nerve with the second stimulation lead being electrically connectable to the first stimulation lead.

(106) With this in mind, FIG. **5**B includes a side view schematically representing an example device **771**. In some examples, device **770** may comprise at some of substantially the same features and attributes as device **700** in FIG. **5**A, except with device **771** providing a first stimulation lead **770** adapted to be in stimulating relation to the ansa cervicalis-related nerve **515**R and to be implanted in an initial implantation procedure instead of first implanting a stimulation lead for the hypoglossal nerve.

(107) As shown in FIG. **5**B, the first stimulation lead **770** comprises a distal portion **772** supporting a stimulation element **713** (including a linear array of electrodes **716**) and a proximal portion **774** in electrical connection with the IPG **533** via header **735**. The first stimulation lead **770** also comprises an intermediate portion **773** which includes port interface **750** (as in FIG. **5**A) to receive a proximal portion **784** of a second stimulation lead **780** to be in stimulating relation to a hypoglossal nerve **505**R. The second stimulation lead **780** comprises a distal portion **782** supporting stimulation element **710** (including a linear array of electrodes **716**). As in other example stimulation leads throughout the present disclosure, the respective stimulation elements **710**, **713** in example device **771** of FIG. **5**B can comprise a wide variety of types of electrode configurations (e.g. cuff, paddle, axial array, etc.).

(108) In operation, the first stimulation lead **770** is implanted for treating sleep disordered breathing via stimulation of the ansa cervicalis-related nerve **515**R. After some period of time elapsing, such as a determination that the neurostimulation therapy is unsatisfactory, a second implant procedure may be performed to implant the second stimulation lead **780** for neurostimulation of the hypoglossal nerve **505**R (or **505**L). As part of this second implant procedure, the proximal portion **784** of the second stimulation lead **780** is electrically (and mechanically) connected to the port **752** of port interface **750**, as represented via directional arrow C in FIG. **5**B.

(109) As further represented in FIG. **4** (in context with FIG. **5**B), in order to add the second stimulation lead **780**, tunneling T**2** may be performed from the location of the port interface **750** of lead **770** (at or near implant access-incision **609**B in FIGS. **4**, **5**B) to the intended implant location of the stimulation element **710** on second stimulation lead **780** at implant access-incision **609**A (FIG. **4**).

(110) Via the example arrangement provided via example device **771** in FIG. **5**B, neurostimulation therapy can be conveniently expanded to include additional nerves when desired, such as to address a change in a patient's underlying condition, to enhance therapy, etc.

(111) Additional example implementations of implanting multiple stimulation element(s) for

second/type of nerve or a second side of the body are described below in, but not limited to, at least FIGS. **10**A-**10**C.

(112) It will be understood that the various leads, stimulation elements, port interfaces, etc. described in association with at least FIGS. **3-5**B may be secured with sutures and/or a wide variety of anchors. FIG. **6**A-**6**B provide example implementations of just some such anchors, while other types of anchors are described in association with at least FIGS. **22**A-**23** and **27**A-**30**B. It will be further understood that the example anchors in FIGS. **6**A-**6**B (e.g. wings, holes for tissue growth, tines, barbs, etc.) may employed on any of the various stimulation elements, leads, etc. as appropriate. Moreover, in some examples, some of the anchor features (e.g. suture-friendly surfaces, wings, holes for sutures, holes for tissue growth, tines, barbs, etc.) may be incorporated into implantable structures, such as a port interface (e.g. **750** in FIGS. **5**A-**5**B, **1070** in FIG. **7**, and the like) to facilitate their anchoring relative to non-nerve tissues and structures to stabilize the respective element within the patient's body. In this regard, it also will be understood that such anchoring may occur via at least some of the non-nerve tissues and structures later detailed in association with at least FIGS. **22**A-**23**.

(113) FIG. **6**A is side view schematically representing an elongate suture anchor element **800**, which comprises a body **811** and a linear array of protrusions **812** to facilitate securing the anchor element, via sutures, relative to a non-nerve tissue. As further shown in FIG. **6**A, the anchor element **800** may be fixed on a portion of a lead **814** or slidable movable along the portion of the lead **814** to be secured.

(114) FIG. **6**B is a side view schematically representing an anchor element **830**, which comprises a body **831** and pair of wings **832** extending perpendicular outward from body **831** to facilitate securing the anchor element **830**, via sutures, relative to a non-nerve tissue. As further shown in FIG. **6**B, the anchor element **830** may be fixed on a portion of a lead **814** or slidable movable along the portion of the lead **814** to be secured.

(115) Upon securing the anchor element **800** or **830**, the lead **814** becomes secured relative to non-nerve tissue within the patient's body. It will be understood that similar types of anchor features may be incorporated into portions of a lead, such as the various example port interfaces (e.g. FIG. **5**A, **5**B, **7**A, etc.) described in several examples of the present disclosure.

(116) FIG. **7**A is a diagram including a top view schematically representing an example arrangement **1000** including an IPG **533**, bifurcated port interface **1070**, and removably insertable stimulation leads **1080**, **1081**. In some examples, the example arrangement **1000** comprises at least some of substantially the same features and attributes as the example arrangements described in association with at least FIGS. **1-6**B, at least with respect to providing for flexibility in a sequence or timing of implanting the respective stimulation leads **1080**, **1081** according to patient conditions, anatomy encountered during implantation, changing health over time, etc.

(117) As shown in FIG. **7**A, a lead support portion **1060** includes a proximal portion **1064**, which is electrically connectable to an IPG **533** via header **735** and a distal portion **1062**, which supports a bifurcated port interface **1070**. The port interface **1070** comprises two spaced apart prongs **1072**A, **1072**B which diverge from each other, with each prong **1072**A, **1072**B comprising a connection port **1075** to removably receive electrical (and mechanical) connection from a proximal portion **1084** of stimulation leads **1080**,**1081**. Each stimulation lead **1080**, **1081** comprises a distal portion **1082** supporting a respective stimulation element **710**, **713**, each of which comprise a linear array of electrodes **716**. As in other examples, the stimulation elements **710**, **713** can take a wide variety of electrode configurations (e.g. cuff, paddle, etc.) other than the axial array depicted in FIG. **7**A. In some examples, the IPG **533** and lead support portion **1060** (including port interface **1070**) may be implanted to support the concurrent implantation of both stimulation leads **1080**, **1081**. However, in some examples, in a manner similar to that previously described in association with at least FIGS. **5**A, **5**B, in some examples, just one of the stimulation leads **1080**, **1081** may be implanted in an initial implantation procedure to be in stimulating relation to a first nerve (e.g. hypoglossal nerve or

ansa cervicalis-related nerve). At a later point in time, the other respective one of the stimulation leads **1080**, **1081** may be implanted to be in stimulating relation to a second nerve (e.g. hypoglossal nerve or ansa cervicalis-related nerve). In such arrangements, the port interface **1070** conveniently permits selective addition of the second stimulation lead (e.g. **1080** or **1081**) during the second implant procedure by insertion of the proximal portion **1084** of the respective stimulation lead.
(118) In some examples, in a manner similar to the port interfaces depicted in FIGS. **5**A-**5**B, the port interface **1070** may be implanted and/or accessed via an implant access-incision, like implant access-incision **609**B in a head-and-neck portion **520** in FIGS. **4**-**5**B.
(119) In some examples, implantation of port interface **1070** and/or stimulation leads **1080**, **1081** may be facilitated via use of tunneling tool **1100** schematically represented in FIG. **7**B. As shown in FIG. **7**B, the tunneling tool **1110** comprises a proximal main portion **1102**, which supports diverging portions **1104**A, **1104**B, from which extend spaced apart prongs **1106**A, **1106**B. The prongs **1106**A are insertable into, and may be advanced through, subcutaneous tissue to form tunnels for implantation of stimulation leads and related structures.
(120) In some examples, the tunneling tool **1090** can be employed with example lead arrangements other than shown in FIG. **7**A, and in which two different tunnels are to be formed subcutaneously to provide path for implantation of leads, stimulation elements, etc. It will be further understood that prongs **1106**A, **1106**B may have lengths with differ from each other, and may have tips which are steerable in some examples.
(121) FIG. **8** is a diagram including a top view schematically representing an example arrangement **1130** including a stimulation lead **1140**. In some examples, the stimulation lead **1140** may comprise at least some of substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least FIGS. **1**-**7**B.
(122) As shown in FIG. **8**, in some examples, the stimulation lead **1140** comprises a proximal support portion **1144** electrically (and mechanically) connectable to an IPG **533** via header **735**, while a distal portion **1142** comprises a bifurcated pair of distal stimulation portions **1164**A, **1164**B. While just a portion of the distal stimulation portions **1164**A, **1164**B are shown for illustrative simplicity, it will be understood that each distal stimulation portion **1164**A, **1164**B may support a stimulation element, such as stimulation elements **510**A, **510**B, **513**A, **513**B, **710**, or **713** etc. as described throughout the previously described examples or such as some of the later described stimulation elements.
(123) FIG. **9** is diagram including a front view schematically representing an example arrangement **1150** including an example device and/or example method for implantation of stimulation elements **510**A and/or **513**A. In some examples, the example arrangement **1150** may comprise at least some of substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least FIGS. **1**-**8**.
(124) It will be further understood that in some examples the stimulation elements **510**A, **513**A may be supported by respective separate stimulation leads, which are not shown in FIG. **9** for illustrative simplicity.
(125) As shown in FIG. **9**, in order to implant stimulation element **513**A a tunnel T**4** is formed between implant access-incision **609**C and **609**B, and in order to implant stimulation element **510**A, a tunnel T**5** is formed between implant access-incision **609**C and **609**A. In some examples, both tunnels T**4**, T**5** may be made at the time of an initial implant procedure in which both stimulation elements **510**A, **513**A (and their respective stimulation leads) are implanted.
(126) However, in some examples, the respective, representative stimulation elements **510**A, **513**A are implanted at different points in time, with one stimulation element being implanted in an initial implant procedure and the other respective implant procedure being implanted in a separate, later implant procedure. In some such examples, the respective stimulation leads (e.g. supporting stimulation elements **510**A, **513**A) may be electrically connected relative to the IPG **533** directly as

shown in the example arrangement of FIG. **10**A, while in some examples, the proximal portion of such stimulation leads may be connected to the IPG **533** indirectly via a port interface (e.g. **750** in FIGS. **5**A-**5**B).

(127) FIG. **10**A is a diagram including a front view schematically representing an example arrangement **1200** relative to a patient's body, including an example device and/or example method for implantation of stimulation elements **510**A and/or **513**A. In some examples, the example arrangement **1200** may comprise at least some of substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least FIGS. **1-9**.

(128) As shown in FIG. **10**A, two separate stimulation leads **1235**, **1237** may be implanted to position their respective stimulation elements **513**A, **510**A for implantation at target nerve locations. In a manner similar to at least some previously described examples, in some examples both of the stimulation leads **1235**, **1237** may be implanted as part of the same initial implantation procedure, while in some examples one of the respective stimulation leads (e.g. **1235**, **1237**) is implanted in a first implantation procedure, while the other respective lead is implanted in a second separate implantation procedure at a later point in time.

(129) As shown in FIG. **10**A, in this example a proximal portion of the respective stimulation leads are electrically connected to the IPG **533** directly. However, in some examples, a port interface with bifurcation features (e.g. **1070** in FIG. **7**A) near IPG **533** may be employed to connect the proximal ends of the respective leads **1235**, **1237** relative the header **735** of the IPG **533**.

(130) In some examples, the stimulation lead **1235** may support multiple stimulation elements **513**A, as shown in FIG. **10**B, in which the distal portion of the stimulation lead **1235** comprises a bifurcation yielding two different distal prongs **1236**A, **1236**B, each of which support a respective stimulation element **513**B, **513**C. The respective prongs **1236**A, **1236**B have a length suitable to place the different respective stimulations elements **513**B, **513**C at different target nerve locations. For example, one simulation element **513**B may be located a first target nerve location of the ansa cervicalis-related nerve (e.g. **316** in FIG. **2**) and the other simulation element **513**C may be located at different, second target nerve location of the ansa cervicalis-related nerve (e.g. **316** in FIG. **2**).

(131) FIG. **100** is a diagram including a front view schematically representing an example arrangement **1300** relative to a patient's body, including an example device and/or example method for implantation of stimulation elements **510**A and/or **1313**A. In some examples, the example arrangement **1300** may comprise at least some of substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least FIGS. **1-9**.

(132) As shown in FIG. **10**C, the example arrangement **1300** may comprise at least some of substantially the same features and attributes as the example arrangement **1200** in FIG. **10**A and/or **1250** in FIG. **10**B, except with the stimulation element **513**A being implemented as a microstimulator **1313**A such that no stimulation lead extends between the microstimulator **1313**A and the IPG **533**. However, in some examples, the microstimulator **1313**A may be in wireless communication with the IPG **533** to share at least control and/or data signals. In some examples, the microstimulator **1313**A may be in wireless communication with the stimulation element **510**A (or with a communication element formed as a part of stimulation lead **1237**) to share at least control and/or data signals to coordinate the actions of the respective microstimulator **1313**A and stimulation element **510**A relative to each other or relative to other therapy elements (e.g. IPG **533**, sensing, tracking, etc.). In some examples, microstimulator **1313**A may be in wireless communication with a control portion, programmer, and/or user interface external to the patient's body, which are in addition to, or instead of, communication with IPG **533**.

(133) In a manner similar to that described in association with the example arrangement of FIG. **10**A, both the stimulation element **510**A (and stimulation lead **1237**) and the microstimulator **1313**A may both be implanted in the same initial implantation procedure. However, in some

examples one of the respective stimulation element **510**A (and lead **1237**) and the microstimulator **1313**A is to be implanted in an initial implantation procedure, and then the other respective element (e.g. element **510**A or microstimulator **1313**A) may be implanted at a later time in a separate second implantation procedure. The second implanted element may be employed to enhance the neurostimulation therapy already established via the initial implantation procedure. As noted elsewhere, in some examples the example arrangement **1300** may be understood as being representative for the implantation of left and/or right sides of the patient's body and for implantation to provide stimulation in relation to any of the nerves identified within the present disclosure for increasing or maintaining upper airway patency or for other noted purposes.

(134) In some examples, the situation may be reversed in which the microstimulator **1313**A is implanted in stimulating relation to the hypoglossal nerve **505**R and a stimulation element **513**A (FIG. **10**A) is implanted in stimulating to the ansa cervicalis-related nerve **515**R

(135) In general terms, the microstimulator **1313**A comprises power and circuitry in a compact package to permit stimulation of an upper airway patency-related tissue (e.g. nerve **515**R) via at least one stimulation element located on a housing of the microstimulator or extending from the housing of the microstimulator. The microstimulator **1313**A also may comprise a sensing element(s). In some examples, the microstimulator **1313**A may comprise at least some of substantially the same features and attributes as described in Rondoni et al, MICROSTIMULATION SLEEP DISORDERED BREATHING (SDB) THERAPY DEVICE, published as WO 2017/087681 on May 26, 2017 and published as US 2020-0254249 on Aug. 13, 2020, and which is hereby incorporated by reference.

(136) FIG. **11**A is a diagram including a front view schematically representing an example arrangement **1350** relative to a patient's body, including an example device and/or example method for implantation of microstimulators **1360**A and/or **1313**A relative to respective target nerves **505**R, **515**R. In some examples, the example arrangement **1350** may comprise at least some of substantially the same features and attributes as, an example implementation of, and/or be usable with the example arrangements described in association with at least FIGS. **1-9**. Accordingly, in some examples, the target nerves **505**R, **515**R may comprise a hypoglossal nerve **505**R and an ansa cervicalis-related nerve **515**R, respectively.

(137) As shown in FIG. **11**A, the example arrangement **1350** may comprise at least some of substantially the same features and attributes as the example arrangement **1300** in FIG. **10**B, except with the stimulation element **510**A being implemented as a microstimulator **1360**A such that no IPG is present and no stimulation lead extends between the microstimulator **1360**A and an IPG **533**. In some examples, the microstimulator **1360**A may be in wireless communication with the microstimulator **1313**A to share at least control and/or data signals to coordinate the actions of the respective microstimulators **1313**A, **1360**A relative to each other or relative to other therapy elements (e.g. sensing, tracking, etc.). In some examples, both of the microstimulators **1360**A, **1313**A may be in wireless communication with a control portion, programmer, and/or user interface external to the patient's body.

(138) In a manner similar to that described in association with the example arrangement of FIG. **10**B, both the microstimulator **1360**A and the microstimulator **1313**A may both be implanted in the same initial implantation procedure, such as via respective implant access-incisions **609**A, **609**B shown in several previously described examples. However, in some examples one of the respective microstimulators **1360**A, **1313**A may be implanted in an initial implantation procedure, and then the other respective microstimulator (e.g. **1360**A or **1313**A) may be implanted at a later time in a separate second implantation procedure. The second implanted element may be employed to enhance the neurostimulation therapy already established via the initial implantation procedure.

(139) In some examples, the respective microstimulators **1313**A, **1360**A may be implanted via a single implant access-incision of the type shown in FIG. **13**, where some maneuvering may be used (relative to the single access-incision) to place the respective microstimulators **1313**A, **1360**A

adjacent their respective target nerves **515**R, **505**R.

(140) FIG. **11**B is a diagram including a front view schematically representing an example arrangement **1400** relative to a patient's body **510**, including an example device and/or example method for implantation of a single microstimulator **1413**A in a head-and-neck region **520**. In some examples, the example arrangement **1400** may comprise at least some of substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least FIGS. **1-9**.

(141) As shown in FIG. **11**B, the microstimulator **1413**A comprises a power/control element **1417** and a pair of stimulation elements **1414**A, **1416**A, which are positioned into stimulating relation to the respective nerves **505**R, **515**R. Each stimulation element **1414**A, **1416**A extends from the power/control element **1417** via a respective lead **1414**B, **1416**B. The power/control element **1417** may operate in a manner similar to an IPG **533**, but is miniaturized to a smaller scale within a considerably smaller housing.

(142) In some examples, FIG. **11**B depicts the respective stimulation elements **1414**A, **1416**A as an array of electrodes (e.g. **716** in FIGS. **5**A, **5**B, **7**A), which may take the form of a small paddle, axial array of ring electrodes, or other electrode configuration. In some examples, cuff or partial cuff configurations may be employed, as well as pigtail configurations. Each respective stimulation element **1414**A, **1416**A may comprise its own anchor elements (e.g. tines, barbs, suture hole, and the like) or a separate anchor element may be used to secure the stimulation element **1414**A, **1416**A relative to the respective nerve **505**R, **515**R and/or relative to an adjacent non-nerve structure.

(143) In some examples, the microstimulator **1413**A (and associated stimulation elements **1414**A, **1416**A) may be implanted via a single implant access-incision of the type shown in FIG. **13**. In some such examples, some minor tunneling and/or maneuvering is used to place the respective stimulation elements **1414**A, **1416**A adjacent their respective target nerves **505**R, **515**R.

(144) In some examples, the microstimulator **1413**A may be deployed at locations within the head-and-neck region **520** in which the first and second target nerve locations are within close proximity to each other. For instance, the microstimulator **1413**A (including stimulation elements **1414**A, **1416**A) may be implanted as one of the example arrangements **2101** or **2401** in the example method in FIGS. **16-17** (or FIGS. **16**, **18-20**) such that a single device (e.g. **1413**A) in the head-and-neck region **520** may serve to stimulate two different nerves (e.g. **505**R, **515**R), such as the portion **307** of hypoglossal nerve **305** (FIGS. **16-17**) and the portion **329**A of the ansa cervicalis-related nerve **315** (FIG. **16-17**).

(145) With regard to the various stimulation elements, leads, etc. described in association with at least FIGS. **1-11**B, it will be understood that such example arrangements, methods of implantation, etc. may be use to implement sensing elements, where the sensing elements may take the place of the respective stimulation elements and/or where the stimulation elements also may act as or carry sensing elements. At least some additional aspects of sensing are further described later throughout various examples of the present disclosure.

(146) Moreover, with regard to the various example arrangements depicted in at least FIGS. **3-11**B, stimulation elements were shown to be in stimulating relation to a right side of the patient's body, such as at hypoglossal nerve **505**R and/or the ansa cervicalis-related nerve **515**R. However, it will be understood that such examples are intended to be representative of implantation, therapy, etc. for a left side of the patient's body and/or for bilateral implantation of such stimulation elements, leads, etc. Moreover, it will be further understood that the example arrangements in FIGS. **3-11**B also may implemented according to the various example implementations represented in association with the example arrangement in FIG. **2**.

(147) Moreover, with regard to at least the example arrangements of FIGS. **10**A-**11**B, it will be further understood that the various stimulation elements, lead, and/or microstimulators may be secured within the patient's body relative to a non-nerve structure or tissue via at least some of the various example anchor elements provided throughout the examples of the present disclosure, such

as at least FIGS. **6**A-**6**B, **22**A-**23**, and/or **27**A-**30**B.
(148) FIG. **12** is a diagram including a front view schematically representing an example arrangement **1600** relative to a patient's body **510**, including an example device and/or example method for implantation of a stimulation element **513**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **1600** may comprise at least some of substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least some of FIGS. **1-11**B.
(149) In some examples, the example arrangement **1600** comprises implantation of a stimulation element **513**A and IPG **533** via a single implant access-incision **609**B. The stimulation element **513**A is implanted to be in stimulating relation to ansa cervicalis-related nerve **515**R, and is electrically (and mechanically) connected to the IPG **533** via a stimulation lead, which is omitted for illustrative clarity. In some examples, the IPG **533** may be implanted and positioned in a region, such as the upper portion of a pectoral region **532** or the head-and-neck portion **520**, in relatively close proximity to the stimulation element **513**A. This arrangement may enable the use of shorter stimulation leads, reduce an amount of subcutaneous invasion, etc. By utilizing a single implant access-incision **609**B to implant all the elements of the example arrangement **1600**, the implantation procedure may be completed faster and in a less invasive manner for the patient.
(150) FIG. **13** is a diagram including a front view schematically representing an example arrangement **1700** relative to a patient's body **510**, including an example device and/or example method for implantation of a stimulation element **1313**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **1700** may comprise at least some of substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least some of FIGS. **1-11**B.
(151) In one particular example, the example arrangement **1700** may comprise at least some of substantially the same features and attributes as the example arrangement **1600** in FIG. **12**, except with stimulation element **513**A being replaced with a microstimulator **1313**A (e.g. **10**A) and with the omission of IPG **533**. In one aspect, this example arrangement **1700** significantly simplifies the implantation procedure by using a single implant access-incision and a single stimulation element, which includes its own power elements, control circuitry, etc. while embodied as a microstimulator **1313**A. In some examples, the microstimulator **1313**A may comprise a linear array of electrodes **716** on its exterior housing to provide stimulation element(s) and/or sensing capabilities. However, it will be understood that the microstimulator **1313**A may provide other electrode configurations.
(152) With regard to either example arrangement **1600**, **1700** in FIGS. **12**, **13**, it will be understood that such example arrangements may be implemented on just one side or both sides of the patient's body **510**. Moreover, it will be understood that such a single implantation procedure via a single implant access-incision **609**B may later be supplemented by additional implant access-incisions to implant a second stimulation element (including a stimulation lead) or microstimulator in a second, separate implant procedure, such as in the example implementations described in association with at least some of FIGS. **3-11**.
(153) With regard to both of the example arrangements depicted in FIGS. **12-13**, it will be understood that at least some aspects of such example arrangements may be applied to implantation of a stimulation element at other nerves such as, but not limited to, the hypoglossal nerve or other nerves.
(154) FIG. **14**A is a diagram including a front view schematically representing an example arrangement **1800** relative to a patient's body **510**, including an example device and/or example method for implantation of a stimulation element **1810**A in stimulating relation to a hypoglossal nerve **505**R and a stimulation element **1813**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **1800** may comprise at least some of

substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least some of FIGS. **1-13**.
(155) In particular, as shown in FIG. **14**A in some examples the example arrangement **1800** may be implanted in a single implantation procedure via a single implant access-incision **609**C in a manner similar to that described for example arrangement **1600** in FIG. **12**, except including an additional stimulation element **1810**A to stimulate the hypoglossal nerve **505**R and with both stimulation elements **1810**A, **1813**A carried on a single stimulation lead **1837**. In some examples, the single implant access-incision may be in pectoral region **532** as shown by indicator **609**C in FIG. **14** or may be in the head-and-neck portion **520**, such as via an implant access-incision **609**B shown in some other example FIGS.
(156) As shown in FIG. **14**A, via single implant access-incision **609**C, tunneling (T**6**) may performed between the implant access-incision **609**C and the target stimulation location of the respective stimulation element **1810**A (to be in stimulating relation to the hypoglossal nerve **505**R) and/or of the stimulation element **1813**A to be in stimulating relation to the ansa cervicalis-related nerve **515**R.
(157) As further shown in FIG. **14**A, via implant access-incision **609**C, IPG **533** is implanted subcutaneously and the stimulation lead **1837** is inserted and advanced through the tunnel T**6** until the respective stimulation elements **1810**A, **1813**A are positioned in stimulating relation to the respective hypoglossal nerve **505**R and ansa cervicalis-related nerve **515**R, as represented in FIG. **14**A.
(158) FIG. **14**A also illustrates that the respective stimulation elements **1810**A, **1813**A may be implemented as a linear array of electrodes **716**, which may facilitate appropriate nerve capture by adjusting the linear position of the array relative to the nerve target. In some instances, this example arrangement of electrodes **716** may sometimes be referred to as an axially-arranged electrode array, axial array, axial lead, and the like terminology. However, it will be understood that in some examples, one or both stimulation elements **1810**A, **1813**A may comprise a different electrode configuration, such as but not limited to some of the example electrode configurations described in association with at least FIGS. **24**A-**30**B.
(159) FIG. **14**B is a diagram including a front view schematically representing an example arrangement **6300** relative to a patient's body **510**, including an example device for, and/or example method of, implantation of a lead **6337**A including a stimulation element **6310**A in stimulating relation to a hypoglossal nerve **505**R and a stimulation element **6313**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **6300** may comprise at least some of substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least some of FIGS. **14**A and/or **1-13**.
(160) As shown in FIG. **14**B, the various components of the example arrangement **6300** are implantable via implant-access incision **609**C (in a manner similar to the arrangement in FIG. **14**A) and via implant-access incision **609**D. In some examples upon forming implant-access incision **609**C, IPG **533** may be implanted subcutaneously, such as in a subcutaneous pocket within pectoral region **532**.
(161) With regard to the various examples of the present disclosure, an implant-access incision comprises a type, size and/or shape of incision adapted to permit subcutaneous implantation of an implantable medical element, such as a stimulation element, sensing element, etc. An implant-access incision is in contrast to a non-implant-access incision, which may be an incision for purposes other than implanting an implantable medical element, such as a stimulation element and/or sensing element.
(162) Upon forming implant-access incision **609**D, a distal portion of lead **6337**A can be implanted subcutaneously, which may include several aspects. As further shown in FIG. **14**B, the lead **6337**A may comprise a proximal portion **6339**A, a body portion **6338**A, and first and second distal

portions **6346**, **6344**A, which extend from body portion **6338**A via a junction **6340**. In some instances, the lead **6337**A may sometimes be referred to as comprising a bifurcated lead at least to the extent that the junction **6340** extending from lead body portion **6338**A is shaped to result in bifurcation of the respective first and second distal portions **6346**, **6344**A from the lead body portion **6338**A.

(163) However, in some examples, the first distal lead portion **6346** of lead **6337**A may be considered a continuance of body portion **6338**A and second distal lead portion **6344**A may be considered as an extension from body portion **6338**A via junction **6340**. Moreover, in some examples, junction **6340** may be formed to cause the second distal portion **6344**A to extend in an opposite orientation from first distal portion **6346** (and stimulation portion **6310**A). In some such examples, the junction **6340** and the portions of the first distal portion **6346**, and second distal portion **6344**A which meet at junction **6340** may be formed as a resilient structure and/or materials so as to bias the second distal portion **6344**A to extend in the opposite orientation from first distal portion **6346** (or vice versa). In some examples, the second distal portion **6344**A may be considered to extend in generally the same orientation as body portion **6338**A of lead **6337**A, at least relative to the orientation of first distal portion **6346** (including stimulation portion **6310**A). As shown in FIG. **14**B, each of stimulation portion **6310**A, **6313**A comprises a linear array of spaced apart electrodes **716** (e.g. ring electrodes or split-ring electrodes), which may be considered an axial arrangement of electrodes **716**. It will be further understood that the electrodes **716** may comprise shapes other than rings, and the stimulation portion **6310**A, **6313**A may comprise other arrangements, such as paddle electrodes, etc. In some examples, the particular arrangement (e.g. number, shape, spacing, orientation, etc.) of electrodes on stimulation portion **6310**A may be different from the particular arrangement of electrodes on stimulation portion **6313**A.

(164) In one aspect, the first distal portion **6346** of lead **6337**A may be implanted subcutaneously, via implant-access incision **609**D, and advanced until stimulation portion **6310**A is in stimulating relation to nerve **505**R. In particular, first distal portion **6346** (including stimulation portion **6310**A) may have a length which is sufficiently short such that first distal portion **6346** (including stimulation portion **6310**A) may be implanted with little or no tunneling from implant-access incision **609**D. Stated differently, the location of implant-access incision **609**D may be selected in sufficiently close proximity to the target stimulation location along nerve **505**R such that little or no tunneling (from implant-access incision **609**D to the target stimulation site) is performed to implant first distal portion **6346** (including stimulation portion **6310**A) in stimulating relation to (a target stimulation location) of nerve **505**R. In some examples, the implant-access incision **609**D may comprise a different location from implant-access incision **609**A (FIGS. **4**, **9**, etc.) in which implant-access incision **609**D is closer to the more distal portions of the nerve **505**R. However, in some examples, the implant-access incision **609**D may correspond to the location of implant-access incision **609**A (e.g. in FIGS. **4**, **9**, etc.).

(165) In some examples of the first distal portion **6346**, the stimulation portion **6310**A may have a length which comprises at least about 50 percent, 60 percent, 70 percent, or 80 percent of the length of the entire first distal portion **6346** extending from junction **6340**. In some examples, this length relationship may sometimes be expressed as the stimulation portion **6310**A having a length comprising a substantial majority of the entire length of the first distal portion **6346**.

(166) In another aspect, prior to implanting a second distal portion **6344**A of lead **6337**A, tunneling (as represented by arrow T**7**) may be performed from implant-access incision **609**D toward nerve **515**R. Thereafter, the second distal portion **6344**A (including stimulation portion **6313**A) may be advanced via the tunnel to place stimulation portion in stimulating relation to nerve **515**R. In some examples of the second distal portion **6366**, the stimulation portion **6313**A may have a length which comprises about 10 percent, 15 percent, 20 percent, 25 percent, or 30 percent of the length of the entire second distal portion **6366** extending from the junction **6340**. Stated differently, the length of the entire second distal portion **6366** extending from the junction **6340** may comprise

several multiples of a length of the stimulation portion **6313**A of the second distal portion.

(167) In one aspect, a proximal end of the body portion **6338** of lead **6337**A is to be implanted to extend toward and into connection with IPG **533**. However, in some examples, tunneling is first performed between the implant-access incision **609**D and implant-access incision **609**C to establish a tunnel (i.e. pathway), as represented by arrow T**8**). It will be understood that the tunneling may be performed starting at either implant-access incision **609**C, **609**D. With the tunnel in place, the proximal portion **6339** is inserted and advanced through implant-access incision **609**D toward IPG **533** until the body portion **6338** extends from the implant-access incision **609**D to implant-access incision **609**C, at which the proximal portion **6339** of lead **6337**A may be further maneuvered to be electrically and mechanically connected to the IPG **533**.

(168) It will be further understood that, in some examples, the particular sequence in which the various aspects of implantation (e.g. first distal portion **6346**, second distal branch **6344**A, body portion **6338**A, IPG **533**) are performed may vary depending on the circumstances, preferences, etc.

(169) As shown later in at least FIGS. **14**E, **14**F, etc. in some examples in which the particular distal portion **6346** or **6344**A (including its stimulation portion) of the lead **6337**A is relatively short and little or no tunneling is performed, the stimulation portion (e.g. **6310**A, **6313**A) may comprise a cuff electrode.

(170) In some examples, as shown in FIG. **14**C, the implant-access incision **609**D (FIG. **14**B) is formed between the mandible bone **6330** and the hyoid bone **6332** so as to place the first distal portion **6346** (including stimulation portion **6310**A) of lead **6337**A in close proximity to at least some portions of the hypoglossal nerve **505**R. In some such examples, the particular implant-access incision **609**D is selected to place the stimulation **6310**A at or near the more distal portions of the hypoglossal nerve **505**R, such as those portions unlikely to innervate retrusor muscles (e.g. styloglossus) of the tongue and likely to innervate protrusor muscles (e.g. genioglossus, geniohyoid) of the tongue/airway. In some examples of these more distal locations, the target stimulation location of the hypoglossal nerve **505**R may be in close proximity to muscle portions innervated by the hypoglossal nerve **505**R, such as being in close proximity to nerve endings of the protrusor-related fibers, fascicles, etc. of the hypoglossal nerve which are more diffusely distributed (vs. well-defined nerve branches) within portions of the genioglossus muscle.

(171) Among other features, the example arrangement **6300** in FIG. **14**B may simplify and expedite a surgical implant procedure at least to the extent that the implant-access incision **609**D may conveniently enable relatively simple implantation of the stimulation portion **6310**A for nerve **505**R, while also including a convenient delivery pathway from the implant-access incision **609**D to the implant site for the stimulation portion **6313**A for nerve **515**R.

(172) FIG. **14**BB is a diagram schematically representing an example arrangement **6347** comprising at least some of substantially the same features and attributes as the example arrangement **6300** in FIG. **14**B, except with at least some portions of the body portion **6338**B and/or distal lead portion **6344**B comprising variable length features (e.g. sigmoid shape, sinusoidal shape, other) which may provide strain relief, among other properties. Accordingly, as shown in FIG. **14**BB, the body portion **6338**B of lead **6337**B extending between the IPG **533** and the junction **6340** (near implant-access incision **609**D) comprises at least one segment including variable length features (e.g. a sigmoid shape, sinusoidal shape, etc.) incorporated into the flexible, resilient structure of the body portion **6338**B.

(173) As further shown in FIG. **14**BB, in some examples a distal portion of the lead body portion **6338**B and/or junction **6340** of lead **6337**B is anchored relative to a non-nerve tissue, as represented by indicator Z**1**. In some such examples, this anchoring (Z**1**) may be implemented via an anchor, such as but not limited to, the example anchors **800**, **830** in FIGS. **6**A, **6**B or other applicable types of anchors disclosed throughout the present disclosure. In some examples,

(174) In some examples, the lead body portion **6338**B may comprise the sole portion of lead **6337**B

which comprises variable length features (e.g. sigmoid shape, sinusoidal shape, and the like).
(175) As further shown in FIG. **14**BB, the distal lead portion **6344**B of lead **6337**B extending between the junction **6340** (near implant-access incision **609**D) and stimulation portion **6313**A comprises at least one segment including variable length features (e.g. a sigmoid shape, sinusoidal shape, etc.) incorporated into the flexible, resilient structure of the distal portion **6344**B. In some examples a distal end (near stimulation portion **6313**A) or other portion of the distal portion **6344**B is anchored relative to a non-nerve tissue, as represented by a second indicator Z**2**. In some examples, this anchoring (Z**2**) may be the sole anchoring for lead **6337**B or may comprise anchoring in addition to anchoring (Z**1**) near or at junction **6340**, in some examples. The second anchoring (Z**2**) may implemented via an anchor element comprising at least some of substantially the same features and attributes as the anchor element(s) used to implement the first anchoring (Z**2**) or may comprise an anchor element(s) having different features.
(176) In some examples, the lead **6337**B may be viewed as having a stimulation element (e.g. stimulation portion **6310**A, such as an axial electrode array, other) interposed between a distal variable length lead portion (e.g. **6344**B) and a proximal variable length lead portion (e.g. **6338**B).
(177) In some examples, anchoring (e.g. Z**1**, Z**2**) may be implemented at other locations along the length of the lead **6337**B in addition to, or instead of, the anchoring shown in FIG. **14**BB.
(178) It will be understood that the variable length features (e.g. sigmoid, sinusoidal, etc.) of lead body portion **6338**B and distal lead portion **6344**B may be implemented in one or more of the lead, lead portions, etc. of any one of the examples of the present disclosure as desired, with or without anchoring Z**1**, Z**2** (e.g. anchor elements **800** in FIG. **6**A, **830** in FIG. **6**B or other types of anchoring) or at least some of the various example anchoring features disclosed throughout the present disclosure.
(179) FIG. **14**D is a diagram including a front view schematically representing an example arrangement **6350** relative to a patient's body **510**, including an example device and/or example method for implantation of a lead **6357** including a stimulation element **6310**A in stimulating relation to a hypoglossal nerve **505**R and a stimulation element **6313**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **6350** may comprise at least some of substantially the same features and attributes as the example arrangement **6300** in association with FIG. **14**B-**14**C, except with a differently located implant-access incision **609**E and variation in the configuration of first and second distal portions **6364**, **6366** of lead **6357** (relative to the configuration of the first, second distal portions **6346**, **6344**A of lead **6337**A in FIG. **14**B).
(180) As shown in FIG. **14**D, the various components of the example arrangement are implantable via implant-access incision **609**C (in a manner similar to the arrangement in FIG. **14**A) and via implant-access incision **609**E. In some examples upon forming implant-access incision **609**C, IPG **533** may be implanted subcutaneously, such as in a subcutaneous pocket within pectoral region **532**. Upon forming implant-access incision **609**E, a distal portion of lead **6357** can be implanted subcutaneously, which may include several aspects.
(181) As further shown in FIG. **14**D, the lead **6357** may comprise a proximal portion **6339**, a body portion **6358**, and first and second distal portions **6364**, **6366**, which extend from body portion **6358** via a junction **6355**. In some examples, the junction **6355** and at least the portions of the first distal portion **6364**, and second distal portion **6366** which meet at junction **6355** may be formed as a resilient structure and/or materials so as to bias the second distal portion **6366** to generally extend in the opposite orientation from first distal portion **6364** (or vice versa).
(182) In one aspect, a second distal portion **6366** of lead **6357** may be implanted subcutaneously, via implant-access incision **609**E, and advanced until stimulation portion **6313**A is in stimulating relation to nerve **515**R. In particular, like first distal portion **6346** of lead **6337** in FIG. **14**B, the second distal portion **6366** (including stimulation portion **6313**A) of lead **6357** in FIG. **14**D has a length which is sufficiently short such that second distal portion **6366** (including stimulation

portion **6313**A) may be implanted with little or no tunneling from implant-access incision **609**E. Stated differently, the location of implant-access incision **609**E may be selected in sufficiently close proximity to the target stimulation location along nerve **515**R such that little or no tunneling (from implant-access incision **609**E to the target stimulation site) is performed to implant second distal portion **6366** (including stimulation portion **6313**A) to be in stimulating relation to (a target stimulation location) of nerve **515**R.

(183) In some examples of second distal portion **6366**, the stimulation portion **6313**A may have a length which comprises at least about 50 percent, 60 percent, 70 percent, or 80 percent of the length of the entire second distal portion **6366** extending from junction **6355**. In some examples, this length relationship may sometimes be expressed as the stimulation portion **6313**A having a length comprising a substantial majority of the entire length of the second distal portion **6366**.

(184) In another aspect, prior to implanting a first distal portion **6364** of lead **6337**, tunneling (as represented by arrow T**9**) may be performed from implant-access incision **609**E toward nerve **505**R. Thereafter, the first distal portion **6364** (including stimulation portion **6310**A) is advanced via the tunnel T**9** to place stimulation portion **6310**A in stimulating relation to nerve **505**R. In some examples of the first distal portion **6364**, the stimulation portion **6310**A may have a length which comprises about 10 percent, 15 percent, 20 percent, 25 percent, or 30 percent of the length of the entire first distal portion **6364** extending from the junction **6355**. Stated differently, the length of the entire first distal portion **6364** extending from the junction **6355** may comprise several multiples of a length of the stimulation portion **6310**A of the first distal portion **6364**.

(185) In one aspect, the body portion **6358** of lead **6357** is to be implanted to extend toward and into connection with IPG **533**. However, in some examples, tunneling is first performed between the implant-access incision **609**E and implant-access incision **609**C to establish a tunnel (i.e. pathway), as represented by arrow T**10**. It will be understood that the tunneling may be performed starting at either implant-access incision **609**C, **609**E. With the tunnel in place, the proximal portion **6339** of lead **6357** is inserted and advanced through implant-access incision **609**E toward IPG **533** until the body portion **6358** extends from the implant-access incision **609**E to at least implant-access incision **609**C, at which the proximal portion **6339** of lead **6357** may be further maneuvered to be electrically and mechanically connected to the IPG **533**.

(186) It will be further understood that, in some examples, the particular sequence in which the various aspects of implantation (e.g. first distal portion **6364**, second distal branch **6366**, body portion **6358**, IPG **533**) are performed may vary in some instances.

(187) In some examples, the particular location of the implant-access incision **609**E may correspond to a location within the head-and-neck region by which one can directly access a portion of the ansa cervicalis-related nerve corresponding to a desired stimulation location. In some such examples, the implant-access incision **609**E may enable direct access for implantation of a stimulation element (e.g. cuff electrode, stimulation portion, etc.) to place the stimulation element in stimulating relation to one or more of the example stimulation locations A, B, or C, as generally described in association with at least FIGS. **2**, **32**A, **32**C and as described more specifically in association with FIG. **16** (stimulation location A), FIGS. **22**A, **32**B (stimulation location B), and FIG. **32**D (stimulation location C). It will be understood that other stimulation locations of/along the ansa cervicalis-related nerve **316** may be accessed via implant-access incision **609**E. Moreover, in some examples, these aspects associated with implant-access incision **609**E are applicable to implant-access incision **609**B (e.g. FIG. **4**) shown in some other example arrangements in the present disclosure.

(188) FIG. **14**E is a diagram including a front view schematically representing an example arrangement **6400** relative to a patient's body **510**, including an example device for, and/or example method of, implantation of a lead **6437** including a stimulation element **6411**A in stimulating relation to a hypoglossal nerve **505**R and a stimulation element **6313**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **6400** may

comprise at least some of substantially the same features and attributes as the example arrangement **6300** in association with FIG. **14**B-**14**C, except with first distal portion **6346** including a cuff electrode **6411**A in FIG. **14**E instead of the stimulation portion **6310**A (e.g. axial-style array of electrodes **716**) in FIG. **14**B. In substantially all other respects, example arrangement **6400** comprises substantially the same features and attributes as arrangement **6300** in FIG. **14**B. As shown in FIG. **14**E, the lead **6437** may comprise a proximal portion **6339**, a body portion **6438**, and first and second distal portions **6346**, **6344**, which extend from body portion **6438** via the junction **6340**.

(189) In a manner similar to the example arrangement **6300** in FIG. **14**B, in the example arrangement **6400** in FIG. **14**E, the relatively short length of first distal portion **6346** may be particularly beneficial for implanting a cuff electrode **6411**A relative to nerve **505**R in close proximity to the implant-access incision **609**D at least because such cuff electrodes **6411**A are typically not suitable for introduction, advancing, etc. via a tunneled path in the same way that an axial lead (e.g. stimulation portion **6313**A) would be. In a manner similar for the first distal portion **6346** (including stimulation portion **6310**A) in FIG. **14**B, in some examples the cuff electrode **6411**A in FIG. **14**E may have a length which comprises at least about 50 percent, 60 percent, 70 percent, or 80 percent of the length of the entire first distal portion **6346** extending from junction **6340**. In some examples, this length relationship may sometimes be expressed as the cuff electrode **6411**A (e.g. one type of stimulation portion) having a length comprising a substantial majority of the entire length of the first distal portion **6346**. Conversely, in some examples of the second distal portion **6344**, the stimulation portion **6313**A may have a length which comprises about 10 percent, 15 percent, 20 percent, 25 percent, or 30 percent of the length of the entire second distal portion **6344** extending from the junction **6340**. Stated differently, the length of the entire second distal portion **6344** extending from the junction **6340** may comprise several multiples of a length of the stimulation portion **6313**A of the second distal portion **6344**.

(190) FIG. **14**F is a diagram including a front view schematically representing an example arrangement **6450** relative to a patient's body **510**, including an example device for, and/or example method of, implantation of a lead **6457** including a cuff electrode **6414**A in stimulating relation to an ansa cervicalis-related nerve **505**R and a stimulation portion **6313**A in stimulating relation to a hypoglossal nerve **505**R. In some examples, the example arrangement **6450** may comprise at least some of substantially the same features and attributes as the example arrangement **6350** in association with FIG. **14**D, except with second distal portion **6366** of lead **6457** in FIG. **14**F including a cuff electrode **6414**A instead of a stimulation portion **6313**A (e.g. including an axial-style array of electrodes **716**) as in the example arrangement **6350** of FIG. **14**D. In substantially all other respects, example arrangement **6450** comprises substantially the same features and attributes as arrangement **6350** in FIG. **14**D. As shown in FIG. **14**F, the lead **6457** may comprise a proximal portion **6339**, a body portion **6358**, and first and second distal portions **6364**, **6366**, which extend from body portion **6358** via a junction **6355**.

(191) In a manner similar to the example arrangement **6350** in FIG. **14**D, in the example arrangement **6450** in FIG. **14**F, the relatively short length of second distal portion **6366** may be particularly beneficial for implanting a cuff electrode **6414**A relative to nerve **515**R in close proximity to the implant-access incision **609**E at least because such cuff electrodes **6414**A are typically not suitable for introduction, advancing, etc. via a tunneled path in the same way that an axial lead (e.g. stimulation portion **6313**A) would be. In a manner similar for the second distal portion **6366** (including stimulation portion **6313**A) in FIG. **14**D, in some examples the cuff electrode **6414**A in FIG. **14**F may have a length which comprises at least about 50 percent, 60 percent, 70 percent, or 80 percent of the length of the entire second distal portion **6366** extending from junction **6355**. In some examples, this length relationship may sometimes be expressed as the cuff electrode **6414**A (e.g. one type of stimulation portion) having a length comprising a substantial majority of the entire length of the first distal portion **6346**. Conversely, in some examples of the

first distal portion **6364**, the stimulation portion **6310**A may have a length which comprises about 10 percent, 15 percent, 20 percent, 25 percent, or 30 percent of the length of the entire first distal portion **6364** extending from the junction **6355**. Stated differently, the length of the entire first distal portion **6364** extending from the junction **6355** may comprise several multiples of a length of the stimulation portion **6310**A of the first distal portion **6364**.

(192) FIG. **14**G is a diagram including a front view schematically representing an example arrangement **6500** relative to a patient's body **510**, including an example device for, and/or example method of, implantation of a lead **6537** including a stimulation element **6310**A in stimulating relation to a hypoglossal nerve **505**R and a stimulation element **6313**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **6500** may comprise at least some of substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least some of FIGS. **1-14**A.

(193) As shown in FIG. **14**G, the various components of the example arrangement are implantable via a single implant-access incision **609**C (in a manner similar to the arrangement in FIG. **14**A). In some examples upon forming single implant-access incision **609**C, IPG **533** may be implanted subcutaneously, such as in a subcutaneous pocket within pectoral region **532**. As further shown in FIG. **14**G, the lead **6537** may comprise a proximal portion **6539**, and first and second lead body portions **6533**, **6549**, which extend from a junction **6534**, which in turn extends distally from proximal portion **6539**. In some examples, the respective lead body portions **6533**, **6549** may sometimes be referred to as being bifurcated. In some examples, lead body portion **6533** comprises a distal portion **6547** which comprises stimulation portion **6310**A while lead body portion **6549** comprises a distal portion **6548**, which comprises stimulation portion **6313**A.

(194) In another aspect, prior to implanting a lead body portions **6533**, **6549** of lead **6537**, tunneling may be performed from implant-access incision **609**C toward nerve **515**R and nerve **505**R, as represented by array T**11**. Thereafter, the first lead body portion **6533** (including stimulation portion **6310**A) and second lead body portion **6549** (including stimulation portion **6313**A) are advanced via the tunnel (T**11**) to place the stimulation portion **6310**A in stimulating relation to nerve **505**R and to place the stimulation portion **6313**A in stimulating relation to nerve **515**R.

(195) In one aspect, the proximal portion **6539** of lead **6537** is inserted and advanced through implant-access incision **609**C toward already-implanted IPG **533** so that proximal portion **6539** may be electrically and mechanically connected to the IPG **533**.

(196) It will be further understood that, in some examples, the particular sequence in which the various aspects of implantation (e.g. lead body portions **6533**, **6549**, **6539**, IPG **533**) are performed may vary depending on the circumstances, preferences, etc.

(197) Among other features, the example arrangement **6500** provides for single implant-access incision and a single tunnel to thereby simplify and expedite a surgical implantation procedure.

(198) FIG. **14**H is a diagram including a front view schematically representing an example arrangement **6550** relative to a patient's body **510**, including an example stimulation device (e.g. **6552**) for, and/or related example method of, implantation, with the arrangement **6550** including a stimulation portion **6310**A in stimulating relation to a hypoglossal nerve **505**R and a stimulation portion **6313**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **6550** may comprise at least some of substantially the same features and attributes as lead **6337** in FIG. **14**B, except omitting lead body portion **6338** and omitting IPG **533** in pectoral region **532**. In particular, instead of IPG **533**, in some examples stimulation device **6552** comprises a microstimulator **6575** to which the first lead portion **6546** (including stimulation portion **6310**A) and second lead portion **6566** (including stimulation portion **6313**A) are directly connected such that no intermediate or body portion (of an implantable medical device) is interposed between the microstimulator **6575** (which provides stimulation circuitry, power, etc.)

and the lead portions **6566**, **6564** (each of which include an array of stimulation electrodes).
(199) As shown in FIG. **14**H, in some examples the various components of the example stimulation device **6552** are implantable via a single implant-access incision **609**D (in a manner similar to the arrangement in FIG. **14**A) and without forming a second implant-access incision, such as incision **609**C in FIG. **14**B which otherwise would have been used to implant the now omitted IPG **533**. As further shown in FIG. **14**H, in some examples the stimulation device **6552** may comprise the microstimulator **6575**, and first and second lead body portions **6566**, **6564**, which extend from the microstimulator **6575**.
(200) In some examples, and with further reference to previously mentioned FIG. **14**C, the implant-access incision **609**D in FIG. **14**H is formed between the mandible bone **6330** and the hyoid bone **6332** so as to place the first lead portion **6566** (including stimulation portion **6310**A) in close proximity to at least some portions of the hypoglossal nerve **505**R. In some such examples, the particular implant-access incision **609**D is selected to place the stimulation **6310**A at or near the more distal portions of the hypoglossal nerve **505**R, as previously mentioned.
(201) As further shown in FIG. **14**H, in some examples the stimulation device **6552** may be formed, assembled, etc. to cause the respective lead portions **6566** and **6546** to extend outward from a periphery of the housing **6577** of the microstimulator **6575** to be spaced apart from each other by an angle (α). As further shown in FIG. **14**HH, the angle (α) at which such lead portions (as represented by solid lines L**1**, L**2**) will extend outwardly in their spaced apart configuration may be from 0 to 360 degrees. Moreover, the arrow shown in FIG. **14**HH represents just one example in which the two lead portions (L**1**, L**2**) would be spaced apart by 180 degrees such that the respective lead portions **6566**, **6564** may extend from opposite portions (e.g. ends, sides, etc.) of the periphery of the microstimulator housing **6577**. It will be understood that a housing of such an example microstimulator **6575** may comprise a wide variety of shapes, sizes, etc. with the particular obround shape shown in FIG. **14**HH being just one example shape.
(202) With further reference to FIG. **14**HH, in some examples, the particular angle (α) by which lead portions L**1**, L**2** are spaced apart about a periphery of the microstimulator housing **6577** may be fixed. In some such examples, this fixation may arise from the lead portions (e.g. **6546**, **6566** in FIG. **14**H) being formed as a unitary member with the microstimulator housing **6577**. However, in some examples, the lead portions (e.g. **6564**, **6566** in FIG. **14**H) are removably attachable (at or near the time of implantation) relative to the microstimulator housing **6577**, such as via separate connection ports (e.g. P**1**, P**2** shown in FIG. **14**HH) formed in the microstimulator housing **6577** at spaced apart locations which correspond to the angle (α) at which the lead portions **6566**, **6564** are intended to extend outwardly from the microstimulator housing **6577**.
(203) In some examples, multiple connection ports (e.g. P**1**, P**2**) may be adjacent each other on a same side of a microstimulator housing such as (but not limited to) both being in proximity to line A in FIG. **14**HH.
(204) Moreover, while FIGS. **14**H and **14**HH depicts just two lead portions **6566**, **6546** (or L**1**, L**2**) extending from the microstimulator housing **6577**, it will be understood that in some examples, more than two lead portions **6566**, **6546** may extend from the microstimulator housing **6577**. It will be further understood that some examples, some additional lead portions may comprise a sensing lead portion versus a stimulation lead portion. Moreover, in some examples, at least some of the electrodes of a stimulation lead portion may sometimes be used for sensing.
(205) Among other aspects, with the lead portions (e.g. **6566**, **6546**) extending outwardly from the microstimulator housing **6577** from different spaced apart locations about a periphery of the housing **6577** (as shown in FIG. **14**H, **14**HH), this example arrangement may simplify the implantation of a multiple lead stimulation device (e.g. **6552**) because each lead portion (e.g. **6566**, **6564**) will already be biased to extend in an orientation (relative to the microstimulator housing **6577**) to align and position the stimulation portions **6313**A, **6310**A relative to the target nerve **515**R, **505**R. Moreover, this example arrangement **6550** helps make practical an implantation

procedure (FIG. **14**H) using a single implant-access incision at least because the single implant-access incision **609**D is generally interposed (in at least some examples) between the target nerve locations of the respective target nerve locations (e.g. **515**R, **505**R). Accordingly, upon formation of the single implant-access incision (e.g. **609**D) at an intermediate location between the respective target stimulation locations (at **505**R, at **515**R), the implantation of the stimulation device **6552** includes positioning the microstimulator **6575** at intermediate location between the two target stimulation locations (at **505**R, **515**R) such that the lead portions may extend outwardly toward the target stimulation locations in a natural way to simplify advancement of the respective lead portions toward their target stimulation locations. In addition, such arrangements may reduce strain on a lead portion to the extent that maneuvering multiple lead portions extending from a microstimulator **6575** into their desired orientations within the body may induce strain under some circumstances.

(206) In some examples a kit of several different stimulation devices (each including a microstimulator and at least two already connected lead portions L**1**, L**2**) may be offered in which each different stimulation device in the kit comprises lead portions (L**1**, L**2**) which extend from the microstimulator at a different angle (a in FIG. **14**HH) from each other. For example, one stimulation device in the kit may have lead portions (e.g. L**1**, L**2** in FIG. **14**HH) which extend from each other by angle (α) of about 130 degrees while a different stimulation device in the kit may have lead portions (e.g. L**1**, L**2**) which extend from each other by an angle (α) of about 170 degrees. Accordingly, upon embarking on an implantation procedure for a stimulation device, a surgeon may select a stimulation device from the kit with an angle (α) suited to ease implantation of the stimulation device (including a microstimulator and lead portions) in view of the particular target stimulation locations of the nerves for which a stimulation portion (e.g. electrode array, cuff electrode, etc.) is be implanted.

(207) As previously noted in connection with the example arrangement of at least FIG. **14**B, the implant-access incision **609**D may be selected such that at least one lead portion (e.g. **6546**) may be implanted from the implant-access incision **609**D without tunneling.

(208) Conversely, as further shown in FIG. **14**H and in a manner similar to that described in connection with at least FIG. **14**B, **14**E, tunneling (T**7**) may be formed via the implant-access incision **609**D to create a path to advance lead portion **6566** subcutaneously until stimulation portion **6313**A becomes aligned and positioned relative to the ansa cervicalis-related nerve **515**R, as shown in FIG. **14**H.

(209) Moreover, it will be understood that with respect to at least FIGS. **14**F, **14**H-**14**K, in some examples tunneling may be performed in two separate orientations, with a first tunnel to be established for a first lead (including a stimulation portion) for stimulating the hypoglossal nerve **505**R and with a second tunnel for a second lead (including a stimulation portion).

(210) As further shown in FIG. **14**H, in some examples the example arrangement **6550** may comprise a recharge element **6575** for recharging a power supply of the microstimulator **6575**. In some such examples, the recharge element **6575** and/or microstimulator **6575** may comprise at least some of substantially the same features and attributes as the example arrangement **2700** as later described in association with at least FIG. **21**.

(211) With regard to the example arrangement **6550** in FIG. **14**H, in some examples the stimulation portion **6310**A in FIG. **14**H may have a length which comprises at least about 50 percent, 60 percent, 70 percent, or 80 percent of the length of the entire first lead portion **6546** extending from microstimulator **6575**. In some examples, this length relationship may sometimes be expressed as the stimulation portion **6310**A having a length comprising a substantial majority of the entire length of the first lead portion **6546**. Conversely, in some examples of the second lead portion **6566**, the stimulation portion **6313**A may have a length which comprises about 10 percent, 15 percent, 20 percent, 25 percent, or 30 percent of the length of the entire second lead portion **6566** extending from the microstimulator **6575**. Stated differently, the length of the entire second lead portion **6566**

extending from the microstimulator **6575** may comprise several multiples of a length of the stimulation portion **6313**A of the second lead portion **6566**. It will be understood that these same "relative length" relationships are exhibited in the example arrangements described below in relation to at least FIGS. **14**I-**14**K with regard to analogous lead portions and stimulation portions of each example arrangement.

(212) FIG. **14**I is a diagram including a front view schematically representing an example arrangement **6600** relative to a patient's body **510**, including an example stimulation device **6602** for, and/or related example method of, implantation, with the arrangement **6600** including a stimulation portion **6310**A in stimulating relation to a hypoglossal nerve **505**R and a stimulation portion **6313**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **6600** may comprise at least some of substantially the same features and attributes as the example arrangement **6550** in FIG. **14**H, **14**HH, except with the single implant-access incision **609**E having a different location from implant-access incision **609**D (FIG. **14**H) and the roles of the respective lead portions **6666** and **6664** in FIG. **14**I being reversed relative to lead portions **6564** and **6566** in FIG. **14**H.

(213) In particular, implant-access incision **609**E is formed at a location in reasonably close proximity to the ansa cervicalis-related nerve **515**R by which a microstimulator **6575** may be implanted such that lead portion **6666** (including stimulation portion **6313**A) becomes suitably aligned and positionable in stimulation relation to the ansa cervicalis-related nerve **515**R. Meanwhile, after and via tunneling (like T**9** in FIG. **14**D), lead portion **6664** may be advanced subcutaneously via the tunnel until stimulation portion **6310**A is suitably aligned and positioned in stimulating relation to hypoglossal nerve **505**R.

(214) FIG. **14**J is a diagram including a front view schematically representing an example arrangement **6620** relative to a patient's body **510**, including an example stimulation device **6622** for, and/or related example method of, implantation, with the arrangement **6620** including a stimulation portion **6310**A in stimulating relation to a hypoglossal nerve **505**R and a stimulation portion provided as a cuff electrode **6414**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **6620** may comprise at least some of substantially the same features and attributes as the example arrangement **6600** in FIG. **14**I, except with the cuff electrode **6414**A in FIG. **14**J replacing the stimulation portion **6313**A in FIG. **14**I.

(215) At least because the single implant-access incision **609**E is formed at a location in reasonably close proximity to the ansa cervicalis-related nerve **515**R, implanting the cuff electrode **6414**A (on lead portion **6666**) into stimulating relation to the ansa cervicalis-related nerve **515**R may be performed in a relatively simple manner without tunneling. Moreover, this relatively direct access may greatly facilitate implanting a cuff electrode **6414**A, which may include more maneuvering in, around, and among tissues in the surgical work field than simply pushably advancing an axial, cylindrically-shaped stimulation portion. Implantation of the cuff electrode **6414**A may be favored in some instances, such as but not limited to, reliably establishing stimulating relation of a stimulation element (e.g. carrier with electrodes) relative to a nerve which may be challenging (in some patients) to ensure stable positioning of a non-cuff electrode type of stimulation element. For example, directly visualizing the ansa cervicalis-related nerve **316** may better enable probing for/among different branches of the ansa cervicalis-related nerve **316** to identify the stimulation location (e.g. A, B, C in FIG. **2**, or other locations) with the best muscle response to a test stimulation(s). Upon making such identification, the cuff electrode **6414**A may be then placed at the identified location and secured in place to establish reliable chronic implantation and robust stimulating relation to the target stimulation location of the nerve. Among other aspects, using a "direct access" implant-access incision may enhance visualization and probing, which in turn may enable greater flexibility and success in performing implantations in view of the anatomical variations among different patients.

(216) FIG. **14**K is a diagram including a front view schematically representing an example

arrangement **6650** relative to a patient's body **510**, including an example stimulation device **6652** for, and/or related example method of, implantation, with the arrangement **6650** including a stimulation portion (provided as a cuff electrode **6411**A) in stimulating relation to a hypoglossal nerve **505**R and a stimulation portion **6413**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **6650** may comprise at least some of substantially the same features and attributes as the example arrangement **6550** in FIG. **14**H, except with the cuff electrode **6411**A in FIG. **14**K replacing the stimulation portion **6310**A in FIG. **14**H.

(217) At least because the single implant-access incision **609**D is formed at a location in reasonably close proximity to the hypoglossal nerve (e.g. more distal portions thereof), implanting the cuff electrode **6411**A (on lead portion **6666**) into stimulating relation to the hypoglossal nerve **515**R may be performed in a relatively simple manner without tunneling. Moreover, this relatively direct access may greatly facilitate implanting a cuff electrode **6411**A, which may include more maneuvering in, around, and among tissues in the surgical work field than simply pushably advancing an axial, cylindrically-shaped stimulation portion. Implantation of the cuff electrode **6411**A may be favored in some instances, such as but not limited to, reliably establishing stimulating relation of a stimulation element (e.g. carrier with electrodes) relative to a nerve which may be challenging (in at least some patients) to ensure stable positioning of a non-cuff electrode type of stimulation element. For example, directly visualizing the hypoglossal nerve **505**R may better enable probing for/among different branches (and/or distal nerve endings) of the hypoglossal nerve **505**R to identify the stimulation location with the best muscle response to a test stimulation(s). Upon making such identification, the cuff electrode **6411**A may be then placed at the identified location and secured in place to establish reliable chronic implantation and robust stimulating relation to the target stimulation location of the nerve. Among other aspects, using a "direct access" implant-access incision may enhance visualization and probing, which in turn may enable greater flexibility and success in performing implantations in view of the anatomical variations among different patients.

(218) FIGS. **14**L-**14**R relate to at least some example methods of implantation and/or methods of stimulation therapy. For instance, various examples methods of stimulation therapy are described in association with at least FIG. **3**A which may comprise applying electrical stimulation to a left hypoglossal nerve, a right hypoglossal nerve, a left ansa cervicalis-related nerve, and/or a right ansa cervicalis-related nerve. In some such examples, the therapy may be applied unilaterally or bilaterally for the same type of nerve (e.g. just the hypoglossal nerves or just the ansa cervicalis-related nerves), unilaterally for different types of nerves (e.g. stimulating nerves on just the right side of the body or stimulating nerves on just the left side of the body), or bilaterally for different types of nerve (e.g. stimulating the left hypoglossal nerve and the right ansa cervicalis-related nerve, vice versa). With this in mind, FIGS. **14**L-**14**R comprise some example implementations of the example arrangement in FIG. **3**A which are directed to bilateral stimulation of the left and right hypoglossal nerves and unilateral stimulation of a single ansa cervicalis-related nerve (e.g. left or right). In particular, FIGS. **14**L-**14**R relate to at least some examples of methods of implantation and example stimulation devices for implementing the bilateral stimulation of the left and right hypoglossal nerves and unilateral stimulation of a single ansa cervicalis-related nerve (e.g. left or right). Various example methods of stimulation therapy regarding whether (and how) different nerves (e.g. left HGN, right HGN, left AC, and right AC) are stimulated simultaneously, alternately, staggered, sequentially, synchronized, non-synchronized, etc. are provided in at least FIGS. **1**-**3**C, **16**, and **32**A-**50** and/or other various therapy examples throughout the present disclosure. Moreover, the implantation of the various elements of the example stimulation devices in association with at least FIGS. **14**L-**14**R also may be further implemented via at least some of the example arrangements (e.g. devices and methods) described in association with the delivery tools, anchoring elements, lead connectability features, etc. of at least FIGS. **1**-**32**C, **49**B-**51**B, and the like.

(219) While FIGS. **14**L-**14**R illustrate examples in which stimulation may be applied to a right ansa cervicalis-related nerve in combination with bilateral stimulation including the left and right hypoglossal nerves, it will be understood that these examples are equally applicable to example implementations in which stimulation is to be applied a left ansa cervicalis-related nerve in combination with bilateral stimulation including the left and right hypoglossal nerves. Moreover, in some examples, using the similar elements, methods, etc. as described in association with at least FIGS. **14**L-**14**R, some example methods of implantation and/or example stimulation devices comprise bilateral stimulation of a left ansa cervicalis-related nerve (e.g. **515**L) and a right ansa cervicalis-related nerve (e.g. **515**R).

(220) FIG. **14**L is a diagram including a front view schematically representing an example arrangement **3400** including an example device for, and/or example method of, implantation of a stimulation device **3405**. The stimulation device **3405** is adapted for providing bilateral stimulation of a left and right hypoglossal nerve **505**R, **505**L, and unilateral stimulation of an ansa cervicalis-related nerve **515**R (e.g. **316** in FIG. **2**A) in order to maintain and/or restore upper airway patency, such as to treat obstructive sleep apnea and/or other sleep disordered breathing. Accordingly, FIG. **14**L depicts a head-and-neck region **520** of a patient's body **510**, while denoting at least some anatomical landmarks such as a chin **509** and providing dashed line **3401** to distinguish between the right (RIGHT) and left (LEFT) sides of the patient's body **510**.

(221) As shown in FIG. **14**L, the stimulation device **3405** may comprise a stimulation lead **3410** to deliver therapeutic stimulation signals, generated via an implantable pulse generator (IPG) **533** and applied through at least one of a stimulation portion **6310**A, stimulation portion **6310**B and/or a cuff electrode **6414**A. In some examples, the stimulation portions **6310**A, **6310**B may comprise a linear array of spaced apart electrodes (e.g. ring, split ring, and the like), which may sometimes be referred to an axial lead or axial stimulation portion. In some examples, the stimulation lead **3410** may comprise a proximal portion **3412**, body portion **3414**, junction **3418**, lead portion **3416**, and distal lead portions **3420**, **3422**. The proximal portion **3412** of lead **3410** is connectable to the IPG **533**, and the body portion **3414** extends distally from the proximal portion **3412**. The junction **3418** connects the distal lead portions **3420**, **3422** and lead portion **3416** relative to each other and relative to body portion **3414** of lead **3410**. In some examples, the junction **3418** may sometimes be referred to as including or defining a bifurcation point for distal lead portions **3420**, **3422** and lead portion **3416** relative to each other and/or relative to body portion **3414** of lead **3410**.

(222) In some examples, these elements may be implanted via an implant-access incision **609**E, which may comprise the sole implant-access incision via which the elements of the stimulation device **3405** are implanted, in some examples. As noted elsewhere, using a single implant-access incision may reduce surgical complexity, increase patient comfort, reduce procedure time, etc. in at least some examples. As shown in FIG. **14**L, the implant-access incision **609**E is located in close proximity to the ansa cervicalis-related nerve **515**R.

(223) It will be understood that in some examples, more than one implant-access incision may be used.

(224) In one aspect, a method of implantation may comprise forming the implant-access incision **609**E and then introducing and advancing the lead portion **3416** to place the cuff electrode **6414**A into stimulating relation to the ansa cervicalis-related nerve **515**R, such as on one side (e.g. right side) of the body. From the single implant-access incision **609**E, a tunnel T**8** may be formed subcutaneously toward a pertinent portion of the right hypoglossal nerve **505**R, followed by introducing and advancing distal lead portion **3420** to place stimulation portion **6310**A in stimulating relation to a target stimulation portion on the right hypoglossal nerve **505**R. Similarly, also via implant-access incision **609**E, a tunnel T**9** may be formed relative to the left hypoglossal nerve **505**L, followed by introducing and advancing distal lead portion **3422** to place stimulation portion **6310**B in stimulating relation to a target stimulation portion on the left hypoglossal nerve **505**R. Via this arrangement, the respective distal lead portions **3420**, **3422** provide a mechanism by

which bilateral stimulation of the left and right hypoglossal nerves may be delivered. As previously noted elsewhere, in some examples the stimulation portions **6310**A, **6310**B may comprise an axial lead-type of stimulation portion including a linear array of spaced apart electrodes (e.g. ring electrodes, split-ring electrodes, and the like). As further noted elsewhere, each stimulation portion **6310**A, **6310**B and supporting distal lead portion **3420**, **3422** (respectively) may comprise anchor element(s) as described in various examples of the present disclosure, such as but not limited to those in FIGS. **30**B-**32**B.

(225) Among other aspects, a cuff-style electrode **6414**A secured at the ansa cervicalis-related nerve **515**R may help ensure robust engagement of the stimulation electrodes in stimulating relation of the stimulation electrodes relative to nerve **515**R in view of the smaller size of the nerve, in view of the type and location of the nerve **515**, and/or in view of the type, size, etc. of surrounding non-nerve tissues. In other aspects, the axial type stimulation portions **6310**A, **6310**B are more conducive to introduction and advancement through a tunnel (e.g. T**8**, T**9**) than a cuff electrode, and also may help avoid having to make a separate implant-access incision near nerves **505**R, **505**L, thereby helping to implement an implantation procedure with fewer implant-access incisions. By doing so, surgical time and complexity may be reduced, patient comfort increased, etc.

(226) In a further aspect of a method of implantation, from the single implant-access incision **609**E, a tunnel T**10** may be subcutaneously formed to a suitable location at which the IPG **533** may be implanted, such as in the pectoral region **532** of the patient's body. In some examples, an additional implant-access incision may formed in close proximity to the location at which the IPG **533** is to be implanted. With tunnel T**10** formed, via implant-access incision **609**E a body portion **3414** of lead **3410** is introduced and advanced to the implant location of IPG **533** to enable proximal portion **3412** of stimulation lead **3410** to be connected (electrically and mechanically) to the IPG **533**.

(227) While the example arrangement shown in FIG. **14**L provides a bifurcation point near the implant-access incision **609**E, in some examples the stimulation lead may comprise a bifurcation point near the IPG **533** such as (but not limited to) the example implementation in later described FIG. **14**N.

(228) FIG. **14**M is a diagram including a front view schematically representing an example arrangement **3440** including an example device for, and/or example method of, implantation of a stimulation device **3450**. In some examples, the stimulation device **3450** may comprise at least some of substantially the same features and attributes as the stimulation devices, methods, etc. as previously described in association with at least FIG. **14**L, except with the stimulation device **3450** in FIG. **14**M including a microstimulator **6575** (e.g. FIG. **14**H) instead of the IPG **533** in FIG. **14**L (and omitting the body portion **3414** of lead **3410** associated with the IPG **533**).

(229) As shown in FIG. **14**M, in some examples the microstimulator **6575** is implanted at or in close proximity to the target stimulation location of the ansa cervicalis-related nerve **515**R.

(230) Accordingly, with further reference to FIG. **14**M in comparison to FIG. **14**L, the stimulation device **3450** in FIG. **14**M comprises the same distal lead portions **3420**, **3422** and their respective stimulation portions **6310**A, **6310**B for delivering bilateral stimulation to the respective right and left hypoglossal nerves **505**R, **505**L. Moreover, the stimulation device **3450** in FIG. **14**M also comprises the same lead portion **3416** and cuff electrode **6414**A in stimulating relation to the ansa cervicalis-related nerve **515**R.

(231) As further shown in FIG. **14**M, the microstimulator **6575** is directly connected to the respective lead portions **3420**, **3422**, **3416** with distal lead portions **3420**, **3422** having a bifurcation point **3425** at or near a housing of the microstimulator **6575**. In some examples, the lead portion **3416** may extend directly from the microstimulator **6575** as shown in FIG. **14**M or in some examples, may extend from the same bifurcation point **3425** (or junction) as the distal lead portions **3420**, **3422**. As noted elsewhere in relation to some examples of the present disclosure, the microstimulator **6575** may comprise stimulation/control circuitry, a power source (e.g.

rechargeable), and may be in communication with an external power recharging device, element, and the like.
(232) It will be further understood that in some examples, at least some of the stimulation portions (e.g. **6310**A, **6310**B), cuff electrodes (e.g. **6414**A) may be in wireless communication with the microstimulator **6575** such that one or more of the associated lead portions (e.g. supporting a respective stimulation portion or cuff electrode) may be omitted in some examples.
(233) FIG. **14**N is a diagram including a front view schematically representing an example arrangement **3460** including an example device for, and/or example method of, implantation of a stimulation device **3469**. In some examples, the stimulation device **3469** may comprise at least some of substantially the same features and attributes as the stimulation devices, methods, etc. as previously described in association with at least FIG. **14**L, except with the stimulation device **3469** including a stimulation lead array **3470** having bifurcated leads **3464**, **3476** extending from the IPG **533** and distal lead portions **3466**, **3468** of lead **3464** originating from a bifurcation point **3467** much closer to the target stimulation locations at the hypoglossal nerves **505**R, **505**L. As further shown in FIG. **14**N, the stimulation device **3469** comprises a cuff electrode **6414**A supported on a distal portion of lead **3476** for stimulating the ansa cervicalis-related nerve **515**R. The stimulation device **3469** also comprises stimulation portions **6310**A, **63106** (e.g. axial/linear electrode array) on respective distal lead portions **3466**, **3468** for respectively stimulating the hypoglossal nerves **505**R, **505**L on opposite sides (RIGHT, LEFT) of the patient's neck.
(234) With this framework in mind and as shown in FIG. **14**N, one aspect of an example method of implantation may comprise forming the implant-access incision **609**E, and implanting (via the incision **609**E) the cuff electrode **6414**A to be in stimulating relation to the ansa cervicalis-related neve **515**R. Via the implant-access incision **690**E, a tunnel (T**13**) may be formed to an implant location for IPG **533** and lead **3476** may be introduced and advanced via tunnel T**13** to enable proximal portion **3474** of lead **3476** to be connected (electrically and mechanically) to an IPG **533**.
(235) In some examples, an additional implant-access incision may be formed near IPG **533** to facilitate implantation of the IPG **533**, lead **3476**, and/or lead **3464**.
(236) In another aspect of the method of implantation, in some examples, an implant-access incision **609**D may be formed in close proximity to expected target stimulation locations of the hypoglossal nerve **505**R on the same side (RIGHT) of the neck as the implant-access incision **609**E. From the implant-access incision **609**D, a tunnel T**11** is formed to the target stimulation location of the left hypoglossal nerve **505**L. Via implant-access incision **609**D, distal lead portion **3466** (including stimulation portion **6310**A) and distal lead portion **3468** (including stimulation portion **63106**) are introduced and advanced subcutaneously to the target stimulation locations of the respective right and left hypoglossal nerves **505**R, **505**L to yield the chronically implanted configuration of stimulation portions **6310**A, **63106** shown in FIG. **14**N.
(237) In some examples, the implant-access incision **609**D is in close proximity to the target stimulation location of the right hypoglossal nerve **505**R such that little or no tunneling may be used to place distal lead portion **3466** in stimulating relation to the right hypoglossal nerve **505**R.
(238) Via implant-access incisions **609**D and/or **609**E, a tunnel T**12** is formed to enable introduction and advancement of lead **3464** between implant-access incision **606**D and implant-access incision **6009**E. Previously formed tunnel T**13** may be used to further subcutaneously advance the lead **3464** to, and for electrical and mechanical connection of proximal portion **3462** with, the IPG **533** in the pectoral region **532**.
(239) FIG. **14**O is a diagram including a front view schematically representing an example arrangement **3500** including an example device for, and/or example method of, implantation of a stimulation device **3505**. In some examples, the stimulation device **3505** may comprise at least some of substantially the same features and attributes as the stimulation devices, methods, etc., as previously described in association with at least FIGS. **14**L-**14**N, except with the stimulation device **3500** comprising a single stimulation lead **3510** extending from the IPG **533** and bifurcated distal

lead portions **3520**, **3522** originating from a junction **3523** (i.e., a bifurcation point) which may be positioned in close proximity to the target stimulation locations at the hypoglossal nerves **505**R, **505**L. In some examples, a stimulation portion **6310**B (e.g. axial/linear electrode array) is supported on (and by) distal lead portion **3522** for stimulating the left hypoglossal nerve **505**L, while a cuff electrode **6411**A is supported on and by distal lead portion **3520** for stimulating the right hypoglossal nerve **505**R. As further shown in FIG. **14**O, the stimulation device **3550** comprises a cuff electrode **6414**A supported on a lead portion **3516** extending, via a junction **3517** (i.e. bifurcation point), from main portion **3514** of lead **3510** and configured for stimulating the ansa cervicalis-related nerve **515**R.

(240) With this framework in mind and as shown in FIG. **14**O, one aspect of an example method of implantation may comprise forming the implant-access incision **609**E, and implanting (via the incision **609**E) the cuff electrode **6414**A (supported on lead portion **3516**) to be in stimulating relation to the ansa cervicalis-related neve **515**R. Via the implant-access incision **690**E, a tunnel (T**13**) may be formed to an implant location for IPG **533** and lead portion **3514** of lead **3510** may be introduced and advanced via tunnel T**13** to enable proximal portion **3512** of lead **3510** to be connected (electrically and mechanically) to IPG **533**, such as in pectoral region **532**. In some examples, an additional implant-access incision may be formed near IPG **533** to facilitate implantation of the IPG **533**, lead **3476**, and/or lead **3464**.

(241) In another aspect of the method of implantation, in some examples, an implant-access incision **609**D may be formed in close proximity to expected target stimulation locations of the hypoglossal nerve **505**R on the same side (RIGHT) of the neck as the implant-access incision **609**E. From the implant-access incision **609**D, a tunnel T**11** is formed to the target stimulation location of the left hypoglossal nerve **505**L. Via implant-access incision **609**D, distal lead portion **3520** (including cuff electrode **6411**A) and distal lead portion **3522** (including stimulation portion **6310**B) are introduced and advanced subcutaneously to the target stimulation locations of the respective right and left hypoglossal nerves **505**R, **505**L to yield the chronically implanted configuration of stimulation portions **6411**A, **6310**B shown in FIG. **14**O.

(242) In some examples, the implant-access incision **609**D is in close proximity to the target stimulation location of the right hypoglossal nerve **505**R such that little or no tunneling would be used to place distal lead portion **3520**. Via this arrangement, sufficient space is available to implant cuff electrode **6411**A on nerve **505**R. However, as noted above, a non-cuff stimulation portion **6310**B is provided for left hypoglossal nerve **505**L so that the distal lead portion **3522** (including stimulation portion **6310**B) may be delivered to the target stimulation location via tunneling (T**11**) without making an addition implant-access incision on the left side of the patient's neck.

(243) Via implant-access incisions **609**D, **609**E, a tunnel T**12** is formed to enable introduction and advancement of lead portion **3518** between implant-access incision **606**D and implant-access incision **609**E.

(244) Via the example arrangement, a cuff electrode **6414**A is secured relative to the ansa cervicalis-related nerve **515**R and a cuff electrode **6411**A is secured relative to the right hypoglossal nerve **505**R via pertinent implant-access incisions **609**E, **609**D, while capability for bilateral stimulation of left and right hypoglossal nerves **505**L, **505**R is achieved via tunneling (T**11**) from the implant-access incision **609**D. In this way, robust secure implantation of stimulation elements for multi-target therapy may be implemented with generally reduced surgical complexity.

(245) In some examples, the IPG **533** may be omitted and instead a microstimulator (e.g. **6575** in FIG. **14**M) may be implanted, via implant-access incision **609**D, in close proximity to the target stimulation location of the right hypoglossal nerve **505**R such as, but not limited to, the location of junction **3523** of lead **3550**. Similarly, instead of implanting the IPG **533**, a microstimulator (e.g. **6575** in FIG. **14**M) may be implanted, via implant-access incision **609**E, in close proximity to the target stimulation location of the ansa cervicalis-related nerve **515**R such as, but not limited to, the location of junction **3517** of lead **3550**. It will be understood, of course, in these examples that

appropriate modifications would be made for connecting the various lead portions relative to the microstimulator. As noted elsewhere, in some examples the implanted microstimulator may be in wireless communication with at least some of the stimulation portions, cuff electrodes, etc.
(246) FIG. **14**P is a diagram including a front view schematically representing an example arrangement **3540** including an example device for, and/or example method of, implantation of a stimulation device **3545**. In some examples, the stimulation device **3545** may comprise at least some of substantially the same features and attributes as the stimulation devices, methods, etc. as previously described in association with at least FIG. **14**O, except with the stimulation device **3545** in FIG. **14**P comprising a single distal portion **3519** of lead **3510** comprising a paddle electrode **3560** in FIG. **14**P (to achieve bilateral hypoglossal nerve stimulation) instead of the bifurcated distal lead portions **3520**, **3522** in FIG. **14**O.
(247) With this framework in mind and as shown in FIG. **14**P, one aspect of an example method of implantation may comprise forming a tunnel T**14** from the previously described implant-access incision **609**D, and via the tunnel T**14**, introducing and advancing a paddle electrode **3560** subcutaneously to establish the paddle electrode **3560** in a position extending between, and overlapping with, both the target stimulation locations of the left and right hypoglossal nerves **505**L, **505**R as shown in FIG. **14**P. In some examples, the paddle electrode **3560** may comprise a carrier (e.g. body) **3562** supporting a linear array **3566** of electrodes **3568** on a first portion **3564**R of the carrier **3562** and supporting a linear array **3566** of electrodes **3568** on a second portion **3564**L of the carrier **3562**. By providing these linear arrays, one can be assured that at least some electrodes **3568** will become juxtaposed in stimulating relation to target stimulation locations of the respective left and right hypoglossal nerves **505**L, **505**R. As a related aspect, by establishing multiple electrodes **3568** in a juxtaposed position of being in potential stimulating relation to each respective nerve **505**L, **505**R, some example methods of stimulation therapy may include selective stimulation of different multiple fascicles within a nerve, nerve branch, etc. to optimize the intended therapeutic effect, manage fatigue, etc.
(248) In some examples, the arrays **3566** on the left and right portions **3564**L, **3564**R of the paddle electrode **3560** may be sized so that they join to form a single array of electrodes **3568** extending along/across substantially the entire length of the carrier **3562** of the paddle electrode **3560**.
(249) In some examples, the array(s) **3566** of electrodes **3568** may comprise electrodes **3568** which are sized, shaped, and/or arranged to comprise a two dimensional array of electrodes **3568** having rows/columns of spaced apart electrodes **3568**.
(250) In some examples, the particular features of paddle electrode **3560** and associated methods of implantation, methods of therapy, etc. may comprise at least some of substantially the same features and attributes as described in PCT Application PCT/US21/17754, entitled STIMULATION ELECTRODE ASSEMBLIES, SYSTEMS AND METHODS FOR TREATING SLEEP DISORDERED BREATHING, filed Feb. 12, 2021, and filed on Feb. 12, 2022 as a U.S. Section 371 National Stage application having Ser. No. 17/631,982 published as US 2022-0280788 on Sep. 8, 2022, which is hereby incorporated by reference in its entirety.
(251) FIG. **14**Q is a diagram including a front view schematically representing an example arrangement **3570** including an example device for, and/or example method of, implantation of a stimulation device **3575**. In some examples, the stimulation device **3575** may comprise at least some of substantially the same features and attributes as the stimulation devices, methods, etc. as previously described in association with at least FIG. **14**O, except with the stimulation device **3575** in FIG. **14**Q comprising an axial stimulation portion **6310**C for stimulation of the ansa cervicalis-related nerve **515**R instead of the cuff electrode **6414**A in FIG. **14**O and with the axial stimulation portion **6310**C in FIG. **14**Q being supported by a different lead portion **3573**, among other differences. However, similar to the example arrangement in FIG. **14**), the stimulation device **3575** in FIG. **14**Q comprises a stimulation portion **6310**B (e.g. axial/linear electrode array) supported on and by distal lead portion **3577** for stimulating the left hypoglossal nerve **505**L, while a cuff

electrode **6411**A is supported on and by distal lead portion **3578** for stimulating the right hypoglossal nerve **505**R. As further shown in FIG. **14**Q, the stimulation device **3550** comprises a lead portion **3573** extending from a main portion **3574** of lead **3571**, via a junction **3576** (i.e. bifurcation point) of lead **3571** near implant-access incision **690**D, to a nerve stimulation location of the ansa cervicalis-related nerve **515**R to support the axial stimulation portion **6310**C in stimulating relation to the ansa cervicalis-related nerve **515**R.
(252) With this framework in mind and as shown in FIG. **14**Q, one aspect of an example method of implantation may comprise forming the implant-access incision **609**D, and implanting (via the incision **609**D) the cuff electrode **6411**A and stimulation portion **6310**B in a manner similar to that described in association with at least FIG. **14**O. In addition, via the implant-access incision **690**D, a tunnel T**16** may be formed toward the ansa cervicalis-related nerve **515**R and lead portion **3573** of lead **3571** may be introduced and advanced via tunnel T**16** to extend toward a target stimulation location of ansa cervicalis-related nerve **515**R to place stimulation portion **6310**C (similar to **6310**B) in stimulating relation to the ansa cervicalis-related nerve **515**R.
(253) As further shown in FIG. **14**Q, a main lead portion **3574** of lead **3571** extends proximally from junction **3576**. Via the implant-access incision **609**D, a tunnel (T**15**) may be formed to an implant location for IPG **533**, at which an additional implant-access incision **609**C may be formed near IPG **533** to facilitate implantation of the IPG **533** and lead portion **3574**. With this framework, in some examples, lead portion **3574** of lead **3571** may be introduced and advanced via tunnel T**15** to enable proximal portion **3572** of lead **3571** to be connected (electrically and mechanically) to IPG **533**, such as in pectoral region **532**.
(254) In some examples, the junction **3576** may be configured to permit releasable connectability of the various lead portions **3573**, **3577**, **3578** relative to main lead portion **3574** and/or relative to each other. Moreover, in some examples, junction **3576** may be configured to permit releasable connection of main lead portion **3574** relative to the junction **3576**.
(255) Among other aspects, in association with implant-access incision **609**D, the example arrangement **3570** may reduce surgical complexity while providing a way to establish a cuff electrode **6411**A at a right hypoglossal nerve **505**R, an axial stimulation portion **6310**B at a left hypoglossal nerve, and an axial stimulation portion **6310**C at an ansa cervicalis-related nerve **515**R.
(256) FIG. **14**R is a diagram including a front view schematically representing an example arrangement **3580** including an example device for, and/or example method of, implantation of a stimulation device **3582**. In some examples, the stimulation device **3582** may comprise at least some of substantially the same features and attributes as the stimulation devices, methods, etc. as previously described in association with at least FIG. **14**Q, except with the stimulation device **3580** in FIG. **14**R including a microstimulator **6575** instead of an IPG **533** in FIG. **14**Q with the main lead portion **3574** being omitted in addition to IPG **533**.
(257) Accordingly, with further reference to FIG. **14**R in comparison to FIG. **14**Q, the stimulation device **3582** in FIG. **14**R comprises the same distal lead portions **3578**, **3577** and their respective stimulation elements (e.g. cuff electrode **6411**A, stimulation portion **6310**B) for delivering bilateral stimulation to the respective right and left hypoglossal nerves **505**R, **505**L. Moreover, like the stimulation device in FIG. **14**Q, the stimulation device **3582** in FIG. **14**R also retains the same lead portion **3573** and axial stimulation portion **6310**C in stimulating relation to the ansa cervicalis-related nerve **515**R.
(258) As further shown in FIG. **14**R, the microstimulator **6575** is directly connected to the respective lead portions **3578**, **3577**, **3573** with distal lead portions **3578**, **3577** having a bifurcation point (formed by junction **3585**) at or near a housing of the microstimulator **6575**. In some examples, the lead portion **3573** (of stimulation portion **6310**C) may extend directly from the microstimulator **6575** as shown in FIG. **14**R or in some examples, may extend from the same junction **3585** as the distal lead portions **3578**, **3577**. As noted elsewhere in relation to some examples of the present disclosure, the microstimulator **6575** may comprise stimulation/control

circuitry, a power source (e.g. rechargeable), and may be in communication with an external power recharging device, element, and the like.

(259) Among other aspects, the example arrangement **3580** may comprise an example method of implantation which significantly reduces surgical complexity, reduces the time for performing the implantation, increases patient comfort, etc. In some such examples, these features may be achieved, at least in part, because of the single (i.e. sole) implant-access incision **609**D made at or near the target stimulation location of the right hypoglossal nerve **505**R, which simultaneously enables convenient implantation of the cuff electrode **6411**A on the right side of the patient's neck, implantation of the axial stimulation portion **6310**B on the left side of the patient's neck, implantation of the axial stimulation portion **6310**C on the right side of the patient's neck, and implantation of the microstimulator **6575** to support the respective cuff electrodes and stimulation portions of stimulation device **3582**. In a manner similar to that described in association with FIG. **14**Q, the junction **3585** may permit various forms of permanent connection or releasable connection among the various lead portions **3578**, **3577**, **3573** relative to the microstimulator **6575** and/or relative to each other. This aspect also may enhance reduced surgical complexity, shortened procedure time, etc.

(260) FIG. **15**A is a diagram including a front view schematically representing an example arrangement **1830** relative to a patient's body **510**, including an example device and/or example method for implantation of a stimulation element **1810**A in stimulating relation to a hypoglossal nerve **505**R and a stimulation element **1813**A in stimulating relation to an ansa cervicalis-related nerve **515**R. In some examples, the example arrangement **1850** may comprise at least some of substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least some of FIGS. **1-13**.

(261) In particular, as shown in FIG. **15**A in some examples the arrangement **1830** may be implanted in a single implantation procedure via a single implant access-incision **609**C in a manner similar to that described for example arrangement **1800** in FIG. **14**A. However, instead of tunneling as in the example of FIG. **14**A, in the example of FIG. **15**A a stimulation lead **1838** is delivered via access point **1842**, and implanted within, the vasculature to position the stimulation element **1810**A within vein **1832** shown in dashed lines (to be in stimulating relation to the hypoglossal nerve **505**R).

(262) Similarly, a stimulation lead **1837** is delivered via access point **1843**, and implanted within, the vasculature to positon the stimulation element **1813**A within vein **1833** (shown in dashed lines) to be in stimulating relation to the ansa cervicalis-related nerve **515**R. As further described later in association with at least FIGS. **32**A-**32**B, in some examples the applicable vasculature **1855** may comprise veins such as the anterior jugular vein, inferior thyroid vein, superior thyroid vein, external jugular vein, etc.

(263) In some examples, both stimulation elements **1810**A, **1813**A may comprise an axial array of electrodes **716**, which facilitates linear positioning and adjustment to ensure a desired co-extensive location of the electrodes **716** relative to a desired portion of the respective nerves to be stimulated. In some examples, the respective stimulation elements **1810**A, **1813**A may comprise one of the electrode configurations, such as one of the stimulation elements as later described in association with at least FIGS. **25-26** and **29-30**B, which may comprise anchor elements in some examples. In addition, one example implementation of the stimulation leads **1838** and/or **1837** is described in association with FIG. **15**C.

(264) As further shown in FIG. **15**A, IPG **533** may also implanted subcutaneously via the implant access-incision **609**C and electrically connected to a proximal portion of the respective stimulation leads **1838**, **1837**.

(265) FIG. **15**B is a diagram schematically representing an example arrangement **1850**, which may comprise at least some of substantially the same features and attributes as the example arrangement **1800** in FIG. **15**A, except with the two stimulation elements **1810**A, **1813**A are arranged on a

single stimulation lead **1867** instead of separate stimulation leads **1838**, **1837** as in FIG. **15**A. As further shown in FIG. **15**B, via single implant access-incision **609**C, a portion of the stimulation lead **1867** is inserted into the vasculature **1855** at access point **1856** and advanced through the vasculature **1855** until the stimulation element **1810**A at the distal portion of the stimulation lead **1867** is positioned adjacent a first target nerve **505**R. In some examples, the first target nerve **1805**R may comprise a hypoglossal nerve (e.g. **505**R), while in some examples the first target nerve **1805**R may comprise some portion of the ansa cervicalis-related nerve **315** (FIG. **2**) or yet another nerve. Upon such positioning, the second stimulation element **1813**A, which is located more proximally on the same stimulation lead **1867** will become positioned within the vasculature **1855** adjacent to, and in stimulating relation to, a second nerve target **1815**R. In some examples, the second target nerve may comprise some portion of the ansa-cervicalis related nerve **515**R or other nerve.

(266) FIG. **15**C is a side view schematically representing an example arrangement **1870** including example stimulation lead **1872**, which may comprise one example implementation of the stimulation leads **1837**, **1838** in FIG. **15**A or of the stimulation lead **1867** in FIG. **15**B. As shown in FIG. **15**C, the stimulation lead **1872** comprises a stimulation element **1880**A including an array (e.g. axial) of electrodes **716** and supported by a distal portion **1874** of a lead body **1875**, with a proximal portion (not shown) of the lead body **1875** adapted for electrical and mechanical connection to IPG **533** (e.g. via a header **735**).

(267) As further shown in FIG. **15**C, the lead body **1875** defines an interior lumen **1877**, which extends through a length of the lead body **1875** and of the stimulation element **1880**A. The lumen **1877** is sized and shaped to enable the stimulation lead body **1875** and stimulation element **1880**A to be slidably advanced and maneuverable over a guide wire **1879**, or stylet or other guiding element. Via this arrangement, in order to implant the stimulation lead **1872** in a manner shown like that in FIG. **15**A or **15**B, the guide wire **1879** may first be inserted into, and advanced through, the vasculature (e.g. **1832**, **1833** in FIG. **15**A) until a distal portion of the guide wire **1879** is positioned just beyond the most distal target stimulation location, such as the hypoglossal nerve **505**R in FIG. **15**A. Next, an open end of the lumen **1877** at the distal portion **1881** of the stimulation lead **1872** (including the stimulation element **1880**A) is slidably mounted onto a proximal end of the guide wire **1879** in the region of the implant access-incision **609**C (e.g. FIG. **15**A, **15**B). The stimulation lead **1872** is then slidably advanced through the vasculature (e.g. **1832**, **1833** in FIG. **15**A or **1855** in FIG. **15**B) until the distal portion **1881** of the stimulation lead **1872** is positioned such that the stimulation element **1880**A (e.g. an example implementation of element **1810**A in FIG. **15**A) is in stimulating relation to the target nerve, such as hypoglossal nerve **505**R for stimulation lead **1838** in FIG. **15**A. It will be understood that the example arrangement **1870** is likewise applicable to the stimulation lead **1837** in FIG. **15**A and/or stimulation lead **1867** in FIG. **15**B.

(268) Moreover, in some examples more than one transvascular (e.g. transvenous) stimulation lead and/or more than one branches of such transvascular stimulation leads may be implanted to provide stimulation of multiple stimulation targets of the ansa cervicalis-related nerve and/or other upper airway patency-related tissues.

(269) FIG. **16** is a diagram including a side view schematically representing an example arrangement **2000** including example devices and/or example methods for stimulating a portion of the ansa cervicalis-related nerve **316** (FIG. **2**) and/or a portion of the hypoglossal nerve **305**. In one example implementation, an example stimulation arrangement **2101** is deployed as further described in association with at least FIGS. **17-19**. In one example implementation, a stimulation arrangement **2401** includes separate stimulation elements **2410**, **2420** as further described in association with at least FIG. **20**.

(270) FIG. **17** is a diagram including a side view schematically representing an example arrangement **2100**, which may comprise one example implementation of the example stimulation element **2101** in FIG. **16**. In some examples, the example arrangement **2100** in FIG. **17** comprises a

stimulation element **2109** in stimulating relation to both a hypoglossal nerve **305** and portion **317** of the ansa cervicalis-related nerve **316** to implement an example method for treating sleep disordered breathing, such as via increasing and/or maintaining upper airway patency. In some examples, the example arrangement **2100** may comprise at least some of substantially the same features and attributes as, comprise an example implementation of, and/or be usable with the example arrangements described in association with at least some of FIGS. **1-16**.

(271) As shown in FIG. **17**, the stimulation element **2101** represented in FIG. **16** may comprise a paddle electrode **2109** including an array **2125** of spaced apart electrodes **2126**, which are disposed on body **2120**. While not shown for illustrative clarity, it will be understood that in some examples the paddle electrode **2109** may be supported on a distal portion of a stimulation lead or in some examples may form part of a microstimulator which omits a stimulation lead of the type connected to an IPG (e.g. **533**). With this in mind, one microstimulator which may comprise one example implementation of stimulation element **2101** may comprise at least some of substantially the same features and attributes as the microstimulator **1413**A, as previously described in association with at least FIG. **11**B.

(272) With further reference to the paddle electrode **2109**, the body **2120** and array **2125** of electrodes **2126** are sized and shaped such that when the paddle electrode **2109** is juxtaposed with a pair of nerves, one or both of the respective nerves may be stimulated as desired. In some such examples, the nerves may comprise a hypoglossal nerve **305** and portion **317** of an ansa cervicalis-related nerve **316**. In particular, in the example implementation shown in FIG. **17**, the stimulation element **2109** is positioned proximal to a junction **311** (FIG. **16**) at which superior root **325** of the ansa cervicalis-related nerve **315** diverges from a proximal portion **307** of the hypoglossal nerve **305**. It will be understood that in this context, the term “proximal portion” of the hypoglossal nerve **305** is with specific regard to the junction **311**.

(273) Via this example arrangement **2100** in FIG. **17**, an example method of treating sleep disordered breathing may comprise stimulating one or both of the respective hypoglossal nerve **305** and the ansa cervicalis-related nerve **316** to increase and/or maintain upper airway patency. In one aspect, this arrangement may comprise use of selective steering of the stimulation signals to capture particular fascicles (e.g. motor) within each respective bundles of nerves **305**, **316** at least because, at this particular location, both the hypoglossal nerve **305** and portion **329**A of the ansa cervicalis-related nerve **316** include some non-targeted fibers (e.g. innervating retractor muscles of the tongue for the HGN) among the targeted nerve fibers, such as those nerve fibers (of the HGN) innervating protrusor muscles of the tongue and/or those nerve fibers (of the ACN) innervating the sternothyroid muscles and/or sternohyoid muscles (as an example).

(274) At the particular stimulation location in the example arrangement **2000** in FIG. **16** (including example arrangement **2101** or **2401**), stimulation is to be applied to a main trunk of the hypoglossal nerve **305** and portion **329**A of the ansa cervicalis-related nerve **316**. In some examples, this stimulation location may provide sufficient space and an anatomical environment to enable placement of a paddle electrode (FIG. **17**) or cuff electrode(s) (FIGS. **18-20**), such as but not limited to a single implant-access incision adjacent the hypoglossal nerve **305** (e.g. main trunk). Via such arrangements, a single electrode arrangement as in FIGS. **17**, **18-19**, and/or **20** is able to provide stimulation to both the hypoglossal nerve **305** and the ansa cervicalis-related nerve **316** (via portion **329**A). Moreover, this stimulation location may enable use of a larger size (e.g. diameter) cuff electrodes or larger paddle electrodes, which are easier to handle and may provide for more robust chronic implantation than if such cuff electrodes or paddle electrodes are implanted in relation to small diameter nerves.

(275) FIG. **18** is a diagram including a sectional view schematically representing an example arrangement **2200** including an example device and/or example method of providing stimulation to two different types of nerves for increasing and/or maintaining upper airway patency. In some examples, the example arrangement **2200** comprises one example implementation of the example

arrangement **2101** in FIG. **16** to provide stimulation to one of, or both, the hypoglossal nerve **305** and the ansa cervicalis-related nerve **316** (FIG. **16**). As shown in FIG. **18**, in some examples the example arrangement **2200** may comprise cuff electrode **2230**, which comprises a cylindrically shaped body **2231** defining a lumen **2233** to at least partially enclose or encircle the respective nerves **305**, **316**. As shown in FIG. **18**, the body **2231** may comprise a slit or re-closable opening **2235** to permit placing the cuff electrode **2230** about the nerve(s) **305**, **315** and re-closure of the wall of the body **2231** about the nerves. While not shown for illustrative simplicity, in some examples the cuff electrode **2230** may comprise overlapping flange members to enhance releasably securing the cuff electrode about the nerves **305**, **316**. Moreover, in some examples, the cuff electrode **2230** comprises an array of circumferentially spaced apart electrodes **2236** exposed on an interior surface **2237** to be in stimulating relation to the respective nerves **305**, **316**. Via various combinations of the electrodes **2236** and selectable parameters (e.g. amplitude, pulse width, current, frequency, duty cycle, sequence of activation, etc.) of stimulation signal, various fascicles **309** within the hypoglossal nerve **305** and/or various fascicles **313** within the ansa cervicalis-related nerve **316** may be targeted to effect desired stimulation of at least motor fibers to increase and/or maintain upper airway patency. In some such examples, the various nerves **305**, **316** (and their various fascicles) may be stimulated according to at least some of the stimulation patterns as described in association with at least FIGS. **33**A-**37**D.
(276) FIG. **19** is a side view schematically representing the cuff electrode **2230** in FIG. **18**, which further illustrates various features and attributes of the cuff electrode **2230**. For instance, FIG. **19** illustrates one example configuration of the electrodes **2236** when arranged in an array in which the electrodes **2236** extend in a spaced apart manner axially along a length of the body **2231** of cuff electrode **2230** and extend in a spaced apart manner circumferentially about the interior surface **2237** (FIG. **18**) of the body **2231** of cuff electrode **2230**.
(277) FIG. **20** is a sectional view schematically representing an example arrangement **2413** which comprises one example implementation of the example arrangement **2401** in FIG. **16**. As shown in FIG. **20**, in some examples the example arrangement **2413** comprises a first cuff electrode **2411**, which may comprise one example implementation of stimulation element **2410** in FIG. **16** and may comprise a second cuff electrode **2421**, which may comprise one example implementation of stimulation element **2420** in FIG. **16**. As shown in FIG. **20**, each cuff electrode **2411**, **2421** comprises at least some of substantially the same features and attributes as the cuff electrode **2230** in FIG. **18**, except being sized to at least partially encircle and enclose just one nerve, such as nerves **305**, **316** respectively instead of two nerves as in FIG. **18**. Accordingly, the features of cuff electrodes **2411**, **2421** are identified via similar reference elements as in FIG. **18**.
(278) Via this example arrangement **2411**, stimulation of each nerve **305**, **316** is applied via separate cuff electrodes **2411**, **2421** in a side-by-side arrangement, which may simplify at least some aspects of selectively stimulating certain fascicles within each respective nerve **305**, **316** relating to controlling upper airway patency and related physiologic functions.
(279) In some example implementations, the cuff electrodes **2230**, **2411**, and/or **2421** may comprise at least some of substantially the same features and attributes as described in Bonde et al, SELF EXPANDING ELECTRODE CUFF, issued as U.S. Pat. No. 9,227,053 on Jan. 5, 2016, in Bonde et al, SELF EXPANDING ELECTRODE CUFF, issued as U.S. Pat. No. 8,340,785 on Dec. 25, 2012, in Johnson et al, NERVE CUFF, issued as U.S. Pat. No. 8,934,992 on Jan. 13, 2015, and in Rondoni et al, CUFF ELECTRODE, published as WO 2019/032890, on Feb. 14, 2019, and later published as U.S. 2020/0230412 on Jul. 23, 2020, and which are all hereby incorporated by reference in their entirety.
(280) In some examples, the cuff electrodes of FIGS. **18-20** may be employed in other example arrangements of the present disclosure and are not limited to use solely in the anatomical and physiologic context presented in relation to FIGS. **18-20**. Accordingly, in any example of the present disclosure calling for a stimulation element in which a cuff electrode may be a suitable

example implementation, such stimulation elements may comprise one of the cuff electrodes in FIGS. **18-20** or in later described FIGS. **26**A-**26**B.
(281) In some examples, the stimulation location for example stimulation electrode arrangements **2101**, **2401** in FIG. **16** may correspond to the stimulation location “A” in the example arrangements later described in association with at least FIG. **32**C, in which stimulation at location “A” may be implemented via an intravascular approach (e.g. transvenous) through the interior jugular vein **4250**, in some examples.
(282) FIG. **21** is a side view schematically representing an example arrangement **2700** including a stimulation element **513**A and a passive receiver **2725** to obtain and provide power and control signals to the stimulation element **513**A. In this arrangement, the stimulation element **513**A is a standalone element without a stimulation lead (such as connected to an IPG **533**) and is positioned in stimulating relation to the ansa cervicalis-related nerve **316**. However, in other respects, the stimulation element **513**A may comprise at least some of substantially the same features and attributes as stimulation elements (and related arrangements) described in association with various example stimulation elements described throughout the present disclosure. It will be further understood that the same example arrangement also may be implemented relative to the hypoglossal nerve **505**R in addition to, or instead of, being implemented relative to the ansa cervicalis-related nerve **316**.
(283) As further shown in FIG. **21**, the passive receiver **2725** may be connected (e.g. via wires **2727**) to the stimulation element **513**A and positioned adjacent an external surface **2732** of the patient's body, such as in the head-and-neck region **520**. In some such examples, the example arrangement **2700** may comprise an externally located power-control element **2735** to provide power and/or control signals to the stimulation element **513**A, via wireless communication with the passive receiver **2725**. In some examples, whether also embodied as a sensing element or not so embodied, the power-control element **2735** can receive sensed data from the stimulation element **513**A via the passive receiver **2725**.
(284) FIG. **22**A is a diagram including a side view schematically representing an example arrangement **2900** including a stimulation element **513**A in stimulating relation to the ansa cervicalis-related nerve **316** (e.g. at superior root **325**) and a supporting stimulation lead **2917** anchored relative to non-nerve structure **2929** (e.g. tissue). In some examples, the stimulation element **513**A and/or stimulation lead **2917** may comprise at least some of substantially the same features and attributes as stimulation elements (and related arrangements) described in association with various examples described in association with at least FIGS. **1-21**.
(285) It will be understood that the particular location of the stimulation element **513**A in FIG. **22**A is merely representative of many different positions of the ansa cervicalis-related nerve **316** at which the stimulation element(s) **513**A may be located.
(286) As shown in FIG. **22**A, the stimulation lead **2917** includes a distal portion **2919** which may be formed into a strain relief loop or portion extending between the stimulation element **513**A and the anchor element **2927**, with the anchor element **2927** secured to the non-nerve structure **2929** in order to secure the stimulation lead **2917** thereto. A lead body **2921** of the stimulation lead **2917** extends proximally from the anchor element **2927**.
(287) As further shown in FIG. **22**B, box **2950** schematically represents at least some of the non-nerve structures **2929** (in FIG. **22**A) to which the anchor element **2927** may anchor a portion of the stimulation lead **2917**. In some examples, such non-nerve structures may comprise an omohyoid tendon, a hyoid bone, a clavicle, a sternum (including the manubrium), a trachea, a digastric tendon, and/or other non-nerve structures. Moreover, such non-nerve structures may be used for anchoring a stimulation lead, port interface (e.g. FIGS. **5**A, **5**B, **7**A, and the like), stimulation element, etc. relative to an upper airway patency-related tissue, whether in relation to the example of FIG. **22**A, **23** and/or other examples throughout the present disclosure.
(288) FIG. **23** is a diagram including a side view schematically representing an example

arrangement **3100** which comprises at least some of substantially the same features and attributes as the example arrangement **2900** in FIG. **22**A-**22**B, except with a distal portion of a stimulation lead **3117** including a pre-formed strain relief segment **3119** between the anchor element **2927** and the stimulation element **513**A. The pre-formed strain relief segment **3119**, shown within the dashed lines, may comprise any flexible, resilient shape (e.g. sigmoid, other) which helps to relieve strain on the stimulation element **513**A in its fixed position relative to a nerve or muscle to be stimulated, such as strain occurring during movement of the neck and/or other body movements.

(289) It will be understood that the anchoring arrangements (e.g. anchor element, non-nerve structures, strain relief segments, etc.) described in association with at least FIGS. **22**A-**23** may be implemented in various forms with any of the stimulation elements, stimulation leads, port interfaces, sensing leads, etc. as described throughout the various examples of the present disclosure.

(290) FIG. **24**A-**26**B provide a series of illustrations of various example stimulation elements. In some examples, the various stimulation elements described in FIGS. **24**-**26**B may comprise at least some of substantially the same features and attributes as, may be example implementations of, and/or may be consistent with the stimulation elements (and related arrangements) described in association with various example stimulation elements described throughout the present disclosure.

(291) FIG. **24**A is top plan view schematically representing example stimulation element **3200**. As shown in FIG. **24**A, example stimulation element **3200** comprises a paddle electrode **3210** comprising a paddle-style body **3212** on which a linear array **3214** of electrodes **3216** are located. In some examples, the array **3214** may comprise a two-dimension array of electrodes **3216**. The paddle electrode **3210** is supported by, and extends from, a distal portion **3220** of a stimulation lead **3222**.

(292) As further shown in the side view of FIG. **24**B, the paddle electrode **3210** may be positioned in stimulating relation to a nerve **3228**, such as a hypoglossal nerve (e.g. **505**R) or ansa cervicalis-related nerve (e.g. **513**R) or other nerve related to increasing and/or maintaining upper airway patency. The paddle electrode **3210** may be secured in pressing contact with the nerve **3228** or may be secured in close proximity to, but spaced apart from, the nerve **3228**.

(293) FIG. **25**A is a side plan view schematically representing an example arrangement **3241** in which stimulation element **3240** is in stimulating relation to a nerve **3228** (like in FIG. **24**B). In some examples, the stimulation element **3240** may comprise at least some of substantially the same features and attributes as stimulation element **3210**, except with electrodes **3216** being arranged in a linear array **3245** of spaced apart ring electrodes **3246**.

(294) FIG. **25**B is a side plan view schematically representing an example arrangement **3261** in which stimulation element **3260** is adapted to be in stimulating relation to a nerve (like nerve **3228** in FIGS. **24**B, **25**A). In some examples, the stimulation element **3260** may comprise at least some of substantially the same features and attributes as stimulation element **3240** in FIG. **25**A, except comprising a linear array **3265** of spaced apart split ring electrodes **3266** instead of ring electrodes **3246** in FIG. **25**A and with body **3242** comprising a generally cylindrical shape.

(295) FIG. **26**A is a side view, and FIG. **26** B is a side view, schematically representing an example arrangement **3300** including a cuff electrode **3330**. In some examples, the cuff electrode **3330** in FIGS. **26**A-**26**B may comprise at least some of substantially the same features and attributes as the cuff electrode **2230** in FIGS. **18**-**19**, except with the cuff electrode **3330** comprising fewer electrodes as shown in FIGS. **26**A, **26**B. In particular, cuff electrode **3330** comprises a bottom row of axially spaced apart electrodes **3336**D and a middle row of circumferentially spaced apart electrodes **3336**A, **3336**B, **3336**D, **3336**C. By employing various combinations of the respective electrodes **3336**A, **3336**B, **3336**C, **3336**D, as well as variations in the stimulation signal as previously described, this electrode configuration may be used to provide selective stimulation and/or stimulation steering of a stimulation signal relative to different fascicles, nerve fibers, etc. within a nerve about which the cuff electrode **3310** is secured.

(296) FIG. **27**A-**31**G are a series of diagrams of various example arrangements of stimulation elements, each of which are equipped with some form of anchoring elements to provide example devices and/or example methods for anchoring a stimulation element within a patient's body relative to a non-nerve tissue. Via such anchoring, the stimulation element may be secured in stimulating relation to a target nerve associated with controlling upper airway patency. In some examples, the various stimulation elements described in FIGS. **27**A-**31**G may comprise at least some of substantially the same features and attributes as, may be example implementations of, and/or may be consistent with the stimulation elements (and related arrangements) described in association with various example stimulation elements described throughout the present disclosure. Moreover, the various anchor elements described in association with FIGS. **27**A-**31**G may be used with at least some of the various previously described (and some later described) example stimulation elements, as appropriate to the context in which they are being implanted.
(297) FIG. **27**A is a top plan view schematically representing an example arrangement **3600** including a paddle electrode **3610** supported on a distal portion **3220** of a stimulation lead **3222**. The paddle electrode **3610** comprises an array **3614** of electrodes **3616** disposed on a body **3612**, with a proximal portion **3619** connected to the stimulation lead **3222**. A distal portion **3618** of the paddle electrode **3610** supports an anchor element **3630** comprising a pair of curved, pointed fingers **3632**, **3633** which diverge from each other and outwardly relative to sides **3611** of the body **3612** of the paddle electrode **3610**. In some examples, the curved fingers **3632**, **3633** are formed of a resilient, flexible material. As shown in FIG. **27**A, the curved pointed fingers **3632**, **3633** are configured to engage non-nerve tissues adjacent to a position at which the paddle electrode **3610** would be positioned in stimulating relation to a nerve (shown in dashed lines), thereby anchoring the paddle electrode **3610** in a desired, therapeutic position.
(298) As further shown in FIG. **27**B, in some example arrangements **3650**, an introducer **3660** (or guide catheter) defining an internal lumen **3662** is provided to facilitate advancement and positioning of the paddle electrode **3610** with curved fingers **3632**, **3633**. The lumen **3662** of the introducer **3660** may maintain the curved fingers **3632**, **3633** in a folded position against opposite sides **3611** of the paddle electrode **3610** until the paddle electrode **3610** is in its desired position. Then, the introducer **3660** may be slidably withdrawn to permit the curved fingers **3632**, **3633** to expand outwardly into the deployment position and deployment shape shown in FIG. **27**A, thereby causing the fingers **3632**, **3633** to engage the surrounding non-nerve tissue.
(299) As shown in FIG. **27**B, the introducer **3660** may comprise a wall **3663** defining a lumen **3662** in which the paddle electrode **3610** may be slidably, releasably inserted in the manner previously described.
(300) FIG. **28** is a top plan view schematically representing an example arrangement **3800** including a paddle electrode **3810** like paddle electrode **3610** in FIG. **27**A, except omitting fingers **3633**, **3632** and instead including holes **3840** about a periphery of a body **3812** of the paddle electrode **3810** to facilitate tissue growth to help anchor the body **3812** of the paddle electrode **3810** relative to non-nerve tissues/structures. However, in some examples, the holes **3840** may be used to secure the paddle electrode **3810**, via sutures, relative to surrounding non-nerve structures/tissues. As shown in FIG. **28**, the paddle electrode **3810** comprises a two-dimensional array **3814** of electrodes **3816**, and may be mounted on a stimulation lead **3222** similar to the arrangement in FIG. **27**A.
(301) FIG. **29**A is a side view schematically representing an example stimulation element **3910** comprising a linear array **3914** of spaced apart ring electrodes **3916** and an anchor element comprising an array **3926** of flexible, resilient tines **3927** extending outward from opposite sides of body **3911** of the stimulation element **3910**. In one aspect, the stimulation element **3910** comprises distal portion **3918** and opposite proximal portion **3919**, which may be supported via a distal portion **3220** of a stimulation lead **3222**, as further shown in FIG. **29**B. When deployed in a desired location, the tines **3927** engage non-nerve tissue to secure the stimulation element in a position to

be in stimulating relation to a target nerve, such as for increasing and/or maintaining upper airway patency.

(302) FIG. **29**B is a diagram including a side view schematically representing an example arrangement **3950** including the example stimulation element **3910** of FIG. **29**A in association with an example introducer **3960** (or guide catheter) for delivering the stimulation element **3910** to a target location. In some examples, the introducer **3960** comprises a wall **3963** defining a lumen **3962** within which the stimulation element **3910** is slidably inserted to cause tines **3927** to fold against sides of the body **3911** of the stimulation element **3910** to prevent their engagement with surrounding non-nerve tissues during delivery of the stimulation element **3910** to a target nerve location. Upon arrival of the stimulation element **3910** at the target nerve location, the introducer **3960** is slidably withdrawn, which releases the tines **3927** to fold outward into their deployment position and shape, such as shown in FIG. **29**A, to engage the surrounding non-nerve tissue to thereby anchor the stimulation element **3910** in stimulating relation to the target nerve location.

(303) FIG. **29**C is a diagram including a side view schematically representing an example arrangement **3980** in which the introducer **3960** comprises a window **3982** formed in wall **3963**. In some examples, the window **3982** may permit at least some electrodes **3916** of the stimulation element **3910** to be exposed to potential target nerve locations so that test stimulation signals may applied while maneuvering the stimulation element **3910**, with tines **3927** in their non-deployed position, during such test maneuvering. In this way, the introducer **3960** allows selective deployment of the anchor tines **3927**, while also permitting application of test stimulation signals via window **3982**.

(304) It will be further understood that a similar style introducer **3960** including window(s) **3982** may be employed as or with at least some other example arrangements (e.g. introducer **3660** in FIG. **27**B) associated with a stimulation element in other examples of the present disclosure in order to facilitate application of test stimulation signals when identifying a target nerve location.

(305) At least the example implementations of FIGS. **30**A-**31**G relate to delivery tools, anchors, and related elements, which may be used as part of an example method of implantation and/or example device for implantation, treatment, etc. Among other attributes, the use of delivery tools as part of an example method of implantation for multi-target therapy may minimize the amount of dissection of tissues (e.g. on a path to and/or near the intended target stimulation site), may minimize a size of an incision in the patient's skin, tissues, etc. while also providing access to multiple target stimulation sites. In some examples, at least some of these features may be implemented or achieved via a single implant-access incision (i.e. sole implant-access incision in the patient's body, in some examples) such as but not limited to at least some of the previously described examples of single implant-access incisions. As a related aspect, these features may reduce surgical implantation time.

(306) FIG. **30**A is a flow diagram schematically representing an example method **2500** of implantation. In some examples, the method **2500** may comprise an example implementation of at least some methods of implantation of the various stimulation leads in at least some of the examples of the present disclosure. For example, at least some aspects of method **2500** may comprise one example implementation of the various examples of implanting a stimulation element (e.g. device, lead, stimulation device, stimulation portion, etc.) as described in association with at least FIGS. **3**-**29**C and **30**B-**32**C and/or one example implementation of the various examples of identifying a target stimulation site in association with at least FIGS. **51**A-**51**B.

(307) As shown at **2512** in FIG. **30**A, method **2500** comprises inserting a probe needle into a patient's body in a region at which at target stimulation location (e.g. nerve) is generally located. The probe needle may comprise at least some conductive elements supported by stimulation-control circuitry for applying a test stimulation signal via the probe needle to the tissue in which the probe needle has been inserted. As further shown at **2514** in FIG. **30**A, via application of the test stimulation signal, one can determine a location at which a stimulation portion of a lead is to be

delivered within the pertinent tissue of the patient's body.

(308) During application of the test stimulation signal via the probe needle, a clinician may observe a muscle response, such as a response of upper airway patency-related muscle(s) to the test stimulation signal. In some examples, the response to be observed may comprise a tongue protrusion (e.g. contraction of the genioglossus muscle innervated by the hypoglossal nerve) and/or contraction of muscles (e.g. sternohyoid, sternothyroid) innervated by an ansa cervicalis-related nerve **316** (e.g. at least FIGS. **2**A, **32**A, other). Accordingly, to the extent that a muscle response is not observed upon application of a test stimulation signal, the user may continue to advance and maneuver the probe needle until a suitable response is observed.

(309) It will be understood that the probe needle may be inserted into more than one location and/or maneuvered carefully within a given area in order to determine the desired target stimulation location with a test stimulation signal being applied at the various locations at which the probe needle is maneuvered. Moreover, in some examples, the probe needle may comprise an elongate flexible needle capable of being flexed in a desired orientation relative to pertinent anatomical structures, tissues, etc. in order to reach the desired target locations.

(310) FIG. **30**B is a side view schematically representing an example probe needle **2550** which may comprise one example implementation of a probe needle used in the example method of FIG. **30**A. As shown in FIG. **30**B, in some examples the probe needle comprises an elongate tubular member **2552** (i.e. sleeve) defining a lumen **2556** via side wall **2555**. The tubular member **2552** may comprise a semi-rigid or flexible, resilient material and extend between a distal end **2554** and a proximal end **2556**. In some examples, a handle portion **2558** may be formed or mounted at or near the proximal end **2556** of the probe needle **2550** to facilitate handling, maneuverability, etc. of the probe needle **2550**.

(311) In some examples, the probe needle **2550** may comprise at least one stimulation test electrode **2560** for applying the test stimulation signal. In some examples, the test electrode **2560** may be in a position spaced apart proximally from the distal end **2554** of the probe needle **2550** by a distance X**1** such that the test electrode **2560** may sometimes be referred to as being set back from the distal tip **2554** of the probe needle **2550**. In some such examples, the setback distance X**1** may generally correspond to a setback distance on a stimulation lead, i.e. a distance between a distal tip of a stimulation lead and a distal end of an electrode array of a stimulation portion on the stimulation lead. In one aspect, the setback position of the test electrode **2560** on the probe needle **2550** may increase a likelihood that, upon full implantation of a stimulation lead based on a method of implantation including the use of the probe needle (e.g. at **2512**, **2514**, **2516** in FIG. **30**A; **2550** in FIG. **30**B), all or most of the electrodes of a stimulation electrode array will coincide positionally with (or be in close proximity to) the target stimulation location identified via the probe needle (e.g. **2550**) from or during the application of test stimulation signals via the probe needle at various potential stimulation locations. In some such examples, the coincidental position may sometimes be referred to as the stimulation electrode array being generally centered on, or generally co-located with, the target stimulation site of the target nerve.

(312) In some examples, the test electrode **2560** of probe needle **2550** may be positioned at the distal end **2554** of probe needle **2550**, e.g. a distal tip of the probe needle **2550** and not setback from the distal end **2554** as shown in FIG. **30**B.

(313) In some examples, in one example implementation of a method of implantation, a determination where and how to insert and advance of a probe needle (e.g. **2550**) may be performed via a visualization method including palpation, such as with respect to known, observable anatomical landmarks and/or user experience. In some examples, in addition to or instead of such palpation, example methods may comprise using visualization provided external image-based monitoring, such as via ultrasound, fluoroscopy, x-ray, etc. In some examples, one example implementation of visualization and/or other forms of guiding a probe needle and other delivery tools, stimulation lead, etc.) within a patient's body during a method of implantation may

comprise at least some of substantially the same features and attributes as described in U.S. Pat. No. 9,888,864, issued Feb. 13, 2018, entitled METHOD AND SYSTEM FOR IDENTIFYING A LOCATION FOR NERVE STIMULATION, and which is hereby incorporated by reference in its entirety.
(314) In some examples, instead of using a probe needle (e.g. **2512**, **2514** in FIG. **30**A, **2550** in FIG. **30**B) to identify a target stimulation site along a nerve, some example implementations of method **2500** in FIG. **30**A may comprise use of the stimulation electrodes on the to-be-implanted stimulation lead to provide the role or function of a probe needle, such as but not limited to aspects **2512**, **2514** in method **2500** (FIG. **30**A). In some such examples, application of a test stimulation signal may be applied via a stimulation portion of a stimulation lead while the stimulation lead is present within a delivery tool (e.g. cannula, hollow insertion needle, etc.) and during insertion and advancement of the delivery tool within or near the pertinent tissue at which the target stimulation location is expected to be identified. In some such examples, instead of using the stimulation electrodes of the stimulation lead, a dedicated test electrode may be present on the stimulation lead in addition to the stimulation electrodes. In some examples, another alternative to use of the probe needle may comprise a delivery tool (e.g. cannula, hollow insertion needle, and the like) which carries one or more test stimulation electrodes to enable application of a test stimulation signal to identify and/or confirm a location of a target stimulation site prior to withdrawal of the delivery tool (which may complete implantation of the stimulation lead).
(315) Once the target location has been identified per **2512**, **2514** of method **2500**, then at **2516** as shown in FIG. **30**A, the method **2500** comprises inserting a guidewire into the patient's body and into and through the probe needle, which is already located at the target stimulation location. With the guide wire in its desired position at the target stimulation location, the probe needle is withdrawn (via the guidewire) from the body.
(316) With this in mind, FIG. **30**B provides a schematic representation of an example guide wire **2570** extending through a lumen **2556** of the probe needle **2550**. It will be understood that the guidewire **2570** may comprise an elongate flexible, resilient element, which may comprise a metal material in some examples, and may comprise a biocompatible outer coating, etc.
(317) As further shown at **2518** in FIG. **30**A, a dilator is advanced over the guidewire to the target stimulation location and at **2520**, a hollow sheath (e.g. introducer, etc.) is advanced over the dilator to the target stimulation location with the dilator being removed thereafter, thereby leaving the hollow sheath in place at the target stimulation location. At **2522**, a stimulation lead (or other type of lead) is inserted into and advanced through the hollow sheath (with or without the guidewire) until the stimulation portion of the lead arrives at the target stimulation location.
(318) In some examples, after the stimulation portion of the lead has been delivered to its location for chronic implantation, the hollow sheath may be removed from the patient's body, which may comprise peeling or breaking the hollow sheath in order to free the hollow sheath from the stimulation lead and from the implantation location within the patient's body.
(319) In some examples, removal of the hollow sheath from the stimulation lead may result in the automatic activation of any anchor elements (e.g. tines, threads, coils, filaments) on the stimulation lead which were being retained in a non-deployed state (e.g. collapsed, covered, etc.) within the delivery tool(s) (e.g. hollow sheath, sleeve, etc.) during delivery of the stimulation element (e.g. electrode array, etc.) of the stimulation lead to the target stimulation location, such as in accordance with method **2500** of FIG. **30**A. With this in mind, various example implementations of delivering a stimulation lead while retaining an anchor element in a non-deployed state and later deploying the anchor element upon withdrawal of a delivery tool are described in association with at least FIGS. **30**B-**31**G.
(320) While some details of example methods of implantation may vary depending on a size, length, shape of an element (e.g. stimulation lead) to be implanted, it will be understood that the example method **2500** of FIG. **30**A provides one example implementation by which at least some

of the example leads, example stimulation devices, etc. of the present disclosure may be implanted, including but not limited to at least some of the leads, stimulation devices of the examples described in association with at least FIGS. **1-29**C, FIGS. **30**C-**30**W, and/or FIGS. **31**A-**32**D.

(321) FIG. **30**C is a diagram including a front view schematically representing an example arrangement **6700** comprising an example stimulation device **6710** which may be used in example methods of implantation, with or without additional tools. In some examples, the stimulation device **6710** may comprise at least some of substantially the same features and attributes of at least some of the example arrangements described in association with at least FIGS. **1-29**C and later described in association with at least FIGS. **30**D-**32**C. As shown in FIG. **30**C, the stimulation device **6710** may comprise a body **6713** which extends between a distal end **6719** and a proximal end **6718**. In some such examples, the proximal end **6718** may be connected to, or extend from, a lead body connectable to a pulse generator or microstimulator, in a manner similar to various example stimulation portions, leads, etc. of the present disclosure.

(322) As further shown in FIG. **30**C, stimulation device **6710** comprises an array **6714** of electrodes **6716**, such as ring electrodes or split-ring electrodes spaced apart from each other axially along a portion of a length of the body **6713** of device **6710**. In some examples, the stimulation device **6710** also comprises an anchoring structure **6725**, which in some examples may comprise an array of tines (or similar elements) **6727**A, **6727**B, **6727**C, **6729**.

(323) While shown in a two-dimensional format in FIG. **30**C, it will be understood that a plurality of tines (e.g. **6727**A, etc.) at a particular location along the body **6713** of stimulation device **6710** may be arranged about a circumference of the body (e.g. cylindrical).

(324) With further reference to FIG. **30**C, tines **6727**A and tines **6727**B are positioned distal to the array **6714** of electrodes **6716**, being between the array **6714** and the distal end **6719** of the body **6713** of the stimulation device **6710**. Meanwhile, in some examples, tines **6727**C and **6729** are positioned proximal to the array **6714** of electrodes **716**, being between the array **6714** and the proximal end **6718** of the body **6713** of stimulation device **6710**. Each of the respective tines (**6727**A, **6727**B, **6727**C, **6729**) are connected to (and extend from) the side(s) **6711** of the body **6713** and extend outwardly at an angle (Ω) from the side(s) **6711**. As further shown in FIG. **30**C, an end **6728** of the tines **6727**A, **6727**B, **6727**C extend in a first orientation (F), with their ends **6728** pointed rearwardly toward the proximal end **6718** of the body **6713** of device **6710**. In contrast, tines **6279** extend in an opposite second orientation (arrow S), with their ends **6728** pointed toward the distal end **6719** of the body **6713** of device **6710**. Accordingly, the tines **6729** have an orientation which are opposite to the orientation of tines **6727**A, **6727**B, **6727**C. As further described later below, this orientation may facilitate robust, stable anchoring of the stimulation device **6710** within a patient's body via the respective tines **6727**A, **6727**B, **6727**C, **6729** engaging non-nerve structures.

(325) In some examples, the tines **6727**A, **6727**B, **6727**C, **6729** are made of a flexible, resilient material and formed relative to the side(s) **6711** of body **6713** such that the tines are biased to extend outward (at an angle Ω) from the side(s) **6711** in the manner shown in FIG. **30**C. From this outwardly extending angle, the tines (e.g. **6727**A, **6727**B, **6727**C, **6729**) also are collapsible (e.g. flexibly bending, folding) toward and/or against the side(s) **6711** of body **6713**, as later shown in at least FIGS. **30**E-**30**G, upon some external force or structure causing such collapse (i.e. bending toward side(s) **6711**). In some examples, at least some of the tines may comprise an elongate, cylindrical member having a cross-sectional shape which is circular or similar. Of course, while still bearing a generally elongate shape, in some examples the tines may comprise different/other cross-sectional shapes such as rectangular, triangular, etc.

(326) In some examples, the stimulation device **6710** may be implanted using a wide variety of tools for delivery, implantation, etc. With this in mind, FIGS. **30**D-**30**G schematically represent example methods to prepare for implantation, and/or execute implantation of, stimulation device **6710**.

(327) FIG. **30**D is diagram including a side view schematically representing an example arrangement **6750** including example stimulation device **6710** in use with a hollow insertion needle **6760** and sleeve **6751** as part of a method of implanting stimulation device **6710**. In some examples, the stimulation device **6710** may comprise at least some of substantially the same features and attributes as the stimulation device **6710** of FIG. **30**C. As further shown in FIG. **30**D, sleeve **6751** may comprise a body defined by sidewall(s) **6753**, which are spaced apart by a distance to slidably fit over and cause temporary collapse (i.e. bending, folding, flexing, etc.) of tines **6729** against side **6711** of body **6713** of stimulation device **6710**, as shown in FIG. **30**D. In some examples, the sleeve **6751** may comprise a slit along its side or other mechanism to enable removably mounting the sleeve **6751** onto the stimulation device **6710** in the region of the “reverse orientation” tines **6729** into the collapsed configuration shown in FIG. **30**D. In some such examples, this removably mounting may comprise a sliding motion as represented via the directional arrow at identifier A to facilitate collapse of tines **6729**.
(328) As further shown in FIG. **30**D, in one aspect the preparation to implant the stimulation device **6710** further comprises slidably inserting the stimulation device **6710** into and within the hollow insertion needle **6760**, beginning with insertion of the distal end **6719** of stimulation device **6710** into the proximal end **6764** of the hollow insertion needle **6760**, as represented by directional arrow B. In some examples, the hollow insertion needle **6760** comprises a lumen **6768** defined by side wall **6765**, with the lumen **6768** extending between the proximal end **6764** and opposite distal end **6762**. The needle **6760** also may include a beveled portion **6763** at distal end **6762** to facilitate penetration of the needle **6760** into pertinent tissues at, and through, which the stimulation device **6710** is to be implanted. At least some example pertinent tissues may comprise tissues such as (but not limited to) subcutaneous tissues including but not limited to muscles like the sternocleidomastoid (SCM), platysma, omohyoid, sternohyoid, sternothyroid, etc.
(329) FIG. **30**E is a diagram including a side view schematically representing the example arrangement **6750** of FIG. **30**C, **30**D including example stimulation device **6710** being fully inserted within the hollow insertion needle **6760** to result in collapse (e.g. bending) of the tines **6727**A, **6727**B, **6727**C relative to (e.g. toward, against, etc.) side **6711** of stimulation device **6710**. Moreover, as shown in FIG. **30**E, the lumen **6768** of the needle **6710** is sized to slidably receive the sleeve **6751**, in its already mounted state over tines **6729** of stimulation device **6710**, within the hollow insertion needle **6760**. Finally, as represented by directional arrow C in FIG. **30**E, sleeve **6751** may be slidably removed out of the proximal end **6764** of needle **6760** and off the “reverse orientation” tines **6729** of the stimulation device **6710** to yield the configuration shown in FIG. **30**F in which the tines **6729** expand outward slightly to be in contact against the sidewall **6765** of needle **6760**.
(330) Accordingly, in this ready-to-be-implanted configuration, hollow insertion needle **6760** is inserted into and through pertinent tissues, as represented via directional arrow D.
(331) As further shown in FIG. **30**G, once the distal end **6719** of stimulation device **6710** has been positioned in its desired location relative to a target stimulation site (e.g. in stimulating relation to a target location along a nerve), then the needle **6760** may be slidably removed from the stimulation device **6710** as represented via directional arrow E. This maneuver causes release of tines **6727**A, **6727**B (from their collapsed position as in FIGS. **30**E, **30**F) into their extended position for engaging surrounding non-nerve tissues to secure the stimulation device **6710** (including the array **6714** of electrodes **6716**) to be in stimulating relation to a target nerve or tissue (e.g. muscle). Further proximal sliding movement of needle **6760** relative to stimulation device **6710** will result in the release of tines **6727**C and of the “reverse orientation” tines **6729** to also engage surrounding tissues so that the stimulation device **6170** will be present in the configuration shown in FIG. **30**C will all tines **6727**A, **6727**B, **6727**C, **6729** engaging surrounding tissues. In one aspect, by including at least one set of “reverse orientation” tines **6729** spaced apart from, and juxtaposed axially, relative to the tines **6727**A, **6727**B, **6727**C, the combination of tines **6727**A, **6727**B, **6727**C

and “reverse orientation” tines **6729** may act to prevent or minimize “ratcheting” which may sometimes occur with some implanted medical elements. In some instances, ratcheting may arise in areas of high motion, such as for implanted medical elements in the neck region at which repeated turning, tilting, flexion, etc. of the head repeated flexes the neck in various orientations. To the extent that an implanted medical element may have tines or other protrusions engaging surrounding non-nerve tissue, these repeated motions of the neck may result in the implanted medical element moving or migrating from its originally implanted position because the tines (or other protrusions) may move or “walk” slightly during or upon such repeated neck motion, thereby resulting in movement of the implanted medical element.

(332) However, the juxtaposition of tines **6729** with tines **6727**A, **6727**B, **6727**C may prevent or minimize such “ratcheting” because such repeated neck motion, with the presence of the “reverse orientation” tines **6729** would tend to cause potential movement of the stimulation device **6170** in a direction or orientation opposite of the movement that might otherwise result from the orientation of tines **6727**A, **6727**B, **6727**C.

(333) FIG. **30**H is a diagram including a front view schematically representing an example arrangement **6771** comprising an example stimulation device **6770** which may be used in example methods of implantation with or without additional tools. In some examples, the stimulation device **6770** may comprise at least some of substantially the same features and attributes as the stimulation device **6710** as described in association with at least FIG. **30**C, except with all tines being positioned proximal to the electrode array **6714** and a lower quantity of tines (e.g. **6727**C) which have a first orientation. However, as in the stimulation device **6710**, the stimulation device **6770** of FIGS. **30**H-**30**J comprises at least one set of opposite second orientation (i.e. “reverse orientation”) tines **6729**, which when juxtaposed with first orientation tines **6727**C, may prevent or minimize ratcheting-type migration of stimulation device **6770** from its original or intended implant location.

(334) As shown in FIG. **30**H, in addition to the features common with stimulation device **6710**, in some examples the stimulation device **6770** may comprise no tines distal to the electrode array **6714**, comprise at least one set of first orientation tines **6727**C adjacent electrode array **6714**, and at least one set of opposite, second orientation (i.e. “reverse orientation”) tines **6729** interposed between the first orientation tines **6727**C and proximal end **6718** of stimulation device **6770**. In examples in which the least one set of first orientation tines **6727**C may comprise more than one set of tines, the tines may be configured similar to the configuration shown in FIG. **30**C in which at least two sets of tines **6727**A, **6727**B are present but are spaced from each other along a portion of a length of the body **6713**. Moreover, in some examples, to the extent at no tines are present distal to the electrode array **6714**, a distal portion **6722** of body **6713** may be reduced in length as shown in FIG. **30**H, as compared to such a distal portion **6722** having a greater length when tines (e.g. **6727**A, **6727**B) are present.

(335) With further reference to FIG. **30**H, like the stimulation device **6710** in FIGS. **30**C-**30**G, a method of implanting stimulation device **6770** may comprise initially collapsing (e.g. bending) the “reverse orientation” tines **6829** against the side **6711** of body **6713** of stimulation device **6770** in order to permit loading of the stimulation device **6770** into and within a hollow insertion needle **6760**. Accordingly, in some examples the example arrangement **6771** may utilize a sleeve like sleeve **6751** in FIG. **30**D to engage and causes collapse (e.g. bending) of “reverse orientation” tines **6729** into a collapsed configuration. In a manner similar to that shown in FIGS. **30**E-**30**F, upon proximal slidable removal of such a sleeve, the resulting configuration of just the stimulation device **6770** within the lumen **6768** of the needle **6760** is shown in FIG. **30**I and in which the “reverse orientation” tines **6729** are in a partially collapsed state as constrained by sidewall **6765** of hollow insertion needle **6760**.

(336) In the configuration shown in FIG. **30**I, with the stimulation device **6770** releasably retained within the hollow insertion needle **6760**, the needle **6760** is inserted into and through pertinent tissues to deliver the stimulation device **6770** (which may comprise a stimulation portion of a

longer lead (not shown for illustrative simplicity)) into stimulating relation to target tissue, such as a target location along a nerve, as represented by directional arrow D.

(337) Once the stimulation device **6770** has been delivered and positioned as desired relative to a target tissue, the needle **6760** is slidably withdrawn from the stimulation device **6770**, as represented by directional arrow E in FIG. **30**J. This maneuver results in the release of tines **6727**C, **6729** from their collapsed state (FIG. **30**I) into an expanded state (FIG. **30**K) so that the tines **6727**C, **6829** may engage surrounding non-nerve tissues to secure the stimulation device **6770** in a robust, stable position in stimulating relation to a target tissue (e.g. nerve).

(338) However, with further reference to FIG. **30**J, when the hollow insertion needle **6760** is in a position just prior to the tines **6727**C, **6729** being released, in some examples the electrode array **6714** may significantly protrude from the distal end **6762** of the hollow needle **6760** such that further maneuvering of the combination of the needle **6760** and stimulation device **6770** may be performed while applying test stimulation signals via the electrode array **6714** to further identify or confirm a location of a desired target stimulation site. In one aspect, the absence of tines distal to the electrode array **6714** and the absence of tines among the electrodes **6716** of array **6714** may facilitate further positioning of the stimulation device **6770** (with support of needle **6760**) without the interference of more distal tines (as in FIGS. **30**C, **30**D, etc.) relative to a target stimulation location.

(339) Accordingly, after any further refinement of identifying or confirming a target stimulation location and upon further slidable removal of needle **6760** from the now implanted stimulation device **6770** (and from the pertinent portion of the patient's body), the tines **6727**C, **6729** of stimulation device **6770** may fully expand to their unrestrained state like that shown in FIG. **30**H, and which in turn engage surrounding non-nerve tissues to secure the stimulation device **6770** in the patient's body at the desired stimulation site.

(340) FIG. **30**K is a diagram including a front view schematically representing an example arrangement comprising an example stimulation device **6790** which may be used in example methods of implantation with or without additional tools. In some examples, the stimulation device **6790** may comprise at least some of substantially the same features and attributes as the stimulation device **6770** as described in association with at least FIG. **30**H-**30**J, except with at least one set of first orientation tines **6727**C being positioned among the electrodes **6716** of electrode array **6714**, such as being interposed between adjacent electrodes **6716**. Via this arrangement, the generally more proximal location of the tines **6727**C, **6729** relative to the electrode array **6714** (e.g. no tines distal to the electrode array) may still permit some maneuverability of the stimulation device **6790** (in a manner similar to that described for stimulation device **6770**) while further identifying and/or confirming a target stimulation site just prior to finalizing a chronic implantation location. Moreover, as in the stimulation device **6770**, the stimulation device **6790** of FIG. **30**K comprises at least one set of opposite second orientation (i.e. “reverse orientation”) tines **6729**, which when juxtaposed with first orientation tines **6727**C, may prevent or minimize ratcheting-type migration of stimulation device **6810** from its original or intended implant location as a result of flexion and motion of the neck and upper body. Finally, as also noted elsewhere regarding some other example implementations, by interposing some tines **6727**C between some electrodes **6716** of array **6716** may enhance robust securing of the electrode array **6714** in close proximity to and stimulating relation to a target stimulation site.

(341) FIG. **30**L is a diagram including a front view schematically representing an example arrangement **6800** comprising an example stimulation device **6810** which may be used in example methods of implantation with or without additional tools. In some examples, the stimulation device **6810** may comprise at least some of substantially the same features and attributes as the stimulation device **6710** as described in association with at least FIG. **30**C, except with all tines being positioned distal to the electrode array **6714** and a lower quantity of tines **6727**A which have a first orientation. However, as in the stimulation device **6710**, the stimulation device **6810** of FIGS. **30**L-

30R comprises at least one set of opposite second orientation (i.e. "reverse orientation") tines **6829**, which when juxtaposed with first orientation tines **6727**A, may prevent or minimize ratcheting-type migration of stimulation device **6810** from its original or intended implant location. However, in some examples, all tines present distal to the electrode array **6714** may have the same orientation.

(342) As shown in FIG. **30**L, in addition to the features common with stimulation device **6710**, in some examples the stimulation device **6810** may comprise no tines proximal to the electrode array **6714**, at least one set of first orientation tines **6727**A adjacent distal end **6719**, and at least one set of opposite, second orientation (i.e. "reverse orientation") tines **6829** interposed between the electrode array **6714** and the first orientation tines **6727**A. In examples in which the least one set of first orientation tines **6727**A may comprise more than one set of tines, the tines may be configured similar to the configuration shown in FIG. **30**C in which at least two sets of tines **6727**A, **6727**B are present but spaced from each other along a portion of a length of the body **6713**.

(343) With further reference to FIG. **30**L, like the stimulation device **6710** in FIGS. **30**C-**30**G, a method of implanting stimulation device **6810** may comprise initially collapsing (e.g. bending) the "reverse orientation" tines **6829** against the side **6711** of body **6713** of stimulation device **6810** in order to permit loading of the stimulation device **6810** into and within a hollow insertion needle **6760**. Accordingly, in some examples the example arrangement **6800** may comprise a sleeve **6830** which is removably mounted relative to the body **6713** of stimulation device **6810** near proximal end **6718** (which may be connected to or extend distally from a lead body) and then slidably advanced, as represented via directional arrow F, over and along the body **6713** of stimulation device **6810** until a distal end **6831** of sleeve **6830** engages and causes collapse (e.g. bending) of "reverse orientation" tines **6829** into a collapsed configuration within lumen **6833** of sleeve **6830**, as shown in FIG. **30**M.

(344) With the stimulation device **6810** and sleeve **6830** in the configuration shown in FIG. **30**M, this combination of elements is slidably inserted, via a proximal end **6764** of the hollow insertion needle **6760**, into and within lumen **6768** of needle **6760**, as represented via directional arrow G (FIG. **30**M) to cause collapse (e.g. bending, rotation, etc.) of the first orientation tines **6727**A relative to the sides **6711** of body **6713** of stimulation device **6810** and insertion and advancement of the already removably-mounted sleeve **6830** within lumen **6768** of needle **6760**. The resulting configuration is shown as example arrangement **6850** in FIG. **30**N.

(345) Thereafter, as represented by directional arrow H in FIG. **30**O, the sleeve **6830** is slidably withdrawn proximally out of needle **6760** via the proximal end **6764** of needle **6760** while stimulation device **6810** is retained within lumen **6768** of needle **6810**. Among other factors, the releasable engagement of tines **6727**A against the sidewall **6765** of needle **6760** help to retain stimulation device **6810** within the lumen **6768** of needle **6760** both during and after slidable removal of sleeve **6830** from stimulation device **6810** and needle **6760**. As the sleeve **6830** is being slidably removed proximally just past the opposite second orientation (i.e. "reverse orientation") tines **6829**, those tines **6829** are released to extend slightly outward while still being constrained by the sidewall **6765** of the needle **6760**, as shown in FIG. **30**O. Upon complete withdrawal of the sleeve **6830** from the stimulation device **6810** and needle **6760**, the resulting configuration of just the stimulation device **6810** within the lumen **6768** of the needle **6760** is shown in FIG. **30**P.

(346) In the configuration shown in FIG. **30**P, with the stimulation device **6810** releasably retained within the needle **6760**, the needle **6760** is inserted into and through pertinent tissues to deliver the stimulation device **6810** (which may comprise a stimulation portion of a longer lead (not shown for illustrative simplicity)) into stimulating relation to target tissue, such as a target location along a nerve, as represented by directional arrow I.

(347) Once the stimulation device **6810** has been delivered and positioned as desired relative to a target tissue, the needle **6760** is slidably withdrawn from the stimulation device **6810**, as represented by directional arrow J in FIG. **30**Q. This maneuver results in the release of tines **6727**A, **6829** from their collapsed state (FIG. **30**P) into an expanded state (FIG. **30**Q) so that the

tines **6727**A, **6829** may engage surrounding non-nerve tissues to secure the stimulation device **6810** relative in a robust, stable position in stimulating relation to a target tissue (e.g. nerve).
(348) Upon further slidable removal of needle **6760** (FIG. **30**Q) from the now implanted stimulation device **6810** (and from the pertinent portion of the patient's body), the stimulation device **6810** remains chronically implanted (in the configuration shown in FIG. **30**R) in the patient's body at the desired stimulation site.
(349) FIG. **30**S is a diagram **6900** including a side view schematically representing an example stimulation device **6910**. In some examples, the stimulation device **6910** comprises at least some of substantially the same features and attributes as the stimulation device **6810** described in association with at least FIGS. **30**L-**30**R, except with stimulation device **6910** comprising anchoring structure **6920** instead of the anchoring arrangement of tines **6727**A, **6829** in the stimulation device **6810** of FIGS. **30**L-**30**R. As in the examples of stimulation devices **6710**, **6810**, the stimulation device **6910** may comprise a distal portion of a stimulation lead body which extends proximally from the proximal end **6718** of the stimulation device **6910**.
(350) As shown in FIG. **30**S, in some examples the anchoring structure **6920** comprises a plurality of anchor elements **6924** which protrude from the sides **6711** of the body **6713** of the stimulation device **6910**. In some examples, the anchor elements **6924** may be grouped into different arrays **6922**A, **6922**B while in some examples, the anchor structure **6920** may comprise a single cluster of elements **6924**.
(351) It will be understood that in some examples, the elements **6924** may extend about an entire periphery (e.g. circumference of body **6713**).
(352) As shown in FIG. **30**S, the anchor structure **6920** is positioned distal to the electrode array **6716**, being between the electrode array **6714** and the distal end **6719** of the body **6713** of the stimulation device **6910**.
(353) In this configuration, the position of the anchor structure **6920** on just one end (e.g. the distal end) of the electrode array **6714** may prevent or minimize “lead elongation”, i.e. elongation of the lead body **6713** which may potentially be caused by muscle movement when anchoring elements (e.g. tines) are present on opposite ends of the electrode array **6714**.
(354) In some examples, the elements **6924** may comprise a filament (e.g. fine thread) which is flexible and resilient, and biased to extend outward from the side **6711** of body **6713**. The filament may be formed of a polymer material, such as but not limited to, nylon, propylene, silk, polyester, trimethylene carbonate, and the like. In some examples, such filaments may be resorbable or may be non-resorbable.
(355) In some examples, each element **6924** may comprise a diameter (or greatest cross-sectional dimension) of about 0.05 to about 0.40 millimeters. In some examples, each element **6924** may comprise a length of about 0.2 to about 2 millimeters. In some examples, each element **6924** may comprise a length about 0.5 percent to about 50 percent of a diameter of the lead body **6710** in the region of the electrode array **6714** and/or at distal end **6719**. In some examples, the anchor structure **6920** may be embodied as a matrix of heterogeneous elements via filaments having pseudo-random sizes, shapes, orientations and/or positions exhibiting more variation than a plurality of identical discrete elements (e.g. **6927** in FIG. **30**T), which may be visually recognizable. Meanwhile, all of the various features of the matrix of heterogeneous elements may not be readily visually recognizable. Among other features, this heterogeneous matrix may enable fixation in both (e.g. opposite) orientations (along length of stimulation portion/lead) and ease deliverability of the lead, lead portions. At least some example implementations of anchor structures **7000**, **7100** comprising a matrix of heterogeneous elements are described later in association with at least FIGS. **30**V-**30**W. In some examples, the heterogeneous elements may sometimes be referred to as heterogeneous fixation elements.
(356) In some examples, the anchor structure **6920** may comprise a plurality of well-defined, discrete elements but with at least some of the discrete elements comprising a size, shape,

orientation, and/or position different from a size, shape, orientation, and/or position of other respective discrete elements of the anchor structure **6920**.

(357) In some examples, the anchor structure **6920** may enhance some example methods of implantation of a stimulation device at least because the respective elements **6924** exhibit a low profile relative to an outer diameter of the body **6713** of the stimulation device **6910** such that the stimulation device **6910** (FIG. **30**S-**30**) can be delivered via hollow insertion needle **6760** without a sleeve (e.g. **6751** in FIG. **30**D, **6830** in FIG. **30**O, etc.) or similar elements while still robustly securing the stimulation device **6910**.

(358) As further shown in the greatly enlarged side view of just one element **6924** in FIG. **30**T, in some examples, at least some (or all) of the elements **6924** may comprise protrusions **6927** on their surfaces, which in some examples may comprise barbs, hooks, or other sharp tipped structures. In some examples, the protrusions **6927** may be present on just a portion of the element **6925**, such as but not limited to a distal portion **6929** of the element **6924**. However, in some examples, the protrusions **6927** may be present on the entire or substantially entire surface of the element **6924**. In yet other examples, groups of protrusions **6927** may be positioned in spaced apart clusters, which are spaced apart from each other along and around the surface of the element **6924**.

(359) It will be further understood that the protrusions **6927** are not strictly limited to structures having a sharp-tip or hook but may comprise structures comprising a rounded edge while including a sticky surface coating or formed as a non-sharp tipped member which can securely engage a surrounding non-nerve tissue in close proximity to a target stimulation site.

(360) With regard to the example stimulation device **6910** in FIGS. **30**S-**30**W, it will be understood that in some examples the anchoring structure **6920** may be located solely proximally of the electrode array **6714** such that no similar anchoring structure **6920** is located distal to the electrode array **6714**.

(361) However, in some examples, a first anchoring structure **6920** may be present distal to the electrode array **6714** as shown in FIGS. **30**O-**30**Q and a second anchoring structure, similar to anchoring structure **6920**, may be present proximal to the electrode array **6714** so that the stimulation device **6910** bears resemblance to the stimulation device **6710** of FIG. **30**C, at least to the extent that some anchoring structure or elements are present on opposite sides of the electrode array **6714**.

(362) FIG. **30**U is a diagram including a side view schematically representing an example arrangement **6950** of an example device and/or example method of implantation including the stimulation device **6910** slidably, removably inserted within the hollow insertion needle **6760**. In some examples, the needle **6760** may comprise at least some of substantially the same features and attributes of the needle **6760** and associated example methods as previously described in association with at least FIGS. **30**C-**30**N.

(363) As shown in FIG. **30**U, upon insertion of stimulation device **6910** within lumen **6768** of hollow insertion needle **6760**, the elements **6924** of anchor structure **6920** become at least partially collapsed against side **6711** of stimulation device **6910**. In a manner similar to previously-described examples, with the stimulation device **6910** carried within the hollow insertion needle **6760**, the combination of these elements are finally positioned within the vicinity of a target stimulation location. Needle **6760** is then withdrawn (represented by directional arrow Q) to leave the stimulation device **6910** in stimulation relation to the target stimulation location and to enable the elements **6924** of anchor structure **6924** to engage surrounding non-nerve tissues to robustly secure the stimulation portion (e.g. electrode array **7614**) in the stimulating relation position.

(364) FIG. **30**V is a diagram including an enlarged top view schematically representing an example anchor structure **7000** formed on, and including as part of the anchor structure, a base **7002**. In some examples, the anchor structure **7000** may comprise an analogous example implementation of the anchor structure **6920** in FIGS. **30**S-**30**U and may comprise at least substantially the same features and attributes as the anchor structure **6920**, particularly with respect to providing a matrix

of heterogeneous elements. However, in some examples, the anchor structure **7000** may have wide applicability to act as an anchor or position-influencing element.

(365) As shown in FIG. **30**V, the anchor structure **7000** may comprise an array **7010** of example heterogeneous elements **7012**, **7013**, **7016** which together may form a matrix, network, or the like which may overlap or otherwise be juxtaposed relative to each other to create a generally traction-favoring surface profile. It will be understood that in some examples, the various heterogeneous elements of array **7010** may be positioned much closer to each other than shown in FIG. **30**V in order to touch, overlap, partially interlock or interfere with each other, etc. so as to increase the frictional properties (e.g. slide-resistance) of the anchor structure or to reduce the frictional properties (e.g. slidability) of the anchor structure, depending on the type, size, orientation, coating, etc. of the particular arrangement of elements of the array **7010**.

(366) In general terms, the various elements of the array **7010** may comprise a flexible, resilient material. However, depending on the goals re slidability or slide-resistance, some elements may be firmer or softer.

(367) In some examples, the particular types, spacing between, orientation, position, relative flexibility, etc. of the heterogeneous elements of the array **7010** may be selected and formed to correspond to a selectable coefficient of kinetic friction to enable a desired bias for controlled slidable movement relative to tissues within a patient's body and/or relative to lumen within a patient's body and/or to correspond to a selectable coefficient of static friction to enable a desired bias to remain statically positioned at a chose location relative to tissues or within a lumen.

(368) In some examples, whether or not expressed formally as a coefficient of kinetic or static friction, the various heterogeneous elements of the array **7010** are selected and formed according to their shape, position, spacing, orientation relative to each other, relative flexibility, etc. to create a desired anchoring effect while still permitting some degree of slidable advancement.

(369) As shown in FIG. **30**V, at least some example shapes (as seen in cross-section from a top view) may comprise elements with shapes which are triangular **7012**, circular **7013**, rectangular **7016**, and the like. The elements also may have different sizes (e.g. S**2**), and spacing (e.g. S**1**) between each other or relative to an edge **7031** (e.g. S**2**) of the base **7002**. In some examples, at least some of the elements of array **7010** may comprise hook-shapes, J-shapes, U-shapes, etc. In some examples, at least some of the elements or the juxtaposed pattern of such elements, may promote tissue in-growth and long term fixation, such as but not limited to, apertures formed in such elements or by the juxtaposition of some of the respective elements.

(370) The various elements also may be organized in directional patterns, such as being in rows aligned in a first orientation (K) or second orientation (L) which are orthogonal to each other, or in other non-orthogonal orientations. Such orientations may be used to effect selectable bias to permit or prevent slidable movement in various directions, which may enhance positioning and/or anchoring of the medical element on which the anchor structure **7000** is located.

(371) In some examples, at least some elements of the array **7010** may be arranged along a periphery **7030** of the base **7002** in a row or other organizational pattern. The elements **7034** may have the same size, shape, positions, etc. or may have sizes, shapes, positions different from each other. By providing this configuration along one or more edges **7031** of the base **7002**, the anchor structure **7010** may influence slidability or slide-resistance in particular directions. In a related aspect, the presence or absence of elements of array **7010** in an interior portion **7040** also may provide analogous influences, with or without the edge-type rows, etc. of such elements.

(372) In some examples, the surface **7040** of the base **7002** and/or the elements of array **7010** also may comprise a coating with desired lubricous and/or frictional qualities, which may be selected to work synergistically with the various shapes, sizes, positions, spacing, orientation, etc. of the elements of array **7010**.

(373) FIG. **30**W is a diagram including an enlarged side view schematically representing an example anchor structure **7100** formed on, and including as part of the anchor structure, a base

7002. In some examples, the anchor structure **7100** may comprise an analogous example implementation of the anchor structure **6920** FIGS. **30**S-**30**U and may comprise at least substantially the same features and attributes as the anchor structure **6920**, particularly with respect to providing a matrix or network of heterogeneous elements. However, in some examples, the anchor structure **7100** may have wide applicability to act as an anchor or position-influencing element.

(374) In some examples, the anchor structure **7100** in FIG. **30**W may comprise at least some of substantially the same features and attributes as anchor structure **7000** in FIG. **30**V.

(375) As shown in FIG. **30**V, the array **7110** of elements comprise different shapes, sizes, positions, spacing, orientations, etc. For example, rectangular elements **7130**A, **7130**B,**7130**C, **7130**D exhibit differing angular orientations (e.g. relative to a horizontal plane through which base **7002** extends), which may sometimes be referred to as being bi-directional or multi-directional. Other elements may comprise spherical shaped elements **7120**A, **7120**B, pyramid-shaped elements **7122**, etc. The respective elements of array **7110** may be formed according to a selectable height (per height arrow H), which may vary from each other as part of a desired effect to promote slidability or slide-resistance, depending on the intended use of the anchor structure and medical element to which is formed/attached. It will be further understood that some shapes, such as the spherical elements **7120**A, **7120**B may be more likely to enhance slidability because of their smooth convex surface while some shapes, such as the pyramid element **7122** or rectangular elements (**7130**A-**7130**D), may enhance slide-resistance, depending on their orientation. In some examples, directional arrow S**4** may represent relative horizontal spacing between elements of array **7010**.

(376) In some examples, the base **7002** may formed in a two-dimensional plate shape such that the anchor structure **7000** or **7100** may be readily formed or attached to a back side of a carrier opposite to an electrode side of a stimulation portion, such as a paddle electrode. However, in some examples, the base may comprise a cylindrical shape such that the elements of array **7010** (FIG. **30**V) and/or array **7110** (FIG. **30**W) may extend circumferentially outward from a cylindrically shaped lead on which the array **7010** or **7110** is formed or attached.

(377) FIG. **31**A is a diagram including a side view schematically representing an example arrangement **8600** including a stimulation element **6710** comprising a linear array **6714** of spaced apart electrodes **6716** (e.g. ring electrodes, split ring electrodes, or other electrodes). The stimulation device **6710** may comprise a distal end **6719** and an opposite proximal end **6718**, which is supportable on (or which extends from) a stimulation lead body **3222**. In some examples, the example arrangement **8600** (including stimulation element **6710**) comprises at least some of substantially the same features and attributes as at least some of the stimulation devices in examples of the present disclosure, as described in association with at least FIGS. **1**-**30**W.

(378) In addition, the example arrangement **8600** comprises an anchor element **8638** in the shape of a spiral or helix, which may be used to anchor a distal end **6719** of stimulation device **6710** relative to non-nerve tissue to thereby secure the stimulation element in stimulating relation to a target nerve location. In some examples, the anchor element **8638** may be formed as a flexible, resilient member to at least partially wrap around a non-nerve structure (e.g. tendon) and/or have its tip **8623** configured to puncture or penetrate a non-nerve structure near the target nerve location.

(379) FIG. **31**B is a diagram including a side view schematically representing an example arrangement **8640** including a stimulation device **6710** and comprising at least some of substantially the same features and attributes as the example arrangement **8600** in FIG. **31**A, except further comprising a dissolvable capsule **8642**. In some examples, the dissolvable capsule **8642** encapsulates the anchor element **8638** prior to and during delivery (e.g. via an implant-access incision, via delivery tools, etc.) of the stimulation device **6710** to its target stimulation location. In one aspect, the capsule **8642** comprises a dissolvable material, which when exposed to fluids and/or the temperature within a patient's body during implantation, will dissolve within a suitable time frame during which the capsule **8642** remains intact at least until the stimulation lead is delivered to

its target stimulation location and a relatively short time thereafter. Via this arrangement, the capsule **8642** may prevent the anchor element **8638** (including tip **8623**) from engaging tissues prior to the stimulation portion (e.g. electrode array **6714**) reaching its target stimulation location. However, after the stimulation device **6710** has been delivered to the intended location, after a short period of time the capsule **8642** dissolves to expose the anchor element **8638** at which time a clinician may rotate the stimulation device **6710** (as represented by directional arrow R**1**) to cause the exposed anchor element **8638** to rotatably engage the surrounding non-nerve tissue to robustly secure the electrode array **6714** in stimulating relation to the target nerve stimulation location.
(380) In some examples, the material forming the dissolvable capsule **8642** may comprise a sugar-based material or other material which dissolves reasonably quickly (but not instantly) when exposed to body fluids, body temperature, etc. within a patient's body. The particular composition of the material may be selected to control or influence the time duration before the capsule **8642** starts and/or completes dissolving within the patient's body.
(381) FIGS. **31**C-**31**D are diagrams including a side view schematically representing an example arrangement **8660** including a stimulation device **6710** and comprising at least some of substantially the same features and attributes as the example arrangement **8600** in FIG. **31**A, except with the anchor element **8638** being selectively moveable from a retracted position shown in FIG. **31**C to an extended position shown in FIG. **31**D.
(382) As shown in FIG. **31**C, the stimulation device **6710** may comprise an array **6714** of spaced apart electrodes **6716** and an anchor structure **8662** positioned distal to the electrode array **6714** and defining a distal portion of the stimulation device **6710**. In some examples, the anchor structure **8662** may comprise a hollow tubular frame portion **8664** including a sidewall **8665** to define a lumen **8667**. The anchor structure **8662** also comprises an anchor element **8638** (e.g. helix, coil, and the like) releasably retained (e.g. temporarily housed) within the lumen **8667** of the tubular frame portion **8664**. In this configuration, the stimulation device **6710** is adapted for insertion into and within delivery tools and/or for advancement within and among tissues of a patient's body while preventing anchor element **8638** from engaging such tissues at least until a stimulation portion (e.g. electrode array **6714**) of the stimulation device **6710** (of a stimulation lead) has been delivered to its target stimulation location at which it will become chronically implanted. Once in that chronic implant location, the anchor element **8638** is released to extend outwardly (e.g. protrude relative to) from the distal end **8624** of the stimulation device **6710** so that the anchor element **8638** may engage surrounding non-nerve tissues and thereby robustly and reliably secure at least the stimulation portion (e.g. electrode array **6714**) in stimulating relation to the identified target stimulation location. In some examples, as shown in FIGS. **31**C-**31**D, the anchor element **8638** may be supported by a rod **8669** or other element such that translational movement (as represented by directional arrow F**1**) and/or rotational movement (as represented by directional arrow R**1**) of rod **8669** (or other element) will cause the extension of the anchor element **8638** into the position shown in FIG. **31**D. In some examples, the rod **8669** may extend through and within a lumen within the body **6713** of the stimulation device **6710** (and supporting lead). In some examples, mechanisms other than rod **8669** may be used to activate and/or otherwise cause movement of the anchor element **8638** from its retracted position (FIG. **31**C) to its extended position (FIG. **31**D). In some such examples, the rod **8669** and/or other mechanisms may be detachable from the anchor element **8638**.
(383) With the anchor element **8638** in its extended position, the user may rotate (e.g. twist) the body **6713** of the stimulation device **6710** (as part of twisting the entire lead supporting the body of the stimulation device **6710**) to cause the anchor element **8638** to securely engage surrounding non-nerve tissue, as mentioned above, which in turn secures the electrode array **6714** in stimulating relation to the identified target stimulation location of a nerve.
(384) In some examples, the tubular frame portion **8662** may comprise a length (LA**1**) which is generally the same as or slightly longer than a length of the anchor element **8638** so that when the

anchor element **8638** is in its retracted position (FIG. **31**C), the anchor element **8638** is prevented from engaging surrounding non-nerve tissues until the anchor element **8638** is moved into its extended position as noted above.

(385) FIG. **31**E is a diagram including a side view schematically representing an example arrangement **8670** comprising an example device for, and/or example method of, implantation of stimulation device **6710**. In some examples, the stimulation device **6710** may comprise at least some of substantially the same features and attributes as the stimulation device in the example arrangement **8600** of at least FIG. **31**A, except including a helical-type anchor element **8672** formed or mounted on a body **6713** of the stimulation device **6710** (FIG. **31**E) instead of an end-mounted anchor element **8638** in FIG. **31**A. In some examples, the anchor element **8672** comprises a helical screw thread extending outward from side **6711** of body **6713** of the stimulation device **6710** with gaps **8675** extending between successive threads **8674** of the anchor element **8672**. In some examples, an utmost distal thread **8676** of the screw thread **8672** terminates proximal to the electrode array **6714**. In some such examples, the utmost distal thread **8676** may terminate in close proximity to the electrode array **6714**, which may in some examples further ensure that the electrode array **6714** remains robustly in stimulating relation to a target stimulation location. The screw thread **8672** is sized and shaped to engage surrounding non-nerve tissues upon a clinician exerting, during implantation, rotation (e.g. twisting) of the body **6713** of the stimulation device **6710** (and its supporting stimulation lead body), as represented by directional arrow R**1**.

(386) In a manner similar to that described with respect to at least some of the previously described examples tines, in some examples the electrodes **6716** of array **6714** may be spaced apart by a distance large enough such at least some threads **8676** may be interposed between adjacent electrodes **6716** of the array **6714**. This arrangement may help further co-locate the anchoring forces (created by the gripping action of the threads **8676** relative to surrounding non-nerve tissue) with the elements (e.g. electrodes **6716**) which are desired to be secured robustly in stimulating relation to a target stimulation location. While not shown in explicitly in FIG. **31**E, at least some threads **8674** also may be located distal to the electrode array **6714**, whether or not some threads **8674** are present proximal to electrode array **6714** and/or interposed among electrodes **6716** of the electrode array **6714**.

(387) FIGS. **31**F-**31**G are diagrams including a side view schematically representing an example arrangement **8680** comprising an example device for, and/or example method of, implantation of stimulation device **6710**. In some examples, the stimulation device **6710** may comprise at least some of substantially the same features and attributes as the stimulation device of at least FIG. **31**A, except with the end-mounted anchor structure **8682** taking the form of a non-helical structure. FIG. **31**F shows the anchor structure **8682** in a collapsed, first state prior to deployment while FIG. **31**G shows the same anchor structure **8682** in an expanded, second state upon deployment within the patient's body in proximity to a target stimulation location for electrode array **6714**.

(388) As further shown in both FIGS. **31**F-**31**G, in some examples the anchor element structure **8682** may comprise multiple elements **8684**, each of which may comprise a base portion **8686**, arm **8688**, and extension **8689**. The base portion **8686** of each element **8684** is mounted to the distal portion **6722** of the stimulation device **6710** and in some examples, the base portion **8686** of the multiple elements **8684** may be connected together or form a common element. In some examples, arm **8688** extends in a generally opposite orientation from base portion **8686** and extension **8689** may extend at some angle relative to the arm **8688**. In general terms, the size, shape, and relative orientations of the base portion **8686**, arm **8688**, and extension **8689** are arranged together so that in a retracted/collapsed state as shown in FIG. **31**F, each element **8682** exhibits a compressed volume which is capable of expanding to a much larger volume as shown in FIG. **31**G. In some examples, each element **8684** may comprise a shape-memory material (e.g. Nitinol, other) such that the anchor structure **8682** can remain in its collapsed/reduced volume state (FIG. **31**F) until the anchor structure **8682** is placed within another tool (e.g. cannula, hollow sheath, sleeve, other)

and/or desired environment (e.g. within the patient's body at desired location) in which expansion of the anchor structure **8682** into its expanded volume (FIG. **31**G) will be appropriate. In particular, once placed within the patient's body, the shape memory material will respond to the increased temperature and transition from the collapsed state (FIG. **31**F) to the expanded state (FIG. **31**G). However, using a delivery tool such as, but not limited to, at least some of the example delivery tools in the various examples of the present disclosure, the anchor structure **8682** may be prevented from fully expanding by the walls (or other elements) of the delivery tool until the delivery tool is within the patient's body at a desired location and the delivery tool is withdrawn, thereby permitting the elements **8684** of anchor structure **8682** to fully expand via a slight divergence of the base portions **8686** (relative to each other), unfolding of the arms **8688** relative to the base portions **8686**, and unfolding of the extensions **8689** relative to the arms **8688** due to the automatic, natural activation of the shape memory features of the respective elements **8684** of anchor structure **8682**.

(389) Among other aspects, in its expanded state (FIG. **31**G), the arms **8688** and extensions **8689** of each element **8684** have an opposite, second orientation (i.e. a rearward orientation) relative to a first orientation (i.e. forward) of the base portion **8686**, such that the arms/extensions **8688**/**8689** effectively form a hook or catch which may further facilitate robust securement of the stimulation device **6710** relative to the surrounding non-nerve tissues.

(390) It will be understood that the shape, size, number, etc. of the elements **8684** of anchor structure **8682** may take a wide variety of forms and that the particular arrangement of the base portion **8686**, arm **8688**, and extension **8689** of elements **8684** as shown in FIGS. **31**F-**31**G is just one example implementation.

(391) Among other aspects, the use of shape memory material to form elements **8684** may enable the anchor structure **8682** to achieve a significantly smaller collapsed volume and a significantly larger, expanded volume than might otherwise be achieved in the absence of the shape memory material, which eases both delivery and deployment, respectively, of the anchor structure **8682**. As shown in FIG. **31**G, in its fully expanded state the anchor structure **8682** may extend significantly further outward from the sides **6711** of the stimulation device **6710** than may otherwise be achievable via at least some other anchors present on a body of a stimulation device, which in turn, may enhance a robust securing of the electrode array **6714** to be in stimulating relation to a target stimulation location.

(392) However, in some examples, anchor structure **8682** may be formed of materials other than a shape memory material with a delivery tool being relied upon to retain the expandable anchor structure **8682** in a primarily collapsed stated within a delivery tool until the stimulation device **6710** is in a location suitable for deployment of the anchor structure **8682** to its fully expanded state within and relative to surrounding non-nerve tissues.

(393) With regard to the various example arrangements (e.g. devices, methods, etc.) throughout the present disclosure which may comprise implanting more than one lead, such as but not limited to, bifurcated leads, it will be understood that at least some of the various example anchors (e.g. tines, filaments, elements, etc.) for securing a stimulation lead may be included on each lead of a bifurcated lead. In one aspect, doing so will secure each lead independently relative to the other lead(s). Among other features, this independent anchoring of multiple leads (e.g. bifurcated leads, other) may permit relative motion of the leads relative to each other while still maintaining robust securement of the stimulation portion of each respective lead at the respective target stimulation location. This arrangement, in turn, may enhance patient comfort.

(394) It will be understood that the various anchors, delivery tools, elements and associated delivery methods as described throughout various examples of the present disclosure with regard to stimulation devices, etc. may be used to deliver, implant, etc. sensing leads and/or other types of leads, appropriately shaped/sized implantable medical elements, etc.

(395) FIG. **32**A is a diagram including a side view schematically representing an example arrangement **4000** including intravascular pathways and/or other access for delivering a stimulation

element to target stimulation locations at the ansa cervicalis-related nerve **316** and/or hypoglossal nerve **305**. In some examples, the example arrangement **4000** may comprise stimulation elements (and associated methods) comprising at least some of substantially the same features and attributes, or an example implementation of, the previously described example arrangements of the present disclosure.
(396) FIG. **32**A depicts the ansa cervicalis-related neve **316** in the same general manner as in at least FIG. **2**, and further depicts the anterior jugular vein **4031**, thyroid vein **4021** (inferior **4025** and superior **4023**), and the sternohyoid muscle **4060**, which overlies (e.g. anterior to) the sternothyroid muscle **4062**. It will be understood that just portions of the above-identified anatomical features as shown in FIG. **32**A for illustrative simplicity and clarity.
(397) As shown in FIG. **32**A, in at least some patients, one or both of the anterior jugular vein **4031** and the inferior thyroid vein **4025** pass near at least some of the portions of the sternothyroid branches (e.g. **342**) of the ansa cervicalis nerve loop **319**. Accordingly, via an example device and/or example method, a stimulation element may be delivered intravascularly via one or both such veins **4031**, **4025** to be in transvascular stimulating relation to a portion of the ansa cervicalis-related nerve **316** in order to increase and/or maintain upper airway patency by causing contraction of the sternothyroid muscle **4062** and/or sternohyoid muscle **4060**.
(398) As further shown in FIG. **32**A, the superior thyroid vein **4023** may pass near a superior root **325** or other portions of the ansa cervicalis-related nerve **316**. Accordingly, in an example device and/or example method, a stimulation element may be delivered intravascularly via the superior thyroid vein **4023** to be positioned adjacent to, and in transvascular stimulating relation to, the superior root **325** of the ansa cervicalis-related nerve **316** in order to increase and/or maintain upper airway patency by causing contraction of at least some muscles (e.g. sternothyroid, sternohyoid, etc.) innervated by the superior root **4023** of the ansa cervicalis-related nerve **316**. As previously noted, among other effects, contraction of such muscles may cause inferior movement of the larynx, which may increase and/or maintain upper airway patency to thereby prevent or ameliorate sleep disordered breathing, such as obstructive sleep apnea.
(399) In some examples, such intravascular delivery (for transvenous stimulation) via the anterior jugular vein **4031** and/or the thyroid vein **4021** may be implemented via at least some of substantially the same features and attributes of the example stimulation elements as previously described in association with at least FIGS. **1-31**, and in particular with respect to at least some of substantially the same features and attributes of the intravascular delivery examples in association with at least FIGS. **15**A-**15**C. For instance, just one or both of stimulation elements **1810**A, **1813**A may be provided for such intravascular delivery and transvenous stimulation via veins **4031**, **4021** in the example arrangement **4000** of FIG. **32**A. Moreover, more than one transvascular (e.g. transvenous) stimulation lead and/or more than one branch of such transvascular stimulation leads may be implanted to provide stimulation of multiple stimulation targets of the ansa cervicalis-related nerve and/or other upper airway patency-related tissues.
(400) In some examples, other portions of the vasculature may be used to intravascularly deliver a stimulation element to be in stimulating relation to a target nerve location, including but not limited to, the ansa cervicalis-related nerve **316** and/or other upper airway patency-related tissues.
(401) FIG. **32**B is a diagram like the diagram in FIG. **32**A, except further schematically representing an example arrangement **4100** in which at least two microstimulators **4113**A, **4113**B are implanted within a head-and-neck region **520** (e.g. also FIG. **11**A). In some examples, the example arrangement **4100** may comprise stimulation elements (and associated methods) comprising at least some of substantially the same features and attributes and/or comprising an example implementation of, the previously described example arrangements of the present disclosure.
(402) As shown in FIG. **32**B, a first microstimulator **4113**A is delivered within and through the vasculature (i.e. intravascularly) to be adjacent to, and in transvascular stimulating relation to, a

target nerve. In one example shown in FIG. **32**B, the blood vessel comprises a superior thyroid vein **4023** and the target nerve comprises a superior root **325** of the ansa cervicalis-related nerve **316**. However, it will be understood that this depiction is merely representative and that the microstimulator **4113**A may be delivered intravascularly within and through other vessels such as, but not limited to, the anterior jugular vein **4031**, external jugular vein, and/or other blood vessels (e.g. superior laryngeal vein). With this in mind, the microstimulator **4113**A may be placed in transvascular (e.g. transvenous) stimulating relation to nerve branches of the ansa cervicalis-related nerve **316** other than the superior root **325**.
(403) Meanwhile, as further shown in FIG. **32**B, a second microstimulator **4113**B is implanted subcutaneously (or percutaneously) to be in stimulating relation to a target nerve location and secured in place relative to a non-nerve tissue **2929** via an anchor element **2927**. In one example depicted in FIG. **32**B, the second microstimulator **4113**B is placed in stimulating relation to nerve branches **342**A, **342**B (of the ansa cervicalis-related nerve **316**) associated with at least the sternothyroid and sternohyoid muscles. However, it will be understood that this depiction is merely representative and that the microstimulator **4113**B may be implanted relative to other nerve branches, roots, etc. of the ansa cervicalis-related nerve **316**. With this in mind, the microstimulator **4113**A may be placed in stimulating relation to nerve branches of the ansa cervicalis-related nerve **316** other than the branches **342**A, **342**B.
(404) Moreover, in some examples, in general terms the microstimulator **4113**B may be secured relative to a non-nerve tissue **2929** and relative to its target nerve locations via at least some of the anchoring elements described in association with at least FIGS. **6**A-**6**B, **22**A-**22**B, and/or **27**A-**31**. As further shown in FIG. **32**B, depending on the particular nerve branch to be stimulated, the anchor element **2927** may comprise at least one the anchor elements (or analogous elements) identified in legend **2950** “Anchor Locations” in FIG. **32**B.
(405) Via the example arrangement **4100**, multiple different nerve locations of the ansa cervicalis-related nerve **316** may be stimulated in a coordinated manner to more fully leverage the physiologic processes associated with a particular goal, such as increasing and/or maintaining upper airway patency. In some such examples, and as noted relative to other examples of the present disclosure, the respective microstimulators **4113**A, **4113**B may communicate (e.g. wirelessly) with each other and/or with a third device which is implanted or external in order to facilitate control, therapy, etc.
(406) Of course, depending on particular patient anatomy or other purposes/goals, just one of several implanted stimulation elements (e.g. microstimulators **4113**A, **4113**B) may be activated to apply stimulation, sensing, etc.
(407) In some examples, one or both of the stimulation elements **4113**A, **4113**B may comprise a cuff electrode, paddle electrode, axial array, etc. (supported by a IPG **533**) may be implanted instead of one or both of the stimulation elements **4113**, **4113**B comprising a microstimulator. Accordingly, one variation example arrangement may comprise the microstimulator **4113**A being intravascularly delivered and implanted relative to some portion of the ansa cervicalis-related nerve **316** (whether at the superior root **325** or elsewhere) and the other stimulation element **4113**B comprising something other than a microstimulator.
(408) As noted elsewhere, it will further understood that in some examples, the general principles associated with the example arrangement **4100** (and other example stimulation arrangements of the present disclosure) may be used to implement the example arrangement in other areas of a patient's body to treat conditions other than sleep disordered breathing. For example, the arrangement **4100** of stimulation elements (e.g. microstimulators **4113**A, **4113**B), anchors, etc. may be deployed within a pelvic region to treat urinary and/or fecal incontinence or other disorders, such as via stimulating the pudenal nerve, which may cause contraction of the external urinary sphincter and/or external anal sphincter. While not shown explicitly in association with FIG. **32**B, it will be understood that associated sensing elements described within the present disclosure (for sensing

physiologic data relative to the condition of interest) may be deployed in association with the various example arrangements for stimulating multiple nerve targets. However, other body regions and/or disorders may be suitable candidates for an example arrangement (e.g. **4100**) in which multiple nerve targets (of a single nerve or of wholly different nerves) are available to be stimulated to treat one type of physiologic behavior.

(409) FIG. **32**C is a diagram including a front view of a patient's anatomy **4201** relating to a hypoglossal nerve **305** and ansa cervicalis-related nerve **316** and schematically representing an example arrangement **4200** of various potential stimulation locations (e.g. at least A, B, C) and example intravascular delivery of stimulation elements for transvascular (e.g. transvenous) stimulation. As shown in FIG. **32**C, the pertinent patient anatomy comprises the ansa cervicalis-related nerve **316**, in context with the hypoglossal nerve **305** and with cranial nerves C**1**, C**2**, C**3**. At least because of the well-documented variances in patient anatomy (among different patients) regarding the ansa cervicalis-related nerve and/or challenges in graphically depicting such structures and their relationship, the schematic representation of the patient anatomy **4201** in FIG. **32**C (and FIG. **32**D) exhibits some differences relative to the schematic representation of the ansa cervicalis-related nerve **316** in FIGS. **2**, **16**, **32**A, **32**B. Nevertheless, as shown in FIG. **32**C, the general position of the example stimulation locations A, B, and C remain consistent at least in terms of the particular muscle groups which are innervated by the portion(s) of the nerve **316** at the example stimulation locations A, B, C as reproduced in FIG. **32**C-**32**D (relative to their depiction in FIGS. **2**, **16**, **32**A-**32**B).

(410) As shown in FIG. **32**C, portion **4229**A of the ansa cervicalis-relate nerve **316** extends anteriorly from a first cranial nerve C**1** with a segment **317** running alongside (e.g. coextensive with) the hypoglossal nerve **305** (indicated via "**305**, **317**") for a length until the ansa cervicalis-related nerve **3165** diverges from the hypoglossal nerve **305** to form a superior root (e.g. **325** in FIG. **2**A) of the ansa cervicalis-related nerve **316**.

(411) As further shown in FIG. **32**C, patient anatomy **4201** comprises an interior jugular vein **4250**, which extends along a superior-inferior orientation and within the context of the ansa cervicalis-related nerve **316**, may comprise a superior portion **4252** and an opposite inferior portion **4254**. Generally parallel to this portion of the interior jugular vein **4250**, the common carotid artery **4240** extends superiorly toward junction **4243**, from which the interior carotid artery **4242** and exterior carotid artery **4244** bifurcate from each other. As shown, an example first stimulation location (dashed lines A, and indicator "**305**, **317**") generally corresponds to the target stimulation location A previously shown in at least FIG. **2**A, **16**, **32**A, **32**B. Moreover, in some examples, the stimulation at location A may be implemented via example stimulation arrangements **2101**, **2401**, which in turn correspond (in some examples) to example stimulation arrangements in FIGS. **17-20**. Various aspects relating to this stimulation location (A, "**305**, **317**") were previously described in association with at least FIGS. **16-20**, at least some of which are equally applicable in relation to the example arrangement **4200** in FIG. **32**C.

(412) Portions **4229**B, **4229**C in FIG. **32**C generally correspond to portions **329**B, **329**C in FIGS. **2**A, **16**, **32**A, etc.

(413) As further shown in FIG. **32**C, in some examples example arrangement **4200** may comprise a stimulation lead **4270** which may be advanced within and through the interior jugular vein **4250** to position a stimulation portion **4213**B in stimulating relation, at location "A" ("**305**, **317**"), to the hypoglossal nerve **304** and portion **317** of the ansa cervicalis-related nerve **316**. In some examples, the stimulation lead **4270** and its delivery, anchoring, etc. may comprise at least some of substantially the same features and attributes as described in association with at least FIGS. **15**A-**15**C, **25**A-**25**B, **29**A-**29**B, **30**A-**31**G, and/or **30**A, **32**A-**32**B. In just one example, the stimulation portion **4213**B may comprise a linear array of spaced apart electrodes **4216** (e.g. ring electrodes, split ring electrodes, etc.) sized, shaped, and/or distributed to enable applying stimulation selectively to the various fibers, fascicles, etc. of the respective hypoglossal nerve **305** (e.g. main

trunk portion) and portion **317** of the ansa cervicalis-related nerve **316** in order to treat sleep disordered breathing (e.g. at least OSA). As noted elsewhere, applying stimulation at this location A activates at least some nerve fibers of the ansa cervicalis-related nerve which innervate the sternothyroid muscles to increase upper airway patency and activates at least some nerve fibers of the hypoglossal nerve which innervate at least protrusor muscles of the tongue to maintain or increase upper airway patency.

(414) In some examples, the stimulation portion **4213**B may be supported on a lead body **4271**. While FIG. **32**C depicts two stimulation portions (e.g. **4213**A, **4213**B) on lead body **4271**, it will be understood that in some examples, the lead **4270** comprises just one stimulation portion **4213**B for stimulating at location A or just one stimulation portion for stimulating at location B, as further described below. In some examples, the stimulation lead **4271** may comprise both stimulation portions **4213**A, **4213**B on lead **4271**, whether just one or both stimulation locations A and B are to be stimulated.

(415) In some examples, prior to chronic intravascular implantation of at least the stimulation portion **4213**B, the position of lead body **4271** and stimulation portion **4213**B may be optimized for desired effective selective stimulation and/or various combinations of electrodes may be selected to determine which combination of electrodes, stimulation protocol (e.g. timing, sequence, etc.), etc. provides the desired effective selective stimulation of the hypoglossal nerve **305** and/or ansa cervicalis-related nerve **316** at portion **317**.

(416) As further shown in FIG. **32**C, example arrangement **4200** may comprise an example second target stimulation location (dashed lines "B") along the ansa cervicalis-related nerve **316**, as was similarly illustrated in FIG. **2**, **16**, etc. In some examples, a cuff electrode or paddle electrode may be implanted at stimulation location B, in accordance with the many examples throughout the present disclosure of implanting such electrodes to be in stimulating relation to the ansa cervicalis-related nerve **316**. As shown in FIG. **32**C, in some examples the stimulation lead **4270** may be delivered intravascularly within and through the interior jugular vein **4250**, as similarly described above, to position stimulation portion **4213**A in close proximity to stimulation location B.

(417) In some examples, when both stimulation portions **4213**B and **4213**A are provided on lead **4270**, then stimulation may be provided solely at stimulation location A, solely at stimulation location B, or at both stimulation locations A and B. As also further described elsewhere, when both stimulation locations A and B may be stimulated, such stimulation may be simultaneous, alternating, staggered, etc., or the stimulation of the respective locations may depend on other parameters such as a collapse pattern, body position, etc., as well as whether or not the hypoglossal nerve is also being stimulated.

(418) In some examples, stimulation lead **4270** may be constructed to comprise just one stimulation portion (either **4213**B or **4213**A) with such single stimulation portion being positioned within the interior jugular vein **4250** in stimulating relation, at location B, to pertinent portions of the ansa cervicalis-related nerve **316**.

(419) With further reference to stimulation location B in FIG. **32**C, this portion of the ansa cervicalis-related nerve **316** may comprise a significantly large number (e.g., most or all) of the motor nerve fibers which innervate the sternothyroid muscles, such that delivering stimulation at location B may yield a robust response and contraction of the sternothyroid muscle(s), which contributes to upper airway patency. Accordingly, in some such examples, non-selective stimulation may be applied, at least with respect to the nerve fibers at location B innervating the sternothyroid muscles.

(420) As further shown in FIG. **32**C, example arrangement **4200** may comprise an example third target stimulation location (dashed lines "C") at portion **324** along the ansa cervicalis-related nerve **316**. In some examples, a cuff electrode or paddle electrode may be implanted at stimulation location C, in accordance with the many examples throughout the present disclosure of implanting such electrodes to be in stimulating relation to the ansa cervicalis-related nerve **316**, such as at

portion **324**.

(421) As further shown in FIG. **32**D, an example arrangement **4300** may comprise a stimulation lead **4280** to delivered intravascularly within and through the interior jugular vein **4250**, and then within and through a middle thyroid vein **4260**, which branches (at **4251**) off from the interior jugular vein **425**. Via such intravascular delivery, the example arrangement results in positioning a stimulation portion **4213**A of stimulation lead **4280** in close proximity to, and in stimulating relation to, stimulation location C along the portion **324** of the ansa cervicalis-related nerve **316**. As further shown in FIG. **32**D, a main body portion **4282** of lead **4280** may extend within and through the interior jugular vein **4250** while a distal portion **4283** of lead **4280** extends within and through the middle thyroid vein **4260**.

(422) At this stimulation location C, portion **324** of the ansa cervicalis-related nerve **316** may comprise a significantly large number (e.g., most or all) of the motor nerve fibers which innervate the sternothyroid muscles, such that delivering stimulation at location C may yield a robust response and contraction of the sternothyroid muscle(s), which may contributes to upper airway patency. Accordingly, in some such examples, non-selective stimulation may be applied, at least with respect to the nerve fibers at location C innervating the sternothyroid muscles.

(423) In some examples, a single/same type of electrode arrangement (e.g. cuff electrode, etc.) may be implanted at each of the respective stimulation locations A, B, and C or just one or two of such locations. However, in some examples, different types of electrode arrangements may be implanted among the respective stimulation locations A, B, and C. In one non-limiting example, a cuff electrode may be implanted at location A while an axial-style stimulation portion may be intravascularly delivered for applying stimulation at location B. Other combinations will be apparent.

(424) In a manner consistent with several examples throughout the present disclosure, any one of electrode arrangements (e.g. cuff electrode, paddle electrode, axial electrode array, etc.) in FIGS. **32**A-**32**D may be embodied as part of a microstimulator instead of being connected to, supported by, etc. a lead in connection with an IPG (e.g. **533**). In some such examples, the microstimulator may be implanted subcutaneously or intravascularly, such as but not limited to example methods and devices described throughout various examples of the present disclosure.

(425) FIGS. **33**A-**37**D are a series of diagrams including views which schematically represent various stimulation protocols, including closed loop stimulation patterns and/or open loop stimulation patterns, including stimulation of at least the hypoglossal nerve and/or the ansa cervicalis-related nerve **316**. In some examples, the stimulation implemented via the various stimulation protocols may be implemented via at least some of substantially the same features and attributes of the various example stimulation arrangements as previously described in association with at least FIGS. **1**-**32**D and/or as of the various later described example arrangements involving sensing, control, etc. Accordingly, unless specifically noted otherwise the various example stimulation protocols may be applicable for unilateral stimulation or bilateral stimulation of the respective targeted nerves. Moreover, in some examples, the stimulation pattern of one of the example stimulation protocols (as described in FIGS. **33**A-**37**D) for a given nerve (e.g. HGN) may be switched and applied to another nerve (e.g. ACN), or vice versa.

(426) FIG. **33**A is a diagram **5000** schematically representing an example respiratory waveform **5010** and a series of stimulation protocols **5030**, **5050**, **5070**, each of which include a stimulation pattern for a hypoglossal nerve (HGN) and an ansa cervicalis-related nerve (ACN) **316**. Among other things, FIG. **33**A provides an example respiratory waveform **5010**, including an inspiratory phase **5012** having duration INP, active expiratory phase **5014** having duration EA, and expiratory pause **5016** having duration EP. Together, these phases comprise an entire respiratory cycle **5011** having a duration (e.g. respiratory period) of R. This respiratory cycle **5011** is repeated, as represented in successive frames A, B, C, D, E, and so on. It will be understood that the respiratory cycles **5011** depicted in each frame A-E of FIG. **33**A are depicted as being identical, but in reality

there may be variations in the respiratory cycle from breath-to-breath, and each patient may exhibit some variances in their respiratory waveform from other patients. Moreover, for illustrative simplicity, the respiratory waveforms shown in FIGS. **33**A-**37**D do not purport to depict disruptions to the respiratory waveform, which correspond to sleep disordered breathing, signal imperfections, etc.

(427) As shown in FIG. **33**A, one example stimulation protocol **5030** comprises an example first stimulation pattern **5031** for stimulating a hypoglossal nerve (HGN) and an example second stimulation pattern **5041** for stimulating an ansa cervicalis-related nerve (ACN).

(428) The first stimulation pattern **5031** to stimulate the hypoglossal nerve (HGN) comprises a stimulation cycle **5035** including a stimulation period **5032** and a non-stimulation period **5034**, with the stimulation cycle **5035** being repeated through successive frames A, B, C, D, E and so on. As shown for the first stimulation cycle **5035**, the stimulation pattern **5031** includes the stimulation period **5032** comprising an amplitude of N**1** during the inspiratory phase **5012** and the subsequent non-stimulation period **5034** having an amplitude of zero during the expiratory phases **5014**, **5016**. In one aspect, this stimulation pattern **5031** may sometimes be referred to as being synchronous with the inspiratory phase (**5012**) of the patient's respiratory cycles (e.g. breathing pattern). In another aspect, this stimulation pattern **5031** may sometimes be referred to as being a closed loop stimulation pattern in that sensed respiratory information (i.e. sensed feedback) is used to time the stimulation period **5032** to coincide with the inspiratory phase (**5012**) of the patient's respiratory cycles (e.g. breathing pattern).

(429) As further shown in FIG. **33**A, the second stimulation pattern **5041** comprises a stimulation cycle **5046** including a stimulation period **5043** and a non-stimulation period **5045** which lasts through two respiratory cycles **5011** (e.g. two frames). This stimulation cycle **5046** is repeated through pairs of frames A and B, C and D, and so on.

(430) As shown for the first stimulation cycle **5046**, the second stimulation pattern **5041** (for the ACN) includes the stimulation period **5043** comprising an amplitude of P**1** during the inspiratory phase **5012** and the subsequent non-stimulation period **5045** having an amplitude of zero during the expiratory phases **5014**, **5016**, and the entire subsequent respiratory cycle (e.g. frame B). It will be noted that the amplitude P**1** for stimulation of the ACN **315** may comprise a value different than the amplitude N**1** for stimulation of the hypoglossal nerve. In one aspect, this stimulation pattern **5046** may sometimes be referred to as being periodically synchronous with the inspiratory phase (**5012**) of the patient's respiratory cycles (e.g. breathing pattern) to the extent that when stimulation is applied in some respiratory cycles (e.g. periodically in frames A, C, E), the stimulation coincides with the inspiratory phase **5012** of the patient's respiratory cycle **5011**. It may be further observed that when stimulation is applied to the ACN **316** per stimulation pattern **5041**, it is applied synchronous with stimulation of the hypoglossal nerve.

(431) By providing stimulation to the ansa cervicalis-related nerve (e.g. **316** in FIG. **2**) according to the second stimulation pattern **5041**, the action of the stimulation of the hypoglossal nerve (per first stimulation pattern **5031**) to increase and/or maintain upper airway patency is supplemented while also looking to prevent or minimize fatigue to the ansa cervicalis-relate nerve (ACN) **316** and/or its associated targeted muscles by providing stimulation to the ACN **316** every other breath (i.e. respiratory cycle). As previously noted, for at least some patients, some patient positions, etc., providing stimulation to the ansa cervicalis-related nerve **316** (in addition to stimulating the hypoglossal nerve) may help increase and/or maintain upper airway patency because certain patients may have a particular anatomical features, certain co-morbidities, etc.

(432) FIG. **33**A also depicts an example stimulation protocol **5050** in which the second stimulation pattern **5061** of the ACN **316** is substantially the same as in the example stimulation pattern **5041** of protocol **5030**, but a first stimulation pattern **5051** of the hypoglossal nerve provides for stimulation every other respiratory cycle, illustrated as occurring in frames A, C, E, and so on. In one aspect, this stimulation pattern may act to prevent or minimize fatigue of the hypoglossal nerve

and/or genioglossus muscle. In some examples, the alternating stimulation periods in pattern **5051** (for the HGN) are offset from the stimulation periods in pattern **5061** (for the ACN **316**). However, in some examples, the alternating HGN stimulation periods (e.g. frames A, C, E) in pattern **5051** may be shifted so that they are applied to coincide with (i.e. be synchronous with) the alternating stimulation periods (e.g. frames B, D, etc.) in stimulation pattern **5061** for the ACN **316**.
(433) FIG. **33**A also depicts an example stimulation protocol **5070** in which the second stimulation pattern **5041** of the ACN **316** is substantially the same as in the example stimulation pattern **5041** of protocol **5030**, but in a first stimulation pattern **5071** (for the hypoglossal nerve), an amplitude of stimulation varies every other respiratory cycle. As shown in FIG. **33**A, in the first stimulation pattern **5071** of the stimulation protocol **5070**, the amplitude of the stimulation period **5032** in frames B, D, etc. for the hypoglossal nerve comprises N**1** while the amplitude of the stimulation period **5072** in frames A, C, E, etc. for the hypoglossal nerve comprises N**2**, which is less than the amplitude N**1**. In some examples, N**2** may be substantially less (e.g. 50% less, 25% less, etc.) than amplitude N**1**. In one aspect, this stimulation pattern may act to prevent or minimize fatigue of the hypoglossal nerve and/or genioglossus muscle.
(434) Moreover, as further shown in FIG. **33**A, in this example stimulation protocol **5070** the respiratory cycles (e.g. frames A, C, E) for which a lower amplitude N**2** of stimulation is applied to the hypoglossal nerve is timed to coincide with stimulation periods **5043** by which stimulation is applied to the ansa cervicalis-related nerve. In one aspect, this arrangement times the stimulation of the ACN **316** to supplement the stimulation of the hypoglossal nerve when the amplitude of the HGN stimulation is lower, such that the stimulation of the ACN **316** may help increase and/or maintain upper airway patency during such respiratory cycles. Moreover, this stimulation protocol **5070** still provides for alternating stimulation periods for the ACN **316** to also help minimize or manage potential fatigue of the ACN **316** and/or associated targeted muscles. Moreover, while not shown in FIG. **33**A, it will be understood that in some examples, the amplitude P**1** of the stimulation periods **5043** for stimulating the ACN **316** also may be reduced to a lower amplitude in at least some respiratory cycles to further minimize or manage potential fatigue issues.
(435) It will be further noted that the examples shown in FIG. **33**A in which a stimulation period is applied every other respiratory cycle (e.g. **5041**, **5051**, **5061**), whether for the hypoglossal nerve or the ansa cervicalis-related nerve, are also representative for some further examples in which a stimulation period may be applied every third respiratory cycle or every fourth respiratory cycle, and so on. Similarly, the examples in which a reduced amplitude of stimulation is applied every other respiratory cycle (e.g. **5071**) are also representative for some further examples in which a reduced amplitude, stimulation period may be applied every third respiratory cycle or every fourth respiratory cycle, and so on, whether for the hypoglossal nerve or for the ansa cervicalis-related nerve.
(436) With regard to the reduced stimulation amplitude (e.g. N**2**) in the stimulation pattern **5071**, in some examples an intensity of the applied stimulation also can be reduced via adjusting other stimulation parameters (i.e. other than amplitude) such that the reduced stimulation amplitude in the pattern **5071** in FIG. **33**A (or in the pattern **5211** in FIG. **34**) also may be generally representative of reducing or adjusting other stimulation parameters to reduce an intensity of stimulation to a particular nerve and/or at a particular stimulation location.
(437) FIG. **33**B is a diagram schematically representing further example stimulation protocols **5130**, **5150**. FIG. **33**B illustrates a respiratory waveform **5010**, which has the substantially the same features and attributes as the respiratory waveform **5010** as in FIG. **33**A, except with FIG. **33**B depicting a greater number of respiratory cycles **5011** than in FIG. **33**A.
(438) As shown in FIG. **33**B, one example stimulation protocol **5130** comprises an example first stimulation pattern **5131** for stimulating a hypoglossal nerve (HGN) and an example second stimulation pattern **5141** for stimulating an ansa cervicalis-related nerve (ACN).
(439) The example second stimulation pattern **5141** to stimulate the ansa cervicalis-related nerve

(ACN) comprises an example stimulation cycle **5135** which extends over five successive frames (e.g. A, B, C, D, E) including a stimulation period **5143** (e.g. 4 frames) and a non-stimulation period **5145** (e.g. 1 frame), with the stimulation cycle **5146** being repeated. As shown for the first stimulation cycle **5146**, the stimulation pattern **5141** includes the stimulation period **5143** comprising an amplitude of P**1** and the subsequent non-stimulation period **5145** having an amplitude of zero during the entire respiratory cycle **5011** in frame E. In one aspect, this stimulation pattern **5141** may sometimes be referred to as being synchronous relative to at least the inspiratory phase (**5012**) of the patient's respiratory cycles (e.g. breathing pattern), at least to the extent that when the stimulation period **5143** begins, it coincides with a beginning of an inspiratory period of a respiratory cycle. On the other hand, in some respects the stimulation pattern **5141** may be considered as being asynchronous solely relative to an inspiratory phase **5012** of the respiratory cycles **5011**, at least to the extent that the stimulation is maintained through the entire respiratory cycle (e.g. **5011**) of several consecutive respiratory cycles, such that stimulation is not discontinued at the conclusion of each inspiratory phase in each respective respiratory cycle **5011** during which stimulation is being applied.

(440) In another aspect, this stimulation pattern **5141** may sometimes be referred to as being a closed loop stimulation pattern in that sensed respiratory information (i.e. sensed feedback) is used to time the beginning of the stimulation period **5143** to coincide with the beginning of the inspiratory phase (**5012**) of the patient's respiratory cycles (e.g. breathing pattern) and the sensed respiratory information is used to time the termination of the stimulation period **5134** to coincide with an end of the expiratory phase of the last respiratory cycle **5011** (e.g. frame E) in the stimulation cycle **5035**.

(441) As further shown in FIG. **33**B, the first stimulation pattern **5131** comprises a stimulation cycle **5135** which generally corresponds to the number of respiratory cycles (**5011**) of the stimulation cycle **5146** for the second stimulation pattern **5141**. Each stimulation cycle **5135** of the first stimulation pattern **5131** includes a stimulation period **5032** and a non-stimulation period **5037**. In general terms, the non-stimulation period **5037** of the first stimulation pattern **5131** has a duration generally matching the duration of the stimulation period **5143** of the second stimulation pattern **5141**. Via this arrangement, stimulation is withheld (i.e. does not occur) from the hypoglossal nerve (HGN) during periods (e.g. such as several respiratory cycles) (e.g. **5037** in FIG. **33**B) in which stimulation is being applied to the ansa cervicalis-related nerve (ACN) (e.g. **5143** in FIG. **33**B). Conversely, in the same example, as noted below stimulation is applied (i.e. does occur) to the hypoglossal nerve (HGN) (e.g. **5032** in FIG. **33**B) during at least a portion of the period(s) in which stimulation is withheld (i.e. is not being applied to) from the ansa cervicalis-related nerve (ACN) (e.g. **5145** in FIG. **33**B).

(442) In some examples, the stimulation period **5032** has a duration corresponding to a duration of an inspiratory phase **5012** as shown in FIG. **33**B. In some examples, the stimulation period **5032** can be shorter or longer than the inspiratory phase **5012**. Accordingly, in some such examples, the stimulation period **5032** may have a duration corresponding to the duration R of the respiratory cycle **5011** (e.g. single frame E). However, in some examples, the stimulation period **5032** in first stimulation pattern **5131** may have longer durations.

(443) For at least the example in FIG. **33**B, this stimulation cycle **5135** of the first stimulation pattern **5131** is repeated, along with the stimulation cycle **5146** of the second stimulation pattern **5141**, for five respiratory cycles at a time, and repeated.

(444) As shown for the first stimulation cycle **5146**, the second stimulation pattern **5141** (for the ACN) includes the stimulation period **5143** comprising an amplitude of P**1** during the inspiratory phase **5012** and the subsequent non-stimulation period **5145** having an amplitude of zero during the entire respiratory cycle **5011** (e.g. frame E). It will be noted that in some examples the amplitude P**1** for stimulation of the ACN **316** may comprise a value different than the amplitude N**1** for stimulation of the hypoglossal nerve (HGN).

(445) In some examples, a duration of the stimulation cycle **5146** of the second stimulation pattern **5141** may be longer or shorter than shown in FIG. **33**B. In some such examples, a duration of the stimulation period (e.g. **5143**) of a stimulation cycle **5146** in the second stimulation pattern **5141** may be significantly longer such as up to a dozen respiratory cycles (e.g. 1 minute), or even two dozen respiratory cycles (e.g. 2 minutes). Meanwhile, the non-stimulation period **5145** may be some multiple of respiratory cycles.
(446) In general terms, in some examples of the stimulation protocol **5130** in FIG. **33**B, a duty cycle (e.g. percentage of stimulation to non-stimulation) for stimulation may comprise between about 60 to about 90 percent. In some examples, the duty cycle may comprise between about 65 to about 85 percent, while in some examples, the duty cycle may comprise between about 70 percent and about 80 percent. In the particular example shown in FIG. **33**B, the duty cycle is about 80 percent per a ratio of a stimulation period **5143** of four respiratory cycles **5011** to one non-stimulation period **5145** of one respiratory cycle **5011**. While in the example of FIG. **33**B, the stimulation period **5143** comprises a discrete multiple of respiratory cycles (**5011**), in some examples, the stimulation period **5143** may have a duration not corresponding to a discrete multiple of respiratory cycles **5011**.
(447) In some examples, the HGN stimulation period **5032** may have a duration in which the HGN stimulation is applied continuously for a period which is at least as long as, or longer than, a non-stimulation period **5145** (e.g. rest period) of the second stimulation pattern **5141** of the ansa cervicalis-related nerve. In some such examples, in which the ACN non-stimulation period **5143** has a duration no more than 10 seconds, the HGN stimulation period **5032** may be applied continuously during the ACN rest period **5145**.
(448) In some examples, by providing stimulation to the ansa cervicalis-related nerve (e.g. **316** in FIG. **2**) according to the longer duty cycles per the second stimulation pattern **5141**, reasonable patient comfort may be achieved while generally maintaining or increasing upper airway patency while stimulating the hypoglossal nerve (HGN) just periodically when the ansa cervicalis-related nerve (ACN) is resting. In some examples, stimulating the ansa cervicalis-related nerve as a primary target for longer periods of time may result in generally maintaining a stiffer upper airway in a more open position. This effect may be achieved via stimulating portions innervating the sternothyroid and/or sternohyoid muscles, which pull the larynx inferiorly. This arrangement may enhance patient comfort (while maintaining upper airway patency) at least because the contraction of the upper airway muscles innervated by the ansa cervicalis-related nerve may result in more a diffuse sensation than the more discrete, recognizable protrusion of the tongue. This arrangement also may yield a more effective therapy (in at least some patients) because, with the upper airway already being in a more open configuration due to the stimulation of the ansa cervicalis-related nerve, then the tongue need not be moved as far in order to restore or maintain upper airway patency, and therefore less stimulation of the hypoglossal nerve may be applied, which in turn may enhance patient comfort.
(449) For at least some patients, some patient positions, etc., providing stimulation to the ansa cervicalis-related nerve **316** as a primary target (with periodic supplemental stimulation of the hypoglossal nerve) may help increase and/or maintain upper airway patency generally and/or because certain patients may have particular anatomical features, certain co-morbidities, etc. more therapeutically responsive to the ansa cervicalis-related nerve as a primary target.
(450) FIG. **33**B also schematically represents an example stimulation protocol **5150** comprising stimulation of both the ansa cervicalis-related nerve (ACN) and the hypoglossal nerve (HGN). In some examples, the stimulation protocol **5150** may comprise at least some of substantially the same features and attributes as stimulation protocol **5130** in FIG. **33**B, except with stimulation protocol **5150** in FIG. **33**B comprising HGN stimulation which may occur during ACN stimulation instead of occurring during an ACN rest period. Accordingly, as shown in FIG. **33**B, the stimulation protocol **5150** may comprise a second stimulation pattern **5141** for the ansa cervicalis-related nerve

as described for stimulation protocol **5130**, including the variations thereof. Meanwhile, the stimulation protocol **5150** may comprise a first stimulation pattern **5161** for the hypoglossal nerve (HGN) comprising HGN stimulation periods **5132** and HGN non-stimulation periods **5137**. In some examples, the stimulation protocol **5150** may comprise a series of repeating stimulation cycles in which the HGN stimulation period **5132** occurs at regular intervals, and is timed to occur during an ACN stimulation period **5143** as shown in FIG. **33**B. For instance, the HGN stimulation period **5132** may occur every other ACN stimulation period **5143**, may occur every third ACN stimulation period **5143**, and so on.

(451) However, in some examples, the HGN stimulation period **5132** may not occur at regular intervals, but may still be implemented during ACN stimulation periods **5143**. In other words, the HGN stimulation does not occur during an ACN rest period **5145**. In some such examples, the occurrence of the HGN stimulation period **5132** may be pseudo-random (e.g. one type of open loop stimulation). In some example implementations, the pseudo-random HGN stimulation may be implemented without sensing respiration or without using sensed respiration information, In some such examples, a frequency, duration, etc. of the HGN stimulation period **5132** may be selected to ensure a high likelihood that at least some of the pseudo-random HGN stimulation periods **5132** will overlap with at least some of the inspiratory phases of the respiratory cycles **5011**.

(452) In some examples, the occurrence of the HGN stimulation period **5132** (in stimulation protocol **5150**) may occur upon detection that more upper airway patency is warranted, and therefore some HGN stimulation is desirable and will be implemented. In some such examples, the HGN stimulation is timed to be applied simultaneous with an ACN stimulation period **5143** and the HGN stimulation may be applied during and/or overlapping with an inspiratory phase of the respiratory cycle, in some examples.

(453) FIG. **34** is diagram schematically representing the same respiratory waveform **5010** as in FIG. **33**A and two different example stimulation protocols **5210**, **5250**.

(454) As shown in FIG. **34**, the example stimulation protocol **5210** comprises a second stimulation pattern **5061** (to stimulate the ansa cervicalis-related nerve) having substantially the same features as stimulation pattern **5041** as in FIG. **33**A and a first stimulation pattern **5211** (to stimulate the hypoglossal nerve) comprising features and attributes like stimulation pattern **5031** in FIG. **33**A, except with each stimulation period **5072** having a reduced amplitude N**2** (of some selectable value). In one aspect, this example stimulation protocol also may help reduce fatigue for the hypoglossal nerve (and/or associate genioglossus muscle) while the stimulation of the ansa cervicalis-related nerve (ACN) **316** can compensate for the reduced amplitude of the hypoglossal nerve stimulation signal, such that the concomitant stimulation patterns **5211**, **5061** helps to increase and/or maintain upper airway patency.

(455) As further shown in FIG. **34**, an example stimulation protocol **5250** comprises a second stimulation pattern **5061** having substantially the same features as stimulation pattern **5061** as in FIG. **33**A for application to a portion (e.g. ACN**2**) of the ansa cervicalis-related nerve **316**, and a first stimulation pattern **5251** comprising features and attributes like stimulation pattern **5051** in FIG. **33**A, except with that the stimulation pattern **5251** is applied to a portion (e.g. ACN**1**) of the ansa cervicalis-related nerve **316** instead of the hypoglossal nerve. Accordingly, in the example stimulation protocol **5250**, one stimulation pattern **5251** is applied to a first portion ACN**1** (i.e. target stimulation location) of the ansa cervicalis-related nerve **316** and the other stimulation pattern **5061** is applied to a different, second portion ACN**2** of the ansa cervicalis-related nerve **316**. By applying the stimulation in an alternating pattern to different portions (e.g. ACN**1**, ACN**2**) of the ansa cervicalis-related nerve **316**, the example stimulation protocol may enhance upper airway patency by leveraging different mechanisms of action to increase and/or maintain upper airway patency, while also helping to manage potential fatigue of the ansa cervicalis-related nerve **316** that could possibly be associated with a single target stimulation location.

(456) In some examples, the example stimulation protocol **5250** may comprise three or more

different stimulation patterns corresponding to three or more different portions of the ansa cervicalis-related nerve **316**.

(457) In some examples, the example stimulation protocol **5250** also may be enhanced via also applying stimulation to the hypoglossal nerve in addition to the two (or more) different portions of the ansa cervicalis-related nerve. Such hypoglossal nerve stimulation may be applied via any one of the example stimulation patterns described in association with at least FIGS. **33**A-**37**D and/or other suitable stimulation patterns.

(458) FIG. **35** is a diagram schematically representing the same respiratory waveform **5010** as in FIG. **34** and an example stimulation protocol **5310**. As shown in FIG. **35**, the stimulation protocol **5310** comprises a first stimulation protocol **5211** comprising substantially the same features and attributes as stimulation protocol **5211** in FIG. **34** in which the stimulation applied to the hypoglossal nerve comprises a reduced amplitude N**2** applied in a stimulation period **5072** of each stimulation cycle (e.g. each frame A, B, C, etc.). The stimulation protocol **5310** of FIG. **35** also comprises a second stimulation protocol **5311** which also comprises a reduced stimulation amplitude (P**2**) for each stimulation period **5313**, except with the stimulation being applied to the ansa cervicalis-related nerve ACN. In some examples, the amplitude N**2** comprises a value different than a value of the reduced amplitude P**2**.

(459) In a manner analogous to other example stimulation protocols, it is believed that the example stimulation protocol **5310** in FIG. **35** may enhance increasing and/or maintaining upper airway patency and/or may enhance fatigue management of target stimulation locations of the nerves, muscles, etc.

(460) FIG. **36**A is a diagram **5500** schematically representing the same respiratory waveform **5010** as in FIG. **34** and an example stimulation protocol **5510**, which comprises a plurality of stimulation patterns **5551**, **5563**, **5581**, **5561** involving both the left and right hypoglossal nerves and both the left and right ansa cervicalis-related nerves. While FIG. **36**A depicts a particular stimulation pattern for each particular nerve (L and R), it will be understood that the example stimulation protocol **5510** in FIG. **36**A is also generally representative of applying different stimulation patterns to the left patient side and right patient side of a particular nerve. Moreover, in some examples, one, two or three of the stimulation patterns (e.g. **5551**, **5563**, **5581**, **5561**) may be omitted entirely or for just a selectable period of time.

(461) As shown in FIG. **36**A, the example stimulation pattern **5551** is to be applied to a first hypoglossal nerve (e.g. patient left side, HGN L) and may comprise substantially the same features and attributes as stimulation pattern **5051** in FIG. **33**A in which stimulation period **5032** occurs in frames A, C, E (e.g. every other respiratory cycle). Meanwhile, the example stimulation pattern **5563** in FIG. **36**A is to be applied to a second hypoglossal nerve (e.g. patient right side, HGN R) and may comprise substantially the same features and attributes as the stimulation pattern **5051** in FIG. **33**A, except with the stimulation periods **5032** being applied in respiratory cycles (e.g. frames B, D) in which no stimulation is applied to the respiratory cycle such that stimulation is alternated between the left and right hypoglossal nerves.

(462) As further shown in FIG. **36**A, the example stimulation pattern **5581** is to be applied to a first ansa cervicalis-related nerve (e.g. patient left side, ACN L) and may comprise substantially the same features and attributes as stimulation pattern **5041** of stimulation protocol **5070** in FIG. **33**A in which stimulation is applied in frames A, C, E (e.g. every other respiratory cycle). Meanwhile, the example stimulation pattern **5561** in FIG. **36**A is to be applied to a second ansa cervicalis-related nerve (e.g. patient right side, ACN R) and may comprise substantially the same features and attributes as the stimulation pattern **5061** in FIG. **33**A, i.e. with the stimulation periods **5032** being applied in respiratory cycles (e.g. frames B, D), so as to be alternating with respect to stimulation of the left ACN

(463) As apparent from the foregoing example stimulation protocols and the example stimulation devices and methods as previously described in various examples throughout the present

disclosure, adjustments to the stimulation patterns **5551**, **5563**, **5581**, **5561** may be made regarding applying the stimulation to various nerves and left and right sides as desired to achieve the desired increase or maintenance of upper airway patency.

(464) FIG. **36**B is a diagram **5700** schematically representing the same respiratory waveform **5010** as in FIG. **34**, and example stimulation protocols **5710**, **5800**.

(465) As shown in FIG. **36**B, the example stimulation protocol **5710** comprises a first stimulation pattern **5750** to stimulate a hypoglossal nerve (HGN) and a second stimulation pattern **5760** to stimulate the ansa cervicalis-related nerve (ACN). The first stimulation pattern **5750** comprises a stimulation cycle, including a stimulation period **5752** and subsequent non-stimulation period **5754**, with the stimulation cycle repeating itself. In some examples, the stimulation period **5752** comprises a duration greater than a duration of the non-stimulation period **5754**. In some examples, the stimulation period **5752** comprises a duration greater than a duration (INSP) of the inspiratory phase **5012** of the respiratory cycle **5011**. Moreover, in some examples, the stimulation period **5752** is not synchronized relative to a sensed inspiratory phase **5012** of the patient's respiratory cycle **5011**, and therefore the first stimulation pattern **5750** may sometimes be referred to as an open loop stimulation pattern. Via such example stimulation pattern **5750**, the stimulation period **5752** may regularly overlap with at least a portion of the inspiratory phase **5012**, despite the lack of synchronization relative to the inspiratory phase **5012**. In the example shown in FIG. **36**B, the stimulation cycle (including the stimulation period **5752** and non-stimulation period **5754**) has a duration less than a duration (R) of the respiratory cycle **5011**.

(466) However, in some examples, the stimulation period **5752** has a duration greater than the duration of the non-stimulation period **5754**, and the stimulation cycle (including the stimulation period **5752** and non-stimulation period **5754**) has a duration greater than a duration (R) of the respiratory cycle **5011**, which also may act to cause the stimulation period **5752** to regularly overlap with an inspiratory phase of the patient's respiratory cycle **5011**.

(467) In some examples, such open loop stimulation may comprise at least some of substantially the same features and attributes as described in Wagner et al, STIMULATION FOR TREATING SLEEP DISORDERED BREATHING, published as U.S. Patent Publication 2018/01176316 on May 3, 2018, issued as U.S. Pat. No. 10,898,709 on Jan. 26, 2021, and hereby incorporated by reference.

(468) With further reference to FIG. **36**B, the second stimulation pattern **5760** of the example stimulation protocol **5710** comprises substantially the same features and attributes as the first stimulation pattern **5750**, except being applied to an ansa cervicalis-related nerve (ACN) instead of to the hypoglossal nerve (HGN). As shown in FIG. **36**B, the second stimulation pattern **5760** comprises stimulation cycles, each including a stimulation period **5761** and non-stimulation period **5763**, which comprise the same durations, relationships, etc. as the stimulation period **5752** and non-stimulation period **5754** of stimulation pattern **5750**, in some examples. Accordingly, as shown in FIG. **36**B, in some examples the stimulation period **5761** of the second stimulation pattern **5760** is synchronized with the stimulation period **5752** of the first stimulation pattern **5750**, and the non-stimulation period **5763** of the second stimulation pattern **5760** is synchronized with the non-stimulation period **5754** of the first stimulation pattern **5750**.

(469) However, in some examples, the second stimulation pattern **5760** is not synchronized with the first stimulation pattern **5750** and may be implemented such that the stimulation period **5671** of the second stimulation pattern **5670** is initiated at a point time different than initiation of the stimulation period **5752** of the first stimulation pattern **5750**. Moreover, in some examples, the respective durations of the stimulation and/or non-stimulation periods (e.g. **5761**, **5763**) of the second stimulation pattern **5760** may be different than the respective durations of the stimulation and/or non-stimulation periods (e.g. **5752**, **5754**) of the first stimulation pattern **5750**.

(470) FIG. **36**B also schematically represents an example stimulation protocol **5800** including first stimulation pattern **5750** to stimulate a hypoglossal nerve (HGN) and a second stimulation pattern

5790 to stimulate an ansa cervicalis-related nerve (ACN). In some examples, the first stimulation pattern **5750** of example stimulation protocol **5800** comprises substantially the same features and attributes as the first stimulation pattern **5750** of example stimulation protocol **5710** in FIG. **36**B.
(471) In some examples, the second stimulation pattern **5790** comprises stimulation cycles with a stimulation period **5791** and non-stimulation period **5793**, with the stimulation period **5791** being synchronized relative to a sensed inspiratory phase **5012** of the patient's respiratory cycle, and in some examples, having a duration generally corresponding to the duration (INSP) of the inspiratory phase. However, in some examples, the stimulation period **5791** is not synchronized relative to a sensed inspiratory phase **5012**. Accordingly, the hypoglossal nerve is stimulated in an open loop (non-synchronized) manner while the ansa cervicalis-related nerve is stimulated in a closed loop (synchronized) manner.
(472) As further shown in FIG. **36**B, regarding stimulation pattern **5790**, the amplitude of the stimulation period alternates between stimulation periods **5791** having a first amplitude P**1** and a second stimulation period **5792** having a second amplitude P**2** less than first amplitude P**1**, which may minimize potential fatigue of the ansa cervicalis-related nerve and/or associated innervated muscles. However, in some examples, the amplitude of the stimulation periods **5791**, **5792** may be the same. In yet other examples, the stimulation period **5791** having amplitude P**1** may be applied in every other stimulation cycle (e.g. frames A, C, E) with no stimulation therebetween (e.g. no stimulation in frames B, D, etc.), in a manner similar to that shown for stimulation pattern **5041** in FIG. **33**A.
(473) In some examples, any of the stimulation patterns of the example stimulation protocols described in association with FIGS. **33**A-**36**B may be modified such that the beginning portion of the stimulation period of the respective stimulation cycles is to slightly precede the start (S) of the inspiratory phase (INSP) to provide at least some pre-inspiratory stimulation, in a manner similar to that shown and described in the example stimulation protocols in at least FIGS. **37**A-**37**B. In some examples, at least some of the features and attributes (e.g. reduced amplitude, applying stimulation every other respiratory cycle, alternating stimulation between different nerves, etc.) described and illustrated in association with FIGS. **33**A-**36**B may be implemented in at least some of the example stimulation patterns of the example stimulation protocols described and illustrated in FIGS. **37**A-**37**D.
(474) FIG. **37**A is a diagram **6000** schematically representing the same respiratory waveform **5010** as in FIG. **34**, and example stimulation protocols **6020**, **6050**. As shown in FIG. **37**A, example stimulation protocol **6020** comprises a first stimulation pattern **6021** to stimulate a hypoglossal nerve (HGN) and a second stimulation pattern **6041** to stimulate an ansa cervicalis-related nerve (ACN). In some examples, the first stimulation pattern **6021** of example stimulation protocol **6020** comprises substantially the same features and attributes as the first stimulation pattern **5031** of example stimulation protocol **5030** in FIG. **33**A, except with the stimulation period **6032** having a longer duration such that a beginning (B) of the stimulation period **6032** precedes the start (S) of the inspiratory phase (INSP) of the patient's respiratory cycle **5010**, while the end (E) of the stimulation period **6021** coincides with the end of the inspiratory phase (INSP), corresponding to the transition (T) between the inspiratory phase (INSP) and expiratory phase (EA, EP). By beginning (B) the stimulation just prior to the start (S) of the inspiratory phase (INSP), the stimulation may ensure upper airway patency prior to the patient starting (S) inspiration.
(475) Consequently, as further shown in FIG. **37**A, the non-stimulation period **6034** of stimulation pattern **6021** has a shorter duration than in the non-stimulation period **5034** of stimulation pattern **5030** in FIG. **33**A so that the subsequent stimulation period **6034** in FIG. **37**A may begin (B) prior to the start (S) of the next inspiratory phase (INSP) of the patient's next breath (i.e. subsequent respiratory cycle **5011**).
(476) As shown in FIG. **37**A, this stimulation cycle **6035** is repeated throughout the first stimulation pattern **6021** such that the stimulation of the hypoglossal nerve is synchronized in a

closed-loop manner relative to the inspiratory phase (INSP) of the patient's respiratory cycles **5011**.
(477) As further shown in FIG. **37**A, the second stimulation pattern **6041** (to stimulate an ansa cervicalis-related nerve) comprises a stimulation period **6043** which begins (B) prior to, and overlaps with, the start (S) of the inspiratory phase (INSP), with the stimulation period **6043** ending (E) during the inspiratory phase (INSP) and prior to the end of the inspiratory phase (INSP) at transition (T). The non-stimulation period **6034** of the stimulation cycle **6035** lasts until just prior to the start (S) of the next inspiratory phase (INSP) at which the next stimulation period **6043** begins (B).
(478) As shown in FIG. **37**A, this stimulation cycle **6046** for the ansa cervicalis-related nerve is repeated throughout the first stimulation pattern **6041** such that the stimulation of the ansa cervicalis-related nerve is synchronized in a closed-loop manner relative to a portion of the inspiratory phase (INSP) of each of the patient's respiratory cycles **5011**.
(479) By providing stimulation period **6043** of the ansa cervicalis-related nerve (ACN) to coincide with just the start (S) of the inspiratory phase (INSP) (but not the entire inspiratory phase), the stimulation periods **6043** enhance increasing and/or maintaining upper airway patency in a complementary, additive manner to the stimulation of the hypoglossal nerve (via stimulation period **6032**) to increase and/or maintain upper airway patency.
(480) For some patients, once upper airway patency has been established prior to, and during, the start (S) of inspiration, further stimulation of the ansa cervicalis-related nerve may be unnecessary. Accordingly, by ending (E) the stimulation period **6043** shortly after the start (S) of the inspiratory phase (INSP), fatigue of the nerve and/or innervated muscles may be minimized or avoided.
(481) In some examples, the example stimulation protocol **6020** may be modified so that the stimulation period **6032** of the stimulation pattern **6021** (to stimulate the hypoglossal nerve) and/or so that the stimulation period **6043** of the stimulation pattern **6041** (to stimulate the ansa cervicalis-related nerve) begins (B) at the start (S) of the inspiratory phase (INSP) instead of beginning (B) prior to the start (S) of the inspiratory phase (INSP), which may be prudent for at least some patients at least some of the time.
(482) As further shown in FIG. **37**A, an example stimulation protocol **6050** comprises a first stimulation pattern **6051** to stimulate a hypoglossal nerve (HGN) and a second stimulation pattern **6061** to stimulate an ansa cervicalis-related nerve (ACN). In some examples, the first stimulation pattern **6051** of example stimulation protocol **6050** comprises substantially the same features and attributes as the first stimulation pattern **6021** of example stimulation protocol **6020** in FIG. **37**A. Moreover, in some examples, the second stimulation pattern **6061** of example stimulation protocol **6050** comprises substantially the same features and attributes as the second stimulation pattern **6041** of example stimulation protocol **6020** in FIG. **37**A, except with the stimulation period **6063** having a longer duration such that the end (E) of the stimulation period **6063** coincides with the end of the inspiratory phase (INSP), corresponding to the transition (T) between the inspiratory phase (INSP) and expiratory phase (EA, EP). By timing the stimulation period **6063** of the ansa cervicalis-related nerve to coincide with the entire inspiratory phase (INSP) (including some stimulation just prior to the start (S) of the inspiratory phase (INSP)), the stimulation may ensure upper airway patency throughout the entire inspiratory phase for at least some patients where such stimulation is prudent.
(483) Consequently, as further shown in FIG. **37**A, the non-stimulation period **6065** of stimulation pattern **6061** has a shorter duration than the non-stimulation period **6045** in the stimulation pattern **6041** in FIG. **37**A so that the subsequent stimulation period **6063** may begin (B) prior to the start (S) of the next inspiratory phase (INSP) of the patient's next breath (i.e. subsequent respiratory cycle **5011**).
(484) As shown in FIG. **37**A, this stimulation cycle **6055** (including stimulation period **6063** and non-stimulation period **6065**) is repeated throughout the second stimulation pattern **6061** such that the stimulation of the ansa cervicalis-related nerve is synchronized in a closed-loop manner relative

to the inspiratory phase (INSP) of the patient's respiratory cycles **5011**.

(485) FIG. **37**B is a diagram **6100** schematically representing the same respiratory waveform **5010** as in FIG. **34**, and example stimulation protocol **6120**. As shown in FIG. **37**B, example stimulation protocol **6120** comprises a first stimulation pattern **6121** to stimulate a hypoglossal nerve (HGN) and a second stimulation pattern **6141** to stimulate an ansa cervicalis-related nerve (ACN). In some examples, the first stimulation pattern **6121** of example stimulation protocol **6120** comprises substantially the same features and attributes as the first stimulation pattern **6021** of example stimulation protocol **6020** in FIG. **37**A, except with the stimulation period **6132** additionally including an increasing ramped portion from a beginning (B**1**) of the stimulation period **6132** (preceding the start (S) of the inspiratory phase (INSP)) in which the amplitude (or intensity) of the HGN stimulation increases from zero (at B**1**) to an amplitude N**1** at B**2**, wherein the amplitude N**1** is maintained through the start (S) of the inspiratory phase (INSP) and thereafter to point (E), at which the stimulation period **6132** terminates, which coincides with the end of the inspiratory phase (INSP), corresponding to the transition (T) between the inspiratory phase (INSP) and expiratory phase (EA, EP). Non-stimulation period **6134** follows the termination of the stimulation period **6132** and extends through a portion of the expiratory phase (including active expiration (EA) and a portion of the expiratory pause EP), until the next stimulation period **6132** begins (B**1**) with the ramped increase in stimulation to point (B**2**) and brief leveling of stimulation amplitude N**1**, prior to the start (S) of inspiration (INSP), as previously described. In one aspect, the ramped beginning portion (B**1** to B**2**) provides a more gradual initiation of HGN stimulation (of the stimulation period), which may be less noticeable to a patient and/or which may be easier on the respective nerves and muscles. Even with this ramped beginning of stimulation, a full (selectable) amplitude of stimulation is achieved (at B**2**) prior to the start (S) of the inspiratory phase (INSP), which may be desirable or prudent for at least reasons described in association with FIG. **37**A.

(486) As shown in FIG. **37**B, this stimulation cycle **6135** for the hypoglossal nerve is repeated throughout the first stimulation pattern **6121** such that the stimulation of the hypoglossal nerve is synchronized in a closed-loop manner relative to a portion of the inspiratory phase (INSP) of each of the patient's respiratory cycles **5011**, while including a ramped beginning portion (B**1** to B**2**).

(487) As further shown in FIG. **37**B, the second stimulation pattern **6141** of example stimulation protocol **6120** comprises substantially the same features and attributes as the second stimulation pattern **6041** of example stimulation protocol **6020** in FIG. **37**A, except with the stimulation period **6143** (in FIG. **37**B) additionally including an increasing ramped portion from a beginning (B**1**) of the stimulation period **6143** (preceding the start (S) of the inspiratory phase (INSP)) in which the amplitude (or intensity) of the ACN stimulation increases from zero (at B**1**) to an amplitude P**1** at B**2**, wherein the amplitude P**1** is maintained through the start (S) of the inspiratory phase (INSP) and beyond to point (E**1**) at which the amplitude (or intensity) of stimulation decreases in a downwardly ramping manner to point (E**2**). At point E**2** the stimulation period **6143** terminates, which coincides with the end of the inspiratory phase (INSP), corresponding to the transition (T) between the inspiratory phase (INSP) and expiratory phase (EA, EP).

(488) In one aspect, the ramped ending portion (E**1** to E**2**) of the stimulation period **6143** provides a more gradual termination of ACN stimulation (in the stimulation period **6143**), which may help to prolong or maintain the patency-inducing effect achieved by the portion of the stimulation period **6143** which precedes, and coincides with the start (S) of the inspiratory phase (INSP). This gradual termination (in each stimulation cycle) also may be less noticeable to a patient and/or which may be easier on the respective nerves and muscles.

(489) Non-stimulation period **6145** of the second stimulation pattern **6141** follows the termination (at E**2**) of the stimulation period **6143** and extends through a portion of the expiratory phase (including active expiration (EA) and a portion of the expiratory pause EP), until the next stimulation period **6143** begins (B**1**) with the ramped increase in stimulation to point (B**2**) and brief constant stimulation amplitude P**1**, prior to the start (S) of inspiration (INSP), as previously

described. In one aspect, the ramped beginning portion (B**1** to B**2**) provides a more gradual initiation of stimulation (of the stimulation period), which may be less noticeable to a patient and/or which may be easier on the respective nerves and muscles. Even with this ramped beginning of stimulation, a full (selectable) amplitude of ACN stimulation is achieved (at B**2**) prior to the start (S) of the inspiratory phase (INSP), which may be desirable or prudent for at least reasons described in association with FIG. **37**A.

(490) In one aspect, the non-stimulation period **6145** of stimulation pattern **6141** in FIG. **37**B has a shorter duration than the non-stimulation period **6045** in the stimulation pattern **6041** in FIG. **37**A such that the subsequent stimulation period **6143** (of stimulation pattern **6141** in FIG. **37**B) may begin (B**1**, B**2**) prior to the start (S) of the next inspiratory phase (INSP) of the patient's next breath (i.e. subsequent respiratory cycle **5011**).

(491) Similarly, as noted above, the more gradual ramped termination of stimulation (E**1** to E**2**) in each stimulation period **6143** in the second stimulation pattern of protocol **6120** in FIG. **37**B may, in some examples: (1) act to prolong the upper airway patency effect of the portion of stimulation (of the ansa cervicalis-related nerve) coinciding with the start (S) of the inspiratory phase (INSP), (2) be less noticeable to a patient; and/or (3) be easier on the respective nerves and muscles.

(492) As shown in FIG. **37**B, this stimulation cycle **6155** for the ansa cervicalis-related nerve (ACN) is repeated throughout the second stimulation pattern **6141** such that the stimulation of the ansa cervicalis-related nerve is synchronized in a closed-loop manner relative to a portion of the inspiratory phase (INSP) of each of the patient's respiratory cycles **5011**, while including a ramped beginning portion (B**1** to B**2**) and a ramped ending portion (E**1** to E**2**).

(493) FIG. **37**C is a diagram **6200** schematically representing the same respiratory waveform **5010** as in FIG. **34**, and example stimulation protocols **6220**, **6250**. Unlike the closed-loop example stimulation protocols in at least FIGS. **37**A-**37**B and **37**D, the example stimulation protocols **6220**, **6250** in FIG. **37**C comprise open-loop stimulation protocols in which the stimulation periods are not synchronized relative to a feature of the respiratory cycle **5011** such as, but not limited to, a feature of the inspiratory phase (INSP) and/or the expiratory phase (EA, EP).

(494) It will be understood that the presence of the respiratory waveform **5010** in FIG. **37**C is merely to provide some context regarding the duration and/or other aspects of the respective stimulation and non-stimulation periods of the respective example stimulation patterns. Accordingly, any seemingly regular correspondence between certain aspects of the respiratory cycles and the respective stimulation periods (and non-stimulation periods) in FIG. **37**C is merely coincidental and present to facilitate illustrative simplicity and clarity, and not intended to convey synchronization of the stimulation patterns relative to the respiratory cycles.

(495) As shown in FIG. **37**C regarding the first stimulation protocol **6220**, the first stimulation pattern **6221** is provided to stimulate the hypoglossal nerve (HGN) while the second stimulation pattern **6241** is provided to stimulate the ansa cervicalis-related nerve (ACN). The first stimulation pattern **6221** comprises a series of stimulation cycles **6235**, with each stimulation cycle **6235** comprising a stimulation period **6232** and a non-stimulation period **6234**. Meanwhile, the second stimulation pattern **6241** of the first stimulation protocol **6220** in FIG. **37**C comprises a series of stimulation cycles **6246**, with each stimulation cycle **6246** comprising a stimulation period **6243** and a non-stimulation period **6245**. For both patterns **6221**, **6241**, the labels B and E identify a beginning and an end, respectively, of the stimulation periods **6232** and **6243**, respectively.

(496) As shown in FIG. **37**C, the stimulation periods **6232** of the first stimulation pattern **6221** (to stimulate the hypoglossal nerve) are offset relative to the stimulation periods **6243** of the second stimulation pattern **6241** (to stimulate the ansa cervicalis-related nerve). In some examples of this arrangement, the stimulation periods **6232** of the first stimulation pattern **6221** (to stimulate the hypoglossal nerve) generally overlap with the non-stimulation periods **6245** of the second stimulation pattern **6241** (to stimulate the ansa cervicalis-related nerve). Via this arrangement, stimulation is always being applied to increase and/or maintain upper airway patency with the

hypoglossal nerve being rested while the ansa cervicalis-related nerve is being stimulated, and vice versa. In other words, at any given time, a stimulation period (e.g. **6232**, **6243**) of one of the stimulation patterns **6221**, **6241** (of the HGN and of the ACN) is being delivered. While the stimulation periods of the different stimulation patterns **6221**, **6241** may slightly overlap each other, when the stimulation is “ON” for a first nerve (e.g. HGN), the stimulation is “OFF” for the second nerve (e.g. ACN) such that the second nerve (e.g. ACN) is resting while the first nerve (e.g. HGN) is being stimulated. In some examples, this arrangement may be applied between a left HGN (first nerve) and a right HGN (second nerve) or between a left ACN (first nerve) and a right ACN (second nerve).

(497) Accordingly, in some such examples stimulation is being applied continuously, although being split between ACN stimulation and HGN stimulation with the ACN stimulation being “ON” one-half of the treatment period and the HGN stimulation being “ON” one-half of the treatment period. In some examples as later described in association with FIG. **37**CC, this paradigm may be extended to additional targets, such that stimulation is applied per a protocol in which stimulation is “ON” continuously yet distributed among two, three, or four targets, such as the left HGN, left ACN, right HGN, right ACN. It will be understood that stimulation being “ON” continuously presumes that the stimulation signal(s) comprise a series of stimulation pulses applied at a high enough frequency for stimulation to be considered relatively continuous. Moreover, it will be further understood that continuous stimulation may correspond to stimulation being “ON” 100 percent of the time. In some such examples, the stimulation may be applied substantially continuously, such as 95, 96, 97, 98, 99 percent of the duration of the treatment period.

(498) Among other effects, this example stimulation protocol **6220** in FIG. **37**C may help minimize or avoid fatigue of a single type of nerve (e.g. the hypoglossal or the ansa cervicalis-related) while providing consistency in neurostimulation to increase and/or maintain upper airway patency. Moreover, because stimulation of one type of nerve may be beneficial under certain conditions (e.g. body position, head-and-neck position, etc.) and another type of nerve may be more beneficial under certain conditions, alternating stimulation between both types of nerves may provide consistent therapy.

(499) In one aspect, the example stimulation protocol may promote consistency in increasing and/or maintaining upper airway patency because of the relative fast duty cycle (e.g. short duration of the stimulation periods and non-stimulation periods) by which each type of nerve is frequently stimulated. For instance, as shown in FIG. **37**C, in some examples multiple stimulation cycles **6235** may occur within a duration of a typical respiratory cycle **5011**. However, in some examples, the duration of the stimulation period **6232** and/or non-stimulation period **6234** may be longer such that one stimulation cycle **6235** may be longer than the duration of a typical respiratory cycle **5011**.

(500) In one aspect, the open loop nature of the example stimulation protocol **6220** enables stimulation therapy without using a sensor or sensed information that might otherwise be used to synchronize the stimulation cycles (e.g. **6235**, **6246**) relative to at least one feature of the patient's respiratory cycles **5011**. Among other effects, this arrangement may simplify implantation and/or other aspects of providing stimulation therapy for sleep disordered breathing or other physiologic conditions.

(501) As further shown in FIG. **37**C, an example open-loop, stimulation protocol **6250** comprises a first stimulation pattern **6251** for stimulating the hypoglossal nerve (HGN) and a second stimulation pattern **6261** for stimulating the ansa cervicalis-related nerve (ACN). The first stimulation pattern **6251** comprises a series of stimulation cycles **6255**, with each stimulation cycle **6255** comprising a stimulation period **6252** and a non-stimulation period **6254**. Meanwhile, the second stimulation pattern **6261** comprises a series of stimulation cycles **6266**, with each stimulation cycle **6266** comprising a stimulation period **6263** and a non-stimulation period **6265**. For both patterns **6251**, **6261**, the labels B and E identify a beginning and an end, respectively, of the stimulation periods **6252** and **6263**, respectively.

(502) As shown in FIG. **37**C, the stimulation periods **6252** of the first stimulation pattern **6251** (to stimulate the hypoglossal nerve) coincide with the stimulation periods **6263** of the second stimulation pattern **6261** (to stimulate the ansa cervicalis-related nerve).

(503) In a manner similar to example stimulation protocol **6220**, the open loop nature of the example stimulation protocol **6250** enables stimulation therapy without using a sensor or sensed information that would otherwise be used to synchronize the stimulation cycles (e.g. **6255**, **6266**) relative to at least one feature of the patient's respiratory cycles **5011**. Among other effects, this arrangement may simplify implantation and/or other aspects of providing stimulation therapy for sleep disordered breathing or other physiologic conditions.

(504) In some examples, a duration of the respective stimulation periods **6252**, **6263** may be longer than shown in FIG. **37**C to generally increase the overall amount of stimulation delivered to the respective hypoglossal and ansa cervicalis-related nerves.

(505) FIG. **37**CC is a diagram **6270** schematically representing the same respiratory waveform **5010** as in FIG. **34**, and example stimulation protocol **6272** comprising at least some of substantially the same features and attributes as example stimulation protocol **6220** of FIG. **37**C (e.g. comprising open-loop stimulation patterns) except with stimulation protocol **6272** explicitly comprising stimulation patterns for up to four targets including the left HGN, left ACN, right HGN, and right ACN. In a manner similar to the stimulation protocol **6220** in FIG. **37**C, the stimulation protocol **6272** comprises a stimulation pattern **6221** (like **6221** in FIG. **37**C) applied to a first nerve (e.g. left HGN “L HGN”) and a stimulation pattern **6241** (like **6241** in FIG. **37**C) applied to a second nerve (e.g. left ACN “L ACN”). Meanwhile, stimulation protocol **6272** comprises a stimulation pattern **6271** for application to a third nerve such as a right HGN (R HGN) and comprising stimulation cycles **6275** like stimulation cycles **6235** (or **6246**) of stimulation patterns **6221**, **6241**. However, the stimulation pattern **6271** is offset from both of the stimulation patterns **6221**, **6241** such that the stimulation periods **6232** of stimulation pattern **6271** for the right HGN (R HGN) are offset relative to the occurrence of the stimulation periods **6232** of both the stimulation patterns **6221** (L HGN) and **6241** (L ACN). Finally, stimulation protocol **6272** comprises a stimulation pattern **6281** for application to a fourth nerve such as a right ACN (R ACN) and comprising stimulation cycles **6286** like stimulation cycles **6235** (or **6246**) of stimulation patterns **6221**, **6241**. However, the stimulation pattern **6281** is offset from all three of the stimulation patterns **6221**, **6241**, **6271** such that the stimulation periods **6232** of stimulation pattern **6281** for the right ACN (R ACN) are offset relative to the occurrence of the stimulation periods **6232** of all three stimulation patterns **6221** (L HGN), **6241** (L ACN), and **6271** (R HGN).

(506) Via this arrangement, stimulation may be applied to all four target nerves to apply stimulation with the stimulation being apportioned roughly into fourths among the four nerves such that about one-fourth of the stimulation over a treatment period is applied to a first nerve (e.g. L HGN or other), one-fourth applied to a second nerve (e.g. L ACN or other), one-fourth applied to a third nerve (e.g. R HGN or other), and one-fourth applied to a fourth nerve (e.g. R ACN).

(507) In some examples in which just three nerves are stimulated (instead of all four), then with stimulation being applied continuously the stimulation is apportioned roughly into thirds such that about one-third of the stimulation over a treatment period is applied to a first nerve (e.g. L HGN or other), one-third applied to a second nerve (e.g. L ACN or other), and one-third applied to a third nerve (e.g. R HGN or R CACN as an example).

(508) Via such example implementations, it is believed that a robust, stable arrangement of promoting upper airway patency may be achieved while enhancing patient comfort, managing nerve/muscle fatigue, etc.

(509) It will be understood that in some such examples where just two target nerves are being stimulated, the two target nerves may comprise both hypoglossal nerves (left and right) or may comprise both ansa cervicalis-related nerves (left and right). Moreover, in some examples where just two target nerves are being stimulated, the two target nerves may comprise both just nerves on

one side of the body (e.g. left HGN and left CAN OR right HGN and right ACN).
(510) In some examples, one instance of stimulation protocol **6272** may comprise applying offsetting stimulation patterns for both hypoglossal nerves (L HGN and R HGN) and just one of the ansa cervicalis-related nerves (L CAN OR R ACN). It is believed that stimulation of just one of the ansa cervicalis-related nerves may be adequate or sufficient to open the upper airway generally at least because even just one sided (e.g. left or right) ACN stimulation will sufficiently pull the larynx inferiorly to yield the desired opened/stiffened upper airway. In a complementary manner, both hypoglossal nerves may be stimulated bilaterally (e.g. alternately, other) in a way to enhance patient comfort, adapt to different collapse patterns (e.g. type, degree) per different body positions, etc. Accordingly, in one aspect, the stimulation of just one ansa cervicalis-relate nerve may sometimes be referred to as providing a baseline therapy to open/stiffen the upper airway, while the stimulation of the hypoglossal nerves (left and/or right) may sometimes be referred to as providing a more dynamic aspect to the overall stimulation therapy.
(511) In view of these features, in some examples, in general terms not specific to FIG. **37**CC, the particular arrangement of stimulating both the left and right hypoglossal nerves and just one ansa cervicalis-related nerve may be employed using other specific stimulation patterns which also may be open loop and/or closed loop.
(512) It will be understood that nerves other than the hypoglossal nerve and ansa cervicalis-related nerve which relate to promoting upper airway patency or otherwise treating sleep disordered breathing can be included in a complementary manner as part of the example stimulation protocols of FIG. **37**C or **37**CC.
(513) FIG. **37**D is a diagram **6300** schematically representing the same respiratory waveform **5010** as in FIG. **34**, and example stimulation protocol **6320**. As shown in FIG. **37**D, example stimulation protocol **6320** comprises a first stimulation pattern **6321** to stimulate a phrenic nerve (PHR) and a second stimulation pattern **6341** to stimulate both an ansa cervicalis-related nerve and a hypoglossal nerve (AC/HGN). In some examples, the first stimulation pattern **6321** of example stimulation protocol **6320** comprises substantially the same features and attributes as the first stimulation pattern **5021** of example stimulation protocol **5030** in FIG. **33**A, except with the stimulation signal being applied to a phrenic nerve (PHR) instead of a hypoglossal nerve (HGN). As shown in FIG. **37**D, the first stimulation pattern **6321** comprises a series of stimulation cycles **6335**, with each stimulation cycle **6335** comprising a stimulation period **6332** (like **5032** in FIG. **33**A) and a non-stimulation period **6334** (like **5034** in FIG. **33**A).
(514) In some examples, the second stimulation pattern **6341** of example stimulation protocol **6320** in FIG. **37**D comprises substantially the same features and attributes as the first stimulation pattern **6021** of example stimulation protocol **6020** in FIG. **37**A, except with the stimulation signal being applied to both an ansa cervicalis-related nerve and a hypoglossal nerve (AC/HGN) instead of being applied to solely a hypoglossal nerve (HGN). As shown in FIG. **37**D, the second stimulation pattern **6341** comprises a series of stimulation cycles **6355**, with each stimulation cycle **6355** comprising a stimulation period **6343** (like **6032** in FIG. **37**A) and a non-stimulation period **6345** (like **6034** in FIG. **37**A). As further shown in FIG. **37**D, the stimulation period **6343** in the stimulation cycles **6355** of stimulation pattern **6341** (to stimulate both the respective ansa cervicalis-related and hypoglossal nerves) is generally synchronous with, and coincides with the inspiratory phase (INSP) while also including a brief pre-inspiratory stimulation component, as identified via the stimulation beginning (B) just prior to the start (S) of the inspiratory phase (INSP) of the patient's respiratory cycle **5011** in a manner similar to stimulation pattern **6021** in FIG. **37**A.
(515) With this in mind, the example stimulation protocol **6320** in FIG. **37**D may be implemented to treat instances of multiple-type sleep apnea and/or more complex sleep disordered breathing behaviors. For example, in a multiple-type sleep apnea involving both central and obstructive sleep apnea behaviors, the stimulation pattern **6341** may be applied to prevent or minimize obstructive

sleep apneas while the stimulation pattern **6321** may be applied to prevent or minimize central sleep apneas. In some such examples, by applying the stimulation pattern **6341** including the pre-inspiratory stimulation component, the stimulation protocol **6320** may help ensure patency of the upper airway prior to the inspiratory phase (INSP) and the stimulation of phrenic nerve (via stimulation pattern **6321**) may ensure appropriately timed contraction of the diaphragm as part of the target respiratory cycle.
(516) In some examples, the example stimulation protocol **6320** in FIG. **37**D may be employed in association with at least the example stimulation arrangements described later in association with at least FIG. **55**—regarding stimulation elements, anchors, delivery methods for stimulating a phrenic nerve and/or for stimulating an ansa cervicalis-related nerve. Moreover, it will be understood that such example stimulation arrangements in association with at least FIGS. **55-59**B may be employed in a complementary manner, or separately from, the various example stimulation arrangements for stimulating the hypoglossal nerve.
(517) With regard to the various example stimulation elements, sensing elements, target nerve locations, stimulation element/lead delivery arrangements, etc. described throughout various examples of the present disclosure, it will be understood that in some examples, stimulation protocols other than those described and illustrated in association with FIGS. **33**A-**37**D may be employed.
(518) FIG. **38**A is a flow diagram schematically representing an example arrangement **8000** including an example device for, and/or example method of, using sensed data as feedback to adjust an intensity (e.g. strength) of, or other parameters, aspects, etc. regarding, stimulation of a hypoglossal nerve and/or ansa cervicalis-related nerve. In some examples, the example arrangement **8000** may comprise at least some of substantially the same features and attributes as the example stimulation devices and/or methods previously described examples of the present disclosure, including stimulation protocols, stimulation arrangements, etc. In some examples, the sensing may comprise at least some of the same features and attributes as at least the later described sensing/control examples.
(519) As shown in FIG. **38**A, example arrangement **8000** comprises a stimulator **8010** to generate stimulation signals **8012**, **8014** to deliver stimulation via a hypoglossal (HGN) stimulation element **8020** and via an ansa cervicalis-related nerve (ACN) stimulation element **8030**, respectively. During and/or after such stimulation, data **8022**, **8032** can be sensed from the stimulation elements **8020**, **8030**, such as but not limited to, an impedance between the respective elements **8020**, **8030**. In some examples, a sensor(s) **8034** may supply other or additional data as input **8033**, as further described later.
(520) At **8035**, the sensed data is evaluated (e.g. checked) such as a sensed data evaluator and the value **8037** of the measured parameter(s) (e.g. impedance) is fed to the stimulation settings generator (at **8040**), which in turn generates updated (or maintained) settings **8042**, which are fed to the stimulator **8010**. In some examples, the impedance may indicate a degree of upper airway patency. For example, a smaller cross-sectional upper airway, which reflects less upper airway patency, may be sensed as a lower impedance. Conversely, a larger cross-sectional upper airway, which reflects more upper airway patency, may be sensed as a higher impedance. Accordingly, maximal patency (measured as a higher impedance) may general correspond to periods of stimulation (HGN and/or ACN) or correspond to peak expiration of a respiratory cycle. Meanwhile, minimal patency (measured as a lower impedance) generally corresponds to inspiration, just prior to inspiration, or the onset of stimulation (e.g. HGN and/or ACN).
(521) Among other adjustments, the determination made at **8035** may be used to balance a relative amount of stimulation to be applied via the HGN stimulation element **8020** and/or the ACN stimulation element **8030**. For example, per some example determinations at **8035**, balancing the stimulation comprise applying stimulation solely via the HGN stimulation element **8020**, while per some example determinations at **8035**, balancing the stimulation may comprise applying

stimulation solely via the ACN stimulation element **8030**. Moreover, per some example determinations at **8035**, balancing the stimulation may comprise applying the stimulation as some via the HGN stimulation element **8020** and some stimulation via the ACN stimulation element **8030** while controlling a relative proportion of the stimulation between the respective HGN and ACN nerves. In addition to the extent that bilateral stimulation may be applied among a left HGN nerve, a right HGN nerve, a left ACN nerve, and/or a right ACN nerve, the above-described adjustments may be made among those four nerves.

(522) In some examples, generating the stimulation settings at **8040** may comprise other parameters in addition to, or other than, adjusting which nerves are stimulation (and by how much). Such other parameters may comprise adjusting an intensity or strength of the stimulation at any given nerve (HGN and/or ACN), whereby the strength adjustment may comprise adjustments in amplitude, pulse width, pulse frequency, duty cycle, pulse duration, and the like, at least some of which are described in association with and/or implemented via the stimulation engine **8800** in FIG. **38**D. In some examples, additional adjustments may comprise whether stimulation of the nerves (left HGN, right HGN, left ACN, and/or right ACN) are implemented simultaneously, alternately, in a particular sequence, randomly, and the like.

(523) In some examples, in addition to or instead of using the inputs **8022**, **8032** from the stimulation elements **8020**, **8030** to perform a check of sensed data, an example method/device may comprise a sensed data input **8031** from at least one sensing source **8034**, which may comprise a dedicated sensing element or an element not dedicated to sensing. The sensing source(s) **8034** may comprise at least one of a plurality of sensing modalities, types, etc.

(524) FIG. **38**B is a block diagram schematically representing an example sensed data engine **8700** including a plurality of different physiologic parameters determinable from sensed data which may be used in at least the sensed data check **8035** as part of the example method in FIG. **38**A. As shown in FIG. **38**B, in some example physiologic parameters determinable from sensed data may comprise parameters regarding collapse **8710**, position **8722**, respiration **8724**, disease burden **8726**, sleep **8728**, and other **8729**.

(525) In some examples, the collapse parameter **8710** may comprise further parameters regarding a pattern **8712** and/or a degree **8714** of collapse of the upper airway in the patient's body. In some such examples, the sensing of data regarding a collapse pattern parameter **8712** and/or a degree parameter **8714** may be implemented via at least some of substantially the same features and attributes as later described in association with at least FIGS. **53**A-**53**F by which a pattern, location, and degree of collapse may be determined and characterized so as to use this information as part of the sensed data check.

(526) In some examples, per collapse parameter **8710**, if an antero-posterior collapse is detected (of a sufficient degree), then in some examples the stimulation settings are generated (e.g. **8040** in FIG. **38**A) to implement stimulation therapy via applying stimulation to the hypoglossal nerve solely, initially, or primarily (but not solely). In some examples, applying the stimulation initially may sometimes be referred to as applying stimulation first, i.e. prior to applying stimulation to the ansa cervicalis-related nerve.

(527) In some examples, per collapse parameter **8710**, if a lateral and/or concentric collapse pattern (of a sufficient degree) is detected, then in some examples the stimulation settings are generated (e.g. **8040** in FIG. **38**A) to implement stimulation therapy via applying stimulation solely, initially, or primarily (i.e. not solely) to the ansa cervicalis-related nerve. In some examples, applying the stimulation initially may sometimes be referred to as applying stimulation first, i.e. prior to applying stimulation to the hypoglossal nerve.

(528) In some examples, via the collapse parameter **8710** in FIG. **38**B, one example implementation of the example arrangement **8000** in FIG. **38**A may comprise generating stimulation settings (e.g. **8040** in FIG. **38**A) using the sense data check (e.g. **8035** in FIG. **38**A) based on sensor inputs **8022**, **8032** (e.g. impedance) or sensor input **8033** to automatically assess

the pattern, location, and/or degree of collapse and then select which nerves (e.g. HGN, ACN) are applied, and if both types of nerves are to be stimulated, then select a sequence of stimulation, simultaneous application, proportion, etc. In some examples, such selection may be implemented in cooperation with the relationship parameter **8813** of the stimulation engine **8100** (e.g. FIG. **38**B), as further described below.

(529) In some examples, collapse information may be determined via sensing impedance (**8752** in FIG. **38**C), such as via sensor inputs **8022**, **8032** (via the stimulation elements **8020**, **8030** or other electrodes) or other sensor input **8033**. In some examples, one or more of the other sensor tools (array **8750** in FIG. **38**C) may be used to determine collapse, such as but not limited to sensed accelerometer data (per **8754** in FIG. **38**C), which may be used alone or with sensed impedance. Similarly, sensed acoustic data (**8769** in FIG. **38**C) from an accelerometer or other sources may reveal collapse information, with one non-limiting example including sensing snoring information (e.g. via snoring partner, other).

(530) In some examples, a typical collapse pattern for a given patient may be known prior to implanting a nerve stimulation system such that the sensed data engine **8700** (as supported by memory of the control portion **10500**) may retrieve stored data regarding such collapse pattern(s) for use in initial or ongoing programming of stimulation therapy, adapting the stimulation therapy and/or use in confirming sensing of such collapse patterns. In some such examples, a clinician/other may enter such known collapse information as part of the programming, whether initially or later. This information may be entered via user interface **10520** (FIG. **54**D), clinician programmer **10650** (FIG. **54**E), and/or a patient management tool **10660** (FIG. **54**E) such as (but not limited to) a cloud portal resource **10662** (FIG. **54**E).

(531) Moreover, to the extent that the stimulation therapy may be effective in lessening or preventing the known collapse pattern, then example devices/methods may compare a degree, type, etc. of the stored, known collapse pattern with the currently sensed collapse pattern (or lack thereof) as one way to evaluate the stimulation therapy and potentially determine what, if any, adjustments to stimulation therapy may be warranted. For example, one may evaluate the sensed collapse (e.g. pattern, degree) and adjust how each nerve (e.g. left HGN, right HGN, left ACN, right ACN, and combinations thereof) is to be stimulated such as via various aspects of the relationship parameter **8813** and/or other parameters of the stimulation engine **8800** (FIG. **38**D, relative to available stimulation protocols (e.g. FIGS. **33-37**D) as well as in cooperation with other sensed data (**8700** in FIG. **38**B), sensor tools (array **8750** in FIG. **38**C) or other parameters, factors, engines, methods, as described throughout various examples of the present disclosure.

(532) In some examples, the position parameter **8722** of sensed data engine **8700** in FIG. **38**B for a wide variety of purposes. In some examples, sensed data regarding body position (or posture) may be used to initiate, terminate, and/or adjust therapy stimulation settings, patterns, etc. For instance, in some examples, upon sensing the patient being in a supine position, such as when one may expect a highest likelihood of obstructive sleep apnea for at least some patients, then an example method may comprise delivering stimulation to both a hypoglossal nerve (e.g. left and/or right) and an ansa cervicalis-related nerve (e.g. left and/or right), such as via elements **8020**, **8030** in FIG. **38**A. As previously mentioned, posture and/or activity sensed via an accelerometer may be used with sensed body position for making the preceding determination.

(533) In some examples, upon sensing the patient is lying on their side (e.g. a lateral decubitis position), then an example method may comprise delivering stimulation solely to an ansa cervicalis-related nerve (e.g. left and/or right), such as solely via element **8030** in FIG. **38**A.

(534) In some examples, upon sensing the patient is lying prone, then an example method may comprise delivering stimulation to neither the hypoglossal nerve nor the ansa cervicalis-related nerve per **8020**, **8030** in FIG. **38**A.

(535) At least because some patients may exhibit atypical position-dependent sleep disordered breathing, it will be understood that other stimulation settings may be generated (**8040** in FIG.

38A) than described above.

(536) In some examples, a sensed position (e.g. **8722** in FIG. **38**A for **8035** in FIG. **38**A) may be used to determine or adjust timing of when stimulation is to be applied and/or to adjust which nerve targets (left HGN, right HGN, left ACN, right ACN) are to be stimulated.

(537) With further reference to FIG. **38**B and FIG. **38**A, in some examples, a respiration parameter (**8728** in FIG. **38**B) may comprise the sensed data which is evaluated (**8035** in FIG. **38**A) and on which stimulation settings may be generated (**8040** in FIG. **38**A) to determine which nerves (and at which strength settings, etc.) are to be stimulated (**8020**, **8030** in FIG. **38**A). In some such examples, the sleep parameter **8728** may comprise a sleep state, such as whether the patient is awake or asleep and/or such as the sleep stage of the patient.

(538) With further reference to FIG. **38**B and FIG. **38**A, in some examples, a respiration parameter (**8724** in FIG. **38**B) may comprise the sensed data which is evaluated (**8035** in FIG. **38**A) and on which stimulation settings may be generated (**8040** in FIG. **38**A) to determine which nerves (and at which strength settings, etc.) are to be stimulated (**8020**, **8030** in FIG. **38**A). At least some example implementations of the respiration parameter **8724** as sensed data (**8035**) and/or for other uses were previously described in association with at least FIGS. **3**C and/or are later described in association with at least FIGS. **38**C-**38**D, as well in association with at least FIGS. **40**A-**51**B.

(539) With further reference to FIG. **38**B and FIG. **38**A, in some examples, a disease burden parameter (**8726** in FIG. **38**B) may comprise the sensed data which is evaluated (**8035** in FIG. **38**A) and on which stimulation settings may be generated (**8040** in FIG. **38**A) to determine which nerves (and at which strength settings, etc.) are to be stimulated (**8020**, **8030** in FIG. **38**A). In some such examples, the disease burden parameter **8726** may comprise an indication of a severity (e.g. apnea-hypopnea index—AHI) (e.g. burden) on the patient imposed by the disease (e.g. sleep disordered breathing, such as but not limited to obstructive sleep apnea). It will be understood that in some examples, the disease burden parameter **8726** may comprise burden indications of other diseases, such as cardiac disorders, etc. which may be related to the disease burden imposed by sleep disordered breathing. At least some example implementations of the disease burden parameter **8726** as sensed data (**8035**) and/or for other uses were previously described in association with at least FIGS. **3**C and/or are later described in association with at least FIGS. **38**C-**38**D, as well in association with at least FIGS. **40**A-**51**B.

(540) It will be understood that at least some of the various parameters in the sensed data engine **8700** may be used in a complementary manner in various combinations in methods of stimulation therapy for treating sleep disordered breathing according to examples of the present disclosure.

(541) FIG. **38**C is a block diagram schematically representing example sensor tools array **8750**, which may comprise one or more types of sensors which may comprise example implementations of one or more of the various sensors described as part of one or more of the example arrangements throughout the present disclosure, such as but not limited to the sensors (e.g. **560**, **566**, **568**A, **568**B, other) described in association with at least FIG. **3**C. In some examples, the example sensor tools in array **8750** may comprise sensors to sense impedance **8752**, acceleration (e.g. accelerometer) **8754**, pressure **8756**, EMG **8758**, ECG **8760**, ballistocardiograph (BALLISTO) **8762**, seismocardiograph (SEISMO) **8764**, heart rate (HR) **8766**, and other **8768**.

(542) In some examples, the impedance sensor **8752** in sensor tools array of FIG. **38**C may not be a separate standalone sensor but represent that impedance sensing may be performed via multiple spaced apart electrodes and further represent any associated hardware or aspects of a control portion for performing such impedance sensing.

(543) Like the sensing types, physiologic parameters etc. described in association with FIG. **3**C, sensing parameters regarding ballistocardiograph **8762** or a seismocardiograph sensor may determine respiration, among other physiologic parameters. In some examples, sensing an EMG (parameter **8758**) may be used to determine upper airway patency, tongue protrusion, responsiveness to stimulation therapy, and the like, and thereby be used for timing in applying

stimulation, determining effectiveness in the delivered stimulation therapy. As related aspects, the sensed EMG information may provide respiratory phase information, stimulation response information, and the like. In some examples, methods of sensing an EMG (parameter **8758**) or using sensed EMG information may further comprise at least the example implementations of method **8250** in FIG. **51**B, method **8400** in FIG. **44**A, and/or other examples of the present disclosure including EMG sensing or sensed EMG information.

(544) With regard to sensing various parameters such as (but not limited to) respiration (**8724** in FIG. **38**B), sleep (**8728** in FIG. **38**B), and/or position/posture (**8722** in FIG. **38**B) via an accelerometer **8754** (FIG. **38**C), in some examples such sensing may comprise at least some of substantially the same features and attributes as described in: (1) U.S. 2019/0160282, published May 30, 2019, and titled ACCELEROMETER-BASED SENSING FOR SLEEP DISORDERED BREATHING (SDB) CARE; (2) WO/2021/016558, published Jan. 28, 2021, titled SLEEP DETECTION FOR SLEEP DISORDERED BREATHING (SDB) CARE, and filed on Sep. 4, 2020 as a 371 National Stage Application, Ser. No. 16/978,470, published as U.S. 2023-0095780 on Mar. 30, 2023; (3) WO/2021/016562 published on Jan. 28, 2021, titled RESPIRATION DETECTION, and filed on Sep. 2, 2020 as a 371 National Stage Application, Ser. No. 16/977,664, published as U.S. 2023-0119173 on Apr. 20, 2023; and/or (4) WO/2021/016536 published Jan. 28, 2021, titled SYSTEMS AND METHODS FOR OPERATING AN IMPLANTABLE MEDICAL DEVICE BASED ON POSTURE DETECTION, and filed on Sep. 4, 2020, as a 371 National Stage Application, Ser. No. 16/978,275, published as U.S. 2021-0268279 on Sep. 9, 2021, each of which are hereby incorporated by reference in their entirety.

(545) FIG. **38**D is a block diagram schematically representing an example stimulation engine **8800**. In some examples, the stimulation engine **8800** may comprise one example implementation of control portion **10000** in FIG. **54**B as a whole, and/or of instructions **10511** of control portion **10000** In some examples, the stimulation engine **8800** may comprise a treatment period parameter **8801**, closed loop parameter **8802**, an open loop parameter **8804**, an auto-titration parameter **8805**, a ramp parameter **8806**, a titration parameter **8808**, a patient control parameter **8810**, a breath-by-breath parameter **8812**, a relationship (e.g. nerve stimulation relationship) parameter **8813**, and/or a strength limits parameter **8820**. In some examples, the strength limits parameter **8820** may further comprise a threshold parameter **8822**, a therapeutic parameter **8824**, a comfort parameter **8826**, and a sleep disturbance parameter **8828**.

(546) In some examples, stimulation may be delivered according to a treatment period, which may comprise a period of time beginning with the patient turning on the therapy device and ending with the patient turning off the device. In some examples, the treatment period may comprise a selectable, predetermined start time (e.g. 10 p.m.) and selectable, predetermined stop time (e.g. 6 a.m.). In some examples, the treatment period may comprise a period of time between an auto-detected initiation of sleep and auto-detected awake-from-sleep time. With this in mind, the treatment period corresponds to a period during which a patient is sleeping such that the stimulation of the upper airway patency-related nerve and/or central sleep apnea-related nerve is generally not perceived by the patient and so that the stimulation coincides with the patient behavior (e.g. sleeping) during which the sleep disordered breathing behavior (e.g. central or obstructive sleep apnea) would be expected to occur.

(547) To avoid enabling stimulation prior to the patient falling asleep, in some examples stimulation can be enabled after expiration of a timer started by the patient (to enable therapy with a remote control), or enabled automatically via sleep stage detection. To avoid continuing stimulation after the patient wakes, stimulation can be disabled by the patient using a remote control, or automatically via sleep stage detection. Accordingly, in at least some examples, these periods may be considered to be outside of the treatment period or may be considered as a startup portion and wind down portion, respectively, of a treatment period.

(548) In some examples, stimulation of an upper airway patency-related nerve may be performed

via open loop stimulation, such as per open loop parameter **8804** of stimulation engine **8800** in FIG. **38**D. In some examples, the open loop stimulation may refer to performing stimulation without use of any sensory feedback of any kind relative to the stimulation.

(549) In some examples, the open loop stimulation may refer to stimulation performed without use of sensory feedback by which timing of the stimulation (e.g. synchronization) would otherwise be determined relative to respiratory information (e.g. respiratory cycles). However, in some such examples, some sensory feedback may be utilized to determine, in general, whether the patient should receive stimulation based on a severity of sleep apnea behavior.

(550) Conversely, in some examples and as previously described in relation to at least several examples, stimulation of an upper airway patency-related nerve may be performed via closed loop stimulation, such as per closed loop parameter **8802** of stimulation engine **8800** in FIG. **38**D. In some examples, the closed loop stimulation may refer to performing stimulation at least partially based on sensory feedback regarding parameters of the stimulation and/or effects of the stimulation.

(551) In some examples, stimulation of one nerve (left or right) may be performed via open loop stimulation (e.g. without use of sensed physiologic data for timing application of the stimulation) while stimulation of another nerve (left or right) may be performed via closed loop stimulation, which may use sensed physiologic data for timing application of the stimulation.

(552) In some examples, the closed loop stimulation (per parameter **8802**) may comprise stimulation performed via use of sensory feedback by which timing of the stimulation (e.g. synchronization) is determined relative to respiratory information, such as but not limited to respiratory cycle information, which may comprise onset, offset, duration, morphology, etc. of the respiratory cycles. In some examples, the respiration information excludes (i.e. is without) tracking a respiratory volume and/or respiratory rate. In some examples, stimulation based on such synchronization may be delivered throughout a treatment period or throughout substantially the entire treatment period. In some examples, such stimulation may be delivered just during a portion or portions of a treatment period.

(553) In some examples of "synchronization", the stimulation relative to the inspiratory phase may extend to a pre-inspiratory period and/or a post-inspiratory phase. For instance, in some such examples, a beginning of the synchronization may occur at a point in each respiratory cycle which is just prior to an onset of the inspiratory phase. In some examples, this point may be about 200 milliseconds, or 300 milliseconds prior to an onset of the inspiratory phase.

(554) In some examples in which the stimulation is synchronous with at least a portion of the inspiratory phase, the upper airway muscles are contracted via the stimulation to ensure they are open at the time the respiratory drive controlled by the central nervous system initiates an inspiration (inhalation). In some such examples, in combination with the stimulation occurring during the inspiratory phase, example implementation of the above-noted pre-inspiratory stimulation helps to ensure that the upper airway is open before the negative pressure of inspiration within the respiratory system is applied via the diaphragm of the patient's body. In one aspect, this example arrangement may minimize the chance of constriction or collapse of the upper airway, which might otherwise occur if flow of the upper airway flow were too limited prior to the full force of inspiration occurring. In some examples, such stimulation may relate to stimulation of the hypoglossal nerve, the ansa cervicalis-related nerve, other nerves relating to upper airway patency, and/or stimulating various combinations of such nerves including left and/right options.

(555) In some such examples, the stimulation of the upper airway patency-related nerve may be synchronized to occur with at least a portion of the expiratory period.

(556) With regard to at least some of the example methods of treating sleep apnea as previously described throughout the present disclosure, at least some such methods may comprise performing the delivery of stimulation to the upper airway patency-related first nerve without synchronizing such stimulation relative to a portion of a respiratory cycle. In some instances, such methods may

sometimes be referred to as the previously described open loop stimulation.

(557) In some examples, the term “without synchronizing” may refer to performing the stimulation independently of timing of a respiratory cycle. In some examples, the term “without synchronizing” may refer to performing the stimulation being aware of respiratory information but without necessarily triggering the initiation of stimulation relative to a specific portion of a respiratory cycle or without causing the stimulation to coincide with a specific portion (e.g. inspiratory phase) of respiratory cycle.

(558) In some examples, in this context the term “without synchronizing” may refer to performing stimulation upon the detection of sleep disordered breathing behavior (e.g. obstructive sleep apnea events) but without necessarily triggering the initiation of stimulation relative to a specific portion of a respiratory cycle or without causing the stimulation to coincide with the inspiratory phase. At least some such examples may be described in Wagner et al., STIMULATION FOR TREATING SLEEP DISORDERED BREATHING, issued as U.S. Pat. No. 10,898,709 on 1/26/2021, and which is incorporated by reference herein in its entirety.

(559) In some examples, while open loop stimulation may be performed continuously without regarding to timing of respiratory information (e.g. inspiratory phase, expiratory phase, etc.) such an example method and/or device may still comprise sensing information for diagnostic data and/or to determine whether (and by how much) the continuous stimulation should be adjusted. For instance, via such sensing, it may be determined that the number of sleep disordered breathing (SDB) events are too numerous (e.g. an elevated AHI) and therefore the intensity (e.g. amplitude, frequency, pulse width, etc.) of the continuous stimulation should be increased or that the number of SDB events are relatively low such that the intensity of the continuous stimulation can be decreased while still providing therapeutic stimulation. It will be understood that via such sensing, other SDB-related information may be determined which may be used for diagnostic purposes and/or used to determine adjustments to an intensity (e.g. strength) of stimulation, initiating stimulation, and/or terminating stimulation to treat sleep disordered breathing.

(560) Some non-limiting examples of such devices and methods to recognize and detect the various features and patterns associated with respiratory effort and flow limitations include, but are not limited to Christopherson, U.S. Pat. No. 8,938,299 Issued on Jan. 30, 2015, titled SYSTEM FOR TREATING SLEEP DISORDERED BREATHING (SDB) (formerly published as PCT Publication WO/2010/059839, titled A METHOD OF TREATING SLEEP APNEA, published on May 27, 2010); Christopherson U.S. Pat. No. 5,944,680, titled RESPIRATORY EFFORT DETECTION METHOD AND APPARATUS; and Testerman U.S. Pat. No. 5,522,862, titled METHOD AND APPARATUS FOR TREATING OBSTRUCTIVE SLEEP APNEA, each of which is hereby incorporated by reference herein in their entirety.

(561) Moreover, in some examples various stimulation methods may be applied to treat obstructive sleep apnea, which include but are not limited to: Ni et al. U.S. 2019/0009093, published on Jan. 10, 2019, titled METHOD AND SYSTEM FOR SELECTING A STIMULATION PROTOCOL BASED ON SENSED RESPIRATORY EFFORT (previously published as WO 2013/023218, SYSTEM FOR SELECTING A STIMULATION PROTOCOL BASED ON SENSED RESPIRATORY EFFORT); Christopherson et al. U.S. Pat. No. 8,938,299, SYSTEM FOR TREATING SLEEP DISORDERED BREATHING, issued Jan. 20, 2015; and Wagner et al., U.S. 2018/0117316, STIMULATION FOR TREATING SLEEP DISORDERED BREATHING, published on May 3, 2018, issued as U.S. Pat. No. 10,898,709 on Jan. 26, 2021, and previously published as WO 2016/149344, on Sep. 22, 2016), each of which is hereby incorporated by reference herein in its entirety.

(562) In some examples, at least the respective closed loop and open loop parameters **8802**, **8804** of the stimulation engine **8800** in FIG. **38**D may be implemented via, and/or comprise an example implementation of, the example method **8260** of FIG. **40**D, the example stimulation protocols of FIGS. **33**A-**37**D, and at least some of the example methods, protocols, etc. described in association

with at least FIGS. **38**A-**39**, and/or **40**B-**51**B. In some examples, one example implementation of the closed loop parameter **8802** is later described as example arrangement **8100** in association with at least FIG. **39**.

(563) In some examples, the stimulation engine **8000** may comprise an auto-titration parameter **8805** by which stimulation settings may be automatically adjusted based on some sensed information and other criteria, such patient parameters, clinician parameters, and/or other parameters. One example implementation of the auto-titration parameter **8805** is later described as example arrangement **8100** in association with at least FIG. **39**, and at least some of the substantially the same features and attributes of the example methods, protocols, etc. described in association with at least FIGS. **40**B-**51**B.

(564) In some examples, per patient control parameter **8810** of the stimulation engine **8800** and a patient remote control (e.g. **572** in FIG. **3**B) in communication with an implantable medical device (e.g. IMD **570** in FIG. **3**B, such as IPG **533** (e.g. FIG. **3**A) or microstimulator **6575**), a patient may disable or enable stimulation of either the hypoglossal nerve (e.g. **505**R or **505**L) and/or the ansa cervicalis-related nerve **316** (e.g. **515**R, **515**L). Similarly, per patient control parameter **8810**, the patient may control stimulation strength (e.g. amplitude, other) via inputs on the patient remote control for each nerve independently of the other respective nerves. Stated differently, the patient remote control **572** (FIG. **3**B), as supported by patient control parameter **8810** of stimulation engine **8800**, enables a patient to have full independent control over stimulation of different nerves, such as the hypoglossal nerve (left and/or right) and the ansa cervicalis-related nerve (left and/or right). In some examples, a patient app (**10630** in FIG. **54**E) may cooperate with the remote control (**572** in FIG. **3**B; **10530** in FIG. **54**C, **10640** in FIG. **54**E) to implement the patient's control over stimulation.

(565) In some examples, the patient remote control **572** (FIG. **3**B), per patient control parameter **8810** of stimulation engine **8800** (FIG. **38**D) may enable a patient to have some limited control (e.g. amplitude, other) over stimulation of a first nerve but not a second nerve, except to turn off stimulation for both nerves. In some such examples, a clinician may adjust which nerves the patient has control over (and the degree of control for each nerve), such as via clinician programmer, clinician portal, etc. as described in FIGS. **54**A-**54**E.

(566) For instance, in some examples, the patient control parameter **8810** may be used to permit the patient to control stimulation parameters (e.g. amplitude) for the hypoglossal nerve but not for the ansa cervicalis-related nerve. In this arrangement, the patient control parameter **8810** is used to permit some patient control (within manufacturer or clinician limits) over a nerve for which the patient may be more sensitive to different stimulation strengths in relation to patient comfort. Conversely, the patient control parameter **8810** then does not offer such patient adjustments to stimulation strength of a second nerve (e.g. ansa cervicalis-related nerve) because the patient generally experiences more overall comfort with such stimulation such that patient adjustability of stimulation strength for that second nerve may be unnecessary in most instances.

(567) In some examples, in association with at least titration parameter **8808**, stimulation may be applied by a breath-by-breath parameter **8812**, by which stimulation is delivered on an alternating basis to a first nerve (e.g. one of hypoglossal nerve and ansa cervicalis-related nerve) and then a second nerve (e.g. the other one of the hypoglossal nerve and the ansa cervicalis-related nerve), the first nerve, the second nerve, and so on. By doing so, one can evaluate the effect of each instance of stimulation and adjust stimulation strength, targets along a particular nerve, whether to favor stimulation of one nerve more than another etc. At least some example implementations of switching stimulation between two different respective nerves on a breath-by-breath basis (via parameter **8812**) are provided in association with at least FIGS. **33**A-**36**A, including alternating patterns, every third breath patterns, etc.

(568) In some examples, the stimulation engine **8800** comprises a ramp parameter **8806** by which increases in stimulation strength (e.g. amplitude, other) during titration of stimulation settings

and/or as inputs of patient remote control adjustments of stimulation therapy (within limits set by clinician and/or manufacturer) are implemented as a ramped increase (versus a step change increase) and/or ramped decrease.

(569) In some examples of implementing ramp parameter **8806**, such as when stimulation is applied to both an ansa cervicalis-related nerve and a hypoglossal nerve simultaneously, increases in stimulation strength (e.g. amplitude, other) may be implemented simultaneously by a ramped increase in a first stimulation signal for the ansa cervicalis-related nerve (e.g. left and/or right) and a ramped increase in a second stimulation signal for the hypoglossal nerve (e.g. left and/or right). However, because of the different nerves, potentially different stimulation thresholds, etc. the ramped increase for one nerve (e.g. HGN or ACN) may not necessarily comprise the same slope, etc. as the ramped increase for the other nerve.

(570) In some example implementations of ramp parameter **8806**, whether for titration and/or patient control inputs, increases in stimulation strength may be implemented as ramped increases among different nerves on an alternating basis, such as a partial ramped increase of a first nerve (e.g. ansa cervicalis-related), followed by a partial ramped increase of a second nerve (e.g. hypoglossal nerve), followed by a further partial ramped increase of the first nerve, and so on. In some such examples, these gradual increases may be implemented as incremental changes in strength (e.g. amplitude, other) or conversely, if stimulation strength is to be decreased, then such changes may be made in decrements. Via these arrangements, stimulation strength can be increased gradually among all the different nerves to be stimulated, thereby providing time to observe the effects of the respective ramped increase for the different nerves, which in turn may help prevent too quickly increasing overall stimulation strength for multiple nerves.

(571) In some examples, the ramp parameter **8806** may be implemented in a complementary manner with method **8220** in FIG. **40**C.

(572) In some examples, per titration parameter **8808**, stimulation settings (e.g. amplitude, pulse width, duty cycle, frequency, etc.) may be determined, adjusted, etc. for each nerve to be stimulated and/or in a comprehensive manner for the multiple nerves. In some examples, per the titration parameter **8808**, the stimulation settings of the ansa cervicalis-related nerve (e.g. left and/or right) may be titrated to reach a functional threshold (parameter **8822** in FIG. **38**D) corresponding to a minimum level at which stimulation causes contraction (versus subcontraction) of the muscle(s) innervated by the particular targeted portion of the ansa cervicalis-related nerve. In particular, in some examples, delivering stimulation to an upper airway patency related nerve is to cause contraction of upper airway patency-related muscles. In some such examples, the contraction comprises a suprathreshold stimulation, which is in contrast to a subthreshold stimulation (e.g. mere tone) of such muscles. In one aspect, a suprathreshold intensity level corresponds to a stimulation energy greater than the nerve excitation threshold, such that the suprathreshold stimulation may provide for upper-airway clearance (i.e. patency) and obstructive sleep apnea therapy efficacy.

(573) Meanwhile, the stimulation settings of the hypoglossal nerve may be titrated to a level greater than a functional contraction threshold (for the hypoglossal nerve) to an optimal therapeutic level (parameter **8824** in FIG. **38**D) and/or to a comfort level (parameter **8826** in FIG. **38**D). In some such examples, this arrangement may enhance selection of appropriate stimulation settings at least because the tongue (innervated by the hypoglossal nerve) may exhibit more sensitivity regarding selection and adjustment of a comfort level than muscles associated with the ansa cervicalis-related nerve. In some such examples, at least some of the foregoing aspects of the titration parameter **8808** may be implemented in a complementary manner with the ramp parameter **8806** of stimulation engine **8800** of FIG. **38**D.

(574) In some examples, per the titration parameter **8808** of stimulation engine **8800** in FIG. **38**D, different stimulation settings may be established relative to a comfort limit (per parameter **8826** in FIG. **38**D) and/or a sleep disturbance limit (per parameter **8828** in FIG. **38**D), which may be in

addition to the previously described threshold limit (per parameter **8822**) and/or therapeutic parameter **8824**.
(575) In some examples, titrating for appropriate settings per the titration parameter may comprise determining at least the comfort limit (**8826** in FIG. **38**D) and/or sleep disturbance limit (**8828** in FIG. **38**D) by starting the titration process below a functional threshold (**8822** in FIG. **38**D) or at the functional threshold (**8822** in FIG. **38**D). Titrating toward and up to the respective limits (e.g. comfort, sleep disturbance, etc.) may comprise incrementing stimulation strength in steps or a ramped manner as described above with respect to at least the previously described ramp parameter **8806** in FIG. **38**D.
(576) In some examples, the threshold parameter **8822** in the stimulation engine **8800** of FIG. **38**D and impedance sensor **8752** in sensor tools array **8570** in FIG. **38**C may be used to implement an automatic threshold detection function by which a functional threshold (e.g. an amplitude at which muscle contraction occurs) for electrical stimulation of an upper airway patency-related nerve may be automatically determined. Such determination may replace and/or supplement more cumbersome clinician-intensive titration techniques, which may include a protocol such as a clinician programming an IPG with a stimulation setting (e.g. amplitude), the clinician applying a test stimulation, observing a patient response, the clinician further increasing the stimulating setting and the clinician programming the IPG with the updated settings, testing stimulation, and so on.
(577) In some examples, in general terms and per threshold parameter **8822**, the automatic threshold detection function may automatically determine a functional threshold based on sensed impedance information (e.g. per sensor **8752** in FIG. **38**C) in response to test stimulations automatically applied at different stimulation strengths (e.g. amplitude expressed as voltage). In one aspect, sensing impedance in relation to the upper airway may be implemented via a plurality of electrodes spaced apart from each other and at least some of which are in proximity to the upper airway, including upper airway muscles and related tissue including (but not limited to) the genioglossus muscle (e.g. tongue). The spaced apart electrodes may comprise sensing electrodes (i.e. dedicated to sensing) and/or stimulation electrodes, which also may be used for sensing in some instances. The electrodes may be supported on a non-electrically conductive carrier to form a stimulation element, such as a cuff body to form a cuff electrode, such as a paddle to form a paddle electrode, such as an axial lead body to form an axial electrode lead/array, and the like. In some examples, at least one or more of the spaced apart electrodes may be present on a housing (e.g. case) of an IPG (**533** in FIG. **2**) or of a microstimulator. Several examples of each of these electrode arrangements of stimulation elements are described and illustrated throughout many examples of the present disclosure.
(578) In another aspect, upon implantation of such example stimulation elements (and supporting IPG or microstimulator), then multiple spaced apart electrodes become positioned relative to pertinent tissues which move in response to electrical stimulation of an upper airway patency-related nerve (e.g. the hypoglossal nerve, ansa cervicalis-related nerve, and the like), which may then be sensed as a changed impedance. Among the implanted multiple spaced apart electrodes, at least some of the electrodes may be used for applying stimulation while at least some of the electrodes may be used for sensing the changes in impedance. In some examples, some of the electrodes may be used for both stimulation and sensing.
(579) In some examples, some of the implanted multiple spaced apart electrodes may comprise an array of electrodes on a first implanted stimulation element and some of the multiple spaced apart electrodes may comprise an array of electrodes on a second implanted stimulation element. In some examples, the first and second stimulation elements may be implanted in spaced apart locations on the same side of the body or in some examples may be implanted on opposite sides (e.g. left and right) of the body to assume a spaced apart relationship. Moreover, in some examples even the electrodes of an array of electrodes of a stimulation element (such as an axial lead array of electrodes, array of electrodes on a cuff electrode or on a paddle electrode, etc.) are spaced apart

from each other and may be used to sense a change in impedance. In each instance, implanting the stimulation elements in stimulating relation to an upper airway patency-related nerve necessarily places the electrodes of such implanted stimulation elements in sufficient proximity to potentially responsive muscles/tissues (e.g. those muscles/tissues which will move in response to electrical stimulation of the target upper airway patency-related nerve) such that various combinations of the spaced apart electrodes of the implanted stimulation elements may effectively form an array of impedance sensing electrodes, i.e. an effective impedance sensing array.

(580) In some examples, sensing a change in impedance (via the effective array of impedance sensing electrodes) may be representative of opening/stiffening of the upper airway generally in response to ACN stimulation while in some examples, sensing a change in impedance (via the effective array) may be representative of protrusion of the tongue in response to hypoglossal nerve stimulation. In some examples, sensing a change in impedance (via the effective array) may be representative of both a general change in the opening of the upper airway and movement of the tongue relative to the opening in the upper airway, which may both occur in response to a combination of ACN stimulation and HGN stimulation.

(581) In some examples, a change in the sensed impedance may result from contraction of pertinent muscles and/or result from directional movement of the pertinent muscles, such as protrusion of the tongue which causes movement of the electrodes used for sensing.

(582) With this framework in mind and per the impedance sensor **8752** (FIG. **38**C) and the threshold parameter **8822** (FIG. **38**D), an IPG or microstimulator may automatically apply test stimulation signals to a target nerve and then automatically sense an impedance via the effective array of impedance sensing electrodes. If no change in impedance is sensed upon application of the test stimulation signal, the functional threshold has not been met. Accordingly, the IPG (or microstimulator) automatically makes an incremental increase in a strength setting (e.g. amplitude identified as a voltage setting) and again delivers electrical stimulation to the target nerve, and further senses the impedance via the effective array. This process is automatically repeated iteratively until one of the automatic, incremental increases in the stimulation strength setting results in a changed impedance (which is automatically sensed) of a magnitude indicative of muscular contraction associated with tongue protrusion and/or opening/stiffening of the upper airway tissues. At this point, it may be concluded that a functional threshold for electrical stimulation of an upper airway patency-related nerve has been established.

(583) In some examples, the stimulation and impedance sensing may be occur simultaneously or on an interleaved basis. In some examples, stimulation may be applied on a first side of the body and impedance sensing may be performed on an opposite second side of the body. In some examples, both the stimulation and the impedance sensing may be performed on a single/same side of the body.

(584) In some examples, per at least the threshold parameter **8822**, the automatic threshold detection function may comprise monitoring the functional threshold over time as a diagnostic on system performance.

(585) It will be understood that, in some examples and per at least parameter **8822**, the automatic threshold detection function may be applied via a plurality of different stimulation electrode configurations which may be used to apply the test stimulation signal, with at least some of the different stimulation electrode configurations corresponding to different target nerve locations, such as but not limited to a left HGN, right HGN, left ACN, right ACN, etc.

(586) In some examples, per threshold parameter **8822**, in addition to or instead of sensing impedance, the automatic threshold detection function may be implemented via a signal from an accelerometer, such as but not limited to, a signal from an implanted accelerometer where the accelerometer is used to detect the change resulting from muscle contraction or movement of the tongue or other physiologic effects exhibited from the functional threshold being met. In some examples, the implanted accelerometer may be incorporated within a microstimulator implanted

within the neck region or as the on-board sensor **560** of the IPG in FIG. **3**C.

(587) In some examples, the stimulation engine **8800** comprises a relationship parameter **8813** (e.g. nerve stimulation relationship) which may facilitate a relationship by which multiple different nerves are stimulated relative to each other. In some examples, multiple different nerves may comprise two different types of nerves, such as the hypoglossal nerve, the ansa cervicalis-related nerve, the phrenic nerve, etc. In some examples, multiple different nerves also can comprise a left nerve (e.g. left HGN, left ACN, right phrenic, etc.) and a right nerve (e.g. right HGN, right ACN, right phrenic, etc.).

(588) In some examples, per the relationship parameter **8813**, the stimulation engine **8800** may track and/or control a sequence of stimulation, i.e. within a time frame or cycle (e.g. stimulation cycle), which nerve (e.g. hypoglossal, ansa cervicalis-related, phrenic, other) is stimulated first, which stimulated second, etc. Similarly, the relationship parameter **8813** may track and/or control that multiple different nerves be stimulated simultaneously, alternately, or in a staggered manner. In some examples, the relationship parameter **8813** may track and/or control which nerve(s) of an array of nerves are being stimulated, such as whether stimulation is to be applied solely to one type of nerve, applied to at least two types of nerves, applied to the full array of nerves, applied to both left and right nerves of a type of nerve, etc. Of course, specifying which nerve is to be stimulated (or not stimulated) may further depend on a timing of the stimulation of one nerve relative to another.

(589) In some example implementations of the relationship parameter **8813**, a stimulation protocol may be implemented in which, for a given stimulation cycle, an ansa cervicalis-related nerve (ACN) stimulation is started first or prior to starting stimulation of the hypoglossal nerve (HGN) and the ACN stimulation continues as HGN stimulation commences such that stimulation of both the hypoglossal and ansa cervicalis-related nerves continues (within the stimulation cycle) simultaneously once stimulation has started for both. For instance, as mentioned elsewhere herein, stimulating the ansa cervicalis-related nerve tends to stiffen the upper airway and increase the size of the opening of the upper airway by action, at least, of the sternothyroid muscles and/or sternohyoid muscles pulling down on the thyroid/larynx. Among other aspects, in the absence of such stimulation of the ansa cervicalis-related nerve, some OSA patients may exhibit a collapse of the lateral walls of the upper airway, which may contribute to antero-posterior collapse of the upper airway. Meanwhile, stimulation of the hypoglossal nerve may cause protrusion of the tongue to move the tongue out of the upper airway opening, but such protrusion does not otherwise generally contribute to stiffening/opening of the upper airway in the manner noted above in relation to stimulation of the ansa cervicalis-related nerve. Accordingly, in some example implementations of the present disclosure, such as per the relationship parameter **8813** of the stimulation engine **8800** (FIG. **38**D), stimulating the ansa cervicalis-related nerve prior to the hypoglossal nerve (within a stimulation cycle of a stimulation pattern of repeating stimulation cycles) may increase and/or generally maintain a size of the opening of the upper airway such that a desired patency can be achieved with less stimulation (e.g. degree of stimulation, duration of stimulation, etc.) of the hypoglossal nerve because the tongue need not be moved as much to achieve the desired degree of patency.

(590) In some examples, this particular sequence of stimulating the ansa cervicalis-related nerve prior to the hypoglossal nerve may be implemented via the relationship parameter **8813** in cooperation with (at least) the ramp parameter **8806** of the stimulation engine **8800** by which stimulation of the ansa cervicalis-related nerve may be ramped up to about 60 to about 70 percent of a therapeutic stimulation strength for that nerve prior to stimulating the hypoglossal nerve. This arrangement may significantly establish opening/stiffening of the upper airway prior to causing a portion of the tongue to move out of the opening of the upper airway via tongue protrusion form stimulation of the hypoglossal nerve.

(591) In some such examples, as the stimulation period of a stimulation cycle is to be terminated,

the stimulation pattern may comprise terminating stimulation of the hypoglossal nerve prior to terminating (such as but not limited to a ramped decrease) stimulation of the ansa cervicalis-related nerve. This arrangement may help to prolong patency because the stimulation of the ansa cervicalis-related nerve produces a more significant overall effect on patency than merely moving the tongue out of the way via hypoglossal nerve stimulation.

(592) At least some example implementations of the nerve relationship parameter **8813** are described in association with at least some of the examples of the present disclosure such as, but not limited to, the examples associated with at least FIGS. **1-3**C, **16-20**, **32**A-**32**C, **33-51**B, etc. In some such example implementations, the relationship parameter **8813** may be implemented in cooperation with the any one or more of the parameters of the sensed data engine **8700**, any one or more of the types, modalities, etc. of sensor tools array **8750**, and/or any one or more of the parameters, engines, etc. of the stimulation engine **8800**. In just one example, a particular expression of the relationship parameter **8813** (e.g. which nerves are stimulated, their sequence or simultaneous, etc.) may depend on sensed data, such as a sensed type or degree of collapse (e.g. **8710** in FIG. **38**B), may depend on sensed body position (e.g. **8722** in FIG. **38**B), may depend on sensed respiration (e.g. **8724** in FIG. **38**B), may depend on sensed disease burden (e.g. **8726** in FIG. **38**B), etc.

(593) FIG. **39** is a flow diagram schematically representing an example arrangement **8100** including an example device and/or example method to automatically select between stimulation of only the hypoglossal nerve (at **8110**) or both the hypoglossal nerve and the ansa cervicalis-related nerve (at **8115**). In some examples, the example arrangement **8100** may comprise at least some of substantially the same features and attributes as the example stimulation devices and/or methods in the previously described examples of the present disclosure, including stimulation protocols, stimulation arrangements, etc. In some examples, the sensing in example arrangement **8100** may comprise at least some of the same features and attributes as the later described sensing/control examples.

(594) As shown at **8110** in FIG. **39**, stimulation may be applied to the hypoglossal nerve (e.g. left and/or right) or stimulation may be applied to both the hypoglossal nerve (HGN) and the ansa cervicalis-related nerve (ACN) (e.g. left and/or right), as shown at **8115**. Upon such stimulation of these tissues to increase and/or maintain upper airway patency, a resulting effect **8112**, **8117** on the patient's physiology and breathing behavior may be sensed at **8135**. This sensed information is fed (at **8137**) to a control portion or other element (at **8140**) to adjust the therapy settings (at **8140**), which may be implemented automatically (at **8142**) or manually via a patient resource (**8143**) and/or clinician resource (**8145**). These respective resources **8143**, **8145** may be dedicated programmers or non-dedicated programmers (e.g. smart phone, web portal, etc.). When implemented automatically, the adjustment may occur within an implantable pulse generator or other resource by which the stimulation settings and signal are implemented.

(595) Upon any adjustment (or lack thereof) to the settings, and as further shown in FIG. **39**, an output **1847** is fed to a determination or query (at **1848**) whether a sensed severity index (e.g. an Apnea-Hypopnea Index AHI) is greater than a selectable quantity N (e.g. threshold). If the answer is YES, then via path **8149**A, the example arrangement **8100** implements stimulation of both the hypoglossal nerve (HGN) and the ansa cervicalis-related nerve (ACN) (at **8115**) in order to provide more aggressive therapy for treating sleep disordered breathing. However, if the answer to the query at **1848** is NO, then via path **8149**B, the example arrangement **8100** implements stimulation of solely the hypoglossal nerve (at **8110**) to maintain, increase, or decrease the therapy within a range which can be met by the hypoglossal nerve (HGN) without stimulation of the ansa cervicalis-related nerve (ACN).

(596) Via this example arrangement **8100**, efficacious therapy can be achieved while balancing stimulation of multiple different nerves.

(597) In some examples, multiple stimulation locations of the ansa cervicalis-related nerve may be

included in such determinations, at least with regard to their effectiveness in promoting therapy.

(598) While not shown in FIG. **39**, it will be understood that in some examples, the element **8110** may comprise solely stimulation of the ansa cervicalis-related nerve instead of solely stimulating the hypoglossal nerve such that activation of block **8115** provides supplemental stimulation via the hypoglossal nerve (instead of via the ansa cervicalis-related nerve) so that both the ansa cervicalis-related nerve and the hypoglossal nerve would be stimulated in block **8115** in this example.

(599) The example arrangement in FIG. **39** provides just one example within the present disclosure of using multiple stimulation elements, which are already implanted within the patient's body and positioned among multiple nerve targets (e.g. multiple targets on the ansa cervicalis-related nerve, the hypoglossal nerve, other nerves) in a method of therapy in which one or more such stimulation elements are selectively included (e.g. added) in the stimulation therapy or one or more such stimulation elements are selectively excluded (e.g. removed) from the stimulation therapy. The selective inclusion or selective exclusion of the respective nerve targets (via a corresponding already implanted stimulation element) may be based on one parameter or a plurality of parameters. In some examples, the parameter(s) may be selectable for their inclusion or exclusion as affecting the stimulation therapy and/or a value, criteria, threshold associated with those parameters may be selectable as well. In some examples, these features can be implemented with at least some externally located stimulation elements.

(600) With these general principles of at least some examples of the present disclosure in mind, FIG. **40**A schematically represents an example implementation of the example arrangement **8100** (FIG. **39**) including a method **8180** (or aspect of an example device) of adding or subtracting a nerve target from among multiple nerve targets based on a body position/posture and/or other parameters. For example, the method may comprise sensing (e.g. at **8135** in FIG. **39**) the body position or posture, and if it were sensed that a patient moved to a supine position, then an additional nerve target may be included in the stimulation therapy to enhance increasing or maintaining upper airway patency. Similarly, if the patient moved out of a supine position into a different body position (lateral decubitis), then at least one nerve target may be excluded (e.g. temporarily, selectively, etc.) from the stimulation therapy if/when such stimulation of the extra nerve target is no longer prudent or helpful. It will be understood that, in some examples, this example arrangement is applicable to changes in other body position/postures and/or applicable to changes in parameters other than body position/posture.

(601) Per the foregoing description regarding the example arrangement in FIG. **39**, in some examples this method (or aspect of an example device) at **8180** may comprise adding stimulation (via an already implanted stimulation element) of the ansa cervicalis-related nerve where stimulation of the hypoglossal nerve was already being implemented. It will be understood that a similar method may comprise removing stimulation (via an already implanted stimulation element) of the ansa cervicalis-related nerve, where stimulation of the hypoglossal nerve was already included or implemented as part of the stimulation therapy.

(602) In some examples, this method (at **8180** in FIG. **40**A) may comprise adding stimulation (via an already implanted stimulation element) of the hypoglossal nerve, where stimulation of the ansa cervicalis-related nerve was implemented as part of the therapy. It will be understood that a similar method may comprise removing stimulation (via an already implanted stimulation element) of the hypoglossal nerve, where stimulation of the ansa cervicalis-related nerve was already included or implemented as part of the stimulation therapy.

(603) In some examples, this method (at **8180** in FIG. **40**A) may comprise adding stimulation (via an already implanted stimulation element) of a different, second nerve target of the ansa cervicalis-related nerve, where stimulation of a first nerve target of the ansa cervicalis-related nerve was already included or implemented as part of the stimulation therapy. It will be understood that a similar method may comprise removing stimulation (via an already implanted stimulation element) of a different, second nerve target of the ansa cervicalis-related nerve, where stimulation of a first

nerve target of the ansa cervicalis-related nerve was already included or implemented as part of the stimulation therapy.

(604) In some examples, this method (at **8180** in FIG. **40**A) may comprise adding stimulation (via an already implanted stimulation element) of a different, second nerve target (other than the ansa cervicalis-related nerve or hypoglossal nerve), where stimulation of the ansa cervicalis-related nerve and/or the hypoglossal nerve was already included or implemented as part of the stimulation therapy. It will be understood that a similar method may comprise removing stimulation (via an already implanted stimulation element) of the different, second nerve target (other than the ansa cervicalis-related nerve or hypoglossal nerve), where stimulation of the ansa cervicalis-related nerve and/or hypoglossal nerve was already included or implemented as part of the stimulation therapy.

(605) In some examples, this method (at **8180** in FIG. **40**A) may comprise adding stimulation (via an already implanted stimulation element) of a different, second nerve target (other than the ansa cervicalis-related nerve and/or hypoglossal nerve), where stimulation of the ansa cervicalis-related nerve and/or the hypoglossal nerve was already included or implemented as part of the stimulation therapy. It will be understood that a similar method may comprise removing stimulation (via an already implanted stimulation element) of the different, second nerve target (other than the ansa cervicalis-related nerve and/or hypoglossal nerve), where stimulation of the ansa cervicalis-related nerve and/or hypoglossal nerve was already included or implemented as part of the stimulation therapy.

(606) In some examples, this method (at **8180** in FIG. **40**A) may comprise adding stimulation (via an already implanted stimulation element) of the ansa cervicalis-related nerve and/or hypoglossal nerve, where stimulation of another nerve target (e.g. other than the ansa cervicalis-related nerve and/or hypoglossal nerve) was already included or implemented as part of the stimulation therapy. It will be understood that a similar method may comprise removing stimulation (via an already implanted stimulation element) of the ansa cervicalis-related nerve or hypoglossal nerve), where stimulation of another nerve target (e.g. other than the ansa cervicalis-related nerve and/or hypoglossal nerve) was already included or implemented as part of the stimulation therapy.

(607) In some examples, the inclusion or exclusion of a nerve target to stimulation therapy may be based on parameters (or physiologic conditions) other than body position or posture, with such parameters generally affecting upper airway patency and/or sleep disordered breathing.

(608) It will further understood that these principles associated with examples of the present disclosure, including but not limited to the example arrangements in FIGS. **39**, **40**A, may be applicable to other nerve targets for other physiologic conditions, which may in the head-and-neck region or may be in areas of the body other than the head-and-neck region. For example, as noted elsewhere herein, these principles may be applied to stimulation of multiple nerve targets in the pelvic region including, but not limited to, the pudenal nerve for treating pelvic disorders such as (but not limited to) urinary and/or fecal incontinence issues, which may involve (but is not limited to) the external urinary sphincter and/or external anal sphincter.

(609) Moreover, while the foregoing principles of the example arrangements associated with at least FIGS. **39-40**A relate to at least some examples comprising multiple, available nerve targets associated with already implanted stimulation elements, it will be understood that these principles may be extended to the addition of a not currently available nerve target by implanting a stimulation element in a second, separate implant procedure to make a desired nerve target available for stimulation therapy in order to implement the example arrangements associated with at least FIGS. **39-40**A. At least the example arrangements associated with at least FIGS. **5**A-**5**B provide some examples by which a nerve target may be made available (for selective inclusion or selective exclusion per the example arrangements in association with at least FIGS. **39-40**A) via implanting a stimulation element via a second, separate implant procedure which occurs some period of time after an initial/original implant procedure for one or more original nerve targets.

(610) In some examples, the preceding example methods and/or example devices associated with at least FIGS. **38-40**A and/or the following example methods and/or example devices described in association with at least FIGS. **40**B-**60** may be implemented via at least some of the features and attributes (such as, but not limited to, the stimulation arrangements, stimulation elements, leads, stimulation protocols, etc.) of the example arrangements described in association with at least FIGS. **1-37**D.
(611) FIG. **40**B is a block diagram schematically representing an example arrangement at **8200** including an example method and/or example device for sensing and/or managing fatigue. In some examples, the example arrangement **8200** in association with FIG. **40**B may be implemented in coordination with, and/or as part of the example arrangements previously described in association with at least FIGS. **39-40**A. As shown at **8200** in FIG. **40**B, in some examples the method comprises identifying fatigue of a target nerve and/or its innervated muscle and adjusting a stimulation parameter to assess potential fatigue and/or to decrease identified fatigue.
(612) In some examples, identifying the fatigue may be performed as part of a closed loop sensing system to determine when stimulation is becoming less effective. In some examples, an accelerometer or EMG may be used to directly measure motion, which may be indicative of reduced therapy effectiveness. In some examples, the accelerometer may already be implanted in the patient. In some examples, a respiratory sensor may be used to detect increasing instances of obstruction, which may be indicative of reduced therapy effectiveness. For patients for which the stimulation was previously effective in increasing or maintaining upper airway patency, sensing the change in motion or increasing obstruction may be indicative of fatigue (of nerves and/or muscles) from stimulation therapy.
(613) In some examples, an intensity (e.g. a duty cycle or other parameter) of stimulation therapy may be decreased or stimulation therapy may be temporarily paused, and then it may be observed if the stimulation becomes more effective, which then may be indicative of fatigue (of the stimulated nerves and/or associated muscles).
(614) In some examples, adjusting the stimulation parameter may comprise switching nerve targets for implementing the stimulation therapy and/or adding (or removing) a nerve target from among multiple nerve targets for implementing the stimulation therapy. In some examples, such switching, adding, or removing may be implemented via at least some aspects of the example arrangements described in association with at least FIGS. **39-40**A.
(615) FIG. **40**C is a block diagram schematically representing an example arrangement at **8220** including an example method and/or example device for increasing amplitudes of stimulation therapy to therapeutically effectively levels when multiple nerve targets are included as part of the stimulation therapy. In some examples, the example arrangement **8220** in association with FIG. **40**C may be implemented in coordination with, and/or as part of the example arrangements previously described in association with at least FIGS. **39-40**A and/or other example arrangements of the present disclosure.
(616) As shown at **8220** in FIG. **40**C, in some examples the method comprises determining and implementing a therapeutic stimulation intensity via increasing an amplitude of stimulation among multiple nerve targets until a threshold(s) is met. In some examples, this method (at **8220**) may comprise a first determination/implementation protocol including increasing a stimulation amplitude applied to a first nerve target until a first threshold is met, and then increasing a stimulation amplitude applied to a second nerve target until a second threshold is met. In some examples, the first protocol is repeated.
(617) In some examples, this method (at **8220**) may comprise a second determination/implementation protocol including increasing stimulation amplitude at first and second nerve targets by equal amounts until a first threshold is met, and then thereafter increasing stimulation amplitude solely at just one of the respective first and second nerve targets.
(618) In some examples, this method (at **8220**) may comprise a third determination/implementation

protocol including increasing stimulation amplitude solely at a first nerve target while maintaining the same amplitude at a second nerve target.

(619) In some examples, this method (at **8220**) may comprise a fourth determination/implementation protocol including increasing stimulation amplitude alternately at a first nerve target and a second nerve target.

(620) In some examples, this method (at **8220**) may comprise a fifth determination/implementation protocol includes combining at least some aspects of the respective first, second, third, and fourth determination/implementation protocols.

(621) It will be understood that the general principles associated with the example arrangement at **8220** in FIG. **40**C may be applied to more than two nerve targets.

(622) FIG. **40**D is a block diagram schematically representing an example arrangement at **8260** including an example method and/or example device for applying stimulation according to a respiratory phase parameter and/or without a respiratory phase parameter, with such stimulation providing therapy to increase or maintain upper airway patency to treat sleep disordered breathing. In some examples, this method may comprise applying stimulation based on detecting a fiducial of an expiratory phase of a respiratory cycle, while in some examples, applying stimulation is based on detecting a fiducial of an inspiratory phase of a respiratory cycle. Such fiducials may comprise an onset, offset, peak, etc. . . . . In some examples, the method may comprise applying stimulation based on detecting both of a fiducial of an inspiratory phase of a respiratory cycle and a fiducial of an expiratory phase of a respiratory cycle.

(623) In some examples, basing the application of stimulation with regard to a respiratory phase parameter may comprise triggering the application of stimulation based on such sensed respiratory phase parameter.

(624) In some examples, basing the application of stimulation with regard to a respiratory phase parameter may comprise causing a stimulation period of the stimulation signal to coincide with at least a portion of a respiratory phase, such as the inspiratory phase and/or expiratory phase. In some such examples, this type of stimulation may sometimes be referred to as the stimulation being synchronous with at least a portion of a respiratory phase. Moreover, in some of these examples, causing the stimulation period to coincide with at least a portion of a respiratory phase may comprise making the stimulation coincide solely with a single type of respiratory phase, e.g. the inspiratory phase. In some examples, such arrangements may comprise the stimulation coinciding with a brief pre-inspiratory phase of the respiratory cycle, which corresponds to a latter portion of an expiratory pause (e.g. respiratory pause).

(625) At least some examples of these arrangements are illustrated in association with at least some of the stimulation protocols in FIGS. **37**A, **37**B, and **37**D.

(626) In some examples, the method at **8260** in FIG. **40**D comprises applying stimulation without regard to any sensed respiratory phase parameter, and may be referred to as open loop-based stimulation in some instances. However, in some such open loop examples, while sensed respiratory phase information is not used to trigger stimulation and/or is not used to make stimulation synchronous with some portion of a respiratory phase, the method(s) do permit sensed respiratory information to be used generally regarding therapy. For examples, some example arrangements may comprise sensing respiratory information to determine whether the open loop stimulation is effective, e.g. is the patient experiencing fewer sleep disordered breathing events. If the stimulation is not as effective as desired, this sensed information may be used to determine how to adjust the stimulation therapy. Accordingly, a distinction may be drawn between using sensed respiratory information for triggering or synchronizing stimulation relative to each respiratory cycle versus using sensed respiratory information more generally to determine the effectiveness of the stimulation therapy and adjustments thereto.

(627) In some examples, the method at **8260** in FIG. **40** may comprise applying stimulation using a combination of both closed loop (e.g. sensing respiratory phase parameter) and open loop (e.g. not

using sensed respiratory phase parameter) arrangements.

(628) FIG. **41**A is a flow diagram schematically representing an example arrangement including an example method **8280** (and/or example device) for sensing, via at least one sensing element, at least one sleep disordered breathing-related parameter. In some examples, the at least one sleep disordered breathing-related parameter comprises at least one of a respiratory phase parameter, an apnea-hypopnea index, a patient comfort parameter, an arousal index, and an upper airway collapse pattern. In some examples, stimulation can be modulated based on at least one of the respiratory phase parameter, an apnea-hypopnea index, a patient comfort parameter, a patient sleeping position, and an upper airway collapse pattern. In some examples, the patient comfort parameter may comprise an arousal index.

(629) In some examples, the stimulation may be synchronized relative to the sensed respiratory phase parameter such as, but not limited to being synchronized relative to a sensed inspiratory phase of the patient's respiratory cycle. At least some example implementations of such synchronization are illustrated in association with the example stimulation patterns in association with at least FIGS. **32**A-**37**D and FIG. **40**D.

(630) In some examples, as shown at **8284** in FIG. **41**B an example method may comprise automatically titrating a stimulation parameter based on sensing at least one of the respiratory phase parameter, the AHI parameter, the patient comfort parameter, the patient sleeping position, and the upper airway collapse pattern. In some such examples, sensing the upper airway collapse pattern may comprises determining the upper airway collapse pattern via sensing a bioimpedance in relation to the neck region of the patient's body.

(631) In relation to at least method **8284** in FIG. **41**B, some example methods may comprise identifying a type of the upper airway collapse pattern via: (1) identifying at least one of a value of, a change in value of, and a location of the sensed bioimpedance along the upper airway; and (2) implementing the automatically titration of the stimulation parameter based on the identified value, identified change in value or identified location. In some examples, the type of upper airway collapse pattern may comprise at least one of anterior-posterior collapse, concentric collapse, lateral collapse, or composite collapse, which are illustrated schematically in FIGS. **53**A-**53**C, while FIG. **53**D also schematically illustrates collapse locations along the upper airway.

(632) In some examples, this sensed information about a type, degree, and/or location of collapse pattern may be used as feedback (sensed data) to determine (e.g. initiate, terminate, select, adjust) stimulating settings to stimulate a hypoglossal nerve (e.g. left and/or right) and/or an ansa cervicalis-related nerve (e.g. left and/or right), as further described in association with at least FIGS. **38**A-**38**D.

(633) It will be understood that sensing information about a collapse pattern may comprise identifying which muscles (and their associated nerves) of the upper airway are involved in a particular collapse pattern according to its type, location, degree, such that stimulation of such identified muscles and nerves (including which multiple potential nerve targets) may be used to prevent or minimize such collapse patterns.

(634) In further relation to at least the example method **8284** in FIG. **41**B, in some examples the sensing a collapse pattern (e.g. type, location, degree) may be implemented via an implantable, removably insertable array of spaced apart electrodes. In some such examples, the removably insertable array may be inserted into and through a nose or mouth of the patient, with one example implementation illustrated in FIG. **52**C. However, in some examples, the sensing of collapse patterns (e.g. their type, location, and degree) may be implemented via an externally mountable array of spaced apart electrodes conformable to an exterior portion of the neck overlying the region of upper airway collapse, with on example implementation illustrated in FIGS. **52**A-**52**B.

(635) As shown at **8286** in FIG. **41**B**2**, some example methods may comprise sensing respiratory information as respiratory phase information, and implementing the automatic titration of a stimulation parameter (per method **8284** in FIG. **41**B) based on the sensed respiratory phase

parameter. As shown at **8288** in FIG. **41**B**3**, in some examples the stimulation parameter may comprise at least one of: (1) at least one of a change in an amplitude, frequency, pulse width, duty cycle of stimulation; and (2) selection of at least one stimulation target among a plurality of stimulation targets, which includes the hypoglossal nerve and the ansa cervicalis nerve. As shown at **8290** in FIG. **41**B**4**, in some examples, the method comprises titrating, with the plurality of stimulation targets comprising a left and/or right hypoglossal nerve, a left and/or right ansa cervicalis-related nerve (including different target stimulation branches), and/or other nerve targets. In some examples, the other stimulation targets may further comprise a glossopharyngeal nerve, a superior laryngeal nerve, a superior cervical ganglion nerve, and a chemoreceptor (e.g. in close proximity to the carotid body). In some such examples, the combination of the glossopharyngeal nerve and the superior laryngeal nerve comprises a stimulation target. One example implementation of this example combination is described and illustrated in association with FIG. **60**.

(636) As shown at **8300** in FIG. **41**C, one example method comprises receiving a patient adjustment, as a single input, to a stimulation parameter, and implementing the patient adjustment by automatically adjusting a stimulation energy among a plurality of stimulation sites including at least one of a hypoglossal nerve site, an ansa cervicalis-related nerve site, and a second non-hypoglossal nerve site. In some such examples, the single input may sometimes be referred to as a single control element, such as a single button on a patient remote control app, whether embodied in a dedicated device or non-dedicated device (e.g. smart phone, tablet, watch, etc.)

(637) In some such examples, the method at **8300** may comprise implementing the automatically balancing of the stimulation among the different stimulation sites.

(638) In some such examples methods, the automatically balancing comprises: (1) initiating and maintaining the stimulation via stimulation of a first one of the respective stimulation sites; and (2) adding, at a later point in time, stimulation of at least one respective second stimulation sites of the plurality of stimulation sites.

(639) As shown at **8320** in FIG. **41**D, one example method comprises implementing the automatic balancing based on sensing at least one of a respiratory phase parameter, an AHI parameter, a patient comfort parameter, a patient sleeping position parameter, and an upper airway collapse pattern parameter.

(640) In some examples, the first one of the respective stimulation sites comprises a hypoglossal nerve and the second respective stimulation site comprises an ansa cervicalis-related nerve.

(641) In some examples, the automatic balancing may comprise a relative percentage among two stimulation sites, such as among the hypoglossal nerve and the ansa cervicalis-related nerve or among two different stimulation locations of the ansa cervicalis nerve. The relative percentage between the multiple stimulation sites may vary over time, and may be periodically adjusted or continually adjusted. In some examples, the relative percentage may be a 50/50 relative percentage, which may be implemented via alternating the stimulation between the two different sites or by applying the stimulation in a non-alternating manner such as applying each stimulation at 50 percent of full stimulation to achieve the 50/50 relative percentage.

(642) In some examples, during the automatic balancing, the first one of the respective stimulation site comprises an ansa cervicalis-related nerve and the second respective stimulation site comprises a hypoglossal nerve.

(643) In some examples, in performing the adjustment according to a patient comfort parameter, the patient comfort parameter may comprise an arousal index.

(644) As shown at **8330** in FIG. **42**A, in some examples a method may comprise: (1) receiving a patient adjustment to a stimulation parameter upon a change in the patient comfort parameter, the stimulation parameter comprising an overall stimulation energy; and (2) implementing the patient adjustment by redistributing the overall stimulation energy among an increased number of stimulation sites. In some such examples, the redistribution may be implemented while maintaining

the same overall energy as before the patient adjustment. In some such examples, the patient adjustment may comprise a single input parameter, such as the previously described "single knob" patient input on a patient remote control, patient app, and the like.

(645) With further reference to the method **8330** in FIG. **42**A, in some examples, the patient adjustment may comprise a request to decrease stimulation. In some such examples, the patient adjustment may implemented by decreasing some aspect(s) of stimulation while maintaining the overall stimulation energy as before the patient request to decrease the stimulation. Via this arrangement, the patient has some control over patient comfort and may experience increased patient comfort, while the example device may still provide efficacious stimulation therapy via a modified stimulation protocol/settings.

(646) As shown at **8350** in FIG. **42**B, in some examples a method may comprise automatically adjusting, upon sensing a change in the at least one sleep disordered breathing-related parameter, at least one of a stimulation parameter and a target stimulation location at the upper-airway-patency-related tissue, wherein the target stimulation location comprises at least one of a hypoglossal nerve and an ansa cervicalis-related nerve. In some examples, the at least one sleep disordered breathing-related parameter comprises an AHI parameter. In some examples, the stimulation location of a non-hypoglossal nerve target may comprise, in addition to or instead of, the ansa cervicalis-related nerve, at least one of: glossopharyngeal nerve; a superior laryngeal nerve; a superior cervical ganglion; and a chemoreceptor (e.g. in close proximity to the carotid body). In some such examples, this may comprise implementing the sensing of the increase in the AHI parameter.

(647) In some such examples, the method may comprise, upon determining that the sensed change in the AHI parameter is associated with a change in patient sleeping position, adjusting the stimulation parameter relating to the ansa cervicalis-related nerve.

(648) In some such examples, the method may comprise adjusting of the change in the stimulation parameter via increasing the stimulation parameter when the sensed change in the AHI parameter comprises a sensed increase in the AHI parameter. However, in some examples, a further adjustment may comprise decreasing a stimulation parameter regarding the hypoglossal nerve to increase or maintain patient comfort while adjusting or increasing stimulation parameters of other nerve stimulation targets to achieve the desired increase in stimulation intensity to respond to the sensed increase in the AHI parameter.

(649) In some examples, the increase in the stimulation parameter may comprise a change from a first value to a higher second value. In some examples, the first value may comprise zero, while in some examples, the first value comprises a non-zero value corresponding to tonic stimulation or contraction-type stimulation.

(650) Some example methods and/or example devices comprise sensing. It will be understood that such sensing may be performed even if no stimulation is occurring, even if no stimulation elements are implanted or present externally, etc. However, in some examples, such sensing may be performed in relation to some aspect affecting stimulation.

(651) In some examples, sensing of various physiologic parameters may be implemented via a first sensing element and a second sensing element. Accordingly, as shown at **8370** in FIG. **42**C, in some examples a method may comprise positioning the first sensing element on a first side of the patient's body and positioning the second sensing element on an opposite second side of the patient's body. However, as further illustrated in FIG. **42**C, in some examples a method may comprise positioning the first sensing element on a first side of the patient's body and positioning the second sensing element on the same first side of the patient's body spaced apart from the first sensing element.

(652) In some such examples, the sensing comprises sensing an impedance between the respective first and second sensing elements. Among other example implementations, such impedance sensing may be performed in association with at least the example arrangement **8000** of FIG. **38**A, the impedance sensor **8752** of FIG. **38**C, the collapse parameter **8710** of FIG. **38**B, etc.

(653) As shown at **8380** in FIG. **43**, in some examples a method comprises sensing, via a first stimulation element and a second stimulation element, a sleep disordered breathing-related parameter comprises sensing at least one of respiration and a sleep-disordered-breathing event. In some example implementations, this sensing method may comprise locating the first stimulation element at a first hypoglossal nerve on the first side of the patient's body and the second stimulation element at a second hypoglossal nerve on the second side of the patient's body. However, in some example implementations, such sensing methods may comprise locating the first stimulation element at a first ansa cervicalis-related nerve on the first side of the patient's body and the second stimulation element at a second ansa cervicalis-related nerve on the second side of the patient's body.
(654) In some examples, such sensing methods may comprise: (1) locating the first stimulation element, on the first side of the patient's body, at at least one of the first hypoglossal nerve and a first ansa cervicalis-related nerve; and (2) locating the second stimulation element, on the second side of the patient's body, at at least one of the second hypoglossal nerve and a second ansa cervicalis-related nerve.
(655) In some such examples, the method comprises locating the first stimulation element, on the first side of the patient's body, at the first hypoglossal nerve and locating the second stimulation element on the first side of the patient's body at the first ansa cervicalis-related nerve.
(656) In some such examples, the method comprises locating the second stimulation element, on the second side of the patient's body, at the second hypoglossal nerve and locating the second stimulation element on the second side of the patient's body at the second ansa cervicalis-related nerve.
(657) With further reference to at least the example method **8370** in FIG. **42**C, in some examples the first sensing element comprises a first stimulation element and the second sensing element comprises an electrode on a housing of a pulse generator. In some such examples, the method may comprise positioning the first stimulation element at at least one of: (1) at least one of a first hypoglossal nerve on a first side of the patient's body and a second hypoglossal nerve on an opposite second side of the patient's body; and (2) at least one of a first ansa cervicalis nerve on the first side of the patient's body and a second ansa cervicalis nerve on the second side of the patient's body.
(658) As shown at **8400** in FIG. **44**A, in some examples a method of sensing at least one sleep disordered breathing-related parameter comprises providing (e.g. via implanting within a patient's body) a first sensing element and a second sensing element, and measuring a bioimpedance between the first sensing element implanted relative to the at least one upper-airway-patency-related tissue and the second sensing element implanted relative to at least one of a phrenic nerve and a diaphragm. In some such examples, the first sensing element comprises a first stimulation element and the second sensing element comprises a second stimulation element.
(659) In relation to a method of sensing at least one sleep disordered breathing-related parameter (e.g. **8280** in FIG. **41**A), an upper airway patency-related tissue may comprise an upper-airway-patency-related muscle, and a method of sensing as shown at **8410** in FIG. **44**B may comprise sensing at least one of a motion of the upper-airway-patency-related muscle and an electromyograph (EMG) of the upper-airway-patency-related muscle. In some such examples, the method of sensing comprises performing, via at least one of the sensed motion and the sensed EMG, at least one of: (1) titration of an amplitude of a stimulation signal to be applied via the at least one stimulation element; and (2) determination of a sufficiency of capture of the upper airway patency-related tissue to be stimulated via the stimulation element.
(660) In relation to a method of sensing at least one sleep disordered breathing-related parameter (e.g. **8280** in FIG. **41**A), as shown at **8420** in FIG. **44**C, in some examples a method of sensing comprises determining at least one sleep disordered breathing-related parameter as a respiratory parameter, by sensing via the at least one sensing element at least one of: (1) a motion of an upper

airway patency related muscle of the at least one upper airway patency-related tissue; and (2) a sound of respiration associated with the at least one upper airway patency-related tissue.
(661) In some examples of stimulation, an example method comprises applying a first duty cycle of stimulation to the hypoglossal nerve to be different from a second duty cycle of stimulation for application to the ansa cervicalis-related nerve. In some examples, the first duty cycle is longer than the second duty cycle, while in some examples, the first duty cycle is shorter than the second duty cycle. At least some example implementations of different duty cycles are described in association with the stimulation protocols in at least FIGS. **33**A-**37**D.
(662) With further respect to some stimulation parameters, in some examples, a method of stimulation comprises applying stimulation to the hypoglossal nerve as a phasic stimulation; and applying stimulation to the ansa cervicalis-related nerve as a tonic stimulation. In some such examples, the tonic stimulation comprises a stimulation cycle comprising a stimulation period greater than a non-stimulation period.
(663) As shown at **8430** in FIG. **45**B, in some examples a method of stimulation comprises simultaneously stimulating a hypoglossal nerve and a second upper-airway-patency-related tissue, which may comprise at least one of an ansa cervicalis-related nerve and an infrahyoid strap muscle. In some such examples, the stimulation may be implemented via interleaving a first stimulation signal to stimulate the hypoglossal nerve and a second stimulation signal to stimulate the second upper-airway-patency-related tissue. In some such examples, the method comprises implementing both of the respective first and second stimulation signals via a single pulse generator.
(664) In some such examples, the method comprises implementing the interleaving via applying the stimulation via a first frequency for the hypoglossal nerve and a different second frequency for the second upper-airway-patency-related tissue.
(665) In some such examples, the method may comprise implementing the simultaneous stimulation via time-shifting or may comprise implementing the simultaneous stimulation synchronously.
(666) As shown at **8440** in FIG. **45**B, in some examples, a method comprises stimulating the hypoglossal nerve followed by stimulating the second upper-airway-patency-related tissue, which comprises at least one of the ansa cervicalis-related nerve and/or muscle innervated by the ansa cervicalis-related nerve. In some such examples, the method comprises maintaining the stimulation of the hypoglossal nerve during stimulation of the second upper-airway-patency-related tissue.
(667) In some examples, the method of stimulation may comprise applying a continuous stimulation at a low frequency. In some examples, the method of stimulation comprises applying a non-continuous stimulation including an on period (e.g. stimulation period) and an off period (e.g. non-stimulation period), in which the intensity of stimulation is sufficient to cause muscular contraction.
(668) As shown at **8450** in FIG. **46**A, in some examples a method of stimulation comprises selecting a stimulation target from among a hypoglossal nerve and the different muscle groups associated with the ansa cervicalis-related nerve based on determination of a first parameter. In some examples, the first parameter comprises at least one of a respiratory parameter including respiratory phase information, a patient comfort, a posture, an effectiveness of therapy, device usage, a sleep stage, an upper airway collapse pattern, and AHI. In some such examples, an effectiveness of stimulation therapy may change over the short term or the long term, and therefore selecting among different muscle groups may help maintain effectiveness despite such changes in effectiveness.
(669) As shown at **8460** in FIG. **46**B, in some examples a method of sensing in relation to adjusting stimulation may comprise implementing, via an accelerometer, the determination of at least one of a respiratory phase/pattern, a posture, a body position or activity, effectiveness of therapy as a sleep-disordered breathing (SDB) severity index (e.g. AHI), and a sleep stage. In some such examples, the method may comprise implementing determination of the sleep stage via sensing via

at least one of modalities of: accelerometer; bioimpedance; pressure; pneumotach; and thermistor.
(670) In further relation to method **8460**, in some examples the method comprises implementing the sensing of sleep stage via employing at least some of the modalities to sense at least one of: respiration; posture; body position; body activity; and cardiac information. In further relation to method **8460**, in some examples, a determination of patient comfort may be implemented via at least one of patient feedback and clinician feedback.
(671) In further relation to method **8460**, in some examples the method comprises implementing the selectivity of a stimulation target, from among the hypoglossal nerve and the different muscle groups associated with the ansa cervicalis-related nerve loop, according to the first parameter via clinician titration.
(672) In further relation to method **8450**, and as shown at **8470** in FIG. **47**, in some examples a method of stimulation comprises via application of a data model and according to the first parameter, implementing the selecting of at least one stimulation target from among the hypoglossal nerve and the different muscle groups associated with the ansa cervicalis-related nerve. In some such examples, the method comprises at a first time period prior to the implementing the selecting of at least one stimulation target, constructing the data model via known inputs corresponding to all of the stimulation targets relative to known outputs corresponding to the first parameter. In some examples, the constructed data model comprises a trained data model, which optionally comprises a trained machine learning model. In some examples, the trained machine learning model comprises at least one of an artificial neural network, deep learning model, and a support vector machine. In some examples, the first time period comprises a non-treatment period of the patient.
(673) As shown at **8480** in FIG. **48**, in some examples, a method comprises applying stimulation to each respective stimulation target to determine among the respective stimulation targets a relative degree of upper airway patency, wherein the respective stimulation targets comprise a hypoglossal nerve; an ansa cervicalis-related nerve; and an infrahyoid strap muscle. In some such examples, the stimulation signal may comprise an increasing ramp signal. In some such examples, the method comprises applying the stimulation simultaneously to at least two of the respective stimulation targets.
(674) In some such examples regarding method **8480**, the method may comprise selecting which of the respective stimulation targets to stimulate based on an upper airway collapse pattern. In some such examples, the method comprises classifying a type and a severity of the upper airway collapse pattern based on at least one of: snoring; a plurality of polysomnongraphy (PSG) parameters; a body position; a head-and-neck position; and a sleep stage.
(675) In further relation to at least the method **8480**, in some examples the method comprises implementing the selection of stimulation targets based on the type and the severity of the upper airway collapse pattern.
(676) In further relation to method **8480** and/or the above-described associated methods, one example method comprises at least one of: (1) selecting the stimulation target as a first portion of the ansa cervicalis-related nerve upon the upper airway collapse pattern being predominantly a lateral wall collapse pattern; (2) selecting the stimulation target as a hypoglossal nerve upon the upper airway collapse pattern being predominantly a genioglossal (tongue) collapse pattern; and (3) selecting the stimulation target of an ansa cervicalis-related nerve as a first stimulation target prior to other stimulation targets upon the upper airway collapse pattern being predominantly a palate collapse pattern. In some such examples, the method comprises titrating at least one stimulation parameter of the stimulation for at least one of: each different collapse pattern; and each sleeping position. In such example methods, at least some stimulation parameters comprise an amplitude, a pulse width, a frequency, a duty cycle, and/or a number, a type, and a location of stimulation elements.
(677) As shown at **8490** in FIG. **49**A, in some examples a method comprises (1) establishing (e.g.

implanting stimulation elements within a patient's body at) a plurality of muscle stimulation targets including a genioglossus muscle, a sternohyoid muscle, a sternothyroid muscle, and an omohyoid muscle; and (2) implementing the stimulation via distributing an energy of a stimulation signal among a selectable multiple number of the stimulation targets as the energy of the stimulation signal is increased. In some such examples, the method comprises implementing the stimulation of the muscle targets via at least one of: (1) directly stimulating the respective muscle stimulation target; and (2) stimulating a nerve by which the respective muscle stimulation targets are innervated.

(678) As shown at **8500** in FIG. **49**B, in some examples a method comprises, prior to completion of chronic implantation of a sleep disordered breathing therapy system, stimulating, via a first stimulation element, at least one test stimulation location indicative of a response of an upper-airway-patency-related tissue to the stimulation. In some such examples, the phrase “prior to completion” comprises at least one of pre-operatively and intra-operatively.

(679) In some such examples, the first stimulation element comprises a test stimulation element, while in some examples, the first stimulation element comprises a chronically implantable stimulation element. In some examples, the chronically implantable stimulation element may comprise a stimulation electrode on a lead or a microstimulator.

(680) In some such examples, the test stimulation location corresponds to a target stimulation location for implantation of a chronically implantable stimulation element.

(681) In some such examples, the method comprises implementing the stimulation, via the first stimulation element, with the at least one test stimulation location comprising at least one of a hypoglossal nerve and a non-hypoglossal nerve.

(682) As shown at **8510** in FIG. **50**, in some examples, a method of stimulation to identify a stimulation location comprises: (1) stimulating a hypoglossal nerve and determining a first response of upper airway patency to the stimulation; and (2) after the determination of the first response, stimulating a non-hypoglossal nerve and determining a second response of upper airway patency to the stimulation. In some such examples, the method comprises maintaining the stimulation of the hypoglossal nerve during stimulation of the non-hypoglossal nerve.

(683) As shown at **8520** in FIG. **51**A, in some examples a method comprises identifying a stimulation location by: first stimulating an ansa cervicalis-related nerve and determining a response of upper airway patency to the stimulation; and after the determination of the response, stimulating a hypoglossal nerve and determining a response of upper airway patency to the stimulation. In some such examples, comprising maintaining the stimulation of the ansa cervicalis-related nerve during stimulation of the hypoglossal nerve.

(684) In some examples, the test stimulation location corresponds to a first location not intended for implantation of a chronically implantable stimulation element. In some examples, the first location corresponds to an external location under the tongue.

(685) In some examples, the method comprises determining the response of the upper-airway-patency-related tissue to the stimulation. In some such examples, the method comprises implementing the determining via observing the response. Moreover, in some examples, the method comprises observing the response via imaging at least one of a patency of the upper airway and a response of the upper airway patency-related muscle, and then evaluating the relative patency and/or the response of the upper airway patency-related muscle. In some examples, such imaging may comprise fluoroscopy, such as biplane modalities.

(686) As shown at method **8520** in FIG. **51**B, in some examples, a method of performing test stimulation comprises the first stimulation element comprising a test needle and implementing the stimulation by percutaneously inserting the test needle into the ansa cervicalis nerve, and determining an upper airway response based on the stimulation. In some such examples of the method, determining the response comprises identifying inferior movement of the larynx and implementing the identifying, during the stimulation, via at least one of measuring an EMG

response of the larynx and imaging of the larynx.

(687) FIGS. **52**A and **52**B are diagrams including a front view and side view, respectively, schematically representing an example arrangement **9000** including an example device and/or example method to identify an upper airway collapse pattern. In one aspect, the example arrangement **9000** comprises a device which is externally, releasably mountable to a neck region **522** of a patient's body. As shown in FIGS. **52**A, **52**B, the example arrangement **9000** comprises an array **9020** of spaced apart electrodes **9022** to be located across and around the external surface of a front **525**A and sides **525**B of the patient's neck **522**. In some examples, the electrodes **9022** are arranged in rows **9021**, such that the array **9020** may form a grid (e.g. 4×4, 3×6, 4×5, and so on) via the spaced apart electrodes **9022**. The electrodes **9022** of the array **9020** are spaced apart by a distance suitable to provide the desired level of detail regarding the function, structure, collapse, opening, etc. of the patient's upper airway.

(688) As shown in FIG. **52**A-**52**C, a patient's body comprises a head-and-neck including neck region **522**. The neck region **522** comprises an upper airway **530** (FIGS. **52**A-**52**C) and a nasal airway passage **540** (FIG. **52**C). In some examples, the upper airway **530** comprises at least some of the detailed anatomical features illustrated in association with FIGS. **53**A-**53**F.

(689) As further shown in FIG. **52**A, the electrodes **9022** may be interconnected via wires **9024** or other electrically conductive elements. While not shown for illustrative simplicity, in some examples the example arrangement **9000** may comprise a material, such as a fabric, mesh, etc. which acts as a body to provide stability and ease of handling of the array **9020** of electrodes **9022**. In some examples, the electrodes **9020**, wires **9024**, and/or fabric comprises or carries a releasable adhesive for releasably securing the example arrangement **9000** to an external skin surface of the patient's neck **522**. In addition, or alternatively, straps or other wrapping/securing features may be provided to secure the device relative to the patient's neck **522**.

(690) In some examples, the array **9020** of electrodes **9022** is sized and shaped to cover an external portion of the patient's neck and a portion of the surrounding regions in order to provide sensing information regarding anatomical structures, such as muscles, bones, connective tissues, etc. which define the upper airway of the patient. In some examples, one method may comprise sensing an impedance between at least some of the example electrodes **9022**, which may be indicative of the opening, closing, collapse patterns, etc. of the patient's upper airway. In some such examples, this sensing may be performed while the patient is sleeping to identify their particular collapse pattern when the patient experiences sleep disordered breathing (SDB) events, such as obstructive sleep apnea events. In some such examples, the sensed impedance information may provide a type of collapse, a site of collapse, and/or a degree of collapse with details regarding the type, site and/or degree of collapse shown and described in association with at least FIGS. **53**A-**53**F.

(691) In some examples, sensing the impedance (and/or delivering stimulation in response thereto) regarding the collapse pattern may be performed in association with at least the example arrangements within FIGS. **38**A-**38**D, including impedance sensor **8752** (FIG. **38**C), collapse parameter **8710** (FIG. **38**B), disease burden parameter **8726** (FIG. **38**C), etc.

(692) In some examples, the device and/or method comprises performing such impedance sensing during application of a stimulation signal to the hypoglossal nerve and/or the ansa cervicalis-related nerve in order to determine how, when, and/or where the patient may respond to such stimulation in terms of the upper airway collapse pattern being affected by the stimulation. In some examples, such determinations may be used to identify which single portion or which multiple portions (and a pattern of stimulation among those portions) of the ansa cervicalis-related nerve should be stimulated for a particular patient, or some patients in general, in order to increase and/or maintain upper airway patency. Moreover, some examples methods also comprise performing the stimulation of the ansa cervicalis-related nerve without or with stimulation of the hypoglossal nerve, and vice versa, in order to better understand how combinations of stimulating different types of nerves (or different nerve branches) affect increasing and/or maintaining upper airway patency

during sleep to thereby treat sleep disordered breathing. One example arrangement for delivering stimulation in relation to determining a collapse pattern in relation to sensed impedance is described in association with at least FIGS. **38**A-**38**D.

(693) FIG. **52**C is a diagram including a side view schematically representing an example arrangement **9100** comprising an example device and/or example method for at least identifying an upper airway collapse pattern. In some examples, the example arrangement **9100** may be used to determine at least some of substantially the same information described above in association with at least FIGS. **52**A-**52**B regarding upper airway collapse patterns, causes, stimulation responses thereto, etc.

(694) As shown in FIG. **52**C, in some examples an example sensing device **9110** comprises a handle **9120** which supports an insertable, elongate member **9120**, which comprises a proximal portion **9122** and opposite distal portion **9122**. In some examples, the elongate member **9120** is sized and shaped, and is resiliently, flexible to facilitate some maneuverability and ease of insertion and passage, while exhibiting some rigidity to facilitate advancing the elongate member, which may comprise structural features analogous to tools used in intubation procedures.

(695) In some examples, the insertable elongate member **9120** comprises an array **9130** of spaced apart electrodes **9132** arranged on external surface of body **9134** of the member **9120**, with the array comprising a size and shape suitable to perform impedance sensing of the type, manner, and/or purposes described above with respect to at least FIGS. **52**A-**52**B.

(696) In some examples, the electrodes **9132** may be exposed at or mounted on the external surface of the body **9134** of elongate member **9120**. However, in some examples, the electrodes **9132** may be located below the external surface of the body **9134** but with the type, thickness, etc. of the external surface and/or electrodes **9132** suited to perform the desired impedance sensing.

(697) In some examples, the device **9110** may sometimes be referred to as an internally insertable, electrode impedance sensing array (or variations thereof).

(698) In some examples, the size, shape, and configuration of the device **9110** may be adapted for insertion (I**1**) into and through the mouth **526** of the patient, as shown in FIG. **52**C, to be advanced into the upper airway **530**. However, in some examples, the size, shape, and configuration of the device **9110** may be adapted to perform such sensing via insertion (I**2**) of the device into and through the nose **524** of the patient for passage through the nasal airway/passage **540** and into pertinent portions of the upper airway **530**. With regard to either a nasal entry approach or a mouth entry approach, at least some example devices and/or methods include placing at least some sensing electrodes in close proximity to the velopharnyx or at least within a distance by which the velopharnyx may be sensed, along with other anatomical features of the upper airway.

(699) In some examples, the example devices and/or example methods in FIGS. **52**A, **52**B, and **52**C may be utilized to identify an upper airway collapse pattern at a time prior to completing implantation of a chronically implantable stimulation element in stimulating relation to an upper airway patency-related tissue such as, but not limited to, a hypoglossal nerve and a non-hypoglossal nerve, which comprises the ansa cervicalis-related nerve and/or other nerves, muscles, etc.

(700) FIGS. **53**A-**53**C are a series of diagrams schematically representing at least some different upper airway collapse patterns, including an anterior-posterior collapse pattern (FIG. **53**A), a concentric collapse pattern (FIG. **53**B), and a lateral collapse pattern (FIG. **53**C). In addition to observing such collapse patterns and/or other collapse patterns, at least some aspects of such collapse patterns may be measured, such as via impedance sensing using implanted electrodes (e.g. sensing elements and/or stimulation elements), using externally applied arrays of electrodes, etc. such as described and illustrated in association with at least FIGS. **52**A-**52**C. By determining an upper airway collapse pattern, some example arrangements may determine whether to apply stimulation via a hypoglossal nerve, via an ansa cervicalis-related nerve (including which single or multiple portions thereof to stimulate), via other non-hypoglossal nerve related to upper airway

patency, and/or combinations of these nerves including unilateral and bilateral options.

(701) FIG. **53**E is a block diagram schematically representing an example sorting tool **9260** by which to sort and weigh a location, pattern, and degree of obstruction or patency. As shown in FIG. **53**E, obstruction sorting tool **400** includes functions for location detection **9262**, pattern detection **9270**, and degree detection **9280**. In general terms, the location detection function **9262** operates to identify a site along the upper airway at which an obstruction occurs and which is believed to cause sleep disordered breathing. In one example, the location detection function **9262** includes a velum (soft palate) parameter **9264**, an oropharynx-tongue base parameter **9266**, and an epiglottis/larynx parameter **9268**. Each respective parameter denotes an obstruction identified in the respective physiologic territories of the velum (soft palate), oropharynx-tongue base, and epiglottis which are generally illustrated for an example patient in FIG. **53**D. In one aspect, these distinct physiologic territories define an array of vertical strata within the upper airway. Moreover, each separate physiologic territory (e.g. vertical portion along the upper airway) exhibits a distinct characteristic behavior regarding obstructions and associated impact on breathing during sleep. Accordingly, each physiologic territory responds differently to implantable upper airway stimulation.

(702) With this in mind, the velum (soft palate parameter **9264** denotes obstructions taking place in the level of the region of the velum (soft palate), as illustrated in association with FIG. **53**D. FIG. **53**D is a diagram including a side view schematically representing at least some anatomical features of the upper airway, as well as different sites or levels at which obstruction may occur. By determining a site or location of upper airway collapse, some example arrangements may determine whether to apply stimulation via a hypoglossal nerve, via an ansa cervicalis-related nerve (including which portions thereof to stimulate), via other non-hypoglossal nerve related to upper airway patency, and/or combinations of these nerves including unilateral and bilateral options.

(703) As shown in FIG. **53**D, a diagram **9240** provides a side sectional view (cross hatching omitted for illustrative clarity) of a head and neck region **9242** of a patient. In particular, an upper airway portion **9250** extends from the mouth region **9244** to a neck portion **9254**. The upper airway portion **9250** includes a velum (soft palate) region **9260**, an oropharynx-tongue base region **9262**, and an epiglottis region **9264**. The velum (soft palate) region **9260** includes an area extending below sinus **9261**, and including the soft palate **9260**, approximately to the point at which tip **9248** of the soft palate **9246** meets a portion of tongue **9247** at the back of the mouth **9244**. The oropharynx-tongue base region **9262** extends approximately from the tip of the soft palate **9246** (when in a closed position) along the base **9252** of the tongue **9247** until reaching approximately the tip region of the epiglottis **9254**. The epiglottis-larynx region **9262** extends approximately from the tip of the epiglottis **9254** downwardly to a point above the esophagus **9257**.

(704) As will be understood from FIG. **53**D, each of these respective regions **9260**, **9262**, **9264** within the upper airway correspond the respective velum parameter **9264**, oropharynx-tongue base parameter **9266**, and epiglottis parameter **9268**, respectively of FIG. **53**E.

(705) With further reference to FIG. **53**E, in general terms the pattern detection function **9270** enables detecting and determining a particular pattern of an obstruction of the upper airway portion **9244**. In one example, the pattern detection function **9270** includes an antero-posterior parameter **9272**, a lateral parameter **9274**, a concentric parameter **9276**, and composite parameter **9278**.

(706) The antero-posterior parameter **9272** of pattern detection function **9270** (FIG. **53**E) denotes a collapse of the upper airway that occurs in the antero-posterior orientation, as further illustrated in the diagram **9210** of FIG. **53**A. In FIG. **53**A, arrows **9211** and **9212** indicate one example direction in which the tissue of the upper airway collapses, resulting in the narrowed air passage **9214**. FIG. **53**A is also illustrative of a collapse of the upper airway in the soft palate region **9260**, whether or not the collapse occurs in an antero-posterior orientation. For example, in some instances, the velum (soft palate) region **9260** exhibits a concentric (i.e. circular) pattern of collapse, as shown in diagram **9220** of FIG. **53**B.

(707) The concentric parameter **9276** of pattern detection function **9270** (FIG. **53**E) denotes a

collapse of the upper airway that occurs in a concentric orientation, as further illustrated in the diagram **9220** of FIG. **53**B. In FIG. **53**B, arrows **9222** indicate the direction in which the tissue of the upper airway collapses, resulting in the narrowed air passage **9224**.
(708) The lateral parameter **9230** of pattern detection function **9270** (FIG. **53**E) denotes a collapse of the upper airway that occurs in a lateral orientation, as further illustrated in the diagram **9230** of FIG. **53**C. In FIG. **53**C, arrows **9232** and **9233** indicate the direction in which the tissue of the upper airway collapses, resulting in the narrowed air passage **9235**.
(709) The composite parameter **9278** of pattern detection function **9270** (FIG. **53**E) denotes a collapse of the upper airway portion that occurs via a combination of the other mechanisms (lateral, concentric, antero-posterior) or that is otherwise ill-defined from a geometric viewpoint but that results in a functional obstruction of the upper airway portion.
(710) With further reference to obstruction sorting tool of FIG. **53**E, in general terms the degree detector module **9280** indicates a relative degree of collapse or obstruction of the upper airway portion. In one embodiment, degree detection function **9280** includes a none parameter **9282** a partial parameter **9284**, and a complete parameter **9285**. In some examples, the none parameter **9282** may correspond to a collapse of 25 percent or less, while the partial collapse parameter **9284** may correspond to a collapse of between about 25 to 75%, and the complete collapse parameter **9285** may correspond to a collapse of greater than 75 percent.
(711) It will be understood that various patterns of collapse occur at different levels of the upper airway portion and that the level of the upper airway in which a particular pattern of collapse appears can vary from patient-to-patient.
(712) In some embodiments, obstruction sorting tool **9260** comprises a weighting function **9286** and score function **9287**. In general terms, the weighting function **9286** assigns a weight to each of the location, pattern, and/or degree parameters (FIG. **53**E) as one or more those respective parameters can contribute more heavily to the patient exhibiting sleep disordered breathing or to being more responsive to implantable upper airway stimulation. More particularly, each respective parameter (e.g. antero-posterior **9272**, lateral **9274**, concentric **9276**, composite **9278**) of each respective detection modules (e.g. pattern detection function **9270**) is assigned a weight corresponding to whether or not the patient is eligible for receiving implantable upper airway stimulation. Accordingly, the presence of or lack of a particular pattern of obstruction (or location or degree) will be become part of an overall score (according to score parameter **9287**) for an obstruction vector indicative how likely the patient will respond to therapy via an implantable upper airway stimulation system.
(713) FIG. **53**F is diagram (e.g. chart) **9290** schematically representing an index or scoring tool to sort and weigh a location, pattern, and degree of obstruction or patency for a particular patient. Chart **9290** combines information regarding location (**9262** in FIG. **53**E), pattern (**9270** in FIG. **53**E), and degree (**9280** in FIG. **53**E) into a single informational grid or tool by which the obstruction is documented for a particular patient and by which appropriate stimulation settings may be determined and applied according to the various examples of the present disclosure, such as but not limited to those in association with at least FIG. **41**B, FIGS. **38**A-**38**C, etc.
(714) Accordingly, in some examples, the information sensed and collected via at least FIGS. **53**E-**53**F may be used to determine whether to apply stimulation via a hypoglossal nerve, via an ansa cervicalis-related nerve (including which single portion or multiple portions thereof to stimulate), via other non-hypoglossal nerves related to upper airway patency, and/or combinations of these nerves including unilateral and bilateral options.
(715) FIG. **54**A is a block diagram schematically representing an example care engine **10000**. In some examples, the SDB care engine **10000** may form part of a control portion **10500**, as later described in association with at least FIG. **54**B, such as but not limited to comprising at least part of the instructions **10511**. In some examples, the SDB care engine **10000** may be used to implement at least some of the various example devices and/or example methods of the present

disclosure as previously described in association with FIGS. **1-53**F and/or in later described examples devices and/or methods. In some such examples, the SDB care engine **10000** may comprise a sensing engine **10020** and/or a stimulation engine **10030**. In some examples, the sensing engine **10020** may be implemented via at least some of the sensing elements, sensor types/modalities, etc. as described in association with at least FIG. **3**C, FIGS. **38**A-**39**, and throughout FIGS. **40**A-**53**F. In some examples, the stimulation engine **10030** may be implemented via at least some of the stimulation elements (e.g. stimulation portions, electrode arrays, cuff electrodes, paddle electrodes, axial leads/electrodes, IPG **533**, microstimulators, etc.), example methods, example engines (e.g. **8800** in FIG. **38**D) described throughout the various examples of the present disclosure In some examples, the SDB care engine **10000** (FIG. **54**A) and/or control portion **10500** (FIG. **54**B) may form part of, and/or be in communication with, the example arrangements, stimulation elements, sensing elements, microstimulators, pulse generators, etc. such as a portion of the devices and methods described in association with at least FIGS. **1-53**F and/or the later described examples, such as FIGS. **54**B-**60**.
(716) It will be understood that various functions and parameters of SDB care engine **10000** may be operated interdependently and/or in coordination with each other, in at least some examples.
(717) FIG. **54**B is a block diagram schematically representing an example control portion **10500**. In some examples, control portion **10500** provides one example implementation of a control portion forming a part of, implementing, and/or generally managing the example arrangements, the stimulation elements, sensing elements, microstimulators, pulse generators, control portion, instructions, engines, functions, parameters, and/or methods, as described throughout examples of the present disclosure in association with FIGS. **1-53**F and later FIGS. In some examples, control portion **10500** includes a controller **10502** and a memory **10510**. In general terms, controller **10502** of control portion **10500** comprises at least one processor **10504** and associated memories. The controller **10502** is electrically couplable to, and in communication with, memory **10510** to generate control signals to direct operation of at least some of the example arrangements, stimulation elements, sensing elements, microstimulators, pulse generators, control portion, instructions, engines, functions, parameters, and/or methods, as described throughout examples of the present disclosure. In some examples, these generated control signals include, but are not limited to, employing instructions **10511** stored in memory **10510** to at least direct and manage sleep disordered breathing (SDB) care (e.g. sensing, stimulation, etc.) in the manner described in at least some examples of the present disclosure. In some instances, the controller **10502** or control portion **10500** may sometimes be referred to as being programmed to perform the above-identified actions, functions, etc.
(718) In response to or based upon commands received via a user interface (e.g. user interface **10520** in FIG. **54**C) and/or via machine readable instructions, controller **10502** generates control signals as described above in accordance with at least some of the examples of the present disclosure. In some examples, controller **10502** is embodied in a general purpose computing device while in some examples, controller **10502** is incorporated into or associated with at least some of the example arrangements, stimulation elements, sensing elements, microstimulators, pulse generators, control portion, instructions, engines, functions, parameters, and/or methods, etc. as described throughout examples of the present disclosure.
(719) For purposes of this application, in reference to the controller **10502**, the term “processor” shall mean a presently developed or future developed processor (or processing resources) that executes machine readable instructions contained in a memory or that includes circuitry to perform computations. In some examples, execution of the machine readable instructions, such as those provided via memory **10510** of control portion **10500** cause the processor to perform the above-identified actions, such as operating controller **10502** to implement sleep disordered breathing (SDB) care (e.g. stimulation, sensing, etc.) via the various example implementations as generally described in (or consistent with) at least some examples of the present disclosure. The machine

readable instructions may be loaded in a random access memory (RAM) for execution by the processor from their stored location in a read only memory (ROM), a mass storage device, or some other persistent storage (e.g., non-transitory tangible medium or non-volatile tangible medium), as represented by memory **10510**. The machine readable instructions may include a sequence of instructions, a processor-executable machine learning model, or the like. In some examples, memory **10510** comprises a computer readable tangible medium providing non-volatile storage of the machine readable instructions executable by a process of controller **10502**. In some examples, the computer readable tangible medium may sometimes be referred to as, and/or comprise at least a portion of, a computer program product. In other examples, hard wired circuitry may be used in place of or in combination with machine readable instructions to implement the functions described. For example, controller **10502** may be embodied as part of at least one application-specific integrated circuit (ASIC), at least one field-programmable gate array (FPGA), and/or the like. In at least some examples, the controller **10502** is not limited to any specific combination of hardware circuitry and machine readable instructions, nor limited to any particular source for the machine readable instructions executed by the controller **10502**.

(720) In some examples, control portion **10500** may be entirely implemented within or by a stand-alone device.

(721) In some examples, the control portion **10500** may be partially implemented in one of the example arrangements, stimulation elements, sensing elements, microstimulators, pulse generators, etc. and partially implemented in a computing resource separate from, and independent of, the example arrangements, stimulation elements, sensing elements, microstimulators, pulse generators, etc. but in communication with such example arrangements, etc. For instance, in some examples control portion **10500** may be implemented via a server accessible via the cloud and/or other network pathways. In some examples, the control portion **10500** may be distributed or apportioned among multiple devices or resources such as among a server, an example arrangement, and/or a user interface.

(722) FIG. **54**C is a diagram schematically illustrating at least some example arrangements of a control portion **10528** by which the control portion **10500** (FIG. **54**B) can be implemented, according to one example of the present disclosure. In some examples, control portion **10528** is entirely implemented within or by an IPG assembly **10529**, which has at least some of substantially the same features and attributes as a pulse generator (e.g. IPG **533**, microstimulator, etc.) as previously described throughout the present disclosure. In some examples, control portion **10528** is entirely implemented within or by a remote control **10530** (e.g. a programmer) external to the patient's body, such as a patient control **10532** and/or a physician control **10534**. Patient control **572** in FIG. **3**C may comprise one example implementation of the remote control **10532**. In some examples, the control portion **10500** is partially implemented in the IPG assembly **10529** and partially implemented in the remote control **1530** (at least one of patient control **10532** and physician control **10534**).

(723) In some examples, control portion **10500** includes, and/or is in communication with, a user interface **10520** as shown in FIG. **54**D. In some examples, user interface **10520** forms part or and/or is accessible via a device external to the patient and by which the therapy system may be at least partially controlled and/or monitored. The external device which hosts user interface **10520** may be a patient remote (e.g. **10532** in FIG. **54**C), a physician remote (e.g. **10534** in FIG. **54**C) and/or a clinician portal. In some examples, user interface **10520** comprises a user interface or other display that provides for the simultaneous display, activation, and/or operation of at least some of the example arrangements, stimulation elements, sensing elements, microstimulators, pulse generators, control portion, instructions, engines, functions, parameters, and/or methods, etc., as described in association with FIGS. **1-53**F and/or later described FIGS. In some examples, at least some portions or aspects of the user interface **10520** are provided via a graphical user interface (GUI), and may comprise a display **10524** and input **10522**.

(724) FIG. **54**E is a block diagram **10600** which schematically represents some example implementations by which an implantable device (IMD) **10610** (e.g. IPG **533**, sensors, microstimulators, etc.) may communicate wirelessly with external devices outside the patient. As shown in FIG. **54**E, in some examples, the IMD **10610** may communicate with at least one of patient app **10630** on a mobile device **10620**, a patient remote control **10640**, a clinician programmer **10650**, and a patient management tool **10660**. Patient control **572** in FIG. **3**C may comprise one example implementation of the patient remote control **10640**. The patient management tool **10660** may be implemented via a cloud-based portal **10662**, the patient app **10630**, and/or the patient remote control **10640**. Among other types of data, these communication arrangements enable the IMD **10610** to communicate, display, manage, etc. patient therapy information as well as to allow for adjustment to the various elements, portions, etc. of the example devices and methods if and where needed.
(725) FIGS. **55-59**B are a series of diagrams including views which schematically represent several ways in which stimulation elements may be implanted to enable stimulation of the ansa cervicalis-related nerve and/or a phrenic nerve, such as in situations in which a patient may experience multiple type sleep apnea, including obstructive and central sleep aspects.
(726) Moreover, stimulation via such example arrangements involving the phrenic nerve and/or ansa cervicalis-related nerve may be implemented without stimulation of the hypoglossal nerve in some examples. However, in some examples, the hypoglossal nerve may be stimulated in coordination with stimulation of the phrenic nerve and/or ansa cervicalis-related nerve in order to provide therapy for multiple-type sleep apnea. By providing the option of stimulating the hypoglossal nerve, the ansa cervicalis-related nerve, and/or the phrenic nerve, a greater range of therapeutic stimulation protocols are available to better tailor the stimulation therapy to the particular anatomical features and/or other physiologic traits of a particular patient.
(727) Among other aspects, pacing stimulation of the phrenic nerve also may be used to prevent hyperventilation, which may be implemented as a standalone method or implemented in a complementary manner in association with other example arrangements of the present disclosure to treat sleep disordered breathing (e.g. OSA, CSA, multiple type sleep apnea).
(728) In some example implementations, stimulation of the phrenic nerve, with or without stimulation of the ansa cervicalis-related nerve, may be implemented via at least some of substantially the same features and attributes as described in U.S. Patent Publication 2020/0147376, published on May 14, 2020, and entitled Multiple Type Sleep Apnea, and which is hereby incorporated by reference in its entirety.
(729) With the foregoing in mind, among other potential stimulation sites, FIG. **55** is a diagram including a front view schematically representing patient anatomy **4251** and example arrangement **11000**, which illustrates at least three different general stimulation sites A, B, C, D, by which the phrenic nerve may be stimulated with at least sites A and B providing locations at which a single stimulation arrangement may be used to stimulate the phrenic nerve and/or the ansa cervicalis-related nerve.
(730) Moreover, in some examples in which the phrenic nerve and/or ansa cervicalis-related nerve are to be stimulated, stimulation also may be applied in a complementary manner to the hypoglossal nerve according to at least some of the various examples throughout the present disclosure. In some examples, applying stimulation to the phrenic nerve, ansa cervicalis-related nerve, and hypoglossal nerve may be implemented via the stimulation protocol **6320** in FIG. **37**D. Moreover, at least some of the example arrangement in FIGS. **55-59**B may be used to implement the example method **8400** in FIG. **43** regarding sensing and/or stimulation including the phrenic nerve, to implement at least some features and attributes of the example arrangements in FIGS. **38**A-**51**B regarding sensing and/or stimulation in relation to a phrenic nerve, ansa cervicalis-related nerve, and/or hypoglossal nerve.
(731) As shown in FIG. **55**, the patient anatomy **4251** may comprise an internal jugular vein **4250**

and a subclavian vein **4262**, which branch off an innominate vein **4275**, as well as an external jugular vein **4257**. Patient anatomy **4251** further comprises an ansa cervicalis-related nerve **316** (e.g. FIG. **2**, **16**, etc.), with just some portions identified for illustrative simplicity, such as loop **319**, portion **327**, and portion **342** which innervates at least the sternothyroid muscle(s). As shown in FIG. **55**, at least a portion of loop **319** of the ansa cervicalis-related nerve **316** is draped along/across the internal jugular vein **4250**. Potential stimulation site B is located at the upper portion (e.g. superior) of the loop of the ansa cervicalis-related nerve **316**, which includes portions which innervate the sternothyroid muscles and/or sternohyoid muscles, among other muscles. Meanwhile, potential stimulation site A may be located a lower portion (e.g. inferior) of the loop **319** of the ansa cervicalis-related nerve **316**, which includes the portion **342** innervating at least the sternothyroid muscles.

(732) FIG. **55** also shows cranial nerves C**2**, C**3**, C**4**, C**5**, with at least C**2**, C**3** being nerves from which the ansa cervicalis-related nerve **316** originates.

(733) Moreover, the patient anatomy **4251** depicted in FIG. **55** comprises an array **4270** (i.e. phrenic array) of phrenic nerve portions **4271** (shown in multiple dashed lines), at least some of which correspond to accessory phrenic nerve portions and/or secondary phrenic nerve portions joined to a principal phrenic nerve (not shown). As noted above, electrical stimulation of the phrenic nerve (including phrenic nerve portions) may cause contraction of the diaphragm, which may be used to treat central sleep apnea and/or treat multiple type apnea (including aspects of central sleep apnea).

(734) FIG. **56** is a diagram like FIG. **55** while further schematically representing an example arrangement **11050** including a paddle electrode **11100** implanted in a position juxtaposed relative to, and in stimulating relation to, portions of the ansa cervicalis-related nerve **316** and phrenic nerve portions **4271** (shown as dashed lines). As shown in FIG. **56**, the paddle electrode **11100** comprises a carrier body **11102** which supports an array **11110** of spaced apart electrode contacts **11112**. The array **11110** may comprise a two-dimensional array (e.g. multiple rows and columns) as shown, may comprise a single row of electrode contacts. The electrode array **11110** extends between first and second opposite ends **11105**, **11107** of the carrier body **11102**, and between opposite sides **11109**A, **11109**B.

(735) In some examples, the carrier body **11102** may comprise at least some holes **11116** at the ends **11105**, **11107** and/or sides **11109**A, **11109**B to enable tissue growth and/or suturing to secure the carrier body **11102** relative to surrounding non-nerve tissues to anchor the paddle electrode **11100** securely in stimulating relation to the respective target nerves. In some examples, various tines, barbs, and/or similar elements may be provided on the carrier body **11100** to anchor it in place.

(736) In some examples, the carrier body and electrode array **11110** are sized and shaped to be co-located with multiple nerves, such as portions (e.g. **342**) of the ansa cervicalis-related nerve **316**, phrenic nerve portions **4271**, etc., while accommodating anatomy variation among different patients. Via the electrode array **11110** of paddle electrode **11100**, stimulation may be applied to solely to the ansa cervicalis-relate nerve **316**, solely to the phrenic nerve (e.g. **4271**), or to both of these respective nerves. When applied to both nerves, the stimulation may be applied simultaneously, alternately, in a staggered manner, etc.

(737) FIG. **57** is a diagram including patient anatomy **4251** like FIGS. **55-56** while schematically representing an example arrangement **11200** including a stimulating device **11210** including cuff electrodes **11220**A, **11220**B supported on respective distal lead portions **11224**A, **112246** bifurcated, via junction **11225**, from main lead portion **11227**. In some examples, the stimulation device **11210** may comprise one or more anchor elements **11226** which may be provided at each distal lead portion **11124**A, **112246** as shown in FIG. **57** and/or at main lead portion **11226** for robustly securing the at respective lead portions **11124**A, **111246**, cuff electrodes **11220**A, **11220**B, etc. relative to non-nerve surrounding tissues to maintain the cuff electrodes **11220**A, **112206** in stimulating relation to the target nerves.

(738) As further shown in FIG. **57**, in some examples one cuff electrode **11220**A may be chronically implanted relative to a portion, such as but not limited to portion **342**, of the ansa cervicalis-related nerve **316** while one cuff electrode **11220**B may be chronically implanted relative to one of the phrenic nerve portions generally represented via the dashed lines **4271**. In FIG. **57**, one of the phrenic nerve portions is shown as a solid line **4272** to highlight its being engaged via cuff electrode **11220**B.
(739) The cuff electrode **11220**A, **112206** may comprise a wide variety of cuff designs. In some such examples, the cuff electrodes may comprise at least some of substantially the same features as described for cuff electrodes shown in at least FIGS. **18-20**.
(740) In some examples, the cuff electrodes **11220**A, **11220**B (and associated lead portions) may be delivered via at least some of the paths, access incision, tools, etc. or analogous paths, access incisions, tools, etc. as described in the various examples of the present disclosure, such as but not limited to FIGS. **1-32**D.
(741) FIG. **58** is a diagram including patient anatomy **4251** like FIGS. **55-56** while schematically representing an example arrangement **11300** including a stimulation device **11310** including stimulation portions **11220**A, **11220**B (in an axial lead configuration) supported on respective distal lead portions **11318**A, **11318**B bifurcated from main lead portion **11320**.
(742) As further shown in FIG. **58**, in some examples one stimulation portion **11314**A may be chronically subcutaneously implanted relative to a portion, such as but not limited to portion **342**, of the ansa cervicalis-related nerve **316** while one stimulation portion **11314**B may be chronically subcutaneously implanted relative to one of the phrenic nerve portions generally represented via the dashed lines **4271**. The stimulation portions **11314**A, **11314**B may comprise a wide variety of electrode arrangements in which an array of spaced apart electrode contacts **11316** are positioned on a carrier **11317** to position the multiple electrode contacts in close proximity to the target stimulation locations of a nerve, as shown in FIG. **58**. In a manner similar to other axial-style stimulation portions (e.g. linear array of spaced apart electrode contacts) described throughout examples of the present disclosure, different combinations of the multiple electrode contacts **11316** of one or both stimulation portions **11314**A, **11314**B may be used to stimulate the respective ansa cervicalis-related nerve **316** and phrenic nerve portions **4271**. The different combinations of electrode contacts on each respective stimulation portion may be used to optimize nerve capture during or after final positioning of the respective stimulation portions relative to the respective nerves, or may be used to implement selective stimulation as desired.
(743) In some such examples, the stimulation portions **11314**A, **11314**B may comprise at least some of substantially the same features as described for the various example axial-style (e.g. linear electrode array) stimulation portions and lead, including anchoring, delivery, etc. in association with at least FIGS. **1-32**D. In some examples, the stimulation portions **11314**AA, **11314**B (and associated lead portions) may be delivered via at least some of the paths, access incision, tools, etc. or analogous paths, access incisions, tools, etc. as described in the various examples of the present disclosure, such as but not limited to FIGS. **1-32**D. Among the various types of securing elements, some example implementations may comprise sutures, tines, barbs, holes to induce tissue growth, and the like.
(744) FIG. **59**A is a diagram including patient anatomy **4251** like FIGS. **55-56** while schematically representing an example arrangement **11400** including a stimulation device **11410** including stimulation portions **11414**A, **11414**B (in an axial lead configuration) supported on respective leads **11418**A, **11418**B. While FIG. **58** illustrates separate leads **11418**A, **11418**B, it will be understood that in some examples, these leads may comprise bifurcated lead portions extending from a common, main lead portion in a manner similar to other such bifurcated leads disclosed in various examples of the present disclosure.
(745) As shown in FIG. **59**A, each respective stimulation portion **11414**A, **11414**B is chronically implanted via intravascular delivery. In some examples, the stimulation portion **11414**A is

advanced within and through the vasculature to extend within and through the internal jugular vein **4250** to be in transvenous stimulating relation to portions of the ansa cervicalis-related nerve **316**, while the stimulation portion **11414**B is advanced within and through the vasculature to extend within and through the subclavian vein **4262** to be in transvenous stimulating relation to at least some phrenic nerve portions **4271**.
(746) It will be understood that in some examples, stimulation portion **11414**B may be advanced further to extend within and through a portion of the external jugular vein **4257** to place the stimulation portion **11414**B in transvenous stimulating relation to phrenic nerve portions **4271** (shown in dashed lines) within proximity of the external jugular vein **4257**.
(747) The stimulation portions **11414**A, **11414**B in FIG. **59**A may comprise at least some of substantially the same features as the subcutaneously delivered, stimulation portions **11314**A, **11314**B in FIG. **58**, except being delivered intravascularly for transvenous stimulation.
(748) In some such examples, the stimulation portions **11414**A, **11414**B may comprise at least some of substantially the same features as described for the various example transvenously delivered, axial-style (e.g. linear electrode array) stimulation portions and lead, including anchoring, delivery, etc. in association with at least FIGS. **1-32**D. In some examples, the stimulation portions **11414**AA, **11414**B (and associated lead portions) may be delivered via at least some of the paths, access incision, tools, etc. or analogous paths, access incisions, tools, etc. as described in the various examples of the present disclosure, such as but not limited to FIGS. **1-32**D.
(749) However, in some examples such as at least the examples in FIGS. **59**C-**59**F, the stimulation portion (e.g. array of electrodes) may comprise a more flexible, resilient structure, which functions in part, as a retention element for robustly securing the stimulation portions within the vasculature. In some such examples, the body of the lead and/or carrier (in the region supporting the electrodes of the stimulation portion) may be pre-formed in a shape, size, and/or orientation adapted to promote anchoring or fixation of the stimulation relative to the pertinent anatomical features in which the stimulation portion is to become secured. In some examples, the pre-formed shape may be implemented via a shape memory material. Via such arrangements, because of its flexible resilience, such a stimulation portion may be manipulated from its original shape in order to introduce and advance the stimulation portion and lead within the vasculature (or within a subcutaneous access, delivery path), with the stimulation portion being biased to return as close as possible to its original shape, which in turn helps to secure the stimulation portion in a desired location. In some examples, these above-noted size, shape, and/or orientation features may be implemented in the example arrangement of **11600** of FIG. **59**B.
(750) FIG. **59**B is a diagram including patient anatomy **4251** like FIGS. **55-56** while schematically representing an example arrangement **11600** including a stimulation device **11610** including stimulation portions **11615**A, **11615**B supported on respective lead portions **11618**A, **11618**B. In some examples, the lead portions **11618**A, **11618**B may be completely independent physically, while in some examples the respective lead portions **11618**A, **11618**B may be bifurcated at some point proximally of the stimulation portions **11615**A, **11615**B. Each stimulation portion **11615**A, **11615**B comprises a distal lead portion **11617** (extending from main lead portion **11618**A, **11618**B) on which an array of spaced apart electrodes **11416** is supported. In a manner similar to the example arrangement in FIG. **59**A, the respective stimulation portions **11615**A, **11615**B (and associated supporting leads) are advanced within and through the vasculature to position the stimulation portions **11615**A, **11615**B in transvenous stimulating relation to the respective nerves, such as portions of the ansa cervicalis-related nerve **316** and phrenic nerve portions **4271**. In these positions, the stimulation portions **11615**A, **11615**B may comprise at least some of substantially the same features and attributes for positioning, delivering stimulation therapy, etc. as the stimulation portions in FIG. **59**A, except for the differences noted below.
(751) While the main lead portion **11618**A, **11618**B (for respective stimulation portions **11615**A, **11615**B in FIG. **59**B) may be formed of a flexible, resilient material to implement expected

functions of an implantable medical lead, the distal lead portions **11617** carrying the electrodes **11416** may have a higher degree of flexibility and/or degree of configurability, while still retaining their resilience (e.g. biased to maintain and/or return to shape), in order to permit the electrodes **11416** and distal lead portion **11417** to be manipulated into a position and shape within the vasculature to help secure the stimulation portion **11615**A, **11615**B as desired. In some examples, the stimulation device(s) **11610** may comprise an anchor structure **11619** mounted or formed at an utmost distal end of the stimulation portion **11615**A, **11615**B. It will be understood that, in some examples, at least some features of the anchor structure **11619** may be implemented at locations along the lead **11618**A, **11618**B other than the utmost distal end.

(752) In some examples, the anchor structure **11619** may comprise an array of anchor elements (e.g. tines, barbed elements, etc.) which are flexible and resilient, with such elements sized, shaped, oriented, and/or positioned to frictionally engage non-nerve tissues, such as a sidewall of the vasculature through which the stimulation portion is being advanced and positioned. As shown in FIG. **59**B, the anchor elements of structure **11619** are oriented to permit forward movement (advancing) the stimulation portion **11615**A, **11615**B within and through the vasculature while preventing or hindering movement of the stimulation portion in the opposite direction. In some examples, the anchor structure **11619** may comprise at least some of substantially the same features and attributes as described in association with at least FIGS. **30**A-**31**G and/or other applicable examples of the present disclosure. Among those features, in some examples, some anchor elements can extend in an opposite direction from other anchor elements to prevent ratcheting, other undesired migration, etc.

(753) It will be understood that the at least some of the features and attributes of the retention-style stimulation portions shown and described in FIGS. **59**B-**59**F may be implemented in other various examples of the present disclosure for transvenous stimulation (via intravascular access/delivery) of nerve targets other than the phrenic nerve and/or ansa cervicalis-related nerve **316**, and/or may be implemented in non-transvenous (e.g. subcutaneous, other) stimulation examples, forms of access, delivery, etc. Moreover, as applicable for at least some of the examples of the present disclosure, the features and attributes for transvenous delivery and retention of FIGS. **59**A-**59**F (or non-transvenous) may be applied and implemented for nerve targets (e.g. pelvic, pudendal, cardiac etc.) other than those for treating sleep disordered breathing, and therefore may be applied in regions of the body (e.g. pelvic, other) other than the head-and-neck region and/or pectoral region.

(754) FIG. **59**C is a diagram schematically representing an example stimulation portion **11725**A comprising at least some of substantially the same features and attributes as the stimulation portions **11615**A, **11615**B in FIG. **59**B, except comprising more particular retention features as further described below. Stimulation portion **11725**A of FIG. **59**C may comprise one example implementation of the stimulation portions **11615**A, **11615**B of FIG. **59**B. As shown in FIG. **59**C, the stimulation portion **11725**A comprises a main lead portion **11730** and distal lead portion **11732** supporting an array of spaced apart electrodes **11716**, and anchor structure **11619**. The distal lead portion **11732** is made of a flexible, resilient material, which is pre-formed in a generally sinusoidal or sigmoid shape, which may act to help retain or anchor the stimulation portion **11725**A when manipulated to fit within in a desired intravascular location in close proximity to a transvenous target stimulation location. In addition, this configuration also may provide strain relief for the electrodes, lead, etc. In some examples, the distal lead portion **11732** may comprise shape patterns other than sigmoid or sinusoidal while providing a variable length or variable shape feature.

(755) FIG. **59**D is a diagram schematically representing an example stimulation portion **11735**A comprising at least some of substantially the same features and attributes as the stimulation portions **11615**A, **11615**B in FIG. **59**B, except comprising more particular retention features as further described below. Stimulation portion **11735**A of FIG. **59**D may comprise one example implementation of the stimulation portions **11615**A, **11615**B of FIG. **59**B. As shown in FIG. **59**D, the stimulation portion **11735**A comprises a main lead portion **11740** and distal lead portion **11738**

supporting an array of spaced apart electrodes **11716**, and anchor structure **11619**. The distal lead portion **11738** is made of a flexible, resilient material, which is pre-formed in a generally arcuate shape of a single curvature (e.g. J-shape), which may act to help retain or anchor the stimulation portion **11735**A when manipulated to fit within in a desired intravascular location in close proximity to a transvenous target stimulation location. In some such examples, the particular single curvature shape may be formed to correspond to the anatomical shape (e.g. an arch) in which it will implanted or formed to be slightly incongruent to the anatomical shape (e.g. an arch) in which it will implanted so as to accentuate it retention ability.
(756) FIG. **59**E is a diagram schematically representing an example stimulation portion **11745**A comprising at least some of substantially the same features and attributes as the stimulation portions **11615**A, **11615**B in FIG. **59**B, except comprising more particular retention features as further described below. Stimulation portion **11745**A of FIG. **59**E may comprise one example implementation of the stimulation portions **11615**A, **11615**B of FIG. **59**B. As shown in FIG. **59**E, the stimulation portion **11745**A comprises a main lead portion **11742** and distal lead portion **11748** supporting an array of spaced apart electrodes **11716**, and anchor structure **11619**. It will be understood that the shape of the electrodes **11716** shown in FIGS. **59**E-**59**F is provided for illustrative purposes/simplicity, and that in some examples electrodes **11716** may comprise a more rounded shape than shown in FIGS. **59**E-**59**F.
(757) With further reference to FIG. **59**E, the distal lead portion **11748** is made of a flexible, resilient material, which is pre-formed in a generally helical or spiral shape, which may act to help retain or anchor the stimulation portion **11745**A when manipulated to fit within in a desired intravascular location in close proximity to a transvenous target stimulation location. In some such examples, the particular tightness (or looseness) of the helical pattern may be formed to complement the anatomical shape in which it will implanted or formed to be slightly incongruent relative to the anatomical shape in which it will implanted so as to accentuate it retention ability. In some examples, a diameter (D**20**) of the helical pattern may generally correspond to a diameter (or greatest cross-sectional dimension) of the vasculature (V) in which the stimulation portion **11745**A will be implanted, as further shown in the sectional view of FIG. **59**F taken along lines **59**F-**59**F in FIG. **59**E. In some examples, the diameter of the helical shaped stimulation portion **11745**A prior to implantation may be larger or smaller than the diameter of the vasculature (V) depending on the particular strategic goals for implantation and retention.
(758) FIG. **60** is a diagram schematically representing an example arrangement (e.g. example method and/or example device) providing (e.g. via implanting stimulation elements within the patient's body) for the combination of stimulating the glossopharyngeal nerve and of stimulating the internal branch of the superior laryngeal nerve, such as via intravascular access by the superior laryngeal vein, i.e. the internal branch of the superior thyroid vein. In some examples, the example arrangement **11500** in FIG. **60** may comprise one example implementation of the at least some aspects of the method **8284** described in association with at least FIG. **41**B and/or at least the example arrangement in FIGS. **32**A-**32**B to deliver stimulation therapy for treating sleep disordered breathing. It will be understood that the example arrangement **11500** is FIG. **60** is also illustrative for, and may be applicable, to examples of the present disclosure which identify target nerves other than the hypoglossal nerve and/or ansa cervicalis-related nerve.
(759) As shown in FIG. **60**, in one aspect the example arrangement may comprise a stimulation device **11510** including a stimulation portion **11514**A supported on a distal lead portion **11522** and more proximal lead portion **11520**. In some examples, the stimulation portion **11514**A may comprise at least some of substantially the same features as stimulation portion **11414**A (or **11414**B) in FIG. **59**A, which may comprise at least some of substantially the same features and attributes as the axial-style stimulation portions (and associated leads) described throughout various examples of the present disclosure. As shown in FIG. **60**, the stimulation portion **11514**A may be advanced within and through the vasculature, such as within and through the internal jugular vein

4250, within and through a portion of the superior thyroid vein **4253**A, and then within and through the superior laryngeal vein **4253**B to place the stimulation portion **11514**A into transvenous stimulating relation to an internal branch **4253**C of a superior laryngeal nerve. It will be understood that in some examples, this method of implantation and stimulation of the internal branch **4253**C of the superior laryngeal nerve may be performed, regardless of whether a stimulation portion is provided for stimulating the glossopharyngeal nerve **4253**D. Conversely, in some examples, a stimulation portion **11514**B is provided solely for the glossopharyngeal nerve **4253**D without providing one for the internal branch **4253**C of the superior laryngeal nerve.
(760) As further shown in FIG. **60**, in some examples the example arrangement **11500** may comprise chronically implanting a stimulation portion **11514**B in stimulating relation to the glossopharyngeal nerve **4253**D. In some examples, the stimulation portion **11514**B may be delivered intravascularly for transvenous stimulation of the glossopharyngeal nerve **4253**D or may be delivered via other means (e.g. subcutaneously).
(761) Instead of placing a stimulation portion **11514**A with a lead portion (e.g. **11522**, **11520**) intravascularly as shown in FIG. **60**, in some examples a microstimulator may be placed intravascularly (whether supported on a lead or independent) or subcutaneously in a method like that previously described in association with at least the example arrangement in FIGS. **32**A-**32**B. Similarly, stimulation portion **11514**B (for stimulating the glossopharyngeal nerve **4253**D) also may be implemented as a microstimulator, whether delivered intravascularly for transvenous stimulation or delivered subcutaneously or other means.
(762) Although specific examples have been illustrated and described herein, a variety of alternate and/or equivalent implementations may be substituted for the specific examples shown and described without departing from the scope of the present disclosure. This application is intended to cover any adaptations or variations of the specific examples discussed herein.

Claims

1. A method of therapy for sleep disordered breathing therapy, comprising: stimulating, via at least one stimulation element, at least one upper airway patency-related tissue on a first side of the patient's body, which tissue comprises at least one of a hypoglossal nerve or an ansa cervicalis-related nerve, wherein the at least one stimulation element comprises: a first stimulation element to be in stimulating relation to the hypoglossal nerve; and a second stimulation element to be in stimulating relation to the and cervicalis-related nerve; and performing the stimulating via the second stimulation element to stimulate the ansa cervicalis-related nerve upon a determination of a patient exhibiting symptomatic residual level of apnea-hypopnea index (AHI) despite the stimulation via the first stimulation element of the hypoglossal nerve.
2. The method of claim 1, comprising: positioning the second stimulation element at the ansa cervicalis-related nerve to cause contraction of at least a sternothyroid muscle.
3. The method of claim 1, wherein the second stimulation element comprises a second microstimulator.
4. The method of claim 3, comprising: implanting the second stimulation element at the ansa cervicalis-related nerve in a separate implant procedure after a time period following implantation of the first stimulation element.
5. The method of claim 3, wherein the first stimulation element comprises a first microstimulator.
6. The method of claim 5, comprising: implementing communication between the respective first and second microstimulators to coordinate stimulation among the respective first and second microstimulators.
7. The method of claim 1, comprising at least one of: positioning the first stimulation element on a first side of the patient's body and positioning the second stimulation element on an opposite second side of the patient's body; or positioning the first stimulation element on a first side of the

patient's body and positioning the second stimulation element on the same first side of the patient's body spaced apart from the first stimulation element.

8. The method of claim 1, comprising: coordinating stimulation via the respective first and second stimulation elements with each other.

9. The method of claim 1, comprising: implementing stimulation via the respective first and second stimulation elements at least one of: applying the stimulation synchronized with sensed respiratory phase information; or applying the stimulation without synchronization to respiratory information.

10. The method of claim 1, comprising: implementing the stimulation, via the hypoglossal nerve and via at least one infrahyoid strap muscle innervated by the ansa cervicalis-related nerve, based on determination of a first parameter, the first parameter comprising at least one of: a respiratory parameter including respiratory phase information; a patient comfort parameter; a posture parameter; an effectiveness of therapy parameter; a device usage parameter; a sleep stage parameter; an upper airway collapse pattern parameter; or an apnea-hypopnea index (AHI) parameter.

11. The method of claim 1, comprising: receiving a patient adjustment, as a single input, to the stimulation; and implementing the patient adjustment by automatically adjusting a stimulation energy among a plurality of stimulation sites including at least one of the hypoglossal nerve, the ansa cervicalis-related nerve, and a second non-hypoglossal nerve.

12. The method of claim 1, comprising: anchoring the second stimulation element relative to the ansa cervicalis-related nerve via at least one anchor second relative to at least one of: an omohyoid tendon; a digastric tendon; a trachea; a sternum; a hyoid bone; a clavicle; or a thyroid-related tissue.

13. The method of claim 1, comprising: sensing, via at least one sensing element, at least one sleep disordered breathing-related parameter.

14. The method of claim 13, wherein the at least one sleep-disordered-breathing-related parameter comprises at least one of a respiratory phase parameter, an apnea-hypopnea index (AHI), a patient comfort parameter, an arousal index, a patient sleeping position, or an upper airway collapse pattern, and wherein the stimulating comprises modulating the stimulation based on at least one of the respiratory phase parameter, the apnea-hypopnea index (AHI), the patient comfort parameter, the arousal index, the patient sleeping position, or the upper airway collapse pattern.

15. The method of claim 14, comprising: implementing the modulating via automatically titrating a stimulation parameter based on determining, via the sensing, of at least one of the respiratory phase parameter, the apnea-hypopnea index (AHI), the patient comfort parameter, the arousal index, the patient sleeping position, or the upper airway collapse pattern.

16. The method of claim 14, comprising: automatically adjusting, upon sensing a change in the AHI, at least one of a stimulation parameter and a target stimulation location at the upper airway patency-related tissue, wherein the target stimulation location comprises at least one of the hypoglossal nerve and the ansa cervicalis-related nerve.

17. The method of claim 14, comprising: implementing, via an accelerometer, the determination of at least one of: the respiratory phase parameter; a posture parameter; the patient sleeping position; an effectiveness of therapy parameter; the arousal index; the apnea-hypopnea index (AHI) parameter; or a sleep stage parameter.

18. The method of claim 1, comprising: implanting the second stimulation element upon a determination of a patient exhibiting symptomatic residual AHI despite the stimulation via the first stimulation element of the hypoglossal nerve.